



FOURTH EDITION

LIGHT & HEAVY

VEHICLE
TECHNOLOGY

M J NUNNEY

Light and Heavy
Vehicle Technology

*When theory and practice do not agree, you should examine the facts
to see what is wrong with the theory*

Charles F. Kettering, General Motors (1934)

Light and Heavy Vehicle Technology

Fourth edition

M.J. Nunney

CGIA, MSAE, MIMI



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Preface

The purpose of this new fourth edition of *Light and Heavy Vehicle Technology* remains one of providing readily accessible information, which bridges the gap between the purely basic and the more advanced treatments of the subject. By understanding the reasons behind the design, construction and operation of the many and varied components of modern motor vehicles, the technician should be better equipped to deal with their servicing and overhaul. Some references to past automotive practice have been retained, not only because a technician may still be required to test and repair older vehicles, but also to provide a convenient transition to later practice.

Two entirely new sections of the book provide a topical introduction to alternative power sources and fuels, and battery-electric, hybrid and fuel-cell vehicles. Also, the number of entries in the list of automotive technical abbreviations has now increased to over 200. Finally, as in previous editions of the book, the tradition of including brief historical notes on the development of modern automotive concepts has been continued.

M.J. Nunney

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My thanks to the Rolls-Royce Heritage Trust for their permission to use the Dr Stanley Hooker quote on page 6 General note – the ever-increasing sophistication in the design and construction of modern passenger cars and commercial vehicles, makes it more than ever essential for service personnel to consult the vehicle manufacturer for up-to-date technical information and adjustments data in relation to a particular model, both in the interests of vehicle safety and customer satisfaction.

Automotive technical abbreviations

ABC	Active body control (Mercedes-Benz)	C _w	Coefficient of drag (German) – vehicle aerodynamics
ABS	Anti-Blockier-System (German) – anti-lock braking system	C _x	Coefficient of drag (French) – vehicle aerodynamics
ABDC	After bottom dead centre (engine timing)	DC	Direct current
AC	Alternating current	DERV	Diesel engine road vehicles (fuel)
A/C	Air conditioning	DI	Direct injection
ACL	Automatic chassis lubrication (commercial vehicles)	DIS	Direct ignition system (no distributor)
ACT	Air charge temperature	DISI	Direct injection spark ignition
A/F	Air/fuel ratio	DOHC	Double overhead camshafts
AIR	Air injection reactor (emission control)	DRP	Dynamic rear proportioning (brakes)
ALB	Anti-lock brakes (Honda)	DSC	Dynamic stability control
ARCS	Active roll control system (Citroen)	DSG	Direct shift gearbox (Volkswagen group)
ASD	Automatic slip-control differential	DWB	Double wishbone suspension
ASF	Audi space frame (aluminium body construction)	EBA	Emergency brake assist
ASR	Antriebs-Schlupf-Regelung (German) – anti-slip regulation or traction control	EBFD	Electronic brake force distribution
ATC	Automatic temperature control	EBS	Electronic braking system (air brakes)
ATDC	After top dead centre (engine timing)	ECI	Electronically controlled injection
ATF	Automatic transmission fluid	ECM	Electronic control module
AWD	All-wheel drive (also 4WD)	ECS	Evaporative control system (fuel system)
AWS	All-wheel steering (also 4WS)		Electronically controlled suspension
BAS	Brake assist system	ECT	Engine coolant temperature
BBDC	Before bottom dead centre (engine timing)	ECU	Electronic control unit
BDC	Bottom dead centre (engine timing)	EDC	Electronic diesel control
BEV	Battery-electric vehicle	EFI	Electronic fuel injection
BHP	Brake horsepower	EGR	Exhaust gas recirculation (emission control)
BMEP	Brake mean effective pressure	ELV	End-of-life vehicle (materials recycling)
BOFT	Bearing oil film thickness	EMS	Engine management system
BSFC	Brake specific fuel consumption	EP	Extreme pressure (lubricants)
BTDC	Before top dead centre (engine timing)	EPAS	Electrical power-assisted steering (NSK-RHP)
CAD	Computer aided design	EPHS	Electrically powered hydraulic steering (TRW)
CAFE	Corporate average fuel economy (American)	EPS	Electric power steering
CAG	Computer aided gearshift (Scania)	ESP	Electronic stability programme
CATS	Computer active technology suspension (Jaguar)	ETC	Electronic traction control
CB	Contact-breaker	ETS	Enhanced traction system (General Motors)
CBE	Cab behind engine (commercial vehicles)	EUI	Electronic unit injector (Lucas Diesel)
C _d	Coefficient of drag (vehicle aerodynamics)	EVC	Exhaust valve closed (engine timing)
CD	Capacity discharge (ignition system)	EVO	Exhaust valve open (engine timing)
CFC	Chlorofluorocarbon (refrigerant)	FCEV	Fuel-cell electric vehicle
CGI	Compact graphite iron	FHP	Friction horsepower
CI	Compression ignition (diesel engines)	FWD	Front-wheel drive
CN	Cetane number (diesel fuel ignition rating)	GCW	Gross combination weight (articulated vehicles)
CNG	Compressed natural gas (fuels)	GCWR	Gross combined weight rating (vehicle and trailer)
CO	Carbon monoxide (emission control)	GDI	Gasoline direct injection (Mitsubishi)
CO ₂	Carbon dioxide (global warming)	GRP	Glass reinforced plastics
COE	Cab over engine (commercial vehicles)	GTW	Gross train weight (drawbar vehicles)
CP	Centre of pressure (vehicle aerodynamics)	GV	Governor valve (automatic transmissions)
CR	Compression ratio (engine)	GVW	Gross vehicle weight (rigid vehicles)
CRS	Common rail system (diesel fuel injection)	GVWR	Gross vehicle weight rating
CTX	Continuously variable transaxle (Ford)	GWP	Greenhouse warming potential (refrigerants)
CV	Constant velocity (universal joints)	HC	Hydrocarbons (emission control)
CVT	Continuously variable transmission	HDC	Hill descent control (ABS system)
		HEV	Hybrid-electric vehicle

XII AUTOMOTIVE TECHNICAL ABBREVIATIONS

HFC	Hydrofluorocarbon (refrigerant)	RWD	Rear-wheel drive
HGV	Heavy goods vehicle	SAMT	Semi-automated mechanical transmission (Eaton)
HT	High tension	SBC	Stand-by-control (electronic transmission control ZF)
HUCR	Highest useful compression ratio	SBR	Styrene-butadiene rubber (tyres)
HVAC	Heating, ventilation and air conditioning	SCA	Supplemental coolant additives
IFS	Independent front suspension	SCR	Selective catalytic reduction (emission control)
IHP	Indicated horsepower	SCS	Stop control system (Girling)
INJ	Injection (timing mark)	SEFI	Sequential electronically controlled fuel injection (Ford)
IOE	Inlet over exhaust (obsolete valve layout)	SFC	Specific fuel consumption
IPM	Integrated power module (hybrid electric vehicles)	SFI	Sequential fuel injection
IRS	Independent rear suspension	SG	Spheroidal graphite (high-strength cast iron)
IVC	Inlet valve closed (valve timing)	SI	Spark ignition (petrol engines)
IVO	Inlet valve open (valve timing)	SLA	Short and long arm (American) – suspension linkage
KPI	King-pin inclination (steering)	SOHC	Single overhead camshaft
LCV	Light commercial vehicle	SPI	Single point injection (petrol engines)
LGV	Large goods vehicle	SRS	Supplemental restraint system (airbags)
LI	Load index (tyres)	SUV	Sports utility vehicle
LNG	Liquefied natural gas (fuels)	SV	Side valves (obsolete valve layout)
LPG	Liquid petroleum gas (fuels)	TAC	Thermostatic air cleaner
LS	Leading shoe (drum brakes)	TBI	Throttle body injection (SPI)
LSD	Limited slip differential	TC	Twin carburettors
MAF	Mass air flow (engines)	TCI	Transistorized coil ignition
MAP	Manifold absolute pressure	TCM	Transmission control module
MOFT	Minimum oil film thickness	TCS	Transmission controlled spark (engine intervention system)
MON	Motor octane number (more demanding ON test)	TDC	Top dead centre (engine timing)
MPI	Multi-point injection	TDI	Turbocharged direct injection (diesel engines)
MPV	Multi-purpose vehicle (people carrier)	TEL	Tetra ethyl lead (petrol anti-knock additive)
NO	Nitrogen oxides (emission control)	TML	Tetra methyl lead (as above)
NOAT	Nitrite organic acid technology (coolants)	TPS	Throttle position sensor
NVH	Noise, vibration and harshness (vehicle refinement testing)	TS	Trailing shoe (drum brakes)
OAT	Organic acid technology (coolants)	TV	Throttle valve (engine and automatic transmissions)
OBD	On-board diagnosis	TVS	Thermal vacuum switch (exhaust gas recirculation)
OD	Overdrive	TWC	Three-way catalyst (emission control)
ODP	Ozone depletion potential (refrigerants)	TXV	Thermostatic expansion valve (refrigeration)
OHC	Overhead camshaft	UJ	Universal joint
OHV	Overhead valves	ULEV	Ultra-low emission vehicle
ON	Octane number (petrol anti-knock rating)	VCP	Variable cam phasing (valve timing)
PAS	Power-assisted steering	VCU	Viscous coupling unit (transmission)
PBD	Polybutadiene (tyres)	VDC	Vehicle dynamics control (Bosch)
PCM	Power train control module (engine and transmission)	VGT	Variable-geometry turbocharger
PCV	Positive crankcase ventilation (emission control)	VI	Viscosity index (lubricants)
PEM	Polymer electrolyte membrane (fuel cells) (or proton exchange membrane)	VIP	Vehicle intrusion protection (Toyota)
PFI	Port fuel injection (petrol engines)	VIVT	Variable inlet valve timing
PM	Particulate matter (diesel emission control)	VKPI	Virtual king-pin inclination (steering)
PR	Ply-rating (tyres)	VSC	Vehicle skid control
PSV	Public service vehicle	VTG	Variable turbine geometry (turbocharging)
PTFE	Polytetrafluoroethylene	VTT	Variable twin turbo (turbocharging)
PTO	Power take-off (commercial vehicles)	VVT	Variable valve timing
PVC	Polyvinyl chloride	VVTL	Variable valve timing and lift
PZEV	Partial zero emission vehicle	WOT	Wide-open throttle
RC	Roll-centre (suspension geometry)	ZEV	Zero emission vehicle
RON	Research octane number (less demanding ON test)		
RTV	Room temperature vulcanizing (sealant)		

1 The reciprocating piston petrol engine

1.1 MODERN REQUIREMENTS

General background

The motor vehicle engine is basically a device for converting the internal energy stored in its fuel into mechanical energy. It is classified as an internal combustion engine by virtue of this energy conversion taking place within the engine cylinders.

Since the term 'energy' implies the capacity to perform work, the engine is thus able to propel the vehicle along the road and, within limits, overcome unwanted opposition to its motion arising from rolling friction, gradient resistance and air drag. To facilitate this process the engine is combined with a transmission system, the functioning of which is discussed later.

The vast majority of car engines are of the reciprocating piston type and utilize spark ignition to initiate the combustion process in the cylinders. However, the compression ignition or diesel principle to initiate combustion is increasingly challenging the petrol engine for car applications, especially in Europe. Both petrol and diesel engines operate on the four-stroke principle in which the piston travels one complete stroke for each of the successive events of induction, compression, combustion and exhaust.

The late Laurence Pomeroy, a distinguished motoring historian, once summarized the early history of the motor car as follows: From 1885 to 1895 men struggled to make the car go. From 1896 to 1905 they contrived to make it go properly. Between 1907 and 1915 they succeeded in making it go beautifully! What then are the requirements for the engine of the modern passenger car as reflected in many decades of further development and, not least, in the light of present-day energy conservation and environmental pollution considerations? These requirements can now add up to quite a formidable list. As we pursue our studies into the whys and wherefores of engine construction and operation, it will become evident that although some of the requirements are complementary, others are not, and therefore (as in most engineering) some compromise has generally to be accepted in the final product.

Modern requirements

Optimum performance

With modern advances in engine design it is not particularly difficult to obtain sufficient power to give the car a high top speed, especially since the recent trend towards car bodies of lighter construction and more efficient aerodynamic shape. Today, however, a more important engine requirement than a further increase in top speed is an improved accelerating capability together with better flexibility in the low to middle speed range, or what is sometimes termed 'driveability'. A further

performance requirement of a new engine design is that it must usually allow for possible future increases in cylinder size.

Good fuel economy

The overall aim of improving the fuel economy of cars is to minimize the amounts of crude oil used to provide petrol for their engines, because of constraints imposed by limited petroleum resources and rising costs. Fuel economy may also be made the subject of legislation, as it already is in America, where each manufacturer has to comply with corporate average fuel economy standards (or CAFE standards, as they are generally termed). For these reasons, further engine requirements are those of minimum weight so as to reduce total car weight; improved combustion efficiency, better to utilize the fuel; and reduced friction losses between the working parts.

Low pollution

Since the late 1960s increasingly stringent legislation has been applied to limit the levels of atmospheric pollutants emitted from car engines, especially the American FTP (Federal Test Procedure), the Japanese and later the European Community ECE/EEC test cycles, all of which differ in their requirements and are therefore not directly comparable. In Britain The Road Vehicles (Construction and Use) Regulations are also now such that there is a requirement for every motor vehicle to be so constructed that no avoidable smoke or visible vapour is emitted therefrom, and another that makes it an offence to use a vehicle which emits substances likely to cause damage to property or injury to persons. In general, legislation is concerned with carbon monoxide, which has toxic effects; unburned hydrocarbons, which contribute to atmospheric smog; and nitrogen oxides, which cause irritation to the eyes and lungs, and also combine with water to produce acid rain that destroys vegetation. To reduce these harmful emissions, not only is very careful control of the combustion process required in modern engine design, but also various sophisticated devices may have to be added for after-treatment of the exhaust gases. Of further concern to the environmentalist is the emission of carbon dioxide which, although non-toxic, is nevertheless an unwanted contributor to global warming. This has to the development of systems for deactivating half the number of cylinders on some large capacity V8 and V12 engines, to reduce fuel consumption and therefore the emission of carbon dioxide when full power is not required.

Minimum noise level

Noise is generally defined as unwanted sound. Reducing interior noise makes a car more attractive to the buyer. Reducing exterior noise to socially acceptable limits has been the subject of increasingly stringent legislation in the European Community and other countries since the early 1980s, and in

Britain is included in the Provision of the Motor Vehicles (Construction and Use) Regulations relating to noise. A similar function is performed in America by the EPA (Environmental Protection Agency) noise regulations. Since the engine is an obvious source of noise an important requirement is that its design and installation should minimize noise emission, not only that directly radiated from the engine itself to the exterior, but also that arising from vibrations transmitted through its mounting system to the car body interior.

Easy cold starting

An essential driver requirement of any engine, whether it be of past or present design, is that it should possess good cold starting behaviour and then continue to run without hesitation during the warming-up period. A present-day additional requirement is that the cold starting process should be accomplished with the least emission of polluting exhaust gases and detriment to fuel economy. To monitor the required enrichment of the air and fuel mixture for cold starting, increasingly sophisticated controls were applied first to carburettor automatic choke systems and then later to fuel injection cold start systems. These controls form part of what are now termed 'engine management systems'.

Economic servicing

An important owner requirement of a car is that its engine design should acknowledge the need to reduce servicing costs. This aim may be approached by minimizing the number of items that need periodic attention by a service engineer. For example, the use of hydraulic tappets eliminates altogether the need for adjustment of the valve clearances. It is also promoted by allowing ready access to those items of the engine involved in routine preventive maintenance, such as the drive-belt tensioner, spark plugs, and petrol and oil filters.

Acceptable durability

In order to reduce fuel consumption while still maintaining good car performance, it is now the trend to develop engines of smaller size with relatively higher power output. Furthermore, the installation of a turbo-charger permits an increase in power without imposing a corresponding increase in the size or weight of the engine itself. However, the greater heating effect on certain engine components may require changes to their material specifications and also the addition of an oil cooler. The components of modern engines have therefore tended to become more highly stressed, so that engine testing of ever-increasing severity by the manufacturers is now required to maintain durability in extremes of customer service.

Least weight

Another important design requirement of the modern petrol engine is that it should be made as light as possible. This is because a corresponding reduction in car weight can make significant improvements not only in fuel economy and acceleration capability, but also in general handling and ease of manoeuvring the car. Since reducing engine weight is not always consistent with maintaining durability, the need for adequate testing of the engine components is confirmed. Also special manufacturing techniques may have to be adopted to avoid damage to such items as castings with very thin walls.

Compact size

For the modern car, the manufacturer strives to provide the maximum interior space for the minimum possible exterior dimensions. Thus the trend is inevitably towards having the front wheels driven, with the power unit (engine and transmission) installed transversely between them; the conventional arrangement was to have a longitudinally mounted power unit from which the drive was taken to the rear wheels. It follows that the requirement now is for a more compact engine. This is because the engine length is controlled by the distance available between the steerable front wheels, less that required by the transaxle (combined gearbox and final drive); its width by the distance available between the radiator and the dash structure, less that required by the engine auxiliaries; and its height by the need for a low and sloping bonnet line, which contributes to an efficient aerodynamic body shape.

Economic manufacture

This is clearly a most important requirement for any new design of engine, since putting it into production demands a massive initial investment on the part of the car manufacturer. It is, of course, for this reason that the smaller specialist car manufacturer generally uses an existing engine from a volume producer. For economic manufacture a new design of engine should lend itself as far as possible to existing automatic production processes and require the minimum of special tooling. The cost of materials will be reduced in building a smaller engine, and the construction should be as simple as possible to minimize the number of parts to be assembled and thereby further reduce manufacturing costs. Similarly, to produce a range of large capacity V6 and V8 engines a modular design approach may be adopted, so that their major components can be produced on the same machinery.

Aesthetic appearance

In early years the under-bonnet appearance of high-grade cars of the 1920s and 1930s, such as Bugatti, Hispano-Suiza and Rolls-Royce, was much admired for the elegant proportions and beautiful finish of their engines. More recently manufacturers have recognized the customer appeal of a pleasing under-bonnet appearance. Not so much of the engine itself, which is usually buried deeper within the engine compartment, but of the neat arrangement and smooth contours of the modern comprehensive air intake system and its manifold runners that now lie above the engine.

1.2 ENGINE NOMENCLATURE

To understand the information given in an engine specification table, such as those included in a manufacturer's service manual or published in the motoring press, it is necessary to become familiar with some commonly used terms (Figure 1.1). The 'language' of the reciprocating piston engine is summarized in the following sections.

Top dead centre

The top dead centre (TDC) is of general application in engineering; it is any position of a hinged linkage in which three successive joints lie in a straight line. In the case of a motor

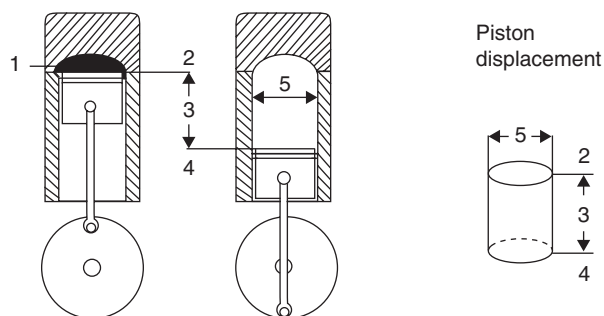


Figure 1.1 Engine nomenclature (Yamaha)

- 1 volume of combustion chamber
- 2 top dead centre (TDC)
- 3 stroke
- 4 bottom dead centre (BDC)
- 5 bore

Piston displacement: volume of gases displaced by the piston as it moves from BDC to TDC.

vehicle engine, top dead centre refers to the position of the crankshaft when the piston has reached its closest point to the cylinder head. This results in the main, big-end and small-end bearings lying in a straight line. A motor vehicle service engineer often needs to establish top dead centre for checking the ignition and valve timing of an engine.

Bottom dead centre

The bottom dead centre (BDC) is, of course, the opposite extreme of crankshaft rotation when the piston has reached its furthest point from the cylinder head.

Piston stroke

In a general engineering sense, the stroke is the movement of a reciprocating component from one end of its travel to the other. In the motor vehicle engine the piston stroke, therefore, is the distance travelled by the piston in its movement from BDC to TDC or, of course, vice versa, and is expressed in millimetres (mm).

Cylinder bore

In engineering practice the term *bore* may refer to a hole through a bushing or pipe, or to the cutting of a large-diameter cylindrical hole, or to an actual measurement of the inside diameter of a hollow cylinder. It is the last named with which we are concerned here, where the bore refers to the inside diameter of the engine cylinder expressed in millimetres (mm).

Piston displacement

This term refers to the volume of cylinder displaced or swept by a single stroke of the piston, and is also referred to as *swept volume*. It is expressed in cubic centimetres (cm³) and may be simply calculated as follows:

$$V_h = \frac{\pi d^2 s}{4000} \text{ cm}^3$$

where V_h is the piston displacement or swept volume (cm³), d is the cylinder bore (mm) and s is the piston stroke (mm).

Engine capacity

Here we are referring to the total piston displacement or the swept volume of all cylinders. For example, if the swept volume of one cylinder of an engine is 375 cm³ and the engine has four cylinders, then the engine capacity is 1500 cm³ or 1.5 litres (1). This can be simply stated as:

$$V_H = V_h z$$

where V_H is the engine capacity (cm³), V_h is the piston displacement (cm³) and z is the number of cylinders.

Stroke/bore ratio

Reference is sometimes made to the stroke/bore ratio of an engine and although the term itself is self-explanatory, its significance deserves further explanation because up to the 1950s the length of the piston stroke almost invariably exceeded the cylinder diameter, whereas until recently the converse situation has usually applied. The reason for the change from so-called *under-square* to *over-square* cylinder proportions from the 1950s onwards, was that for taxation and insurance purposes engines had previously been rated for horse-power by an RAC formula dating from the early years of motoring, which bore little relationship to the actual power they developed. Unfortunately it also meant that to keep the rated horse-power low for taxation purposes, but the actual power high, the engine designer was restricted to small bore and long stroke cylinder proportions. However, the introduction in 1947 of a flat tax on all passenger cars regardless of their engine size, relieved designers of this artificial restraint on cylinder bore size and led to improvements in engine performance.

The advantages claimed at the time for engines with larger bores and shorter strokes included increased size of valves and ports for better engine breathing; lower piston speeds that not only reduced mechanical losses due to friction and therefore improved fuel consumption, but also increased life expectancy of the cylinder bores; greater rigidity for the crankshaft by virtue of a smaller crankthrow (Section 1.8); and last but not least a reduction in engine height for a lower bonnet line. In current practice engines tend to be neither under-square nor over-square but to have a stroke/bore ratio closer to unity, so as better to achieve a compact combustion chamber and reduce harmful exhaust gas emission (Section 3.1).

Mean effective pressure

This term is used because the gas pressure in the cylinder varies from a maximum at the beginning of the power stroke to a minimum near its end. From this value must, of course, be subtracted the mean or average pressures that occur on the non-productive exhaust, induction and compression strokes. Engine mean effective pressure can be expressed in kilonewtons per square metre (kN/m²).

Indicated and brake power

The most important factor about a motor vehicle engine is the rate at which it can do work or, in other words, the power it can develop. It is at this point that we must distinguish between the rate at which it might be expected to work (as calculated from the mean effective pressure in the cylinder, the piston displacement, the number of effective working

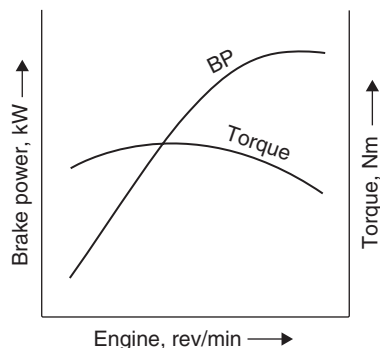


Figure 1.2 Engine power and torque curves

strokes in a given time and the number of cylinders) and the rate at which it actually does work (as measured in practice when the engine is running against a braking device known as a dynamometer).

The significance of this is that the *brake power* delivered at the crankshaft is always less than the *indicated power*, owing to internal friction losses in the engine. A simple expression for calculating in kilowatts (kW) the indicated power of an engine is as follows:

$$P = psAEz$$

where P is the indicated power (kW), p is the mean effective pressure (kN/m^2), s is the piston stroke (m), A is the piston area (m^2), E is the number of effective working strokes per second and z is the number of cylinders. (Since in the four-stroke cycle engine there is only one power stroke for every two complete revolutions of the crankshaft, the number of effective working strokes per second will correspond to one-half the number of engine revolutions per second.)

As mentioned earlier, a dynamometer is used in an engine testing laboratory to measure the brake power (or effective power) of an engine, because it acts as a brake to balance the torque or turning effort at the crankshaft through a range of speeds. A graph of the engine *power curve* can be drawn by plotting brake power values against engine speeds (Figure 1.2). Various standardized test procedures may be adopted in engine testing, such as those established by the American Society of Automotive Engineers (SAE), the German Deutsche Institut für Normung (DIN) and the Italian Commissione tecnica di Unificazione nell'Automobile (CUNA). In an engine specification table only the maximum brake power and corresponding crankshaft speed are usually quoted. For example, the 5.9 litre V twelve-cylinder 48-valve engine used in the high-performance Aston Martin DB9 Volante car is claimed to develop 330 kW at 6000 rev/min.

Engine torque

Also usually included on an engine performance graph is a *torque curve*, which is obtained by plotting crankshaft torque, or turning effort, against engine speed (Figure 1.2). The engine torque, of course, is derived from combustion pressure acting upon the cross-sectional area of the piston, the resulting force from which applies a turning effort to the crankshaft through the connecting rod and crankthrow arrangements (Figure 1.3).

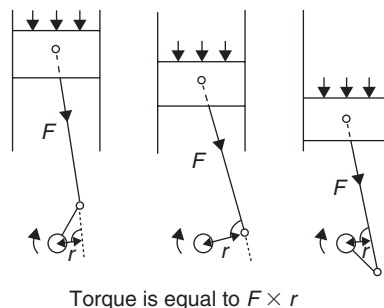


Figure 1.3 Engine torque or turning effort

Engine torque, therefore, may be considered as the force of rotation acting about the crankshaft axis at any given instant.

Engine torque T may be expressed in newton metres (Nm), and generally reaches a peak value at some speed below that at which maximum power is developed; the reason for this is explained at a later stage. An engine that provides good pulling power is typically one in which maximum torque is developed at moderate engine speeds. For example, the 1.4 litre four-cylinder 16-valve K Series engine of advanced construction (Figure 1.30), as originally designed by the Rover Group, developed its peak power of 70 kW at 6250 rev/min and a maximum torque of 124 Nm at 4000 rev/min. At the other extreme a heavy-vehicle diesel engine, such as the 11 litre six-cylinder turbocharged Scania with a peak power of 280 kW at a governed maximum speed of 1900 rev/min, develops a maximum torque of 1660 Nm at 1300 rev/min, which really is pulling power!

1.3 OPERATING PRINCIPLES

The four-stroke petrol engine

As with the various repeating cycles of events in nature, so does the motor vehicle petrol engine need to operate on a constantly repeating cycle known as the four-stroke principle.

It would seem to be generally accepted that the first internal combustion engine to operate successfully on the four-stroke cycle was constructed in 1876 by Nicolaus August Otto (1829–91). This self-taught German engineer was to become one of the most brilliant researchers of his time and also a partner in the firm of Deutz near Cologne, which for many years was the largest manufacturer of internal combustion engines in the world.

Although the Otto engine ran on gas, which was then regarded as a convenient and reliable fuel to use, it nevertheless incorporated the essential ideas that led to the development in 1889 of the first successful liquid-fuelled motor vehicle engine. This was the twin-cylinder Daimler engine, patented and built by the German automotive pioneer Gottlieb Daimler (1834–1900) who, like Otto, had been connected with the Deutz firm. The Daimler engine was subsequently adopted by several other car manufacturers and, in most respects, it can be regarded as the true forerunner of the modern four-stroke petrol engine (Figure 1.4).

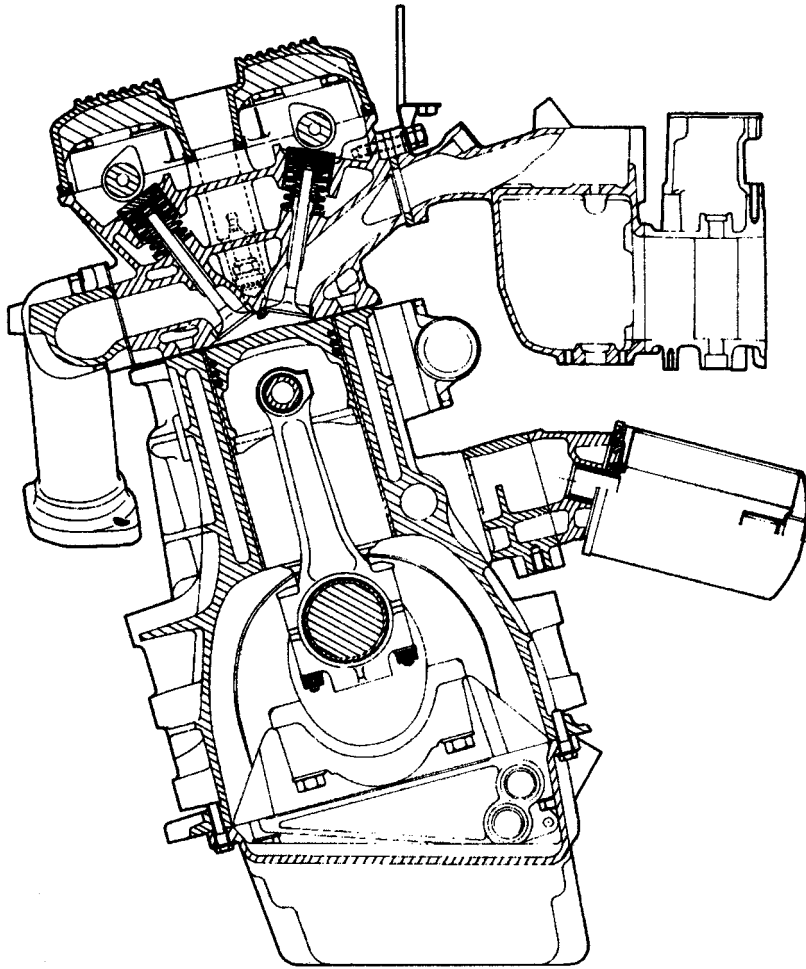


Figure 1.4 Cross-section of a modern four-stroke petrol engine (*Jaguar*)

In this type of engine the following sequence of events is continuously repeated all the time it is running (Figure 1.5):

- 1 The *induction* stroke, during which the combustible charge of air and fuel is taken into the combustion chamber and cylinder, as a result of the partial vacuum or depression created by the retreating piston.
- 2 The *compression* stroke, which serves to raise both the pressure and temperature of the combustible charge as it is compressed into the lesser volume of the combustion chamber by the advancing piston.
- 3 The *power* stroke, immediately preceding which the combustible charge is ignited by the sparking plug and during which the gases expand and perform useful work on the retreating piston.
- 4 The *exhaust* stroke, during which the products of combustion are purged from the cylinder and combustion chamber by the advancing piston, and discharged into the exhaust system.

It thus follows that one complete cycle of operations occupies two complete revolutions of the engine crankshaft. Since energy is necessarily required to perform the initial induction and compression strokes of the engine piston before firing occurs, an electrical starter motor is used for preliminary

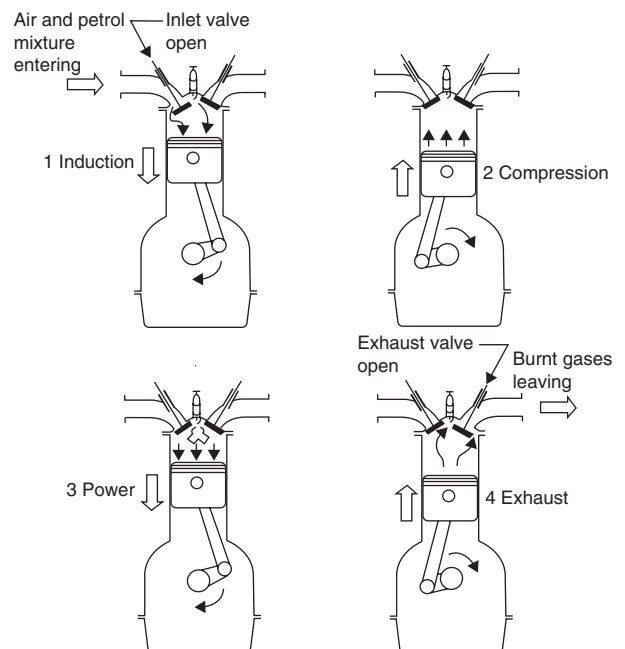


Figure 1.5 The four-stroke petrol engine cycle

cranking of the engine. Once the engine is running the energy required for performing subsequent induction, compression and exhaust strokes is derived from the crankshaft and flywheel system, by virtue of its kinetic energy of rotation. *Kinetic energy* is a term used to express the energy possessed by a body due to its mass and motion. The principle of an engine flywheel is therefore to act as a storage reservoir for rotational kinetic energy, so that it absorbs energy upon being speeded up, and delivers it when slowed down.

In the four-stroke cycle, the functions of admitting the combustible charge before its compression, and releasing the burnt gases after their expansion, are performed by the engine inlet and exhaust valves. The opening and closing of the inlet and exhaust valves are not, in actual practice, timed to coincide exactly with the beginning and ending of the induction and exhaust strokes; nor is the spark timed to occur exactly at the beginning of the power stroke. At a later stage the reasons for these departures in *valve and ignition timing* from the basic four-stroke operating cycle will be made clear.

The two-stroke petrol engine

As its designation implies, the two-stroke petrol engine (Figure 1.6) completes its working cycle in only two strokes of the piston, so that a combustible charge is ignited at each revolution of the crankshaft. Although in its simplest construction the two-stroke petrol engine needs no valves, the induction and exhaust process must be facilitated by a system of *scavenging* or forcible clearing of the cylinder gases. This may either take the form of a separate engine-driven pump, or utilize the motion of the engine piston itself in a sealed crankcase. The flow of gases entering and leaving the cylinder is controlled by the reciprocating movement of the engine piston, which thus acts as a slide valve in conjunction with ports cut in the cylinder wall. Although the two-stroke petrol engine was once favoured in Europe for some small inexpensive passenger cars, it generally became obsolescent because of the difficulty in reducing its harmful exhaust emissions.

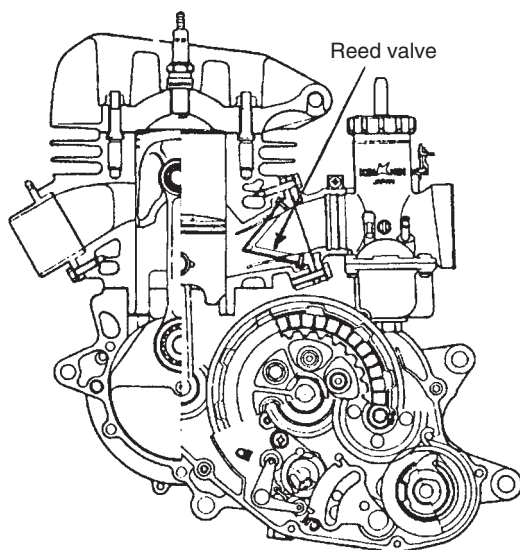


Figure 1.6 Cross-section of a two-stroke petrol engine (Honda)

At the 11th Sir Henry Royce Memorial Lecture in 1966, Cyril Lovesey recalled a remark by another distinguished Rolls-Royce aero engine specialist, Dr Stanley Hooker, who amusingly described a four-stroke engine as one with 'one stroke to produce power and three strokes to wear it out'. It is therefore perhaps not surprising that from time to time attempts are made to revive interest in the two-stroke petrol engine for automotive use, albeit in much more sophisticated forms of which an example will be later described.

It may be of interest to recall that the two-stroke and the four-stroke engine both originated in the late 1870s, so it might reasonably be assumed that both types of engine started out in life with an equal chance of success. The fact that the four-stroke engine became by far the more widely adopted type can probably be explained by its having a greater potential for further development. This is a criterion that can often be applied to rival ideas in all branches of engineering.

The first successful application of the two-stroke cycle of operation to an early gas engine is generally attributed to a Scottish mechanical engineer, Sir Dugald Clerk (1854–1932). It is for this reason, of course, that the two-stroke cycle is sometimes referred to as the *Clerk cycle*. Dugald Clerk, like several other pioneer researchers of the internal combustion engine, was later to achieve high academic distinction, culminating in his election as Fellow of the Royal Society in 1908.

The Clerk engine was scavenged by a separate pumping cylinder. A few early motor vehicle two-stroke petrol engines followed the same principle, but it later became established practice to utilize the underside of the piston in conjunction with a sealed crankchamber to form the scavenge pump. This idea was patented in 1889 by Joseph Day & Son of Bath, England and represented the simplest type of two-cycle engine.

In the two-stroke or Clerk cycle, as applied by Day, the following sequence of events is continuously repeated all the time the engine is running (Figure 1.7):

- 1 The *induction-compression* stroke. A fresh charge of air and fuel is taken into the crankchamber as a result of the depression created below the piston as it advances towards the cylinder head. At the same time, final compression of the charge transferred earlier in the stroke from the crankchamber to the cylinder takes place above the advancing piston.
- 2 The *power-exhaust* stroke. The combustible charge in the cylinder is ignited immediately preceding the power stroke,

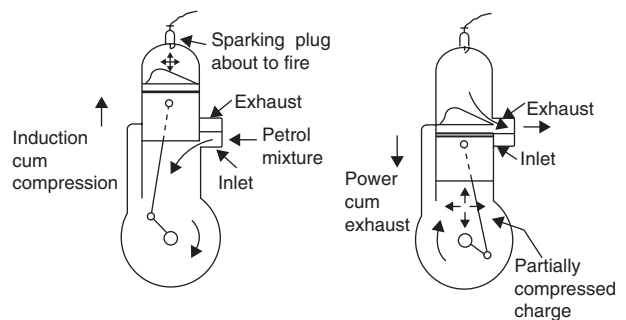


Figure 1.7 The two-stroke petrol engine cycle

during which the gases expand and perform useful work on the retreating piston. At the same time, the previously induced charge trapped beneath the retreating piston is partially compressed. Towards the end of the stroke, the exhaust gases are evacuated from the cylinder, a process that is facilitated by the scavenging action of the new charge transferred from the crankcase.

The uncovering and covering of the cylinder ports of the piston, or *port timing*, is determined by considerations similar to those affecting the valve timing of the four-stroke engine and will be explained at a later stage.

Four-stroke versus two-stroke engines

The following generalizations may be made concerning the relative merits of two-stroke and four-stroke petrol engines in their basic form:

- 1 The two-stroke engine performs twice as many power strokes per cylinder per revolution. In theory at least, this might be expected to produce twice the performance of a four-stroke engine of equivalent size. Unfortunately, this is not realized in practice because of the difficulties encountered in effectively purging the exhaust gases from the cylinder and then filling it completely with a fresh combustible charge. The scavenging efficiency of the basic two-stroke petrol engine is therefore poor.
- 2 In performing twice as many power strokes per revolution, the two-stroke engine can deliver a smoother flow of power, but this may be less true at low engine speeds when irregular firing or 'four-stroking' can result from poor scavenging.
- 3 An obvious practical advantage of the basic two-stroke engine is the mechanical simplicity conferred by its valveless construction, which contributes to a more compact and lighter engine that should be less expensive to make.
- 4 Reduced maintenance requirements might reasonably be expected with the basic two-stroke engine by virtue of point 3. There is, however, the well-known tendency for carbon formation to have a blocking effect on the exhaust ports, which impairs engine performance by reducing scavenging efficiency.
- 5 The fuel consumption of the basic two-stroke engine is adversely affected by the poor cylinder scavenging, which allows part of the fresh charge of air and fuel to escape through the exhaust port before final compression of the charge takes place.
- 6 There is a greater danger of overheating and piston seizure with a two-stroke engine, which can set a limit on the maximum usable performance. It is more difficult to cool satisfactorily, because it does not have the benefit of the second revolution in the four-stroke cycle when no heat is being generated.
- 7 Lubrication of the two-stroke petrol engine is complicated by the need to introduce oil into the fuel supply to constitute what is generally termed a *petroil* mixture. The working parts of the engine are thus lubricated in aerosol fashion by oil in the air and fuel charge, and this tends to increase harmful exhaust emissions. It is for this reason that the basic two-stroke petrol engine is now obsolescent for cars.

Scavenging: further details

Frequent reference has been made to the inherently poor scavenging efficiency of the basic two-stroke petrol engine. The word 'basic' has been used deliberately and is intended to apply to the Day type of early two-stroke engine, which had a deflector-head piston to promote a cross-scavenging effect on the burnt charge leaving the cylinder (Figure 1.8a). This not entirely successful scheme persisted until the mid 1920s, when Dr E. Schnürle of Germany developed an alternative loop-scavenging system. In this the deflector on the piston head is omitted and two transfer ports with angled passages are disposed on either side of, instead of opposite, the exhaust port (Figure 1.8b). The loop-scavenge effect produced is such that before the two streams of fresh charge intermingle, they converge upon the cylinder wall at a point furthest away from the exhaust port, so there is less chance of escape.

The Day type of early two-stroke engine also used what would now be classified as a three-port system of scavenging. This system comprises inlet, transfer and exhaust ports, all in the cylinder wall, and necessarily imposes a restriction on the period during which a fresh charge of air and fuel may enter the crankcase.

To achieve more complete filling of the crankcase, the later two-port system of scavenging is now generally employed. In this system only the transfer and exhaust ports are in the cylinder wall, the inlet port being situated in the crankcase itself and controlled by either an automatic flexible reed valve (Figure 1.9a) or an engine-driven rotary disc valve (Figure 1.9b) which improve the torque and power characteristics respectively of an engine. The two-port system of scavenging thus allows the fresh charge to continue entering the crankcase

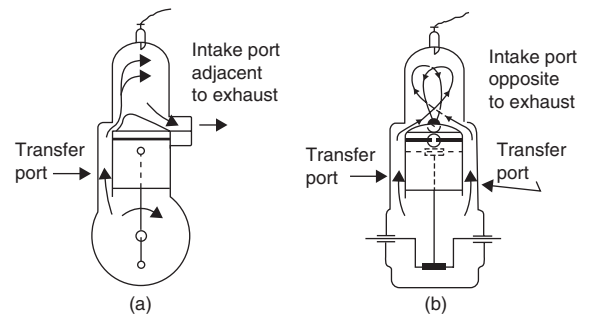


Figure 1.8 Early three-port scavenging systems: (a) cross-scavenging (b) loop-scavenging

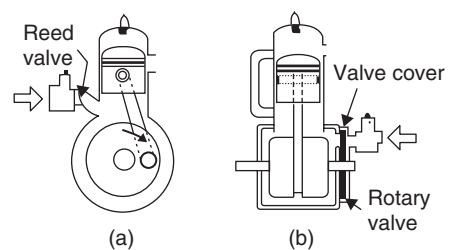


Figure 1.9 Later two-port scavenging systems: (a) reed valve (b) rotary valve (Yamaha)

during the whole, instead of part, of the induction-compression stroke, albeit with a little extra mechanical complication.

Further development of the two-stroke petrol engine

We take as an example the advanced concept of the Toyota S-2 (supercharged two-stroke) engine (Figure 1.10). The development objective of this engine is to deliver greater power more smoothly than a conventional one, and also to confer very high torque at low engine speeds. A Roots-type supercharger or blower (Section 9.2) is used to achieve positive scavenging of the exhaust gases. Unlike the two-stroke petrol engines so far described, the Toyota engine borrows many features from established four-stroke practice. It has a four-valve cylinder head with two valves for intake and two for exhaust, which do of course open and close twice as often in this two-stroke application. The engine is provided with electronically controlled ignition and fuel injection systems, with fuel being injected directly into the cylinders.

The importance of engine compression

Early internal combustion engines were very inefficient because they were provided with a combustible charge that was ignited at atmospheric pressure. However, it was recognized as early as 1838 by William Barnett in the UK that compression of the charge before combustion was advantageous. Nearly 25 years later, a French railway engineer with the splendid name of Alphonse Beau de Rochas was granted a patent in respect of several ideas that related to the practical and economical operation of the internal combustion engine. Among these ideas he stated a requirement for the maximum possible expansion of the cylinder gases during the power stroke, since the cooler they become the more of their energy is transformed into useful work on the retreating piston. To assist in this aim, there was a further requirement for the maximum possible pressure at the beginning of the expansion process, which the motor vehicle service engineer recognizes as the *compression pressure* of an engine. The first successful engine to utilize this principle was, as mentioned earlier, the Otto engine.

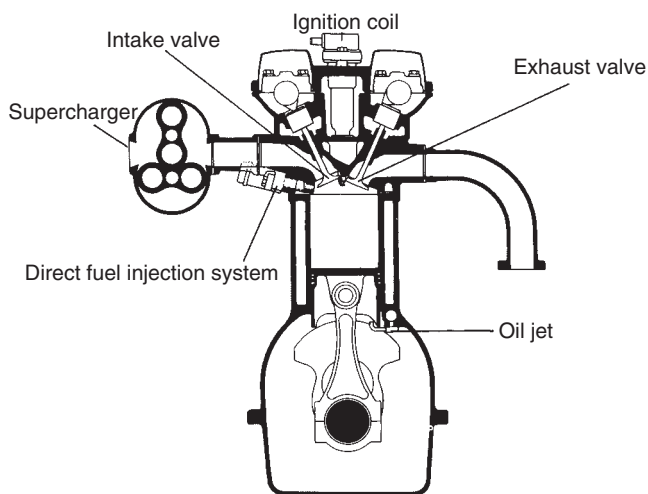


Figure 1.10 Toyota S2 supercharged two-stroke engine

Within certain limitations, which will be better understood at a later stage, a high-compression engine is relatively more efficient than a low-compression one in terms of either improved fuel economy or greater power. This is simply because it better utilizes the internal energy received from its fuel or, in other words, it possesses a higher *thermal efficiency*. It is this particular feature of engine operation that explains the importance of checking compression pressures before attempting to tune an engine in service.

Compression ratio

The extent to which the air and fuel charge in a petrol engine is compressed, prior to the power stroke, is known as the *compression ratio* of an engine. It is calculated as the ratio of the total volume enclosed above the piston at BDC to the volume remaining above the piston at TDC.

The swept volume has already been explained in Section 1.2. A complementary term is *clearance volume*, which is that volume remaining above the piston when it reaches TDC. In some engines the combustion chamber is formed mainly in the piston head (Section 3.3), so that the clearance volume is concentrated within, rather than above, the piston. Hence, the compression ratio (usually abbreviated to CR) may also be expressed as the swept volume plus the clearance volume divided by the clearance volume (Figure 1.11), as follows:

$$\varepsilon = \frac{V_h + V_c}{V_c}$$

where ε is the compression ratio, V_h is the cylinder swept volume (cm^3), and V_c is the combustion space clearance volume (cm^3).

The calculated compression ratios for petrol engines are typically in the range 8:1 to 9.5:1. It should be appreciated, however, that in the petrol engine the calculated compression ratio is realized in practice only when the engine is running with a wide open throttle.

Furthermore, in the case of a two-stroke engine V_h is generally considered as the cylinder volume from the point of exhaust port closing to TDC, which is therefore less than the

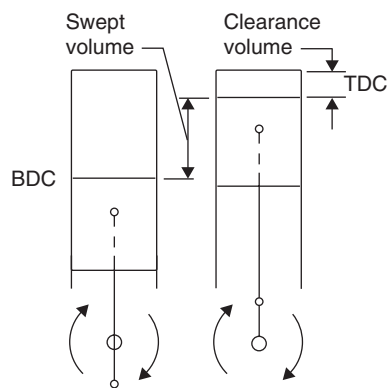


Figure 1.11 Engine compression ratio

swept volume of an equivalent four-stroke engine and gives a lower compression ratio.

1.4 BASIC STRUCTURE AND MECHANISM

It is convenient to introduce the various components of the petrol engine in the following groups:

Cylinder block and crankcase
Piston and rings
Connecting rod and bearings
Crankshaft assembly and bearings
Cylinder head and gasket
Valve train and timing drive
Engine support mountings.

Cylinder block and crankcase

In combination, the cylinder block and crankcase form the main structural component of the engine and perform several important functions, as follows:

- 1 Each cylinder must act not only as a pressure vessel in which the process of combustion can take place, but also as a guide and bearing surface for the piston sliding within it.
- 2 Since the engine cylinders have to be cooled effectively, the cylinder block must also form a jacket to contain the liquid coolant.
- 3 The crankcase provides an enclosure for the crankshaft and various other parts of the engine mechanism and must preserve accurate alignment of their supporting bearings under all operating conditions.
- 4 Pressure conduits in the form of either drilled or cast in ducts must also be incorporated in the crankcase to convey oil to the engine working parts.
- 5 The cylinder block is required to provide external mounting surfaces for various engine auxiliary units.

It has long since been established practice to combine the cylinder block and crankcase into a single unit, this generally being termed *monobloc* construction (Figure 1.12). The historical origins of this form of construction date back to the

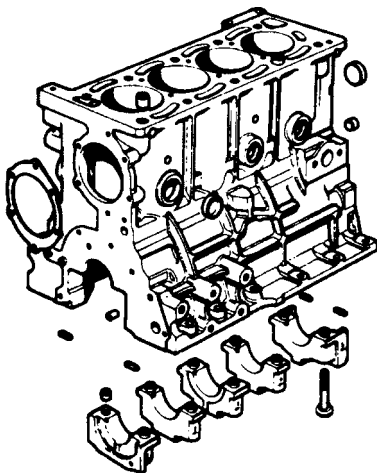


Figure 1.12 Cylinder block and crankcase

early 1920s, when there was a general trend towards simplification of the engine structure. Until that time the cylinder block and crankcase were produced as separate units, in cast iron and aluminium alloy respectively, and then bolted together.

In relation to modern engine design, the monobloc construction provides a very necessary rigid foundation for the engine and reduces manufacturing costs. It should be added, however, that this particular form of construction is not always the best compromise for heavy-duty diesel engines, which will be considered at a later stage.

Piston and rings

The main function of the piston itself is twofold:

- 1 It acts as a moving pressure transmitter, by means of which the force of combustion is impressed upon the crankshaft through the medium of the connecting rod and its bearings.
- 2 By supporting a gudgeon pin the piston provides a guiding function for the small end of the connecting rod.

The piston assumes a *trunk* form to present a sliding bearing surface against the cylinder wall, which thus reacts against the side thrust arising from the angular motion of the connecting rod. Since the piston is a major reciprocating part, it must of necessity be light in weight to minimize the inertia forces created by its changing motion – bearing in mind that the piston momentarily stops at each end of its stroke! Another, perhaps obvious, requirement is that the piston must be able to withstand the heat of combustion and should operate quietly in its cylinder, both during warm-up and at the normal running temperature of the engine.

To perform its sealing function efficiently, the upper part of the piston is encircled by flexible metal sealing rings known as the *piston rings*, of which there are typically three in number for petrol engines (Figure 1.13). In combination the piston rings perform several important functions, as follows:

- 1 The upper *compression rings* must maintain an effective seal against combustion gases leaking past the pistons into the crankcase.
- 2 These rings also provide a means by which surplus heat is transmitted from the piston to the cylinder wall and thence to the cooling jacket.
- 3 The lower *oil control ring* serves to control and effectively distribute the lubricating oil thrown on to the cylinder walls, consistent with maintaining good lubrication and an acceptable oil consumption.

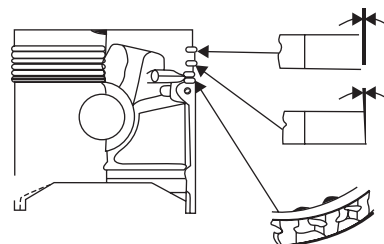


Figure 1.13 Piston and rings (Toyota)

Connecting rod and bearings

The function of the connecting rod and its bearings is to serve as a constraining link between the reciprocating piston and the rotating crankshaft. A fair analogy of this particular conversion of motion is to be found in the pedalling of a cycle, where the knee, calf and foot of the cyclist may be likened to the piston, connecting rod and crankpin of an engine.

The connecting rod is attached at what is termed its *big end* to the crankpin and at its *small end* to the gudgeon pin of the piston. Each connecting rod big-end bearing is divided into two half-liners, so as to make possible its assembly around the crankpin (Figure 1.14). The big-end bearing housing is therefore formed partly by the lower end of the connecting rod and partly by a detachable cap, the two halves being bolted together. No such complication arises in the case of the small-end bearing arrangements, this end of the connecting rod being formed as a continuous eye.

As a consequence of its reciprocating and partly rotating motion, the connecting rod is subjected to appreciable inertia forces. Its detail design must therefore be such as to ensure the maximum rigidity with the minimum weight.

Crankshaft assembly and bearings

The crankshaft represents the final link in the conversion of reciprocating motion at the piston to one of rotation at the flywheel. In the case of the multicylinder engine, the crankshaft has to control the relative motions of the pistons, whilst simultaneously receiving their power impulses.

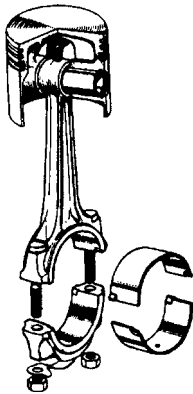


Figure 1.14 Connecting rod and bearings

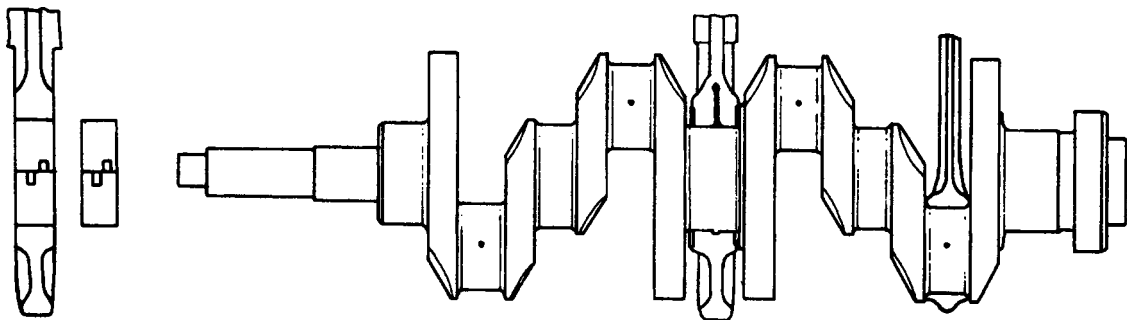


Figure 1.15 Crankshaft and bearings (Alfa-Romeo)

A one-piece construction is most commonly used for the motor vehicle crankshaft, which extends the whole length of the engine and must therefore possess considerable rigidity. The timing drive for the engine valve mechanism is taken from the front end of the crankshaft, as is the pulley and belt drive for the engine auxiliaries, such as the cooling fan and the alternator for the electrical system. Attached to the rear end of the crankshaft is the engine flywheel.

The crankshaft is supported radially in the crankcase by a series of bearings, known as the engine *main bearings*. Each main bearing is divided into two half-liners, similar to the big-end bearings, and again to allow assembly around the journals of the one-piece construction crankshaft (Figure 1.15).

Cylinder head and gasket

The functions of the cylinder head may be listed as follows:

- 1 It must provide a closure or chamber for the upper part of each cylinder, so that the gas pressure created by the combustion process is constrained to act against the piston.
- 2 Associated with function 1 is the need to incorporate a gas porting system with inlet and exhaust valves, as well as a platform upon which to mount their operating mechanism. Provision must also be made for a screwed boss to retain the sparking plug.
- 3 Similar to the cylinder block, the head must form a jacket that allows liquid coolant to circulate over the high-temperature metal surfaces.
- 4 It is required to contribute to the overall rigidity of the engine structure and maintain a uniform clamping pressure on its sealing gasket with the cylinder block.

The sealing gasket is generally known as the *cylinder head gasket* or simply 'head gasket'. In liquid-cooled engines the function of the cylinder head gasket is to seal the combustion chambers and coolant and oil passages at the joint faces of the cylinder block and head. The gasket is therefore specially shaped to conform to these openings, and is also provided with numerous holes through which pass either the studs or the set bolts for attaching the cylinder head to the block (Figure 1.16).

Historically, the cylinder head has not always been made detachable from the cylinder block. It was not until the early 1920s, as mentioned previously, that there was a general trend towards simplification of design which led to the abandonment of the cylinder block with integral head. This change

facilitated both the production and the servicing of the motor vehicle engine, although it did introduce the risk of incorrect tightening down of the cylinder head, which can cause at least joint trouble and at worst distortion of the cylinder bores.

Valve train and timing drive

The overall function of the valve train and timing drive is to provide first for the admission and then the retention of the combustible charge within the cylinder, and finally for the release of the burnt gases from the cylinder, all in synchronism with the motion of the pistons (Figure 1.17).

To perform this sequence of events in accordance with the requirements of the four-stroke cycle, a cam-and-follower mechanism driven at one-half crankshaft speed is used to operate the engine inlet and exhaust valves. There are several different methods of operating the valves from the cam-and-follower mechanism, but in all cases it is necessary for one or more camshafts to be driven from the front end of the crankshaft by what is termed the *timing drive*, and which will be examined in detail at a later stage.

It is, of course, the engine valves themselves that actually perform the functions of admitting the air and fuel charge before its compression, sealing it in the cylinder during compression and combustion, and then releasing the burnt gases

after their expansion. Either single or paired inlet and exhaust valves serve each cylinder, and these are mounted in the cylinder head combustion chambers. Valve springs are fitted to ensure that the motion of the valves and their operating mechanism follows faithfully that intended by the cams, and also to maintain adequate sealing pressure when the valves are closed.

Engine support mountings

The engine is subject to complex vibration effects, which may produce six different free motions or degrees of freedom and combinations of the same. These motions can generate what are termed 'bounce' and 'yaw' about a vertical axis, fore-and-aft movement and roll about a horizontal longitudinal axis, and sideways shake and pitch about a horizontal lateral axis. The three inertia axes intersect at the centre of gravity of the engine (Figure 1.18). For mounting the engine, a resilient anti-vibration system is therefore required, so that the vibratory forces are reduced to the relatively small spring forces transmitted by the support mountings themselves. Apart from supporting the static load of the engine unit and isolating the car structure from engine vibrations, the engine mounts also insulate the engine against vibrations of the car structure and mechanism. By thus minimizing unwanted movements of the engine under all running conditions, any contribution to car shake from this source is reduced.

The engine support mountings almost invariably feature rubber as their spring medium, since this material is highly resilient when loaded in shear (Figure 1.19a). If a rubber block is simply loaded in compression it inevitably bulges sideways (Figure 1.19b), because it is deformable as opposed to being compressible, so the effect is to increase the area of rubber under load and hence reduce its flexibility as a spring. In actual practice, a compromise is often sought by loading the rubber partly in compression and partly in shear by inclining the rubber mountings (Figure 1.19c). A widely used type of support mounting is the sandwich unit, which generally consists of either a rectangular or a circular block of rubber with metal attachment plates bonded to its upper and lower faces. For some applications, one or more metal interleaves may be incorporated in the mount (Figure 1.19d). These serve

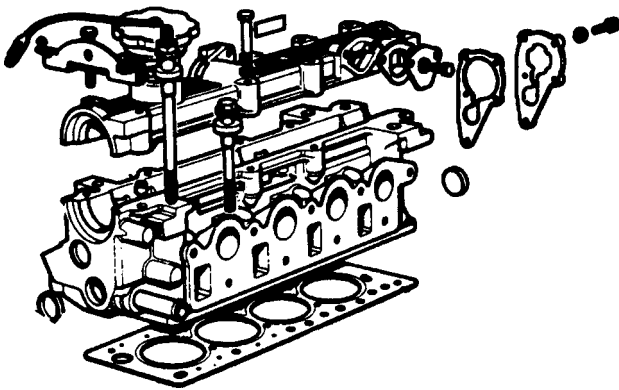


Figure 1.16 Cylinder head and gasket

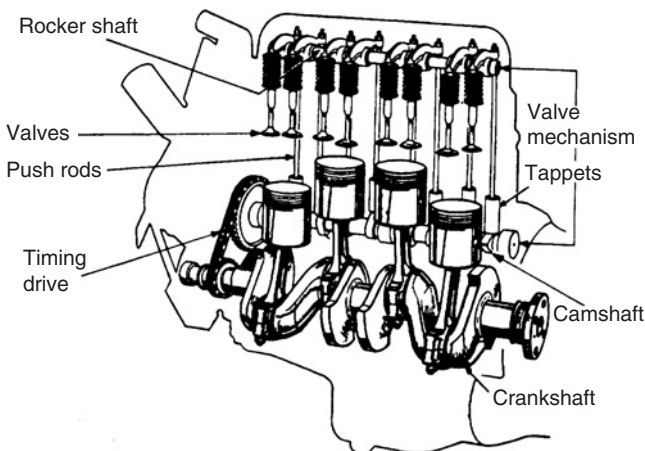


Figure 1.17 Valve train for a push-rod overhead valve engine

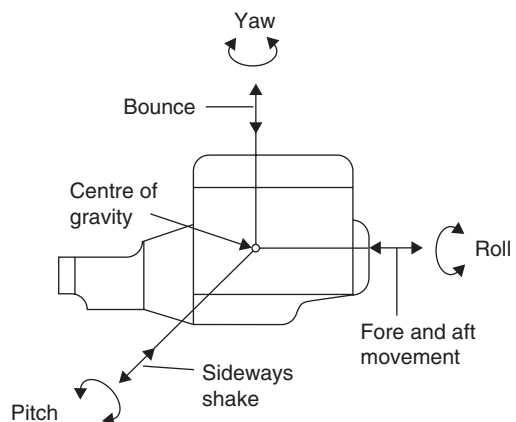


Figure 1.18 Vibratory motions of an engine

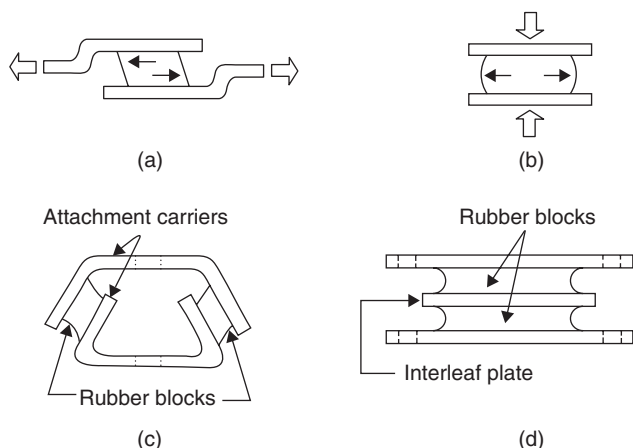


Figure 1.19 Engine support mountings: (a) rubber loaded in shear (b) rubber loaded in compression (c) rubber loaded in combined compression and shear (d) rubber sandwich with interleaf

to restrict sideways bulging of the rubber, thus further reducing its flexibility under compression loading, but leaving its shear loading characteristics unaltered, and they also assist in dissipating the heat of internal friction created by flexing of the rubber. Another advantage of using rubber, rather than steel, for the spring medium is that transmission of sound through the mountings is less, since there is no metal-to-metal path for the sound to travel along. The various types of resilient mounting system employed for longitudinally and transversely disposed engine and transmission units are described in Section 30.1.

1.5 CYLINDER AND CRANKTHROW ARRANGEMENTS

Cylinder number and displacement

Every new engine must be designed with a specific type of service in view, which then determines its general characteristics. Important among these for the car is smooth and efficient operation over a wide range of speeds and loads. Herein lies the explanation why no manufacturer lists a single-cylinder engine and few produce an engine with fewer than four cylinders. The following considerations are pertinent.

Smoothness

With a single-cylinder engine operating on the four-stroke cycle, it will be recalled that only one power impulse occurs for every two revolutions of the crankshaft. The fluctuations in crankshaft torque of a single-cylinder engine would, therefore, be quite unacceptable in motor vehicle operation. Hence the greater number of cylinders used, the shorter will be the interval between the power impulses and the smoother will be the flow of torque from the engine.

Mechanics

It will be evident from the explanation of the factors governing engine power given in Section 1.2 that the power output obtainable from a single-cylinder engine of realistic dimensions and running at a reasonable speed is unlikely to be

sufficient for motor vehicle, as distinct from motorcycle, requirements. This is because a practical limit is set on individual cylinder size by dynamic factors, namely the inertia forces created by accelerating and decelerating the reciprocating masses comprising the piston assembly and the upper portion of the connecting rod.

If an unusually large, and consequently heavy, piston were adopted in a single-cylinder engine intended for high-speed operation, the dynamic effects could be such as to increase the magnitude of the inertia forces to a level that at least would make engine imbalance unacceptable and at worst would prove mechanically destructive. This is partly because the inertia forces are proportional to the cube of the piston mass (e.g. doubling the piston mass will cause the inertia forces to become eight times as great), and partly because they also vary as the square of the engine speed (e.g. doubling the engine speed will cause them to become four times as great).

Temperature

Another problem arising from the use of an unduly large cylinder bore is that cooling of the piston and valves can be seriously impaired, and this may lead to their failure from thermal overstressing. If this is to be avoided it would be necessary to incorporate special features of design, such as oil cooling of the piston, which is often practised in marine diesel engines with very large cylinder bores (although this may still be required for automotive engines with forced induction).

For these reasons it has therefore become established practice for the displacement of an engine to be shared among multiple small cylinders, rather than confined to a single large cylinder. To this must be added the proviso that an excessively large number of cylinders increases the friction losses in an engine, quite apart from the extra complication which makes it more costly to build and maintain. Some authorities now claim that 330 cm³ represents an optimum size of cylinder.

Arrangement of engine cylinders

Once the displacement and number of cylinders have been decided in relation to the required performance characteristics of a new engine, the next consideration is how the cylinders are to be arranged. In cars they may be arranged in three different ways, each with its own advantages and disadvantages.

In-line cylinders

As would be expected, in this arrangement all the cylinders are mounted in a straight line along the crankcase, which confers a degree of mechanical simplicity. Such engines are now produced with any number of cylinders from two to six (Figure 1.20), with four cylinders continuing to represent a very widely used cylinder arrangement (Figures 1.30 and 3.10). The single bank of cylinders may be contained in either a vertical or an inclined plane. The latter type is sometimes referred to as a sloper or slant engine. For this particular mounting of an in-line cylinder engine, the advantages usually claimed include a reduction in overall installation height and improved accessibility for routine servicing.

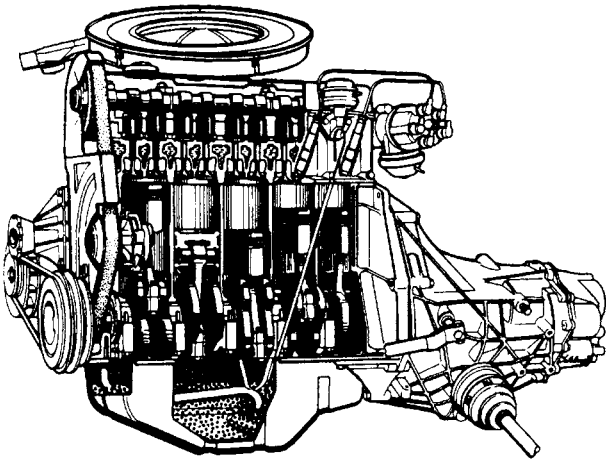


Figure 1.20 An interesting in-line five-cylinder engine (Audi)

With the exception of the in-line two-cylinder or parallel twin engine where both pistons move in unison to obtain even firing intervals, this arrangement of cylinders provides generally satisfactory balance in respect of the reciprocating parts, especially in six-cylinder versions. Apart from space requirements, a mechanical limitation is placed on the acceptable length of an in-line cylinder engine, because of the difficulty in controlling torsional vibrations of the crankshaft. This is a topic that will be discussed at a later stage.

Horizontally opposed cylinders

Horizontally opposed engines have their cylinders mounted on the crankcase in two opposite banks and are sometimes referred to as flat or boxer engines. They are typically produced in two-, four- and six-cylinder versions (Figure 1.21). The main advantages usually claimed for them include inherently good balance of the reciprocating parts, a low centre of gravity, which contributes to car stability, and a short engine structure. It is the latter feature that makes this arrangement of cylinders particularly suitable both for front-wheel-drive and rear-engined cars, since the engine can be mounted either ahead of or behind the driven wheels with the minimum of overhang. By virtue of its low overall height, the horizontally opposed engine can readily allow a sloping bonnet line in front-engined cars and also provide additional space for stowing luggage above it in rear-engined cars. Furthermore, it lends itself admirably to air cooling because with an in-line cylinder arrangement it is difficult to get the rear cylinders to run as cool as the front ones, unless the engine is installed transversely.

The disadvantages associated with horizontally opposed cylinders include the need for lengthy intake manifolds if a central carburettor is used, the duplication of coolant inlet and outlet connections in the case of liquid cooling, and much reduced accessibility for the cylinder heads and valve mechanism. Its greater width can also impose restrictions on the available steering movements of the wheels.

V-formation cylinders

With V engines the cylinders are mounted on the crankcase in two banks set at either a right angle or an acute angle to

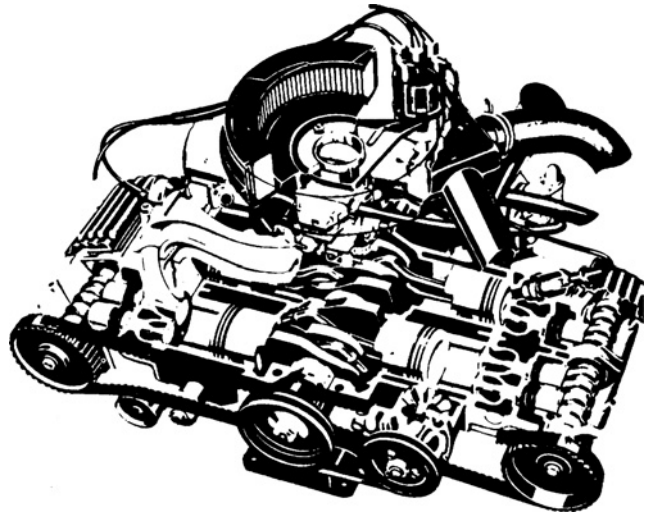


Figure 1.21 A modern horizontally opposed four-cylinder engine (Alfa-Romeo)

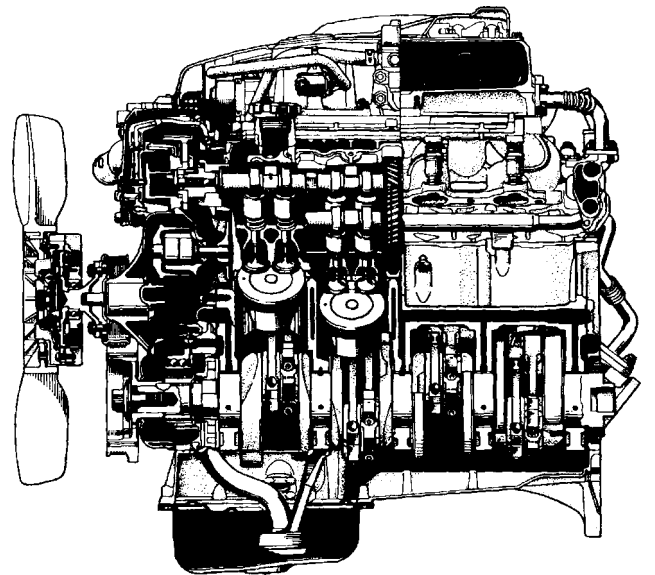


Figure 1.22 A high-grade V eight-cylinder engine (Lexus/Toyota)

each other. They may be produced in four-, six-, eight- and occasionally twelve-cylinder versions (Figure 1.22). Where a V cylinder arrangement has been adopted in preference to mounting the cylinders in line, it has usually been in the interests of providing a more compact and less heavy engine. In particular, the overall length of the engine can be appreciably reduced, so that both the structure and the crankshaft can be made more rigid. The former is thus better able to accept greater combustion loads and the latter is less prone to torsional vibrations. For typically large-displacement V engines, the inherently wider cylinder spacings ensure adequate size of coolant passages, both around the cylinder walls and in the hot exhaust valve regions in the cylinder heads (Figure 1.36).

On the debit side, the V cylinder arrangement generally presents a more difficult balancing problem and also demands a more elaborate intake manifold from a central carburettor or a single-point fuel injection system. In common with horizontally opposed cylinder layouts, V engines tend to be more costly to produce, since there are more surfaces to machine and some duplication of structural features.

However, the V six-cylinder engine has become widely adopted in modern practice for medium class cars with engine capacities, ranging between 2.4 and 3.2 litres. Once described by D.A. Martens of Chevrolet as 'a product of necessity', the V6 engine can provide improved performance and enhanced smoothness when transversely mounted in front-wheel-drive cars, where an in-line four-cylinder engine may also be available as a lower cost option.

W-formation cylinders

A more recent development has been the 'double-V' or 'W' cylinder engine, introduced in eight-, twelve- and sixteen-cylinder versions by the Volkswagen group. Basically the W engine, as it is now generally known, comprises two wide-angle cylinder banks, each of which incorporates two offset rows of cylinders inclined at a much narrower angle. In the early 1990s Volkswagen had introduced a single bank narrow-V six-cylinder engine, which externally resembled an in-line unit, but internally had two offset rows of three cylinders inclined at an angle of only 15°. It therefore followed that if two narrow-V six-cylinder banks could themselves be mounted in a wide-V formation of 72° and share a common crankshaft, a W12 engine then becomes a practicality. Several advantages may be gained from this unorthodox, but albeit complex, layout when compared to a conventional V12 engine. A W12 engine can offer a more compact design and indeed may require no more installation space than a conventional V8 engine. It can also be made less heavy, whilst possessing greater structural rigidity with reduced vibration levels.

A distinguishing feature of the narrow-V type cylinder bank arrangement is that the axes of the cylinder pairs intersect below, instead of coinciding with, the axis of the crankshaft (Figure 1.23). This departure from the operating geometry of a conventional wide-angle V-cylinder layout becomes necessary to accommodate the motion of the pistons as they approach their bottom dead centre positions. Similar consideration applies to the need for either separate, or split-type, staggered crankpins for the cylinder pairs.

Historically, the narrow-V cylinder layout had its origins in piston aero engine practice. It was conceived by the Lancia company in Italy at the beginning of World War I, the aim being to reduce the width of a conventional V-cylinder engine for aircraft installation. Smaller narrow-V four-cylinder engines were subsequently used for many years in the passenger cars made by this company, the Lancia Lambda model of the 1920s being a well-known classic example.

Crankthrow arrangements

The arrangement of the crankthrows in relation to the disposition of the engine cylinders and their number is determined by two sometimes conflicting considerations: acceptable engine balance, and equal firing intervals.

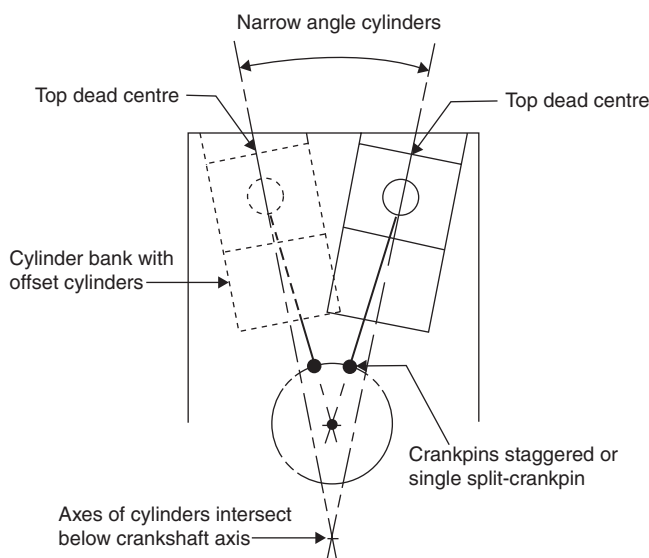


Figure 1.23 Basic arrangement of cylinder pair in narrow-V engine

Acceptable engine balance

Theoretically, a perfectly balanced engine is one which, when running and 'suspended in space' from its centre of gravity, would exhibit no vibratory movements whatsoever. In reality, of course, the reciprocating engine can never be perfectly balanced. Apart from any rotation imbalance, there are also the inevitable torque irregularities, although as stated previously these can be minimized by the use of more than one cylinder.

Multiple cylinders further allow a much better standard of general engine balance, provided that the choice of a particular arrangement and number of cylinders takes into account the presence of what are termed primary and secondary inertia forces. The effect of these forces, which act in the reciprocating sense, will now be explained in simple qualitative terms.

Primary inertia forces These arise from the force that must be applied to accelerate the piston over the first half of its stroke, and similarly from the force developed by the piston as it decelerates over the second half of its stroke (Figure 1.24a). When the piston is around the mid-stroke position it is then moving at the same speed as the crankpin and no inertia force is being generated.

For an engine to be acceptable in practice, the arrangement and number of its cylinders must be so contrived that the primary inertia forces generated in any particular cylinder are directly opposed by those of another cylinder. Where the primary inertia forces cancel one another out in this manner, as for example in an in-line or a horizontally opposed four-cylinder engine with the outer and inner pair of pistons moving in opposite directions, the engine is said to be in *primary balance* (Figure 1.24b).

Secondary inertia forces These are due to the angular variations that occur between the connecting rod and the cylinder axis as the piston performs each stroke. As a consequence of this departure from straight-line motion of the connecting rod, the piston is caused to move more rapidly over the outer

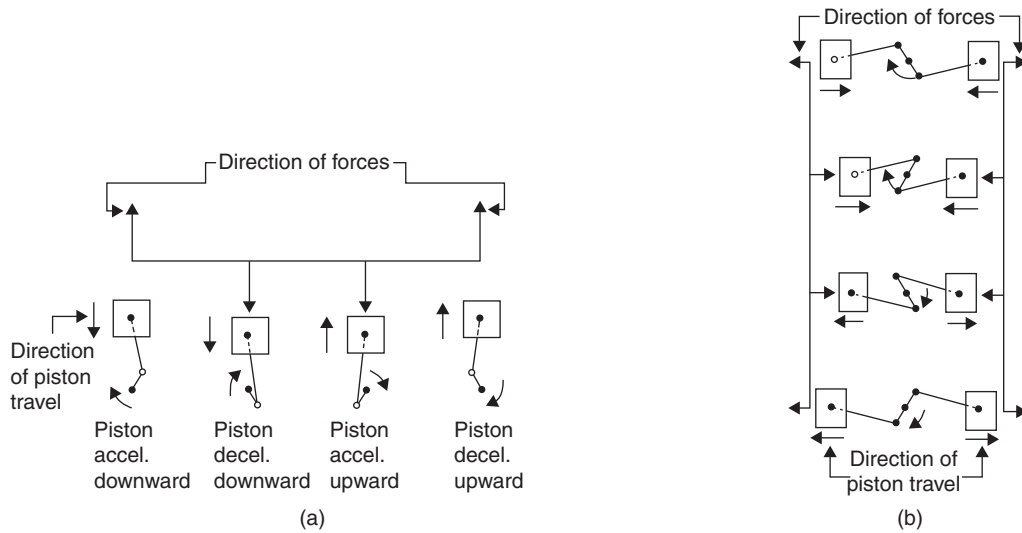


Figure 1.24 Primary inertia forces: (a) unbalanced (single-cylinder) (b) balanced (horizontally opposed cylinders)

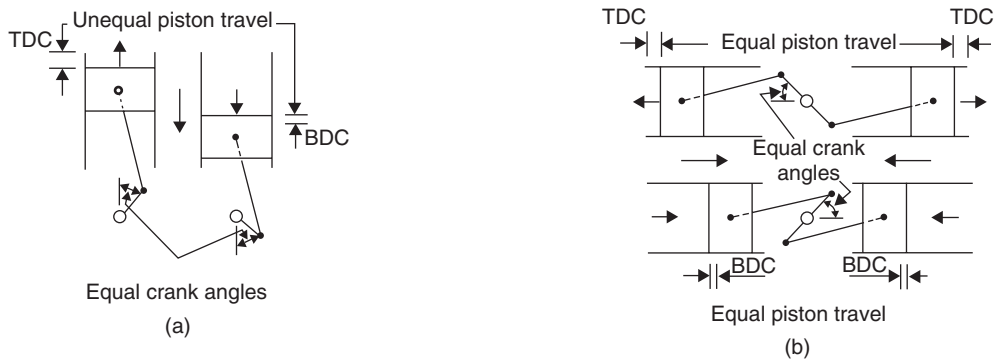


Figure 1.25 Secondary inertia forces: (a) unbalanced (single-cylinder) (b) balanced (horizontally opposed cylinders)

half of its stroke than it does over the inner half. That is, the piston travel at the two ends of the stroke differs for the same angular movements of the crankshaft (Figure 1.25a). The resulting inequality of piston accelerations and decelerations produces corresponding differences in the inertia forces generated. Where these differing inertia forces can be both matched and opposed in direction between one cylinder and another, as for example in a horizontally opposed four-cylinder engine with corresponding pistons in each bank moving over identical parts of their stroke (Figure 1.25b) the engine is said to be in *secondary balance*.

It is not always practicable for the cylinders to be arranged so that secondary balance can be obtained, but fortunately the vibration effects resulting from this type of imbalance are much less severe than those associated with primary imbalance and can usually be minimized by the flexible mounting system of the engine. This is confirmed by the long-established and popular in-line four-cylinder engine, which possesses primary balance but lacks secondary balance. However, the continuing search for greater refinement of running with this type of engine led, in the mid 1970s, to a revival of interest in the use of twin counterbalancing shafts for cancelling out these secondary inertia forces. The modern application of

this system of harmonic balancing will be described in Section 1.8.

Equal firing intervals

The arrangement of the crankthrows is also determined by the requirements for even firing intervals of the cylinders and for spacing the successive power impulses as far apart as possible along the crankshaft, so as to reduce torsional deflections or twisting effects. For any four-stroke engine the firing intervals must, if they are to be even, be equal to 720° divided by the number of cylinders.

For in-line four-cylinder engines the first and fourth crankthrows are therefore indexed on one side of the crankshaft and the second and third throws on the other side (Figure 1.26a). The *firing order* of these engines, numbering from the front, may then be either 1-3-4-2 or 1-2-4-3 at 180° intervals. Similarly, in the case of in-line six-cylinder engines, the crankthrows are spaced in pairs with an angle of 120° between them. Hence, the first and sixth crankthrows are paired, as are the second and fifth, and likewise the third and fourth (Figure 1.26b). The firing order may then be such that no two adjacent cylinders fire in succession; that is, either 1-5-3-6-2-4 or 1-4-2-6-3-5 at, of course, 120° intervals.

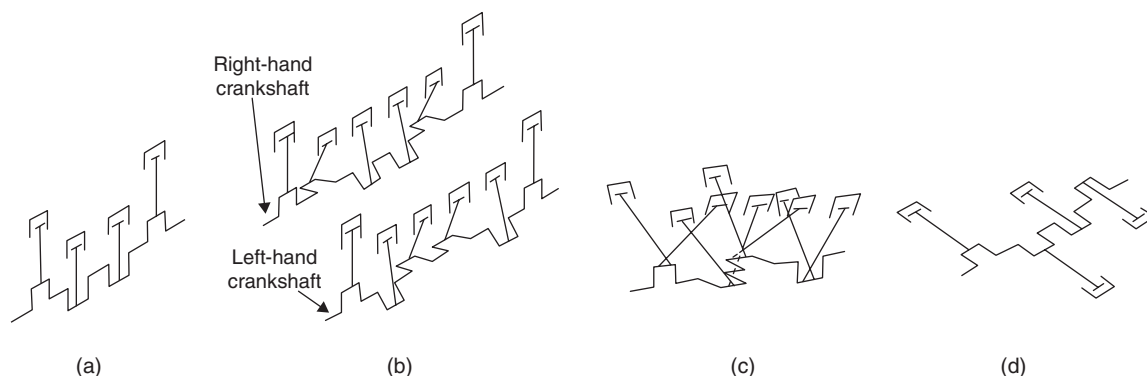


Figure 1.26 Cylinder and crankthrow arrangements: (a) in-line four-cylinder (b) in-line six-cylinder (c) eight-cylinder (d) horizontally opposed four-cylinder

For V and horizontally opposed engines (Figures 1.26c and d), the firing orders follow a sequence similar to those of in-line engines, but the crankthrows are so disposed that firing alternates between the cylinder banks as it proceeds to and fro along the crankshaft.

The V6 engine is nevertheless deserving of special comment. When the Italian Lancia company pioneered the V6 engine layout in 1950, they inclined the two banks of cylinders at 60° to each other, which together with a six-throw crankshaft conferred equally spaced firing intervals of 120° . However, this was not generally true of other early V6 engines, where their designers had adopted an included angle of 90° between the cylinder banks and used a shorter three-throw crankshaft with the three pairs of connecting rods each sharing common, double-length, crankpins. This arrangement resulted in firing intervals of 90° and 150° alternatively instead of even 120° intervals, so although these engines were more economical on fuel than a V8 engine they were not particularly smooth running. To overcome the problem of uneven firing intervals with a 90° V6 engine, Buick in America retained a three-throw crankshaft but ingeniously replaced the common, double-length, crankpins by adjacent single crankpins that were staggered by 15° in opposite directions to produce a so-called 'split-pin' crankshaft. It may be of interest to recall that the reason for adopting a 90° included angle between the cylinder banks of early V6 engines was basically one of production economy. In the case of American V6's it was to utilize tooling that already existed for their (90°) V8 engines, while conversely for European V6's it was to create new tooling that could also be used to meet any future demand for V8 engines.

Some designers of modern even firing V6 engines (even firing especially being desirable where turbocharging is featured) have in fact reverted to a 60° included angle between the cylinder banks, but have retained a three-throw crankshaft supported in four main bearings. The required angular spacing for the crankpins of each throw is then obtained by using what is termed a 'flying arm' (Figure 1.77). This can be likened to a crankweb that does not connect with a main journal and which may also be utilized for balancing purposes.

Cylinder numbering sequence

The numbering sequence for the cylinders of motor vehicle engines has long been defined by the various standards

organizations. In British and European practice, as defined by British Standards BS 5672: 1991 and International Organization for Standardization ISO 1204:1990(E) respectively, the cylinders of an in-line or single bank engine are numbered consecutively as viewed from the flywheel end of the engine (which is in contrast to earlier established practice where the cylinders were numbered consecutively as viewed from the nose of the crankshaft). Both a capital letter and an identical system of numbering is used to designate the cylinders of V and horizontally opposed or double bank engines, the prefix letters A and B denoting the left- and right-hand banks of cylinders respectively as viewed from the flywheel end of the engine. In American practice, the suffix letters R and L are preferred to denote the right- and left-hand banks of cylinders respectively. Owing to the differences that can arise, past and present, the service engineer should always consult the manufacturer's engine specification for confirmation of the cylinder numbering system that applies.

1.6 CYLINDER BLOCK, CRANKCASE AND HEAD

Cylinder block construction

Cylinder block construction takes two forms. *Closed-deck* construction (Figure 1.27a) represents long-established practice and resembles a deep box-like enclosure for the cylinder barrels that also serves as a coolant jacket (Figure 1.28). Transfer ducts are provided in the top face or closed deck of the cylinder block, so as to permit the circulation of coolant to the cylinder head. With the *open-deck* construction (Figure 1.27b) the cylinder barrels are free-standing in that they are attached only to the lower deck of the cylinder block, which in past applications utilized detachable cylinder liners that tended to result in a less rigid construction. By dispensing with a continuous top face, the open-deck construction nevertheless reduces the complexity of the cylinder block casting. Where gravity sand casting is used it facilitates the coring for the mould into which the metal is poured. However, the increasing preference for using aluminium alloy, rather than grey cast iron, for the cylinder block and crankcase of modern lighter weight engines, has led to their manufacture by high-pressure die casting in the interests of economical mass production. Since this method of casting necessarily

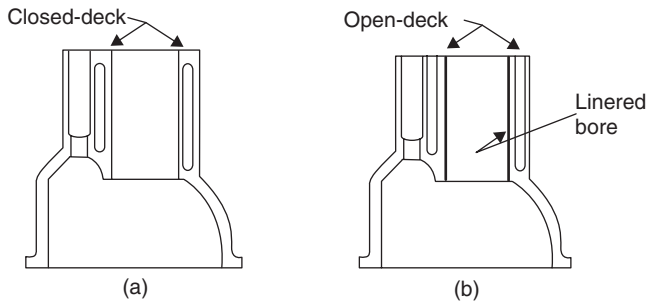


Figure 1.27 Cylinder block construction: (a) closed deck (b) open-deck

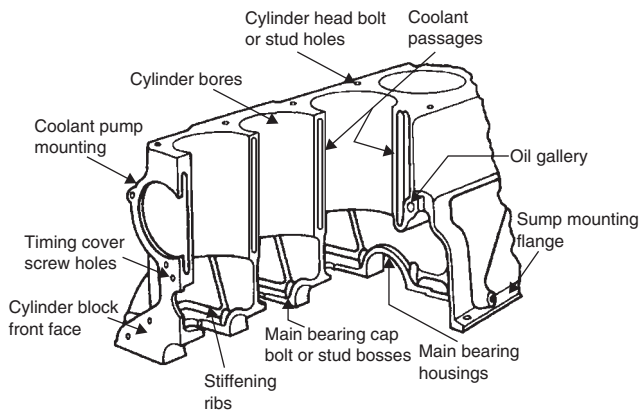


Figure 1.28 Cut-away view of cylinder block

involves the use of steel instead of sand moulds, the need for an open-deck construction to allow withdrawal of the steel cores becomes mandatory. Also, the liners may be cast directly into the cylinder block to restore structural rigidity. An open-deck construction further allows inspection of the coolant jacket for accumulated deposits. To perform this operation in a closed-deck cylinder block requires the addition of detachable cover plates.

In both closed-deck and open-deck cylinder blocks numerous internal supporting bosses must be provided for either the studs, or the setbolts, which clamp the cylinder head to the block. Wherever practicable, these bosses are aligned with the crankcase bulkheads that support the main bearings. This is to secure a direct path of load transference between the cylinder head and main bearing caps and thus minimize bending stresses within the cylinder block and crankcase structure (Figure 1.29). A noteworthy modern example of engine construction where this principle is carried to its logical conclusion can be found in the K Series engine originally developed by the Rover Group (Figure 1.30). In this design ten long bolts pass down through the cylinder block so that they clamp together in sandwich fashion the cylinder head, cylinder block, main bearing ladder and main bearing oil feed rail; all of these layers are located relative to one another by tubular dowels. This type of through-bolt construction therefore makes for a very even distribution of clamping load virtually from top to

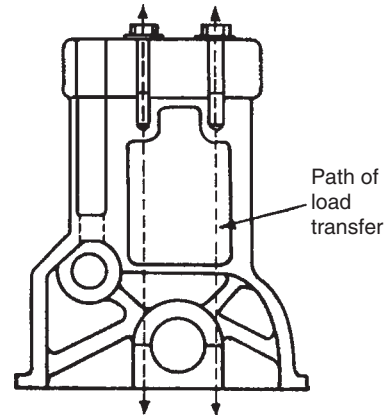


Figure 1.29 Direct load transference between the studs of the cylinder head and main bearing caps

bottom of the engine, which is particularly beneficial in view of its structural material being aluminium alloy.

For closed-deck constructions, the internal supporting bosses for the head studs are arranged symmetrically about the cylinder walls, so that the cylinder head clamping load is as evenly distributed as possible around them. The object is, of course, to avoid any tendency towards cylinder bore distortion, which could lead to blow-by of the combustion gases and increased oil consumption. In the case of open-deck constructions, the supporting bosses for the head studs are usually disposed along the walls of the coolant jacket.

Cylinder bores

Clearly, the cylinder bores constitute the most important feature of the cylinder block. Since they act as a guide and a sealing surface for the sliding piston and rings, their accuracy of machining must be such as to minimize any out-of-roundness and taper effects, and to ensure that they are truly at right angles to both the crankshaft and the top deck of the block.

The cylinder bores must also be given a carefully controlled surface finish, because too rough a surface would cause wear, and too smooth a surface would hinder the running-in process. A suitable surface finish is usually obtained by final honing to give a cross-hatched finish (Figure 1.31), which retains the oil in the bores to lubricate the pistons and so reduces friction losses. The question of the most suitable surface finish for new cylinder bores is one of long standing, and it is perhaps significant that an American engineer once observed that somehow the engine knows how to finish the bore better than we do!

Cylinder bore wear in service may be attributed to various factors, which will be reviewed later in connection with the associated wearing of piston rings.

As the degree of wear varies in different parts of the cylinder, an overall assessment of bore wear is most conveniently made by using a dial gauge equipped with an extension spindle, so that wear readings can be observed outside the cylinder. The gauge is self-aligning in the cylinder bore and in use is rocked over the cylinder axis from side to side, the maximum dial reading corresponding to the diameter being measured.

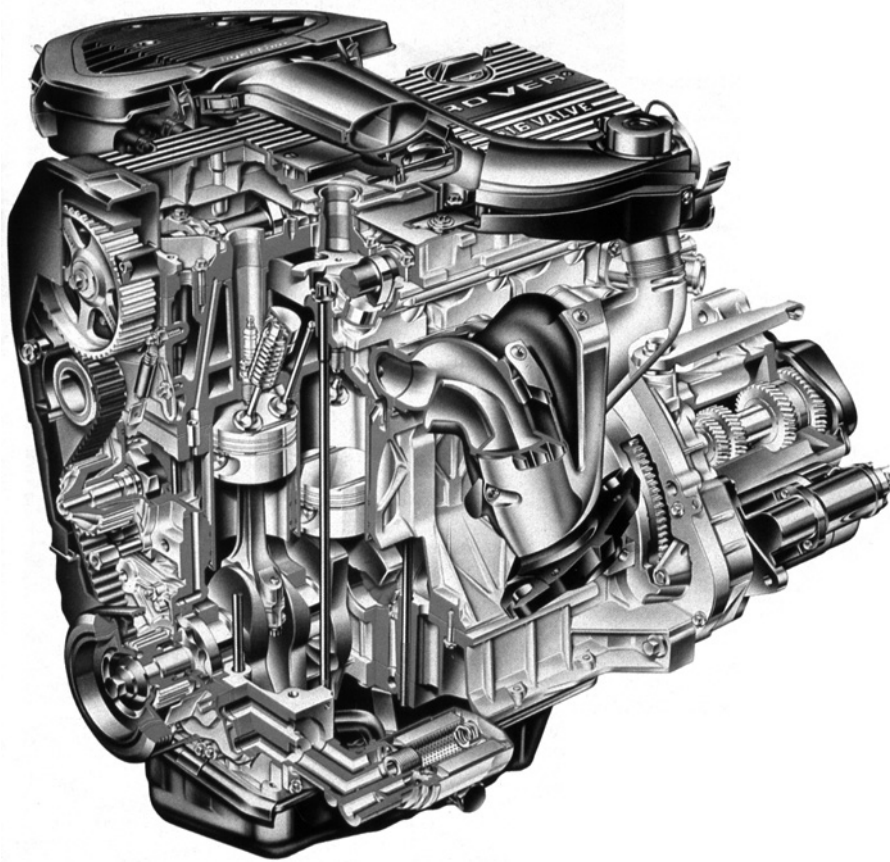


Figure 1.30 A modern example of a through-bolted engine construction

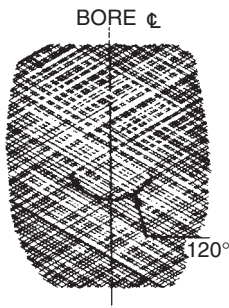


Figure 1.31 Honing pattern for cylinder bore

Crankcase construction

For in-line and V-formation engines, the most commonly used form of crankcase resembles a tunnel structure which extends downwards from the cylinder block (Figure 1.32). The roof is formed by the lower deck of the cylinder block and it is closed off at the base by either a detachable sump or a transmission housing. The crankshaft is underslung in the crankcase and supported by front, intermediate and rear main bearing bulkheads that form a series of crank chambers.

Until recently, it was customary for the side walls or *skirt* of the crankcase to be extended below the axis of the crankshaft, so as to increase the resistance to bending of the structure in the interests of engine smoothness, and also to simplify

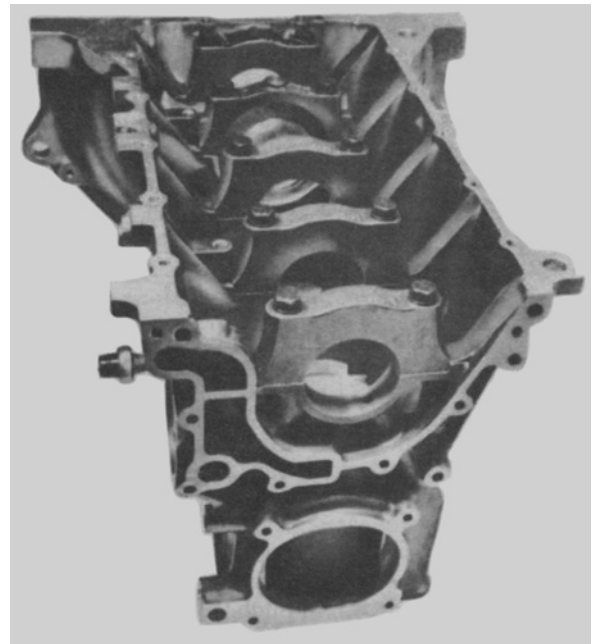


Figure 1.32 Underside view of an aluminium alloy crankcase for a four-cylinder engine with five main bearings (*Chevrolet*)

the attachment of the sump. With the aid of modern computer stress analysis techniques, however, the present trend is to return to the much earlier practice of ending the crankcase side walls at the same level as the crankshaft axis, without sacrificing rigidity and with a consequent saving in weight.

For horizontally opposed engines, the cylinder blocks are again cast integral with the crankcase, which is usually divided on its vertical centre line (Figure 1.33). The crankcase halves are then clamped together by through-bolts on either side of the crankshaft main bearings. A series of bolts is also fitted around the peripheral joint faces of the crankcase, it being usual for the sump to be made integral with each half. Less commonly, a one-piece construction is used with an open underside, detachable main bearing caps and oil sump (Figure 1.33).

Main bearing locations

For in-line and V engines, the upper main bearing halves are carried direct in saddles formed in the crankcase bulkheads, whilst detachable inverted caps of great rigidity accommodate the lower main bearing halves. These bearing caps are usually recessed into the underside of their respective bulkheads and secured to them by either studs or setbolts, which thus support the maximum combustion loads imposed upon the crankshaft (Figure 1.34a). In some designs where the crankcase-to-sump joint face is at the same level as that of the crankshaft axis, the bearing caps may be located laterally by a pair of dowels, since it is no longer expedient to recess them into the lower face of the crankcase (Figure 1.34b).

Of more recent application is a form of crankcase construction that embodies a one-piece main bearing ladder or deck (Figures 1.30 and 1.35). This in effect integrates the

main bearing caps into a single rigid structural element, rather similar to the bed plate construction found in very large diesel engines.

In some later designs of V eight-cylinder engine with aluminium alloy crankcases, the fastening arrangements for the main bearing caps may be duplicated, so that each cap is retained by four nuts and studs threaded vertically into the cylinder block above (Figure 1.36). As a further contribution to crankcase rigidity and bearing support, transverse bolting for the main bearing caps may also be employed (Figure 1.36), a feature that has previously been found in diesel engine practice (Figure 2.6). To complement these more comprehensive main bearing cap fastening arrangements, aircraft style 'bihexagon' or 'twelve-point' headed bolt and nut forms may

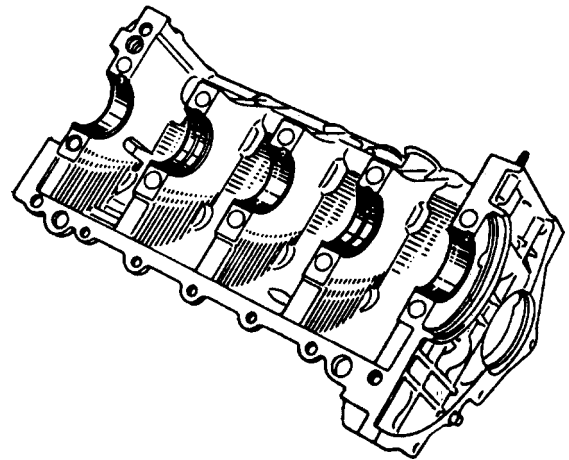


Figure 1.35 A one-piece main bearing deck (*Renault*)

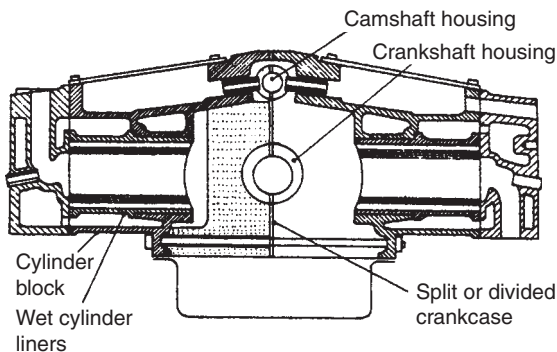


Figure 1.33 Horizontally opposed cylinders with divided crankcase

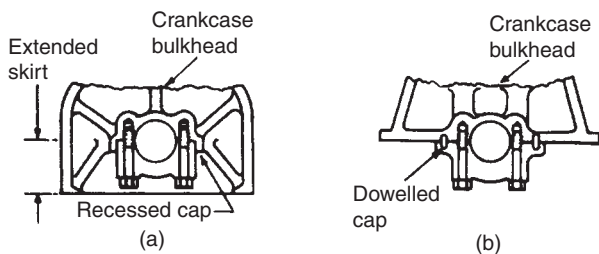


Figure 1.34 Crankcase bulkhead and main bearing cap locations

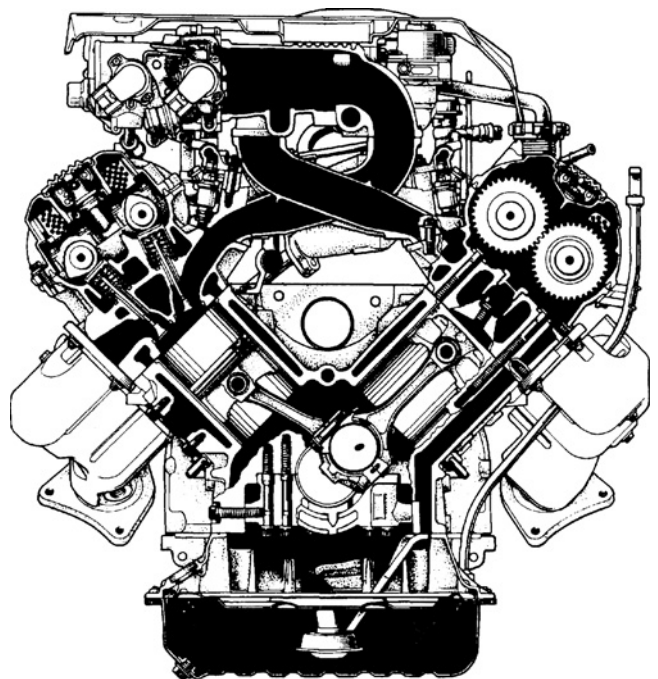


Figure 1.36 Vertical and transverse fastening for the main bearing caps in a high-grade V eight-cylinder engine (*Lexus/Toyota*)

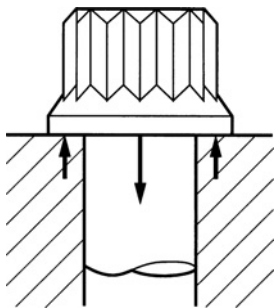


Figure 1.37 Bihexagon or twelve-point headed bolt

be specified (Figures 1.36 and 1.37). The main advantage of the bihexagon pattern is that for a given thread size it saves weight, because its head diameter is much smaller than a corresponding hexagon head, whilst its integral washer face restores an equivalent clamping area. Comparable torque tightening values can still be applied, since the increased number of socket-engaging flats compensate for the reduced turning radius of the bolt head or nut. This form of screw fastener is finding increasing favour among engine designers (Figure 1.30).

With horizontally opposed engines, complementary saddles for the main bearing half-liners are machined in each half of the crankcase. The main bearings cannot, therefore, be changed unless the crankcase halves are taken apart. This drawback of internal inaccessibility has been overcome in a more recent version of the horizontally opposed engine which features a one-piece construction for the cylinders and crankcase, as described in the previous section.

Camshaft bearing locations

A crankcase-mounted camshaft is supported either to one side of, or directly above, the crankshaft, depending on whether an in-line or a V cylinder arrangement is used. With both locations the camshaft bearings are usually carried in webbed extensions of the corresponding main bearing bulkheads. Spanning the underside of the camshaft may be a lubrication trough, which is formed as an integral part of the crankcase. The camshaft followers or tappet barrels slide in bores machined either directly in the material of the crankcase (Figure 1.17), or in bolted-on tappet blocks. In the case of horizontally opposed engines, the bearing bores for a central camshaft are formed similar to those for the main bearings beneath them (Figure 1.33).

Cylinder block and crankcase materials

The combination of cylinder block and crankcase is the single largest and most expensive component of an engine and may be produced from either cast iron or aluminium alloy.

Cast iron is an alloy of iron and carbon; there are two general classifications, known as white cast iron and grey cast iron. Although both varieties have a carbon content between 2.5 and 4.0 per cent, the difference between them is concerned with the condition in which the major portion of the carbon exists in the metal structure. Grey cast iron (so called owing to its grey rather than white appearance when fractured)

is used for cylinder block and crankcase manufacture, because most of the carbon is present as flakes of graphite. This feature not only makes the material more readily machinable, but also provides a satisfactory wear- and corrosion-resistant bearing surface for the cylinder bores. The rigidity of cast iron is such that it exhibits very little tendency towards distortion under the loads and temperatures encountered in the highly stressed engine structure. In addition, it possesses useful sound-damping properties.

Apart from its low cost, an outstanding characteristic of grey cast iron is the ease with which it can be cast into intricate shapes of thin section. Using modern casting techniques, the thickness of the cylinder block walls can therefore be minimized to save weight, which otherwise is the only real disadvantage with the cast iron cylinder block and crankcase.

The term 'aluminium' is generally used to describe not only the very soft and ductile commercially pure variety, but also the numerous aluminium alloys that comprise aluminium with usually more than one element added to it. Only the latter are of interest for cylinder block and crankcase construction, since they can be made harder and more readily machinable than commercial aluminium, which tears badly and poses screw-threading problems. The main attraction of using an aluminium alloy for casting the engine structure is the saving in weight that it affords; its density is about one-third that of cast iron. On the debit side, its strength is about two-thirds that of cast iron. This means that metal sections have to be thickened to compensate for the lower strength, so that in reality the saving in weight is generally nearer to one-half that of the cast iron version.

There are two main classifications for aluminium alloys: those which can be hardened by cold-working processes, and others that can be heat treated to obtain the desired mechanical properties. It is the latter alloys of the aluminium-silicon type that find the widest application for cylinder block and crankcase manufacture, because they retain their strength at moderately high temperatures, possess good casting fluidity and are the most resistant to corrosion.

Although an appreciably lighter construction can be obtained by using aluminium alloy, its wear-resistant properties are less acceptable for the cylinder bores, so that pistons of similar material cannot run directly in them. To overcome this disadvantage, either cast iron cylinder liners, iron-coated pistons or, more recently, silicon-carbide particles dispersed in nickel-plated cylinder bores may be employed. An aluminium alloy engine structure is also less tolerant both of careless handling, especially in respect of screw-thread connections, and of accidental over-heating through loss of coolant.

Historically, aluminium alloys were introduced extensively during the beginning of the aircraft industry. It is perhaps of interest to recall that one of the earliest examples of a cast aluminium crankcase was that used in the engine of the Wright brothers' first aircraft.

Cylinder head construction

In petrol engines the lower deck of the cylinder head contains the cylinder combustion chambers (Figure 1.38). Less commonly, these may be formed either within the piston heads, or by the combination of an inclined top deck for the

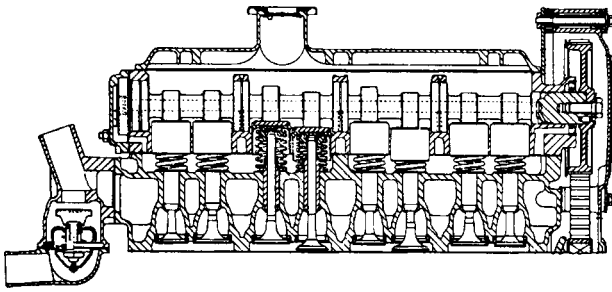


Figure 1.38 Sectional view of a cylinder head (*Fiat*)

cylinder block and specially shaped piston crowns. With the latter arrangements, the valves seat directly in the flat underside of the cylinder head.

The top deck of the cylinder head provides a platform for mounting the overhead valve mechanism and, in the case of some overhead camshaft installations, it may be extended forwards to enclose the upper timing drive system. The walls of the cylinder head form the coolant jacket and provide attachment faces for the intake and exhaust manifolds. An upper continuous flange is formed by the walls for mounting the valve cover and serves to raise the sealing joint face above the level of oil draining from the valve mechanism. The coolant outlet connection at the front part of the cylinder head may also serve as a housing for the thermostat, this being described later. Sandwiched between the upper and lower decks of the cylinder head and surrounded by coolant are the inlet and exhaust valve ports. These are necessarily curved and kept as short as possible, the latter to avoid excessive heat transfer both from the coolant to the induction system and from the exhaust system to the coolant. If one port serves two adjacent cylinders it is said to be *siamesed*.

Valve seats

The valves may seat either directly in the material of the cylinder head or, in harder-wearing rings, inserted therein. Valve seat inserts are usually confined to engines where the cylinder head material is aluminium alloy. Modern valve seat inserts usually take the form of plain rings of greater depth than width, their proportions being such as to confer adequate resistance against distortion (Figure 1.39b). In earlier applications, the more expensive screwed-in type of insert was sometimes used, but it could suffer from inferior heat transfer (Figure 1.39a). For both types it is necessary that they be made an interference fit in the cylinder head. An infrared oven is a feature of the production line for the cylinder head of the Jaguar AJ6 engine, this being used to heat the aluminium alloy cylinder head before the valve seat inserts and valve guides are pressed into place.

Valve seat inserts for aluminium alloy cylinder heads are produced from either a nickel alloy iron casting, or a sintered powder metal the composition of which can be tailor-made to impart the desired hot strength, hardness and corrosion resistance, together with a matching coefficient of expansion. Metal sintering is later explained in Section 1.7 in relation to engine connecting rods.

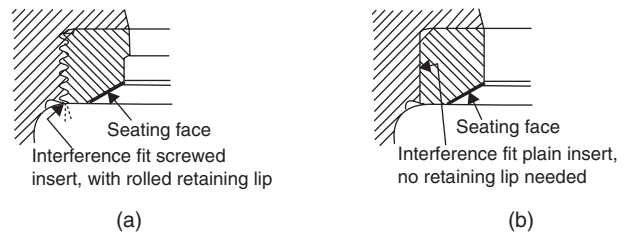


Figure 1.39 Valve seat inserts: (a) early practice (b) later practice

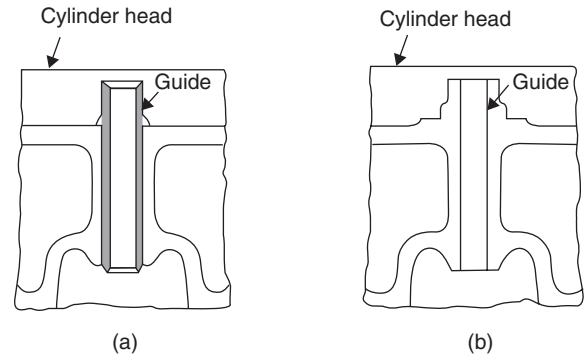


Figure 1.40 Valve guides: (a) removable (b) integral

The use of hard alloy inserts or of local heat treatment to harden the exhaust valve seat areas is also required where cast iron cylinder heads are retained for modern engines designed to run on unleaded petrol. This is because the lubricating effect of lead compounds, formed by the combustion of lead anti-knock additives, previously tended to prevent valve seat recession caused by high temperatures and severe mechanical stress. Although the progressive loss of performance associated with the early stages of exhaust valve recession is hardly likely to be noticed, when the point is reached that valve clearance is lost and one or more valves are prevented from closing properly (Section 1.10), the engine will begin to misfire and potentially serious valve and engine damage can occur.

Valve guides

Coaxial with the valve seatings are the valve guides, which are carried in bosses extending from inside the valve ports to the top deck of the cylinder head. Their length must be such as to present an adequate bearing surface to resist any side loading on the valve stems, and also to provide a ready path of heat transfer from the exhaust valve head. The working surface of the valve guide must possess low friction to minimize resistance to sliding of the valve stem. They are further arranged to project above the level of oil draining from the valve mechanism on to the top deck of the cylinder head.

The valve guides may be either removable from the cylinder head (Figure 1.40a) or, as is sometimes the case in American practice, cast integrally with it (Figure 1.40b). Separate guides are, of course, required with aluminium alloy cylinder heads. Removable guides are always made an interference fit in the cylinder head material, thereby assisting heat transfer from

the valve to the cooling medium, which surrounds their supporting bosses. Grey cast iron is generally the preferred material for the valve guides, although sintered metal may also be used, which is hardened and tempered to match the hardness of the valve stems (as previously mentioned metal sintering is later explained in Section 1.7). It may be of interest to add that cast iron inlet and phosphor-bronze exhaust valve guides were specified at the design stage of the Rolls-Royce V8 engine, the latter material being chosen to avoid valve stem scuffing.

Cylinder head materials

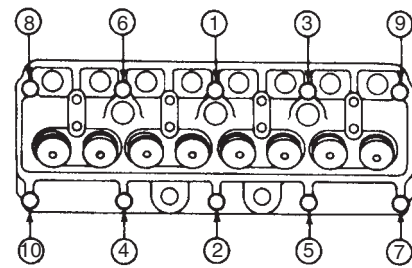
The material used for cylinder head construction is either cast iron or aluminium alloy, as in the case of the cylinder block and crankcase. In favour of the much widely used cast iron cylinder head is its greater rigidity and better noise damping properties. However, apart from effecting a saving in weight, the greater heat conductivity of aluminium alloy is beneficial in maintaining a more uniform temperature throughout the cylinder head. This may sometimes permit the compression ratio of an engine to be raised slightly without incurring detonation or knocking. An aluminium alloy head has often been used in combination with a cast iron block and crankcase, and it is usually considered essential for air-cooled engines to ensure efficient heat transfer.

A service difficulty sometimes encountered with aluminium alloy cylinder heads is that they can prove obstinate to remove if corrosion deposits build up around their fixing studs, and it is for this reason that some designers specify setbolts. Owing to the greater thermal expansion and contraction of an aluminium alloy head, a smoother surface finish is generally specified for its mating face to prevent a ratcheting action against the cylinder head gasket.

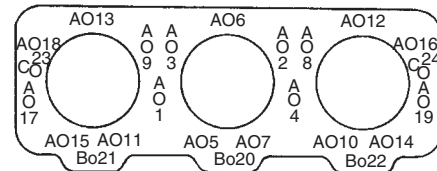
Tightening down the cylinder head

Before carrying out this operation it is always advisable to consult the particular manufacturer's service instructions (Figure 1.41), especially in respect of the following:

- 1 Check whether the screw threads and washer faces require lubrication, and if so the type of lubricant to be used. The point here is that the presence or otherwise of a lubricant on screw-thread assemblies will affect the clamping load they exert for any given value of torque tightness.
- 2 Establish the correct sequence of nut or setbolt tightening, the number of stages in which it is to be achieved and the final torque value to be attained. For improved accuracy in tensioning the cylinder head fastenings a turn-of-nut method (as it is known to structural engineers) may be used in conjunction with initial torque tightening. Following the latter, the setbolts are marked on their heads with paint spots that all face in the same direction; then each nut is given a further specific amount of turn, perhaps in two stages of a quarter-turn each. With the setbolts equally tightened the paint spots will all be facing in the same but of course new direction. Whichever tightening procedure is used, the numerical sequence of tightening specified usually involves starting from the centre and working alternately towards each end.



(a)



(b)

Figure 1.41 Examples of cylinder head nut or bolt tightening sequence: (a) car petrol engine (*Toyota*) (b) commercial vehicle diesel engines (*Gardner*)

Any retightening of the cylinder head after the engine has been run should be done strictly in accordance with the manufacturer's recommendations. Its purpose is to compensate for slight settling of the gasket material after initial exposure to heat and vibration and therefore restore the required sealing pressure. The consequences of incorrect tightening of the cylinder head can prove very expensive to rectify. They may include at least coolant and compression leaks via the head gasket, and at worst the failure of screw-thread fastenings and distortion of the cylinder bores.

Cylinder head gasket

Although the mating faces of the cylinder head and block are machined smooth, flat and parallel, in reality there are always minute surface irregularities and structural deflections to be accommodated. A static seal or head gasket is therefore required, which possesses the necessary degree of both plasticity (pliability) and elasticity (springiness). The effect of tightening down the cylinder head on to the block places the gasket under compression, so that sufficient friction is created between the sealing surfaces to resist extrusion of the gasket by cylinder gas pressure.

However, there is a great deal more to gasket application than this may suggest, because varying amounts of sealing pressure between the cylinder head and block are required at different locations. Sealing against the combustion gases clearly demands far higher unit pressures than are required to seal against the passage of coolant and lubricant through the gasket, while on the other hand very high unit loads are

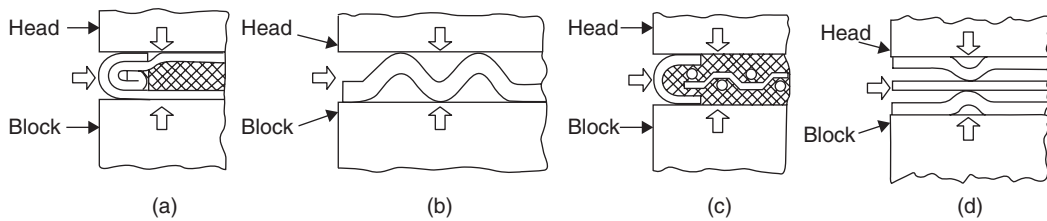


Figure 1.42 Types of cylinder head gasket: (a) metal and asbestos layers (b) embossed steel sheet (c) metal-cored composition materials (d) multi-layer steel

concentrated on those areas immediately surrounding the cylinder head studs or setbolts. The gasket must therefore be designed so that its distribution of sealing pressure conforms to an optimum pattern for any given engine. This requirement has tended to become even more demanding with modern engine design, because not only has there been a trend towards increased peak combustion pressures, but also in the interests of saving weight the structural rigidity of the cylinder block may be compromised, which makes clamping loads more difficult to control.

Apart from maintaining a seal by compensating for the non-uniform loading between the cylinder head and block, the gasket material must also be able to withstand the intense heat of combustion, the penetrating effect of the coolant and the chemical effect of the lubricant. It is for these reasons that the cylinder head gasket has always been regarded as the most critical sealing application on any engine.

Types of cylinder head gasket

In earlier practice, cylinder head gaskets for engines of moderate size and power were traditionally of sandwich construction. They comprised a centre layer of asbestos millboard between two sheets of either copper or tin-plate steel, together with eyelet reinforcements around the combustion chamber openings, which protected the edge of the gasket and increased the sealing pressure in these regions (Figure 1.42a). The embossed steel sheet type of gasket was later introduced for heavier-duty applications, because it better retained cylinder head torque tightening owing to its all-metal construction (Figure 1.42b). A plastics resin coating enabled it to accommodate the minute irregularities of the mating surfaces. By the early 1970s a different type of gasket construction had gained acceptance, which comprised a metal core of tanged or perforated steel sheet faced with fibre composition materials such as asbestos and plastics (Figure 1.42c). In a later version the fibre material facings are bonded to a plain steel sheet to provide a laminated construction, instead of being clinched to the metal core. Eyelet reinforcements around the cylinder bore areas are again used in both forms of composite gasket.

Since exposure to asbestos is now recognized as being a health hazard (and is a topic that will receive further mention in connection with friction materials), the manufacture of non-asbestos cylinder head gaskets has now become established. Gaskets of this type are typically based on a steel core with either aramid fibre, glass fibre or graphite facings with zinc-coated or tin-plated steel beading for protecting their sealing edges.

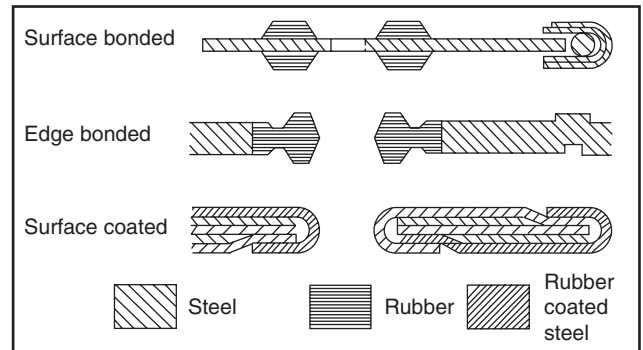


Figure 1.43 Steel and rubber cylinder head gasket constructions (Cooper Payen T&N)

A further development has been the introduction of steel and rubber gasket constructions. Three types of steel and rubber gasket that have been developed by Cooper Payen Ltd are shown in Figure 1.43. In the surface-bonded type a rubber bead is bonded on to a stainless steel core around the oil and water ways. A wire and ring eyelet assembly provides the cylinder bore sealing system. Since the rubber requires relatively little load to seal effectively, the surface-bonded gasket confers effective sealing with moderate overall clamping load. It is specified primarily for petrol engine applications. The edge-bonded type comprises a relatively thick steel plate, its edge being made sufficiently wide to allow the direct bonding of a rubber grommet, which provides sealing against oil and coolant. As this type of gasket is associated with medium to heavy diesel engine applications, the cylinder bore sealing system is generally obtained by deformation of the steel plate between the cylinder head and liner protrusion. The surface-coated type is of laminated construction, which incorporates several laminations of rubber-coated and plain steel, the rubber coating providing interlaminar and surface sealing against oil and coolant. Another feature of this particular construction is that it allows a preferential thickness of the gasket in the cylinder bore area. The various types of elastomer used with these steel and rubber gaskets, in ascending order of temperature requirements, are nitrile butadiene, silicone and fluorocarbon.

Since the late 1990s the multi-layer steel or MLS type of cylinder head gasket has gained widespread application. It is constructed from two or more layers of steel laminate, one or more of which will include embossed sealing beads (Figure 1.42d). The gasket may also receive a special surface treatment. In a typical application for an engine with an aluminium

cylinder block and head, a 0.5 mm (0.02 in) ultra-thin two layer gasket may be specified. The advantages claimed for using this type of cylinder head gasket are those of improved sealing, enhanced reliability and reduced crevice or 'dead' space. This becomes possible because the thinness of the gasket allows it to be made closer fitting around the tops of the cylinder bores. By reducing crevice space here the accumulation of unburned gases and hence their contribution to hydrocarbon emissions is discouraged.

Cylinder head gasket misbehaviour

A cylinder head gasket that is leaking or 'blowing' may do so in various ways, as follows:

Internal leaks Between two adjacent cylinders, as evidenced by weak compression pressure in both; through to the coolant jacket, causing bubbling in the radiator and loss of coolant; and through to the coolant jacket, allowing coolant into the cylinder to cause misfiring and steam issuing from exhaust.

External leaks Outwards from the joint to produce a spitting noise when opening the throttle.

Cylinder liners

During an earlier era of rapid bore wear, cylinder liners were quite widely used either as original equipment or as an overhaul feature. This was because the particular grade of iron from which they were cast centrifugally could be selected for its wear-resistant properties, rather than for the free-flowing characteristics required of an iron for casting the cylinder block and crankcase. However, later developments in the fields of piston ring coatings, lubricating oil formation, oil and air

filtration equipment and cooling system control have all combined to minimize cylinder bore wear, so that the need for detachable liners on this score seldom arises. Cylinder liners are now generally specified either to provide a suitable wear-resistant surface for the cylinders of aluminium alloy engines, or to simplify the production of cast iron engines by permitting an open-deck form of cylinder block.

Dry cylinder liners

A detachable dry liner takes the form of a plain or a flanged sleeve (Figure 1.44a), the entire outer wall of which is maintained in intimate metal-to-metal contact with the cylinder block. This is of closed-deck construction and may be of either cast iron or, less commonly, aluminium alloy.

Non-detachable dry liners have been cast integrally with aluminium alloy cylinder blocks of both closed- and open-deck constructions (Figure 1.44b). In a recent V eight-cylinder engine of aluminium alloy construction produced by General Motors, the cast iron liners are retained in their respective positions while the molten alloy is injected into the die cavity surrounding them. Modern cast-in iron liners have a wall thickness that is typically in the region of 3 mm (0.12 in).

Dry liners generally contribute to the rigidity of the cylinder block, but tend to introduce a barrier to heat flow at the adjoining surfaces. This effect is minimized where the cylinder block is made from aluminium alloy, as a consequence of its good heat conductivity.

Wet cylinder liners

The wet type of liner always takes the form of a flanged sleeve, the outer wall of which is largely exposed to the coolant in the cylinder jacket. It may be incorporated in both closed- and open-deck cylinder block constructions. Clearly, the wet cylinder liner is better cooled than the dry type and can more easily be renewed when worn, as will be explained later. It contributes little to the rigidity of the cylinder block, however, and there is always the possibility that coolant leaks into the crankcase may occur.

Two distinct methods of locating wet liners may be used, according to whether they are being installed in closed- or open-deck cylinder blocks, as follows:

Closed-deck The cylinder liner is provided with a top flange only and is suspended through the coolant jacket from where it is clamped between the cylinder head and the upper deck of the cylinder block (Figure 1.45a).

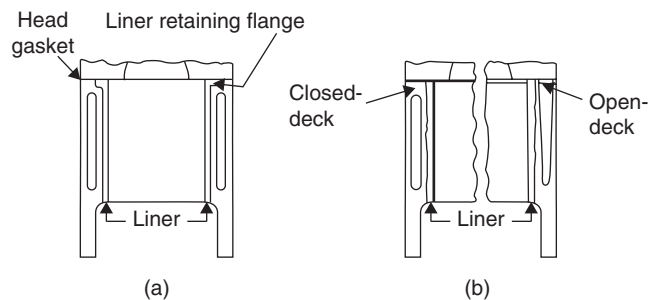


Figure 1.44 Dry cylinder liners

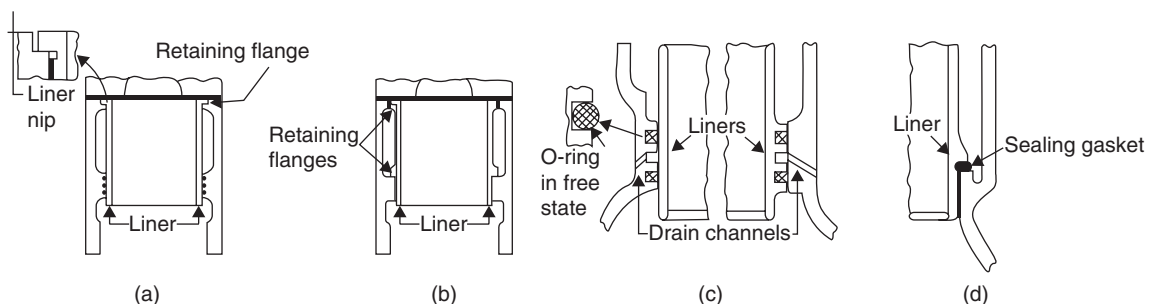


Figure 1.45 Wet cylinder liners in (a) closed-deck and (b) open-deck cylinder blocks. Sealing arrangements for (c) suspended wet liners and (d) wet liner held in compression

Open-deck Here the cylinder liner must be provided with a top and a lower flange and it is held in compression within the coolant jacket between the cylinder head and the lower deck of the cylinder block (Figure 1.45b).

An advantage of the first arrangement is that the cylinder block is relieved of stresses that would otherwise be imposed by the axial expansion of the liner upon heating. With the second arrangement, the less intrusive top flange generally permits better cooling around the upper part of the liner.

For closed-deck cylinder blocks, two oil-resistant (synthetic) rubber O-ring seals encircle the lower part of the liner and are deformed into grooves where it passes freely through the lower deck of the cylinder block. The sealing rings may be grooved into either the cylinder block or the liner itself (Figure 1.45c shows both forms), and a third unfilled groove between them communicates with a drilling in the block that leads to atmosphere. This drilling serves as a drain channel for any coolant and, similarly, oil that may have seeped past the top and bottom sealing rings, respectively.

With open-deck cylinder blocks, a compression sealing gasket is generally used between the flange towards the bottom of the liner and its seating in the lower deck of the block (Figure 1.45d).

In both types of liner installation the cylinder head gasket completes the sealing arrangements for the top end of the liner.

A compromise arrangement of cylinder liner is the so-called 'damp liner', which is part wet and part dry. The upper part of the liner is made thicker in section than the lower part, so that it acts as a wet liner, while the thinner lower part is made a slip fit in the cylinder block and is therefore dry. Since a step is formed between the upper wet and lower dry sections of the liner, it is retained by being nipped between the cylinder head and the abutment in the cylinder block on which the step rests. Although this construction results in a shorter coolant jacket, the direct cooling is nevertheless concentrated where it is most needed in the higher temperature region of the cylinder. There is, in fact, a tendency to shorten the coolant jacket in modern design, because it not only contributes to the rigidity or stiffness of an aluminium alloy cylinder block, but also accelerates the warm-up process to assist emission control.

Cylinder liner installation

It will be evident from the previous descriptions that the dry liner is usually (although not invariably) made an *interference fit* in the cylinder block. Typically, the block is bored out to provide an interference fit of 0.06–0.09 mm (0.0025–0.0035 in) between the cylinder and the liner, which will then need to be lubricated and pressed in under a load of about 2000–3000 kg (2–3 tons) (Figure 1.46). To avoid any possibility of liner bore distortion, the cylinder block studs are usually refitted before the liner bore is honed to final size.

The production method of installing dry liners into the aluminium alloy closed-deck cylinder block of the Jaguar AJ6 engine is of interest, since it does not involve the use of a press. A two-stage infrared oven is used to heat the entire cylinder block for three minutes at a time in each stage, so that when it emerges from the oven the bores have expanded sufficiently for the liners to be slid into position by hand. Immediately all six

liners are in place, the cylinder block enters a special cooling tower where its cooling is rigidly controlled.

In contrast, wet liners are generally made a *slip fit* in the cylinder block (Figure 1.47). A typical cylinder liner-to-block clearance would be 0.05–0.15 mm (0.002–0.006 in). Even so, a manufacturer may recommend that the cylinder block be preheated, so that there is no hindrance to correct insertion of the liners and seals. There is always the danger that if an engine with wet liners is cranked over with the cylinder head removed, the liners could be dragged clear of their locations by the rising pistons. The temporary fitting of retaining clamps on the liners is therefore the safest practice in these circumstances.

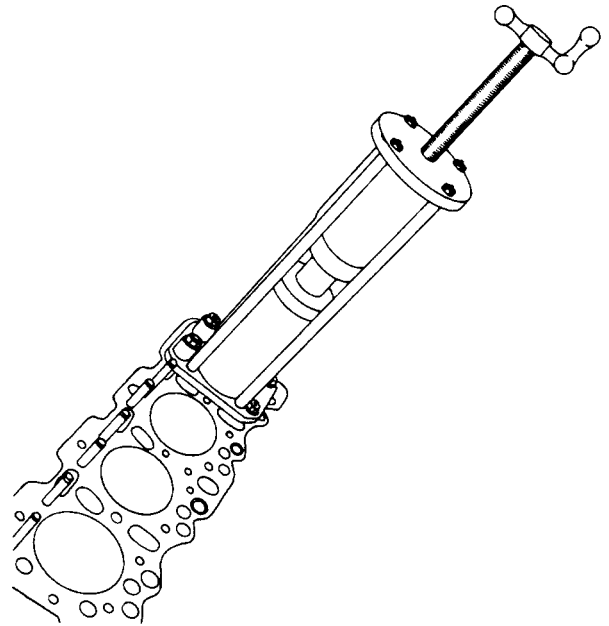


Figure 1.46 Installing a dry cylinder liner with a Flexi-Force hydraulic press (Brown Brothers)

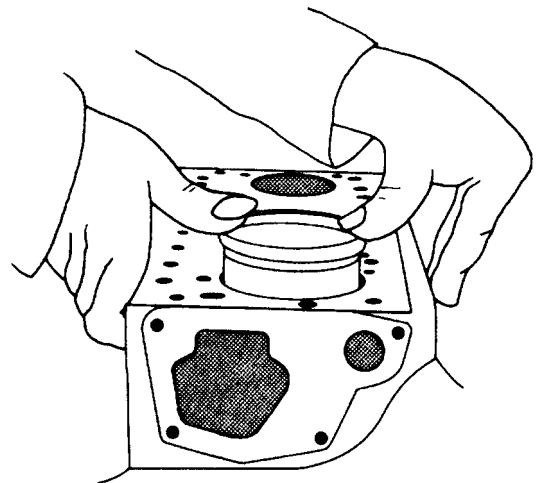


Figure 1.47 Inserting a wet cylinder liner by hand (Perkins)

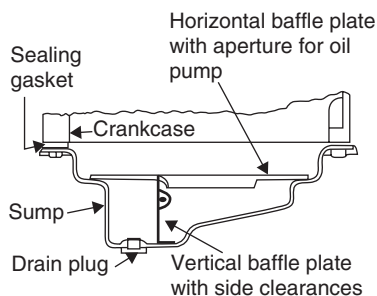


Figure 1.48 Crankcase sump

A final consideration in fitting flanged liners, either wet or dry, is the provision of a small amount of *nip*. This refers to the amount the liner top flange protrudes above the top deck of the cylinder block, so as to promote an efficient gasket seal when the cylinder head is clamped on to it (Figure 1.45a). In practice, a liner nip of 0.05–0.12 mm (0.002–0.005 in) is fairly typical.

Crankcase sump or oil pan

This unit acts as a reservoir to store the oil that is required by the engine lubrication system. It further serves as a vessel in which any sludge, water and metal particles in the oil can settle out, and also provides an opportunity for any entrained air to escape from the oil.

The sump is either of pressed steel construction, or produced from an aluminium alloy. In its latter form it better contributes to the rigidity of the crankcase and also assists with heat dissipation. Attachment of the sump to the crankcase is usually by means of setscrews through mating flanges, between which is sandwiched a flexible packing or gasket.

Baffle plates are normally fitted in the sump to minimize both oil surging and agitation, the former arising from the changing motion of the car and the latter from the oil flung from the crankshaft bearings (Figure 1.48). A screwed plug is incorporated at the lowest point in the sump for draining the oil in service.

In a recent Japanese application the oil sump is of two-piece construction, comprising an upper cast aluminium section that also houses a harmonic balancer system chain driven from the crankshaft, and a lower stamped steel section.

1.7 PISTONS AND CONNECTING RODS

Piston construction and nomenclature

The piston crown may be either flat topped or specially shaped in order to conform to the particular design of combustion chamber of which it forms one wall. Combustion loads are transmitted directly from the crown to the gudgeon pin bosses through intermediate supporting webs, which also facilitate the flow of heat to the encircling piston rings and thence to the cylinder walls. The ring belt immediately below the crown is thus largely relieved of loads that would otherwise tend to deform its grooves. Any closing in of the grooves would, of course, prevent the free radial movement of the rings and thus impair their sealing ability.

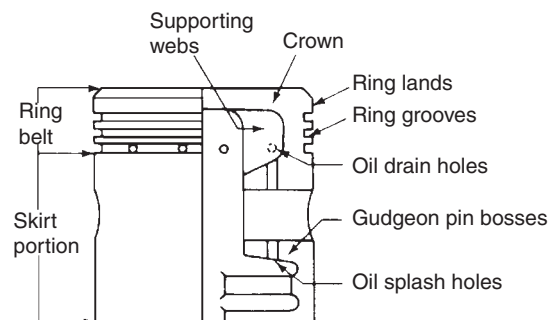


Figure 1.49 Piston construction and nomenclature

The main part of the piston below the ring belt is termed the skirt, and this is made as close fitting as practicable in the cylinder, thereby ensuring quiet operation and the maintenance of the rings at their most favourable attitude to the cylinder wall. Furthermore, the skirt must present an adequate bedding area to the cylinder wall, not only to minimize contact pressure, but also to assist with heat dissipation. It should be appreciated though that the piston skirt is not normally in direct contact with the cylinder wall, but is separated from it by a film of oil. Recessed flats may be provided at the termination of the gudgeon pin holes.

Modern practice favours the use of a *solid skirt* piston of rigid construction, because of the high combustion loads now encountered (Figure 1.49). Basically, its advantages are that it can be made thinner in section to withstand a given loading, so that it affords a saving in weight. It does, however, need a good deal of modification to provide acceptable expansion control of the skirt, as will be explained later. Another type of skirt which was once widely used is that known as the *split skirt* (Figure 1.50a). This incorporated a near-vertical slot extending from the centre of an upper horizontal slot down to the base of the skirt on the non-thrust side of the piston. Here it should be explained that the thrust side of the piston reacts against the side force arising from the angular motion of the connecting rod on the power stroke, while its non-thrust side reacts against the lesser side forces on the compression and exhaust strokes. (The two sides are also known as the major and minor thrust faces.) The split skirt piston was originally introduced to provide quiet running and, by virtue of its skirt flexibility, to accommodate a certain degree of cylinder bore distortion where this was prone to occur.

Piston materials and expansion control

In low-speed engines of early design, the material from which the pistons were made was cast iron to match that of the cylinders. With increasing engine speeds and output, however, it has long since become established practice for the pistons to be cast from aluminium alloy, materials of this type combining lightness in weight with high thermal conductivity. They have a moderate silicon content so that their mechanical strength is better maintained at high operating temperatures, which may now exceed 300°C, whilst their coefficient of thermal expansion is lower than that usually associated with aluminium alloys (Section 2.4). The addition of silicon also increases their resistance to corrosion and wear.

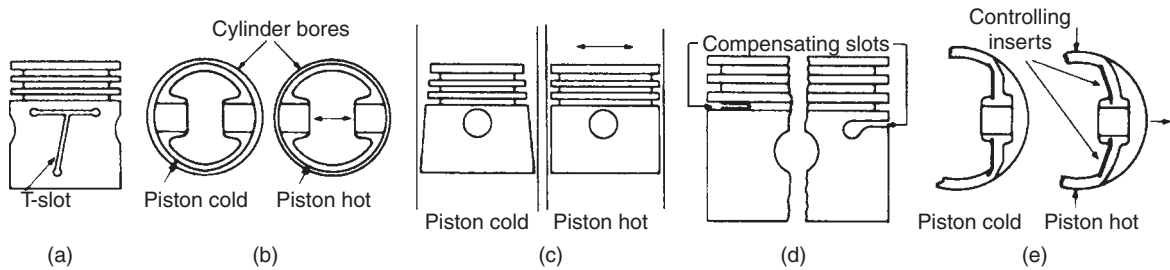


Figure 1.50 Methods of expansion control for pistons. (a) split skirt piston with compensating T-slot. Solid skirt pistons with (b) ovality (c) taper (d) compensating slots (e) controlling inserts

A more recent development has been the *squeeze cast* piston, which was introduced by AE Piston Products in the late 1980s. The squeeze casting technique retains the forming capabilities inherent to the casting process, while demonstrating the strength and integrity of forging. In practice a predetermined amount of molten aluminium is poured into the die, which is then closed and applies pressure to the casting as it solidifies, hence the term squeeze casting.

The associated advantages that accrue from this process include the following:

- 1 More economic use of aluminium, because the risers and runners needed for gravity diecasting are no longer required.
- 2 Porosity is minimized by the application of pressure, which compensates for the normal shrinkage that accompanies cooling and solidification of the molten aluminium.
- 3 Greater refinement of material structure, which arises from the increased heat flow at the interface where the aluminium is squeezed against the die.
- 4 Improved accuracy and surface finish, which again derives from the more intimate contact between the aluminium squeezed against the die.
- 5 Alternative materials can be readily introduced into the casting, such as the expansion control inserts or struts mentioned below and piston ring groove carriers (Section 2.4), with freedom from microporosity and hence improvement in the integrity of the bond.
- 6 A high top ring for better emission control (as explained later) can also be accommodated, by virtue of the enhanced material properties and strength of the squeeze cast piston.

Although the aluminium alloys chosen for the piston material expand less when heated than most other light alloys, they nevertheless expand at nearly twice the rate of the cast iron used for most engine cylinders. In the absence of special expansion control features, this relative difference in expansion rates would result in seizure of the piston when hot, and could be avoided only by tolerating a fitting in the cylinder when cold that would be too slack and would result in piston slap.

A basic method of controlling thermal expansion is to machine the piston to a special form, which is both oval in contour and tapered in profile (Figure 1.50b and 1.50c). The ovality is such that when the piston is cold, the minor axis of the skirt lies in the direction of the gudgeon pin. When the piston is hot the skirt then assumes a circular shape, because

of the greater expansion occurring in the mass of metal comprising the gudgeon pin bosses.

The direction of taper allows for additional clearance when cold near the top of the skirt, since this part ultimately attains a greater running temperature and thus expands more than the cooler running lower portion. Immediately below the ring belt, where the skirt temperature is greatest, the degree of taper may be intensified to produce an overall barrelled shape. Similarly, the clearances around the ring lands are progressively increased towards the piston crown, thereby avoiding contact when hot with the cylinder wall.

In order that the piston can be made as close fitting as possible in the cylinder, either compensating slots or controlling inserts or both may be incorporated in its construction. With split skirt pistons, circumferential expansion of the skirt is simply absorbed by temporary closing in of the near-vertical compensating slot. For solid skirt pistons it is usual for part-circumferential slots to be located within or, in some designs beneath the oil control ring groove (Figure 1.50d). The angular disposition of these horizontal slots is such that the flow of heat from the piston head is diverted from the thrust faces of the skirt, thereby reducing expansion across them.

In modern practice, widespread use continues to be made of controlling inserts or struts, which are stamped from alloy steel and cast parallel with the piston axis and integrally around each gudgeon pin hole. Each insert is pre-heated and placed in the die before pouring; no bond is attempted as they are not intended to add strength to the piston. The inserts are covered with aluminium only on the side away from the piston axis, so that they act in the manner of bimetallic strips when the piston is heated. Since the steel inserts have a lower coefficient of expansion than the aluminium piston, they are compelled to bow outwards and thus restrain expansion of the skirt across the thrust axis (Figure 1.50e). It is perhaps of interest to recall that this principle of controlling thermal expansion of the piston skirt was originally incorporated in the American Nelson-Bohnalite piston of the early 1920s, which was also widely used elsewhere. It was regarded by one prominent automobile engineer of the time as being 'the first piston to have been designed with an intelligent appreciation of the difficulties to be overcome'.

To ensure freedom from piston scuffing on cold starting or hot running conditions, it is important that the surface finish of the piston skirt is oil retentive. Various surface treatments may be applied to aluminium alloy pistons, including tin plating to assist the running-in process, graphite coating to avoid

scuffing under borderline conditions of lubrication, and PTFE coating (Section 13.2) to reduce power losses through friction. Further developments in the surface treatment of pistons for modern high-performance engines include molybdenum spraying of the skirt, and nickel coating of the piston to combat erosion damage where it occurs in engines that are continuously operated close to their knock limit (Section 3.2).

The fitting clearance for a piston must always be sufficient to maintain an oil film during engine operation and thereby prevent scuffing or seizure of the piston in the cylinder bore. It rarely exceeds 0.05 mm (0.002 in) measured at the top of the skirt across the thrust axis, although it is unlikely to be less than this in the case of turbo-charged engines. The fitting clearance should always be in accordance with the manufacturer's specification. If an engine is fitted with wet cylinder liners, it is usual for matched liners and pistons to be made available in service.

Gudgeon pins and their location

The gudgeon pin is the vital mechanical link that hinges the piston to the connecting rod and, although it is of deceptively simple appearance, it must be recognized as being a precision engineered component. This is because it has to satisfy several conflicting requirements; namely, it must combine strength with lightness, be close fitting but with freedom to move, and resist wear without promoting scuffing. The gudgeon pin is of hollow construction and typically produced from a fine-grained plain carbon steel with controlled hardenability. It is lapped to a mirror finish of 0.05–0.10 (2–4 μin). The diameter of the gudgeon pin may be up to 40 per cent of the piston diameter, so that maximum bearing pressure in the piston bosses does not exceed 55 MN/m² (8250 lbf/in²). Under load its ovality and longitudinal bending are not expected to exceed 0.025 mm (0.001 in) and 0.075 mm (0.003 in) respectively. The methods used for gudgeon pin location depend on whether the arrangement is a semi-floating or fully floating one.

Semi-floating gudgeon pin

This is held rigidly in the connecting rod eye and oscillates only in the piston bosses (Figure 1.51a). A semi-floating gudgeon pin is matched by grading to give a clearance of 0.0075–0.0125 mm (0.0003–0.0005 in) in the piston pin holes. Current practice is for the gudgeon pin to be retained by an interference fit, rather than by clamping it in a split eye with a pinch-bolt as in earlier designs. The latter method introduced a discontinuity into the connecting rod eye that

could be a source of weakness. Two advantages of the semi-floating arrangement are that there is less length of gudgeon pin subject to bending loads, and it eliminates one potential source of noise that can otherwise develop in the small-end bearing of a fully-floating arrangement.

Fully-floating gudgeon pin

This is able to oscillate not only in the piston bosses, but also in the connecting rod eye that is bushed for the purpose, as described later. In order to prevent it from escaping sideways and contacting the cylinder wall, the fully-floating gudgeon pin must be located axially in the piston bosses. This is because it is free to oscillate both in the connecting rod eye and the piston bosses at normal operating temperature. Location is usually provided by spring retaining rings or circlips, which are expanded into grooves near the outer end of each boss and thus act as removable shoulders (Figure 1.51b). A fully floating gudgeon pin is matched by grading to give a clearance of 0.0025–0.0075 mm (0.0001–0.0003 in) in the piston pin holes.

Offset gudgeon pin

When the power stroke begins, the piston is forced by a combination of gas pressure and connecting rod angularity to move laterally across the cylinder; this effect is more pronounced with shorter centre distances for the connecting rod, as found in modern engines. The manner in which this movement occurs from the minor to the major thrust side can be such as to cause piston knocking. To minimize noise from this source the axis of the gudgeon pin can be slightly offset from that of the piston and in the direction of the major thrust side. The effect is to tilt the piston during the compression stroke, so that contact is first established between the lower part of the skirt and the major thrust side of the cylinder. At the beginning of the power stroke it then only remains for the upper part of the piston to move across the cylinder to establish full skirt contact with the major thrust side, this movement being beneficially damped by the friction of the rings in their grooves. The amount of gudgeon pin offset used is generally in the region of 1.5 mm (0.06 in) but it can also be influenced by piston clearance and a 'listening session' on the engine development test bed. A stiffer design of piston skirt has sometimes been necessary where an offset gudgeon pin is used.

Types of piston ring and nomenclature

There are two basic types of piston ring used in petrol and diesel engines. These are designated compression and oil control rings, and in each case the nomenclature is most conveniently presented by illustration (Figures 1.52 and 1.56).

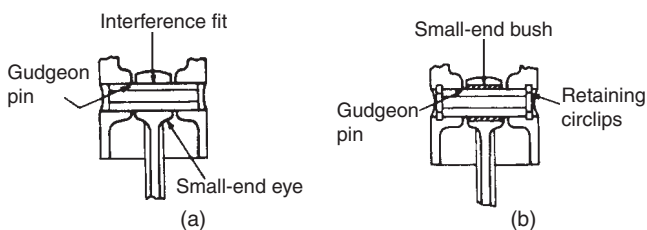


Figure 1.51 Methods of gudgeon pin location: (a) semi-floating (b) fully-floating

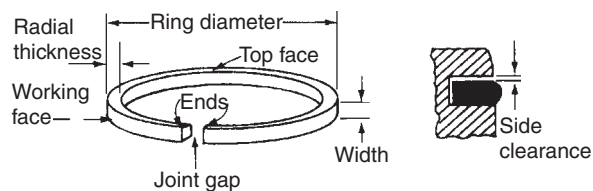


Figure 1.52 Piston ring nomenclature

For modern petrol engines two compression rings and one oil control ring are used on each piston. In earlier practice it was customary for more than three rings to be used. For example, the engines with a fairly low compression ratio used in Rolls-Royce cars of the late 1920s originally had five-ring pistons. The basic reason why fewer rings can now be used is that the higher mean cylinder pressures better augment the sealing action of the rings. Various forms of compression ring may be specified, the differences between them lying in their cross-sectional shapes and the application of wear-resistant surface treatment. Top compression rings are usually of plain rectangular section, their inner and outer edges being slightly chamfered to prevent sticking in the groove (Figure 1.53a). The working surface of the ring may also assume a barrel form, instead of being flat and parallel to the cylinder wall, so that it is better able to accommodate any slight piston rock where the skirt length may be limited (Figure 1.53b).

The second compression ring principally serves to reduce the pressure drop across the top ring, and it can with advantage be made relatively more flexible so as also to assist oil control. Various departures are therefore made from the basic rectangular cross-section, the main object being to compensate for torsional deflection of the ring under combustion pressure, so that top edge contact with the cylinder wall is avoided. Otherwise, the ring tends to pump oil towards the combustion chamber and therefore opposes the action of the oil control ring. To reverse this effect taper-faced and stepped torsional rings are widely used, the two features being combined in some designs (Figures 1.54 and 1.55). Another form of second compression ring with efficient oil control properties is the Napier type, which has a reduced width of taper face with a hook-like undercut below, this being where the bulk of the oil has to be controlled.

The basic slotted type of oil control ring is simply a modification of the plain rectangular-section compression ring. It takes the form of an outward-facing channel section from which collected oil escapes through slots machined radially to the back of the ring (Figure 1.56a). The oil then returns to the sump via communicating holes drilled through the piston wall at the back of the groove. A fairly high radial pressure is exerted against the cylinder wall by virtue of the narrow working surfaces of the ring, so that any tendency for it to ride over the oil film is counteracted.

Oil control requirements have become more exacting for modern engines because the use of high compression ratios increases the depression acting on the rings during the

induction stroke, especially on the overrun with a closed throttle. To improve oil control under these conditions various forms of composite rings are commonly used (Figures 1.56b and c). They generally feature two or more independent working faces, comprising flexible rails with rounded wiping edges, which act in conjunction with a spring expander. This presses the rails axially against the sides of the ring groove and radially against the cylinder wall. Since the flexible rails can act independently of one another, they do not lose their effectiveness in the presence of any piston rock.

Piston ring materials and expansion control

Compression rings and slotted oil control rings are produced from descending grades of cast iron, because this material provides a satisfactory wear-resistant surface and retains its elasticity at high temperatures. Composite oil control rings for petrol engines utilize thin steel rails with chromium-plated

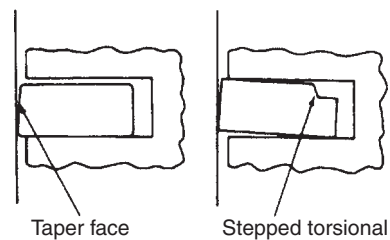


Figure 1.54 Second compression rings

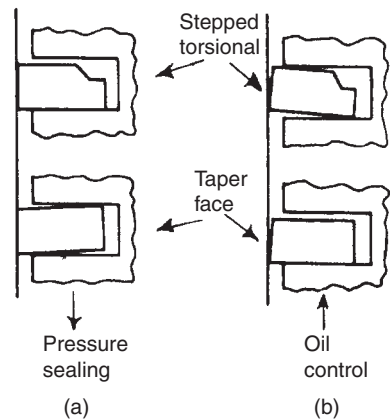


Figure 1.55 Action of second compression rings

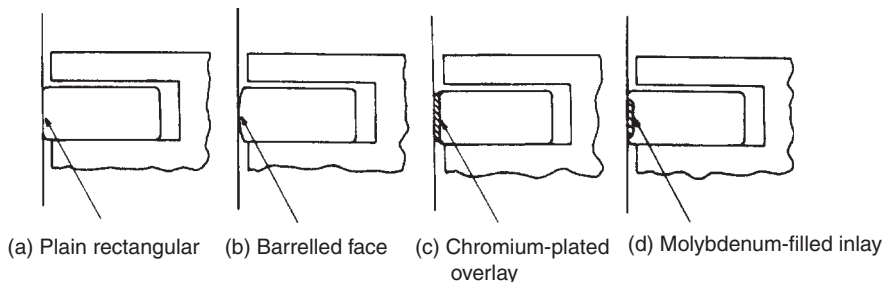


Figure 1.53 Top compression rings

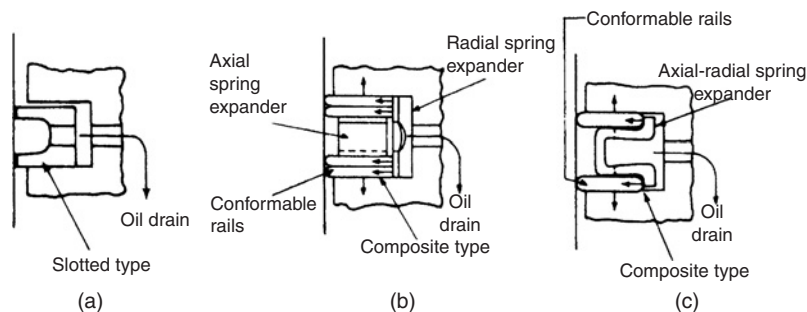


Figure 1.56 Oil control rings: (a) slotted type (b) and (c) composite types

working edges, whilst in the spring-loaded conformable ring used in diesel engines a cast iron slotted ring is expanded by a stainless steel backing spring.

Since the top ring not only is subject to the immediate effects of combustion pressure and temperature, but also must withstand abrasive and corrosive conditions, a wear-resistant coating is now generally applied to its face. This may comprise either a hard chromium-plated overlay or a molybdenum-filled inlay (Figures 1.53c and d). The former treatment was actually first applied to piston rings of military vehicle engines during the African campaign in World War II, and is especially beneficial in resisting abrasive and corrosive wear. Molybdenum inlay was a later development for piston rings, following its introduction on Chevrolet and Cadillac engines in 1963, and is particularly resistant to scuffing wear. Similar benefits are, of course, conferred on the cylinder wall itself.

A piston ring must be provided with a gap at one point on its circumference for three reasons:

- 1 So that it can be expanded over the piston head and then released into its groove.
- 2 To allow it to be compressed into the cylinder and thus exert an initial sealing pressure, which is then greatly augmented by gas pressure when the engine is running.
- 3 To accommodate circumferential expansion of the ring when it is hot.

With regard to point 3, although the coefficients of expansion of the cast iron and alloy steels from which the piston rings are made relate to that of the cylinder material, the circumferential expansion of the rings is greater because they attain a higher mean temperature, otherwise there would be no heat flow from one to the other. It is also for this reason that a piston ring, which maintains a uniformly distributed pressure against the cylinder when cold, is not likely to do so when hot, and this has therefore to be taken into account during manufacture.

The fitting clearances for piston rings are generally as follows:

Piston ring closed gap To compensate for circumferential expansion this should usually be not less than $3/1000$ of the cylinder bore diameter, and is measured with the ring installed in its cylinder. In practice, a gap of between 0.30 and 0.35 mm (0.012 and 0.014 in) is likely to be specified.

Piston ring side clearance To allow for free radial movement of the ring in its groove this should usually be not less than 0.035 mm (0.0015 in) for cylinder diameters of between 50 and 100 mm (2 and 4 in) (Figure 1.52).

Piston ring misbehaviour

This is indicated by loss of cylinder compression, poor oil control and noisy operation of an engine. These complaints can result from wear, sticking or breakage of a ring, although the fault may not necessarily lie with the ring itself. The faults can be summarized as follows:

Abrasive wear This has been much reduced by modern air filtration equipment and closed-circuit crankcase ventilation, which prevent airborne dust particles from entering the cylinders and the crankcase, thereby protecting the compression and oil control rings, respectively. Without some abrasive wear the piston rings would never, of course, bed in at all!

Scuffing wear A definite explanation for this particularly severe type of wear has long been the subject of engineering research (since it can also occur, for example, between gear teeth), but it is generally thought to result from the formation and tearing of tiny welds on the sliding surface of the ring, under the effects of pressure and temperature. If it occurs it does so rather suddenly, typically during the running-in of an engine.

Corrosive wear This is attributed to chemical attack by the products of combustion and occurs when these are able to condense on the ring surfaces at low temperatures. It is now minimized by better regulation of the cooling system, so that the engine warms up more rapidly and takes longer to cool down, and also by applying various surface treatments to the ring face.

Sticking This occurs when the ring groove temperature becomes excessive and causes breakdown of the lubricating oil reaching the ring, so that solid products build up in the groove and prevent free radial movement of the ring. The role played by modern lubricating oils in protecting against this condition will be referred to again later.

Breakage Apart from the breakage that can result from the sticking of a ring, another condition known as ring flutter may cause the breakage of rings in all cylinders if an engine is persistently over-revved. Following experiments made in the late 1940s by Dr P. de K. Dykes at Cambridge University, England, it is now recognized that if the reciprocating inertia

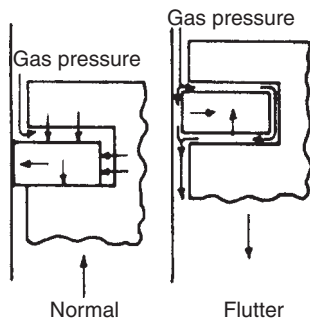


Figure 1.57 Piston ring sealing action

force acting upwards on the ring exceeds the gas pressure that is forcing it downwards and outwards, then during the end of the compression stroke and the beginning of the power stroke the ring will lose contact with the lower side of its groove (Figure 1.57). The consequent release of gas pressure from behind the unseated ring results in it collapsing radially inwards, and this can lead to failure.

Further developments in piston and rings

Since the normal operation of the pistons and their rings accounts for about 75 per cent of the total friction losses in an engine, it is perhaps to be expected that more recent developments have sought to reduce these losses in the interests of improving fuel economy.

To reduce the viscous friction that is caused by shearing of the oil film between the piston and its cylinder during operation, there is a trend towards lessening the contact area of the piston skirt, preferably without shortening the skirt as this could result in undesirable tilting of the piston and unsatisfactory ring behaviour. This conflict has been resolved in the case of the AEconoguide type of piston by specially contouring the skirt so as to form three low-area contact pads on each side.

The advent of direct fuel injection for passenger car petrol engines (Section 6.9) can lead to a requirement for relatively complex shaped piston crowns. Since these tend to raise the centre of gravity of the piston, this could be detrimental to its quiet operation, unless countered by an adequate length of skirt to restore accurate guidance. Hence, the piston can become heavier for this type of application. Also, the presence of a bowl-like depression in the piston crown can lead to a concentration of thermal stresses around the rim of the bowl and any projections therefrom (similar to that encountered in the piston crowns on direct-injection diesel engines). This therefore has to be taken into account during the design and manufacturing processes to avoid crack formation.

A reduction in friction of the piston rings, which receive uncertain lubrication especially when sliding against the upper cylinder wall, can be attributed to the recent Japanese development of the two-ring piston, which carries one plain compression ring and one three-piece oil control ring. Associated advantages include a decrease in piston *compression height*, that is the distance from the gudgeon pin axis to the piston crown, so that not only can the piston be made lighter but also the height of the cylinder block can be reduced. An

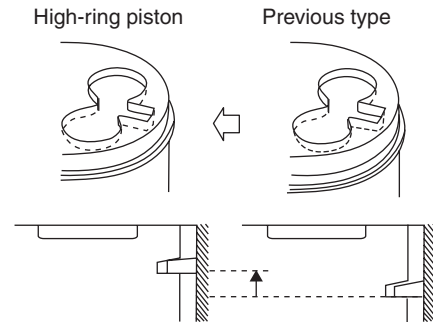


Figure 1.58 Comparison between previous and high-ring pistons (*Toyota*)

improvement in both fuel economy and engine manufacturing costs is therefore made possible.

Other developments in connection with piston rings mainly concern their angular and axial dispositions on the piston and their surface treatment. In the early 1980s a trend began in the use of piston ring grooves that were machined with a very slight upward tilt from the piston axis, thereby creating a similar effect to an internally stepped (or bevelled) piston ring. The purpose of this feature is therefore to ensure that when expansion and bending are taken into account, the ring groove and ring are presented squarely to the cylinder wall and not deflected downwards.

Beginning in the late 1980s it was recognized that by moving the piston rings nearer to the piston crown in both petrol and diesel engines, consistent always with maintaining durability, because it increases mechanical and thermal loading of the top ring groove and ring lands, the efficiency of combustion could be improved for increased power and reduced emissions of hydrocarbons. The reason for this is that the 'dead' space above the top ring and between the piston land and cylinder wall is useless for combustion of the gases compressed therein. Thus, if this space is reduced with what is sometimes called a 'high-ring' piston (Figure 1.58), smaller amounts of unburnt gases are expelled from the exhaust. In a recent example the top ring land height has been optimized at 3.0 mm (0.12 in).

Carbonitriding may be used as a less expensive alternative to chromium plating for cast iron top compression rings. This is a surface hardening treatment that involves the simultaneous diffusion of carbon and nitrogen into the ring material. Top compression rings produced from high alloy steel have also been recently introduced, which can allow ring widths as low as 1 mm (0.04 in) to be achieved, as compared with the more usual 1.5–1.75 mm (0.06–0.07 in) found in modern practice. The advantages associated with this type of compression ring include the following:

- 1 Improved fatigue resistance by virtue of their higher tensile strength.
- 2 Less tendency towards flutter conferred by their reduced inertia.
- 3 Increased conformability to improve control of blow-by and oil consumption.
- 4 Readily incorporated in 'high-ring' pistons for better emission control.

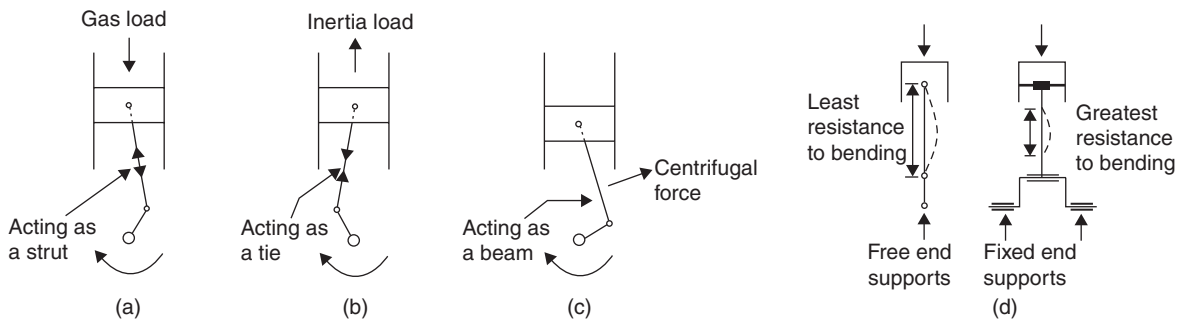


Figure 1.59 Forces acting on the connecting rod: (a) compression (b) tension (c) bending (d) concentration of bending load

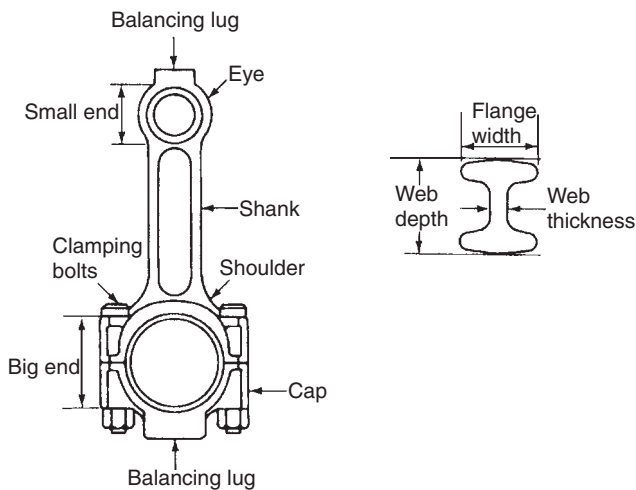


Figure 1.60 Connecting rod assembly

The Nitrolite steel top compression rings produced by Hepworth & Grandage are hardened on all surfaces for reduced side face wear and have smooth blended edges, instead of the chamfers required on chromium-plated rings, to provide improved control of blow-by and oil consumption.

Connecting rods

The nature of the loading on the connecting rod is such that it is subjected to a combination of axial and bending stresses (Figure 1.59), the former arising from reciprocating inertia forces and cylinder gas pressure, and the latter from centrifugal effects. The shank of the connecting rod is provided with an I cross-section for maximum rigidity with minimum weight (Figure 1.60). Since the end supports for the rod are free in the plane of rotation and fixed in the plane containing the crankpin and gudgeon pin axes, the largest dimension of the I-section is disposed in the plane of rotation to resist the greater bending effect therein. The depth of web in the I-section varies in accordance with any taper of the shank, but the flange width and web thickness remain constant over this length. To reduce stress concentrations the big-end arch and the small-end eye are merged very gradually into the shank portion of the connecting rod.

In some designs the big-end bearing parting line is arranged diagonally (Figure 1.61), because otherwise the width of the

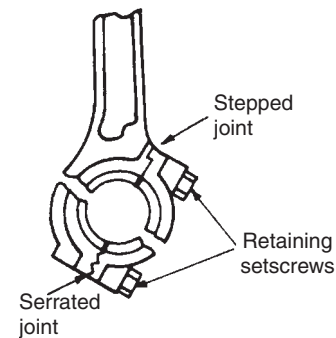


Figure 1.61 Types of angled big end

housing would be such that the connecting rod could not be passed through the cylinder for assembly purposes. To resist the greater tendency for the cap to be displaced sideways relative to the rod, either a serrated or a stepped joint is generally preferred for their mating faces. Hence, the securing setscrews in their clearance holes are relieved of all shear loads. Where the parting line between the rod and cap is arranged at right angles to the axis of the shank, the cap may be secured by either bolts and nuts, studs and nuts, or setscrews. They are produced from high-tensile alloy steel with special care being taken in their detail design to avoid stress-raising corners, which would lower their fatigue resistance. Their clamping load must always be such as to exceed the inertia forces acting on the rod. Reference has already been made to bihexagon or twelve-point headed screw fasteners in Section 1.6 and for the same reason they may be advantageously used to retain the big-end bearing cap (Figures 1.30 and 1.36).

A more recent development in connecting rod fabrication is for a forged sintered metal rod and cap to be produced in one piece and then forcibly separated. Basically, the sintering process involves highly compressing a metal powder, the composition of which can be tailor-made for the application, in a slim I-section mould. Next, the already formed connecting rod is heated to an elevated temperature in a sintering oven, which fuses together the metal particles to complete the process. The manufacturing advantages of what have become known as 'sinter-forged, fracture-split' connecting rods, are generally those of consistency in fabrication, minimal machining requirements and optimum material properties. In production a fully automatic process allows the cap

portion, which is initially defined by a fine groove that corresponds to the parting line, to be split apart from the arch of the rod. This process results in the fractured halves of the rod and cap presenting an ideal mating surface for a perfect fit and improved accuracy of location, as compared to that possible with individually machined parts and, of course, it saves machining time. Although novel to the motor industry in the early 1990s, this type of 'fracture-split' connecting rod construction was originally developed in the late 1960s by the McCulloch Corporation in America, who had investigated various means of improving the structural integrity and fatigue resistance of connecting rods used in their very high speed chain-saw engines.

When a new engine is being developed it is now usual for a component supplier to design the piston and connecting rod as one interacting system, so that the lowest possible reciprocating weight can be attained.

Connecting rod materials

A prime requirement for a connecting rod material is that it should possess a high strength-to-weight ratio. Connecting rods produced from aluminium alloy were used to a limited extent in earlier practice, the primary object being to reduce the loadings on the big-end and main bearings. Also they offered the possibility of being run directly on the crankpins without an intervening separate bearing material. However, their use declined mainly because of an unsatisfactory and unpredictable fatigue life, which made them liable to failure in service. Connecting rods are therefore either forged from a high-strength alloy steel or, as in later American practice, cast from a high-duty iron. With the latter method of manufacture a closer weight tolerance can be maintained than is possible with forgings. However, forged sintered metal connecting rods have more recently found favour for the reason mentioned earlier.

Small-end bearing

Although the gudgeon pin and small-end bearing directly react against the combustion load, they are nevertheless made appreciably smaller in diameter than the crankpin and big-end bearing. This difference in size is not simply to accommodate the small end of the connecting rod within the piston, but can be justified by the small-end bearing benefiting from a much reduced share of the total reciprocating and rotating inertia forces created by the connecting rod and piston.

Where a fully floating type of gudgeon pin is used, a separate bearing bush is pressed into the eye of the connecting rod, its length being such as to limit the maximum bearing pressure to 62 MN/m^2 (9300 lbf/in^2). Since this bearing is difficult to lubricate effectively, because of its oscillating rather than rotating motion, it must possess a high degree of durability. It is now generally of composite construction and comprises a steel backing lined with a hard lead-bronze alloy. The assembly fit of the gudgeon pin in the small-end bush tends to be critical, since too much clearance can produce a small-end tapping noise. Too little clearance when cold can result in the piston being rocked by the angular motion of the connecting rod, thereby causing a temporary piston knocking noise.

Typical pin-to-bush clearances would be $0.005\text{--}0.010 \text{ mm}$ ($0.0002\text{--}0.0004 \text{ in}$) for a high-grade petrol engine and $0.012\text{--}0.030 \text{ mm}$ ($0.0005\text{--}0.0012 \text{ in}$) in the case of a heavy-duty diesel engine.

Big-end bearing and nomenclature

Since the early 1930s it has become established practice for the big-end (and main) bearings to take the form of thin, flexible half-liners, their nomenclature being most conveniently presented by illustration (Figure 1.62). They are of composite construction and consist of a preformed thin steel backing or shell to which is bonded one or more very thin layers of relatively soft bearing material. Multi-layer bearings of this type therefore became known as thin-wall bearings, and were originally introduced by the Cleveland Graphite Bronze Company in America. Before the advent of thin-wall bearings, the housing was often lined direct with the bearing material, which then required boring and hand scraping to achieve a satisfactory fitting. Thin-wall bearings possess several important technical advantages, including a much improved fatigue resistance, a more compact installation and better suitability to mass production requirements. *Fatigue resistance* concerns the ability of the bearing to withstand fluctuating loads at fairly high temperatures, and is the single most important property required of a bearing. It is achieved in the thin-wall bearing because there is less deformation taking place in the much thinner layer of bearing material.

Big-end bearing materials

A heavier-duty bearing material may be selected for the connecting rod big-end bearing than is used for the crankshaft main bearing, because it is subjected to centrifugal loading by the partly rotating motion of the lower part of the connecting rod. This effect is absent in the case of the main bearings supporting a counterbalanced crankshaft. The big-end (and main) bearings must not only possess adequate fatigue resistance, but also provide satisfactory wear qualities. These requirements tend to conflict in practice, since the relatively soft materials that have the best anti-wear properties are

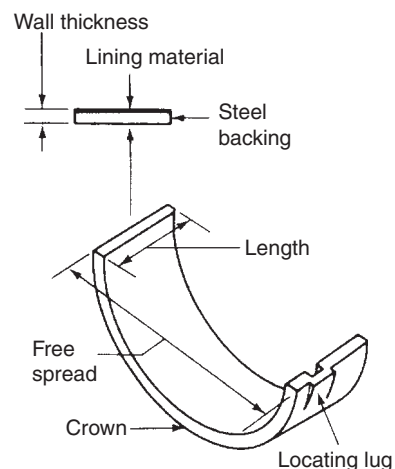


Figure 1.62 Big-end half-liner

usually the least resistant to fatigue. The various bearing lining materials used can be listed as follows:

White metal alloys These are either tin based or lead based and are sometimes referred to as Babbitt metals, although strictly speaking this description should be confined to the tin-based alloy first developed in 1839 by Isaac Babbitt of Massachusetts. They contribute to a low rate of wear on the crankpins because of their good *embeddability*, which means they can readily bury any unfiltered wear particles entering the bearing oil film clearance and thus prevent scoring. The good *conformability* of these alloys also means they can tolerate any slight misalignment and deflections that may affect a bearing. Their load-carrying capacity is, however, limited by a modest fatigue resistance that decreases rapidly with increasing temperatures.

Copper-lead mixtures Harder materials of this type were originally developed to meet the combination of increased loads and higher operating temperatures. They possess a higher fatigue resistance than white metal alloys, but show inferior embeddability and conformability. These disadvantages were largely overcome by plating the lining material with a very thin overlay of a softer lead-based alloy, which also improves its corrosion resistance. It is usual for copper-lead bearings to require surface-hardened crankshaft journals.

Aluminium-tin alloys Bearing alloys containing 20 per cent tin and the balance aluminium, which compared favourably with copper-lead bearings in respect of fatigue resistance, were pioneered by the Glacier Metal Company in the 1950s. Their wear performance is usually such that a plated overlay is seldom required, since they are entirely compatible with both forged steel and the later nodular cast iron crankshafts. Furthermore, their corrosion resistance and thermal conductivity are both high for this type of bearing.

However, in more recent years not only has there been a trend towards reducing the overall length of engines, especially in relation to those transversely mounted, but also power outputs have increased as a result of pressure-charging, multiple valve cylinders and direct fuel injection, all of which have raised combustion pressures. The net effect of these changes has been a trend to shorter and more heavily loaded bearings. To meet this trend aluminium-tin-silicon bearing alloys have been developed to provide improved fatigue strength and anti-seizure characteristics, whilst remaining highly resistant to corrosive attack by the acidic products present in used engine oil. In the case of the aluminium-tin-silicon bearings produced by the Glacier Metal Company, their patented alloy designated Glacier AS124 has a nominal composition of 12 per cent tin and 4 per cent silicon in a matrix of aluminium containing 2 per cent copper.

Location of big-end bearings

Since these bearings are made detachable from the big end of the connecting rod, they must be located in both the rotational and axial senses, as follows:

- 1 To prevent rotational movement, the two half-liners are retained in their housing by an interference fit which, by virtue of maintaining an intimate metal-to-metal contact between liners and housing, also facilitates heat flow from

the bearing. The interference fit is obtained by extending the half-liners a few hundredths of a millimetre beyond their true parting line, so that they are compressed into their housing when the bearing cap is tightened down. This difference in circumferential length between the pair of abutting liners and the closed bore of the big-end housing is known as the bearing *nip* or *crush* (Figure 1.63).

- 2 To prevent axial movement the two half-liners incorporate lugs that register with offset notches in both the connecting rod and its cap (Figure 1.62). The latter is fitted so that its notch faces the same side as the notch in the rod.

Checking bearing liner nip

The following precautions should generally be observed when carrying out this operation, although it is always advisable to consult the particular manufacturer's service instructions:

- 1 Ensure that the bearing seatings are absolutely clean.
- 2 Check that the correct replacement bearing half-liners are being fitted.
- 3 Note that the half-liners have a certain amount of free spread, so that they can be sprung and retained in position during assembly.
- 4 Position the half-liners so that their lugs register correctly with the locating notches in the rod and cap seatings.
- 5 Oil the bearing working surfaces and fit the cap the correct way round.
- 6 Check that the specified bearing nip or crush is present by first tightening the cap nuts or setscrews to the torque value specified by the manufacturer, then slackening one side to finger tightness and inserting a feeler gauge of appropriate thickness between the joint faces of the rod and cap. This procedure may be repeated on the other side. A bearing liner nip in the region of 0.08–0.10 mm (0.003–0.004 in) is fairly typical. Finally, of course, the cap is retightened.

Where a check on the actual bearing clearance is called for, this should usually be about 1/1000 of the crankpin diameter. An insufficient clearance space for the oil film could lead to an excessive rise in bearing operating temperature, which lowers the fatigue resistance of its material. The bearing clearance may conveniently be established by using the proprietary *Plastigage* method. This involves placing cross-wise in the bearing a suitable length of the thread-like plastics material, so that when the bearing cap is replaced and fully tightened down the material flattens out, because it is initially

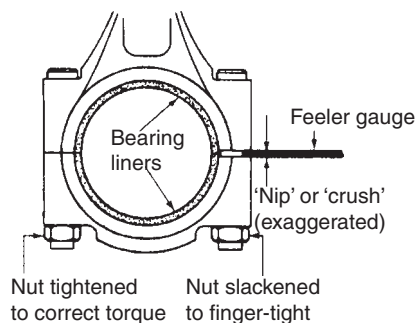


Figure 1.63 Checking bearing liner nip or crush

larger in diameter than the clearance in the bearing. The bearing cap is then removed and the width of the flattened strip of Plastigage is matched against a graduated gauge supplied with the material, so that the corresponding clearance value for the bearing can be checked against that specified by the engine manufacturer. Since the Plastigage material is soluble in oil, care must be taken to wipe clean the bearing surfaces before checking their clearance and, of course, the crankshaft must not be turned during the checking procedure.

Controlling axial movement of the connecting rod

It was once conventional practice to control the axial movement of the connecting rod from the crankpin, so that when the big end came into contact with either shoulder of the crankthrow there still remained a clearance between the small end and the adjacent gudgeon pin boss in the piston. In 1984 the Swedish Volvo company introduced a new method of controlling the axial movement of the connecting rod from the piston instead of the crankshaft, which has now been adopted by other manufacturers in the interests of reducing engine friction losses. Friction is reduced by virtue of the larger-diameter rotating location area being replaced by the smaller-diameter oscillating location area.

1.8 CRANKSHAFT ASSEMBLY AND MAIN BEARINGS

Crankshaft construction and nomenclature

A one-piece, as opposed to a built-up, construction is most commonly used for motor vehicle crankshafts. It consists of a series of crankthrows connected together by the main bearing journals. Each crankthrow is formed by a pair of webs, these being united by the crankpins to which the big ends of the connecting rods are coupled (Figure 1.64). As mentioned in Section 1.5, the angular spacing of the crankthrows is related to engine balance and firing intervals.

The proportions of petrol engine crankshafts are usually such that the crankpin has a diameter of at least 0.60 of the cylinder bore dimension and a length of not less than 0.30 of the pin diameter. Web thickness of the crankthrow is generally in the region of 0.20 of the cylinder bore dimension. The main bearing journal is made larger than that of the crankpin,

with a diameter of up to 0.75 of the cylinder bore dimension and a length of about 0.50 of the journal diameter.

Adequate crankshaft rigidity to resist both bending and twisting is a major requirement for smooth operation. With current short-stroke engines, the proportions of the crankshaft are generally such that in themselves they contribute to greater rigidity. This results from the combination of a smaller crankthrow radius and larger bearing diameters, which permit a beneficial overlap between the main journals and the crankpins (Figure 1.65).

Since the crankshaft is subjected both to bending and to torsional load reversals, it must also be designed to resist failure by fatigue. This condition may be initiated at any point where there is a concentration of stress or, in other words, a heavy loading confined to a very small area. In practice, it may occur at any abrupt change of cross-section, or from the sharp edge of an oil hole or a corner of a keyway. To avoid such stress raisers and therefore extend the fatigue life of the crankshaft, the areas in question are provided with carefully controlled small radii. For example, the corners of each main bearing journal and crankpin may be subject to what is termed 'cold rolling' to a specified fillet radius. This confers a beneficial compressive stress on the crankshaft material. The process of cold rolling basically involves rotating the crankshaft against small hardened steel rollers, which are forced against the corners of the crankshaft journals with a pressure sufficient to cause local plastic deformation and therefore compression of their surface layers. This widely used process actually dates back to 1938, when J.O. Almen of General Motors in America suggested its use to restore the durability of a Chevrolet truck crankshaft following an increase in engine piston stroke.

When the crankshaft is rotating, centrifugal force acting upon each crankthrow and the lower part of its associated connecting rod tends to deflect the crankshaft. Since this deflection is resisted by the main bearings, their loading is correspondingly increased. To reduce these loads, counterbalance weights are either formed integrally with, or separately attached to, the crankthrow webs (Figures 1.66a and b). The former arrangement is now most commonly used, the crankwebs being extended opposite to the crankpin and spread circumferentially.

Crankshaft flywheel

A one-piece construction in cast iron is generally employed for flywheels used in conjunction with friction clutches. For

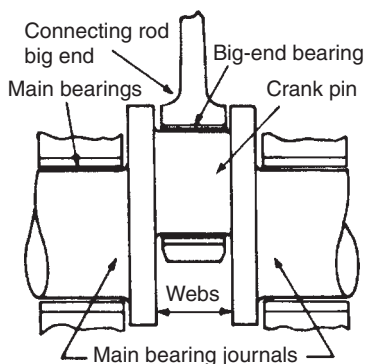


Figure 1.64 Basic crankthrow arrangement

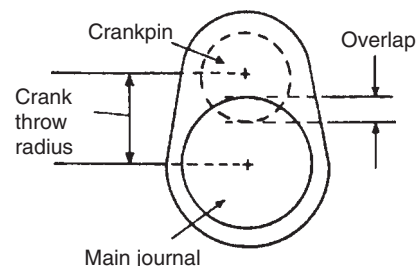


Figure 1.65 Overlap between main journal and crankpin

motor car applications the flywheel is typically of the plain type (Figure 1.67a), although in a few designs a relatively thin disc or *flex-plate* is used to mount a separate cast iron rim (Figure 1.67b). The object of the latter construction is to minimize any disturbance of the flywheel that could result from bending vibration of the crankshaft (Section 1.9) and thus promote smoother running. A similar construction may also be adopted for use with a torque converter automatic transmission, where the flywheel effect derives from the impeller casing, which is mounted from a flex-plate bolted to the end of the crankshaft (Figures 1.67c and 17.6). In the case of some commercial vehicles a pot, instead of a plain, type of flywheel is required to accommodate a heavy-duty, twin-plate, friction clutch (Figures 1.67d and 13.30).

Radial location of the flywheel hub is afforded by a spigot on the rear end of the crankshaft. Owing to its appreciable inertia, the flywheel is located in the rotational sense by dowel pins and clamped firmly to the rear face of the crankshaft by a ring of bolts. The rim of the flywheel provides a mounting

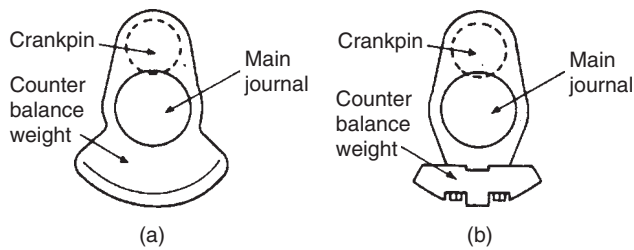


Figure 1.66 Counterbalance weights: (a) integral (b) separately attached

for the starter ring gear and may also bear timing marks for checking the valve and ignition settings, relative to prescribed positions of the crankshaft.

Flywheel with torsional vibration damper

Originally developed in the mid-1980s by Toyota for application to a motor car turbocharged diesel engine, the flywheel with torsional vibration damper or *dual-mass* flywheel as it is now often termed, has in more recent years become increasingly adopted for petrol engines where manufacturers seek additional refinement for the transmission system. The purpose of the dual-mass flywheel is to reduce the extent to which periodic fluctuations in engine torque are passed on to the transmission system, which otherwise create vibration, noise and can lead to wear of components. Typically noticeable with a dual-mass flywheel installation is therefore a reduction in transmission gear noise at low engine speeds. In this context there is a greater opportunity with a modern five-speed and reverse, all-synchromesh, gearbox for light load rattles to occur between the teeth of the more comprehensive train of constant-mesh gears (Section 18.3).

As its name suggests, a dual-mass flywheel basically comprises a two-piece flywheel with an *engine-side mass* and a *transmission-side mass* (Figure 1.68). The latter is supported from the former by an interposed ball-bearing race and its relative oscillatory movements are cushioned by a series of circumferentially spaced compression springs, which are retained in windows shared by the two masses. Frictional resistance to dampen the oscillatory movements between the two masses is supplied in a similar manner to that later

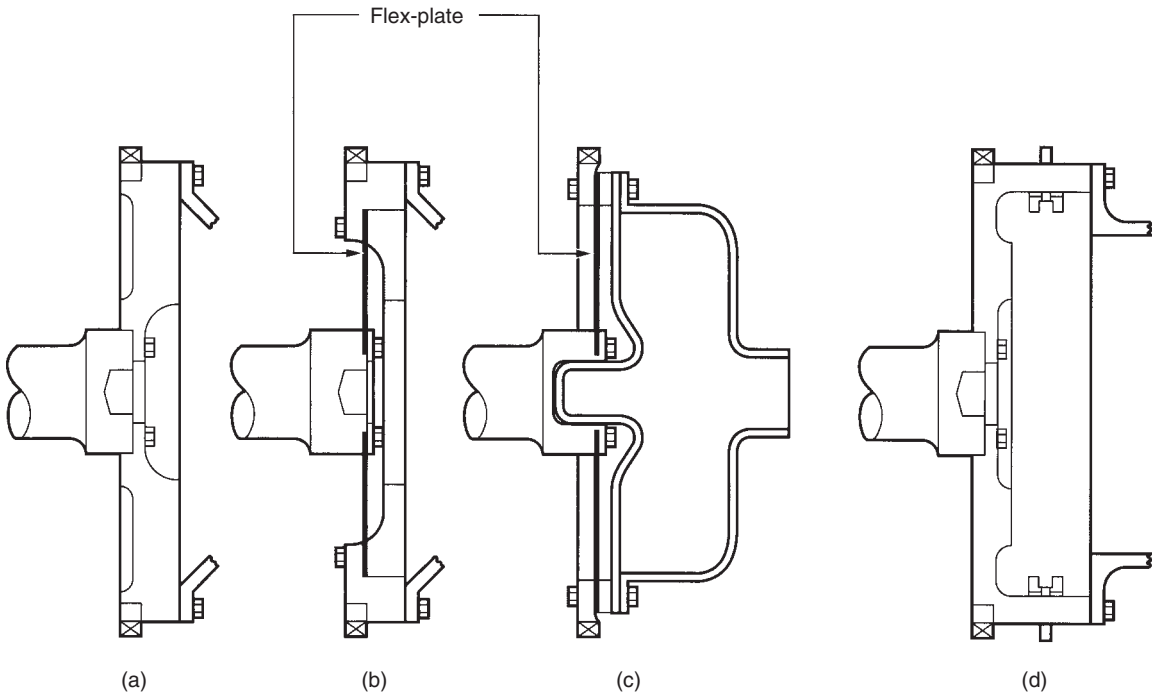


Figure 1.67 Types of engine flywheel

- (a) Plain type for mounting friction clutch
- (b) Flex-plate type for mounting friction clutch
- (c) Flex-plate type for mounting torque converter
- (d) Pot type for mounting heavy-duty friction clutch

described for the centre-plate of a friction clutch (Section 13.3). In an alternative construction hydraulic damping is used between the two masses, fluid-filled chambers with restricted egress and ingress being arranged circumferentially. The changes in volume and restricted fluid movements within the chambers that accompany the relative oscillations between the two masses, therefore, confer the necessary damping properties.

Crankshaft timing wheel and pulley

Ahead of the front main journal of the crankshaft, a cylindrical extension is machined to accept the driving wheel of the timing drive (the various arrangements of which are described at a later stage), an oil flinger (where fitted) and the driving pulley for the belt-driven engine auxiliaries.

These components are made close fitting on the shaft extension or nose and are prevented from turning relative thereto by either a single parallel-faced key, or a series of Woodruff keys, the latter being self-aligning by virtue of their semi-circular form (Figure 1.69).

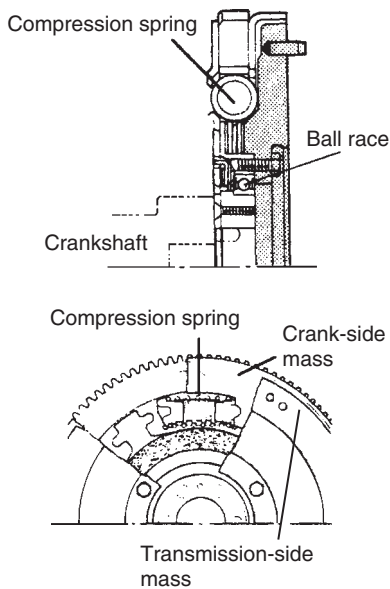


Figure 1.68 Flywheel with torsional damper (Toyota)

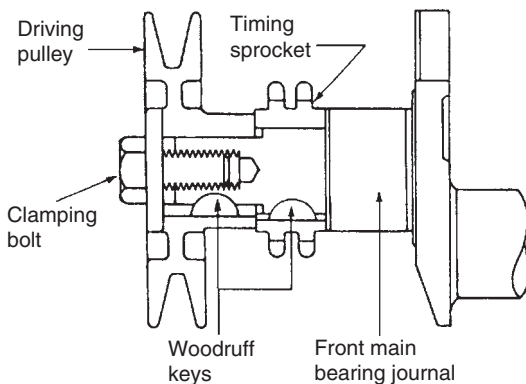


Figure 1.69 Attachment of timing wheel and pulley to nose of crankshaft

The complete assembly of timing wheel, oil flinger and driving pulley are retained endwise on the shaft extension by means of a large setscrew which enters a hole tapped axially in the nose of the shaft and exerts its clamping load via a thick plain washer. In some engines, a driving gear may additionally be sandwiched either between an inner timing wheel and the pulley for a low-mounted oil pump, or between the front main bearing journal and the outer timing wheel for a concentric oil pump (Figure 1.30). In modern engine design a further requirement can be for a crankshaft mounted sprocket wheel, which provides a chain drive to harmonic balancer shafts housed below the crankshaft.

Crankshaft materials

Until the early 1960s petrol engine crankshafts were traditionally forged from high-strength, low-alloy steels, and indeed these are still used for heavy-duty applications. Since then, however, the majority of car manufacturers have favoured the use of crankshafts produced from iron castings of the spheroidal graphite type, or SG iron as it is commonly termed (Figure 1.70).

High-strength cast irons of this type were first developed in the late 1940s, both in Britain and in America, and their distinguishing feature is that the graphite structure takes the form of spheroidal nodules. This feature confers higher strength, better ductility and greater toughness than the flake graphite structure of normal grey cast iron. It is obtained by injecting a trace of magnesium into the iron melt, which causes the graphite flake to gather into little balls or, more technically, spheroidal nodules, that greatly strengthen the grain structure of the material.

The material composition for a high-strength cast iron crankshaft would generally be as follows:

Carbon	3.50–4.20%	Sulphur	0.03% max.
Manganese	0.30–1.00%	Chromium	0.20% max.
Silicon	1.80–2.75%	Nickel	1.00% max.
Phosphorus	0.08% max.	Brinell	
		hardness no.	217–286

The Brinell hardness test, so named after J. A. Brinell who introduced it in 1900, measures the resistance to penetration of a material by a harder one in the form of a steel ball, and gives a useful indication of the tensile strength of the material being tested.

Crankshaft manufacture

A forged crankshaft, like any other forged metal component, is manufactured by a process in which the metal in a more or less plastic, rather than molten, state is forced to flow into the

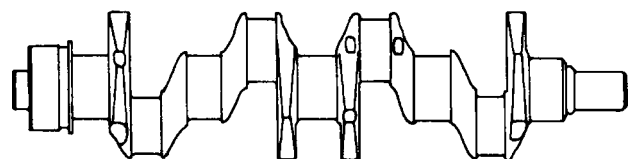


Figure 1.70 General form of a high-strength cast iron crankshaft for a four-cylinder engine

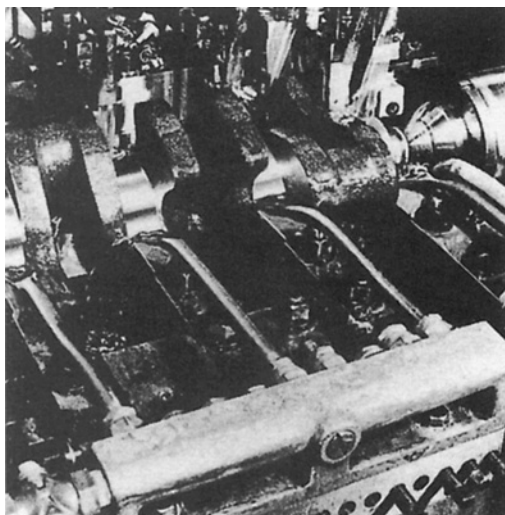
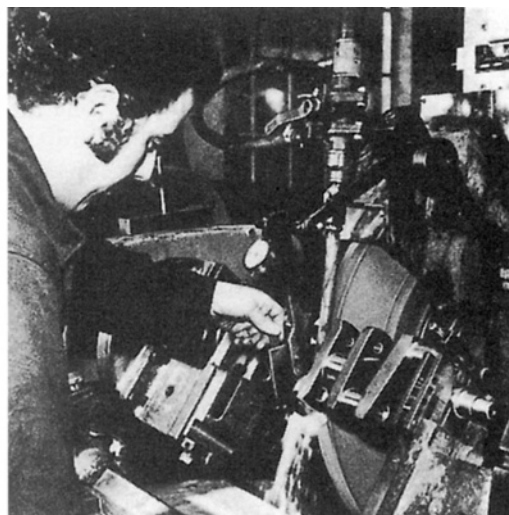


Figure 1.71 Crankshaft turning and grinding (*Laystall*)



desired shape by means of hammering, squeezing and bending. In the case of motor vehicle crankshafts produced in large quantities, the actual shaping is performed by drop-hammer forging in closed dies. The latter are upper and lower blocks of metal in each of which an impression has been formed of the crankshaft.

A cast crankshaft is, in contrast, one that is manufactured by a process in which the metal in a molten state is poured into a mould and allowed to solidify. Motor vehicle crankshafts are not, however, produced in conventional sand moulds, but are cast vertically by the shell moulding process. In this technique, a thin shell-like mould of sand and synthetic resin is made by bringing these materials into contact with a heated metal pattern, the contours of which are exactly reproduced in the shell mould. Two such shells clamped together then form a complete mould.

Among the important advantages offered by the modern shell moulding process is that castings can be produced to much closer tolerances. This in turn reduces the amount of machining required afterwards, and in the case of the crankweb faces can eliminate it altogether.

After heat treatment to remove residual stresses and to give the specified tensile strength of material, usually about 63 kg/mm^2 (40 tons/in^2), the crankshaft must be machined to its final dimensions (Figure 1.71). This involves principally the rough turning, finish grinding and final lapping of the main journals and crankpins. With SG iron crankshafts it is good practice that the bearing journals and pins be final lapped with the crankshaft rotating in the same direction as it does in the engine.

Crankshaft main bearings and nomenclature

In modern high-compression-ratio engines, the number of main bearings employed to support the crankshaft has tended to increase. This is because the crankshaft is subjected to greater bending loads, resulting from the higher peak gas pressures acting upon the pistons. Hence, the crankthrows must receive adequate support from adjacent bearings to

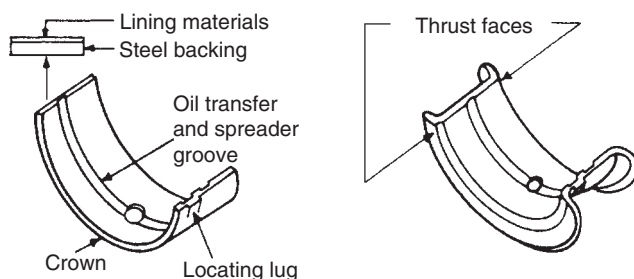


Figure 1.72 Main bearing half-liners

minimize shaft deflections. It is therefore now customary for a main bearing journal to be used at each end of the crankshaft and between each cylinder of in-line engines, and each pair of cylinders in the more compact horizontally opposed and V engines.

The thin-wall main bearings are of similar form to the big-end bearings and must be rigidly supported in the crankcase so as to preserve not only the geometrical truth of their working surfaces but also their correct relative alignment with one another. These requirements must be met to avoid high localized pressures, which may result in destruction of a bearing through breakdown of its oil film with consequent overheating. Thicker than normal main bearing half-liners may be specified for V cylinder engines with aluminium alloy crankcases, so that optimum running clearances are better maintained even though the crankcase and crankshaft differ in their thermal expansion characteristics.

Similarly, the nomenclature of thin-wall main bearings follows that of the big-end bearings with the addition, of course, of an oil groove (Figure 1.72), the purpose of which is explained in Section 4.4. As an alternative to separate crankshaft thrust washers, one pair of the set of main bearing half-liners may be provided with integral flanged ends to serve as thrust faces. Such an arrangement simplifies the production build of an engine but does, however, lack the

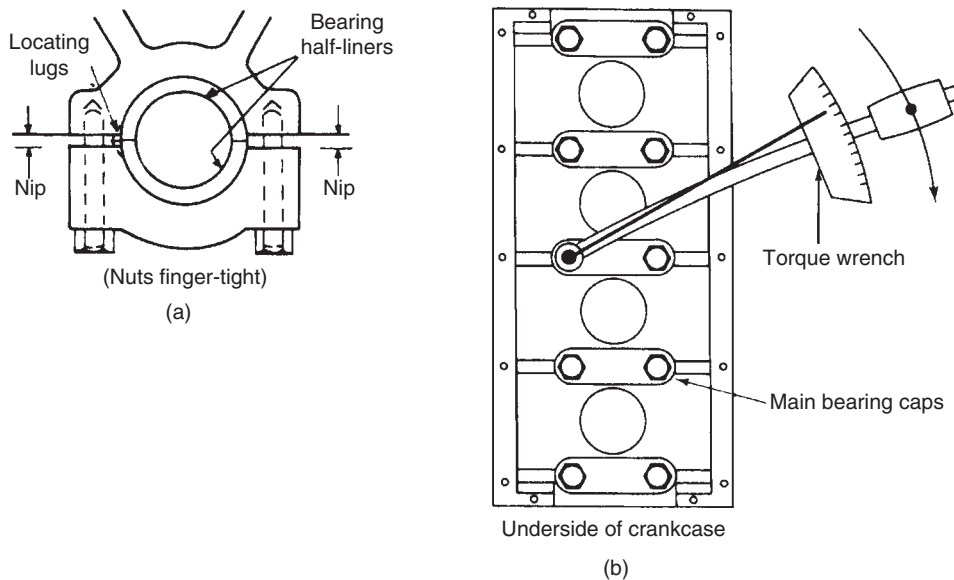


Figure 1.73 Installation of crankshaft: (a) main bearing nip (b) torque tightening the main bearing caps

inherent self-aligning and good heat-conducting properties of separately mounted thrust washers.

Main bearing materials and location

Again, the choice of materials and the methods of location for the main bearing half-liners correspond to those of big-end practice. However, another method of providing lateral location that is sometimes used for heavy-duty main bearings is the dowel-and-hole arrangement. In this method the dowels are fixed securely in the crankcase bearing saddle and cap, and register with circumferentially elongated holes in each half-liner. A disadvantage of this particular method of location is that the presence of the dowel hole disrupts the oil film in the bearing.

Main bearings installation

The fitting precautions to be observed are basically similar to those mentioned earlier for the big-end bearings (Figure 1.73a). It is, of course, necessary to install the crankshaft thrust washers prior to fitting the appropriate main bearing cap, taking care that the oil grooves face the thrust surfaces on the crankshaft. The bearing caps are then torque tightened in sequence to their specified value, starting from the centre and working alternately towards each end (Figure 1.73b). Any undue resistance to rotation of the crankshaft by hand should be checked after finally tightening each bearing cap. The actual main bearing clearances may be checked using the Plastigage method, as previously described for the big-end bearings.

Crankshaft thrust bearings

Before this particular bearing application is discussed, an engineering distinction must be made between two types of bearing. A bearing that is intended to resist a load applied perpendicular to the axis of the shaft is termed a *radial* or

journal bearing. In contrast, a bearing that is intended to resist a load applied along the axis of the shaft is termed an *axial* or *thrust* bearing.

The crankshaft is located axially in the crankcase by plain (as opposed to rolling) thrust bearings which restrain it against endwise movement from loading imposed mainly by the transmission system. This loading may be in a forwards direction during release of a friction clutch and in a rearwards direction when a fluid coupling is in operation (Sections 13 and 15).

Crankshaft thrust bearings are generally of the same composite construction as that employed for the half-liners of the big-end and main bearings, as described earlier, although bronze bearings may also be used. They often take the form of separate semicircular thrust washers, which are installed either in pairs or singly on each side of one of the main bearing housings. Where thrust washers of smaller size are used in pairs, each lower half is keyed to the bearing cap, thus preventing both lower and upper halves from rotating once the cap is fitted (Figure 1.74a). If lower-half thrust washers only are used, rotation is prevented by their upper ends abutting the joint faces of the crankcase bearing saddle (Figure 1.74b). Since the thrust surfaces of the bearings must be separated by an oil film, the washers are provided with grooves or pockets to distribute the oil reaching them from the main bearings they embrace.

To accommodate both the thickness of the oil film and the thermal expansion of the parts concerned, a small clearance is provided on assembly between the thrust bearing surfaces and the adjacent contact faces of the crankshaft. This clearance must, of course, always be less than that existing between the crankweb faces and the sides of all the other main bearings, in order to relieve them of any thrust loads. The required clearance typically falls within the range 0.05–0.15 mm (0.002–0.006 in) and is termed the *crankshaft end float*.

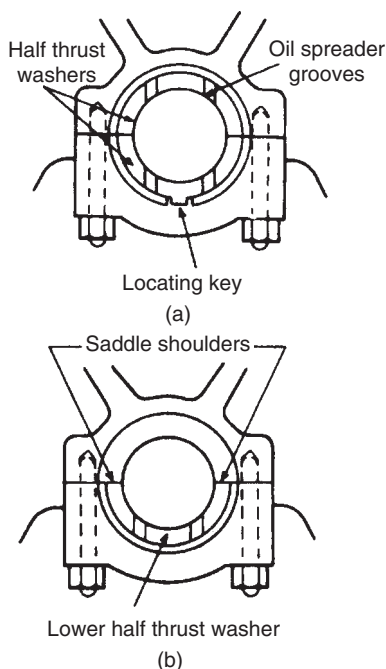


Figure 1.74 Crankshaft thrust bearings: (a) paired (b) single

Crankshaft balancing

In Section 1.5 our attention was directed in simple terms to those factors which make for acceptable overall balance of a reciprocating piston engine. Similarly, we have now to consider the basic principles involved in balancing the crankshaft itself, which compared with such items as the cooling fan and the flywheel is the most difficult part of an engine to correct for imbalance. At the heart of crankshaft balancing is the requirement for counterbalance weights. These are incorporated by the engine designer for one or more of the following purposes.

- Static balancing of the rotating parts
- Partial balancing of the reciprocating parts
- Dynamic balancing of the rotating parts
- Reducing bearing pressures.

Static balancing of the rotating parts

As with all rotating parts used in engineering, the unbalanced rotation of a crankshaft becomes more significant as speed increases. Indeed, the centrifugal forces arising from any such imbalance increase with the square of the engine speed – that is, doubling the speed makes the vibration effect four times worse. It is therefore perhaps stating the obvious that a crankshaft, like a road wheel, must at least be in a condition of *static balance*. This is achieved when the centre of gravity lies on the axis of rotation of the crankshaft, which implies that there is an equal distribution of mass about this axis. To verify this in practice requires the crankshaft to be supported with two of its main journals resting on knife edges; then whatever position the crankshaft is rolled to it should show no tendency to oscillate when coming to rest.

Of all the various arrangements of cylinders used in automotive practice only the single and in-line twin ones require

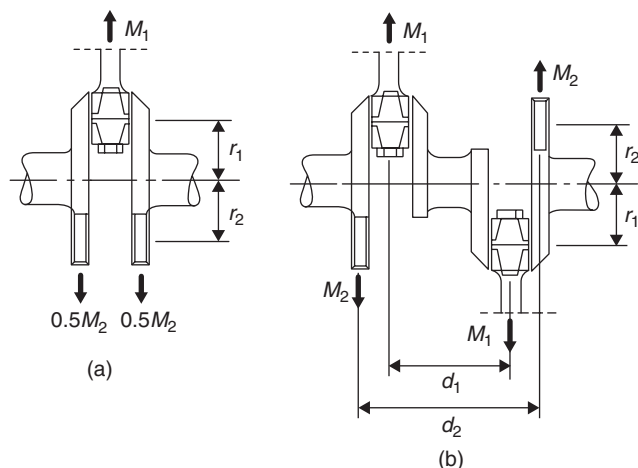


Figure 1.75 Simple examples of force and couple balance in crankthrows:

(a) force balance in single crankthrow when $M_2 = M_1 r_1 / r_2$

(b) force balance in opposed crankthrows when

$$M_1 r_1 d_1 = M_2 r_2 d_2$$

M_1 rotating mass at crankpin

M_2 rotating mass of counterbalance weight

r_1 radius at which M_1 acts

r_2 radius at which M_2 acts

d_1 distance between offset cylinders

d_2 distance between counterbalance weights

the addition of counterbalance weights to correct for inherent lack of crankshaft static balance. This arises simply because the mass of their crankthrows is not distributed symmetrically about the axis of crankshaft rotation. It therefore becomes necessary to provide counterbalance weights on the opposite side of the axis of rotation to the unbalanced crankthrow(s). Since a single counterbalance weight cannot be placed in the same plane as that occupied by the crankpin and connecting rod, the crankwebs on each side are extended in the opposite direction to the crankpin to form a divided counterbalance weight (Figure 1.75a). In actual practice this weight is calculated so as to counterbalance not only the crankthrow but also the lower portion of the connecting rod, which may be considered as revolving with the crankpin.

For all other multicylinder arrangements, where the crankthrows are symmetrically spaced about the crankshaft axis, there is no inherent static imbalance and any correction needed is related to the machining tolerances on the crankshaft assembly. Counterbalance weights may still be required, however, but from other considerations of balance, as we shall presently find.

Partial balancing of the reciprocating parts

A force must be applied to accelerate a piston over the first half of its stroke, and similarly a force is developed by the piston as it decelerates over the second half of its stroke. As explained in Section 1.5, these reciprocating forces are termed primary inertia forces. Again, it is only with single- and in-line twin-cylinder engines that some attempt must be made to compensate for these forces by adding counterbalance weights to the crankshaft, since in all other multicylinder engines they

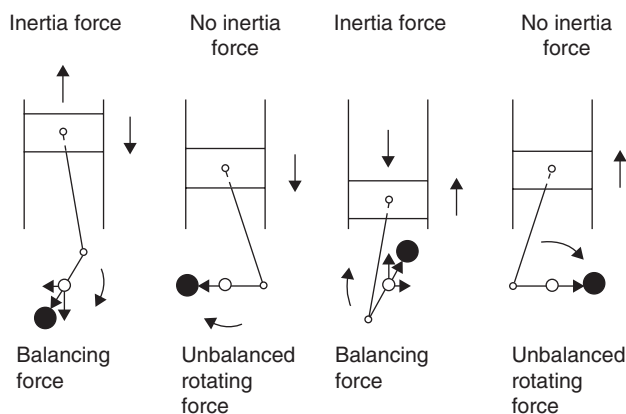


Figure 1.76 Partial balance of reciprocating and rotating parts

are inherently balanced. This is because the arrangement and number of cylinders are so contrived that the primary forces generated in one cylinder are directly in opposition to, and therefore cancelled by, those in another cylinder.

With regard to the rogue single- and in-line twin-cylinder engines, their lack of primary balance can be partially alleviated by increasing the counterbalance weighting referred to in the previous section. The words ‘partially alleviated’ have been deliberately chosen, because a reciprocating mass can never be completely balanced by a rotating mass on the crankshaft owing to the resultant forces acting along different paths (Figure 1.76). Any attempt to counterbalance the reciprocating forces is always made at the expense of introducing an unbalanced rotating force, the centrifugal effect of which becomes the greatest nuisance at right angles to the cylinder axis. A compromise solution is therefore usually sought by increasing the mass of the divided counterbalance weights such that they produce an opposing force equal to one-half the maximum reciprocating force to be cancelled. In other words, the counterbalance weights added to the crankshafts of single- and in-line twin-cylinder engines generally over-correct for static balance and under correct for reciprocating balance.

Dynamic balancing of the rotating parts

Even though a multicylinder engine crankshaft might be in a state of ‘perfect’ static balance, it by no means follows that it is also in a similar state of *dynamic balance*. The reason for this is that in static balancing we are dealing with *forces*, whereas in dynamic balancing we are neutralizing *couples* – that is, opposing forces which are not acting in the same plane. An unbalanced couple can impose a rocking effect on the rotating crankshaft and therefore on the engine as a whole.

A ready impression of the manner in which an unbalanced couple acts may be gained by considering the crankshaft layout of a horizontally opposed twin-cylinder engine. Here the crankthrows lie on opposite sides of the crankshaft rotational axis, so that the symmetrical distribution of their masses allows static balance to be achieved without the addition of counterbalance weights. As a result, however, of each piston connecting to an individual crankthrow, the corresponding cylinders cannot be made truly coaxial. It is the presence of this offset distance between the nearly opposite cylinders which causes

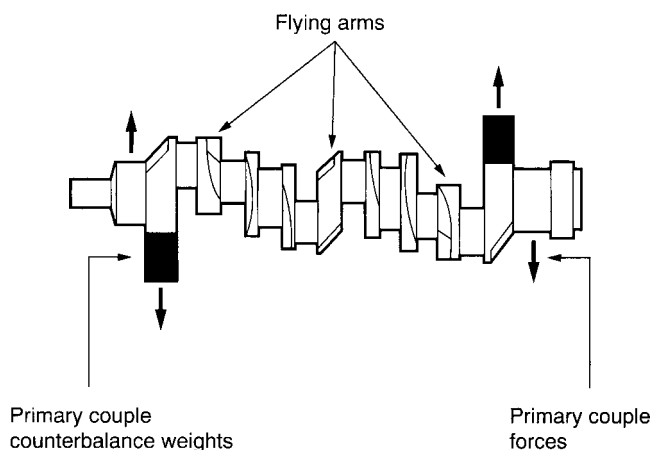


Figure 1.77 Balancing arrangements for a 60° V6 engine crankshaft with four main bearings

the otherwise statically balanced crankthrows to introduce a rocking movement or unbalanced couple on the rotating crankshaft, arising from the centrifugal forces generated.

To attain dynamic balance it is therefore necessary to impose an equal but opposite couple lying in the same plane as the unbalanced one. In practice this can be arranged by extending the end crankwebs in opposite directions to their crankpins to form counterbalance weights (Figure 1.75b). Note that the addition of these counterbalance weights does not destroy the static balance of the crankshaft, so that whereas a statically balanced crankshaft is not necessarily in dynamic balance, a dynamically balanced crankshaft is always in static balance! The requirement for counterbalance weights to achieve dynamic balance of the crankshaft can also arise in other multicylinder engines, notably those with V cylinder arrangements. Taking for example the popular 60° V6 engine, there is an unbalanced primary couple that has to be neutralized by two widely separated and relatively large counterbalance weights, which act in opposition to each other (Figure 1.77). In actual practice a fairly thick counterbalance weight forms part of the foremost web of the crankshaft and a thinner one forms part of the rearmost web, the action of the latter being supplemented by an auxiliary counterbalance weight in the flywheel. A modest unbalanced secondary couple is also present, but this can be comfortably absorbed by the resilience of the engine mountings.

Reducing bearing pressures

This would not be the important balancing consideration that it is but for the fact that a crankshaft cannot be made perfectly rigid. So although a crankshaft might give every appearance of being in ‘perfect’ dynamic balance, the centrifugal forces developed at what may otherwise be inherently balanced crankthrows are still encouraging the shaft to bend as it rotates. In reality, of course, the crankshaft is being restrained from such bending deflection by the main bearings, but in performing this duty it will be clear that their loading and hence working pressures are being increased.

Another reason for adding counterbalance weights to a crankshaft is therefore to counteract bending deflection and thereby reduce bearing pressures. The crankshaft layout of

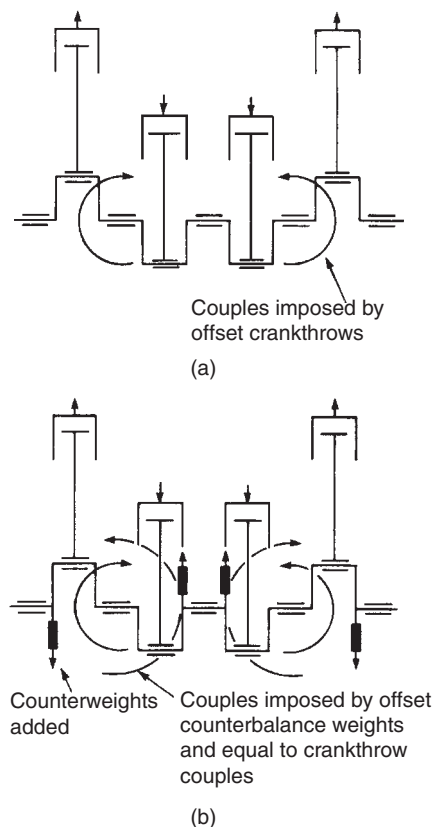


Figure 1.78 Balancing conditions for an in-line four-cylinder engine: (a) dynamic balance (b) dynamic balance with reduced bearing pressures

modern in-line four-cylinder engines affords a good example of this principle being applied in practice. First, it is useful to consider the state of balance of such a crankshaft without the addition of counterbalance weights. Then we find that it possesses inherent static balance, because the two outer crankthrows lie on one side of the crankshaft rotational axis and the two inner ones on the other side. Also we find that although an unbalanced couple acts about the front half of the crankshaft in the same manner as that described earlier for the two crankthrows of a horizontally opposed twin-cylinder engine, this couple is counteracted by a similar one acting about the rear half of the crankshaft (Figure 1.78a). Therefore on the basis of providing a dynamically and by the same token statically balanced crankshaft, the need for counterbalance weight is non-existent; in earlier practice this was how the in-line four-cylinder crankshaft was designed.

With increasing engine speeds, however, the magnitude of the couples acting about each half of the crankshaft, even though self-cancelling, become such that they are trying very much harder to bend the crankshaft. Since this is being resisted by the main bearings, their working pressures are increased. For this reason it is now necessary to provide counterbalance weights for each half of the crankshaft and thereby cancel the individual couples so that bending deflection is minimized (Figure 1.78b).

A disadvantage of adding counterbalance weights to any crankshaft, especially if it is of the longer and more tortuously

shaped six-cylinder kind, is that they lower the resistance to torsional vibration. This topic will be dealt with in Section 1.9.

During manufacture, final imbalance correction in a crankshaft is performed by drilling into the counterbalance weights.

Harmonic balancers

In Section 1.5 it was mentioned that there has been a revival of interest in the use of twin counterbalancing shafts, to cancel out the unbalanced secondary inertia forces of in-line four-cylinder engines. It was the highly esteemed automobile (and aeronautical) pioneer Dr F. W. Lanchester (1869–1946) who first applied this classic solution to the problem when in 1911 he patented what was originally known as an ‘anti-vibrator’, then later as a ‘harmonic balancer’ and in more recent years variously as ‘counterbalancing shafts’ and ‘silent shafts’.

Technically speaking, though, we should perhaps prefer to use the description ‘harmonic balancer’. This is because if we were to consult a treatise on engine balancing, it would become evident that the secondary inertia forces of an in-line four-cylinder engine result from the fact that the common centre of gravity of the four pistons, instead of remaining stationary as would be the case if the connecting rods were infinitely long, has a small harmonic motion of double frequency. It is this unwanted small harmonic motion that must be cancelled or balanced out, and explains the designation ‘harmonic balancer’. The term harmonic may be thought particularly apt, because as in music when two notes sound right and pleasing together (Dr F. W. Lanchester was also an authority on music theory!), so we can contrive agreement between two moving forces.

However, it also seems to follow that a harmonic balancer would be unnecessary if only we could go out and buy ourselves a set of infinitely long connecting rods! Since the notion of infinitely long connecting rods may seem rather obscure to the motor vehicle service engineer, let us now try to understand its significance in simple terms. Referring to Figure 1.79a it will be seen that instead of using a conventional hinged connecting rod between the piston and crank, we have substituted a crank and slotted-bar or ‘Scotch yoke’ mechanism, which was sometimes employed in early steam and pumping engines. Comparing this arrangement with the conventional one shown in Figure 1.79b, it will be noticed that there is no inequality of piston travel towards the dead centres for corresponding angular movements of the crank. In fact by conferring a straight-line, instead of a swinging, motion on the connecting rod we have arrived at the practical equivalent of an infinitely long connecting rod. As a result the piston acquires a simple, as opposed to non-simple, harmonic motion. That is, it now oscillates about its equilibrium or mid-stroke position in such a way that its acceleration (positive or negative) towards this position is directly proportional to its displacement therefrom. So in the case of our in-line four-cylinder engine, the common centre of gravity of the four pistons would remain stationary and there would be no unbalanced secondary inertia forces.

Unfortunately a crank and slotted-bar mechanism would be quite impractical for a high-speed internal combustion engine, because of its unwieldy nature and the difficulty in arranging effective lubrication. We are therefore left with little choice but to use a hinged connecting rod, which imposes

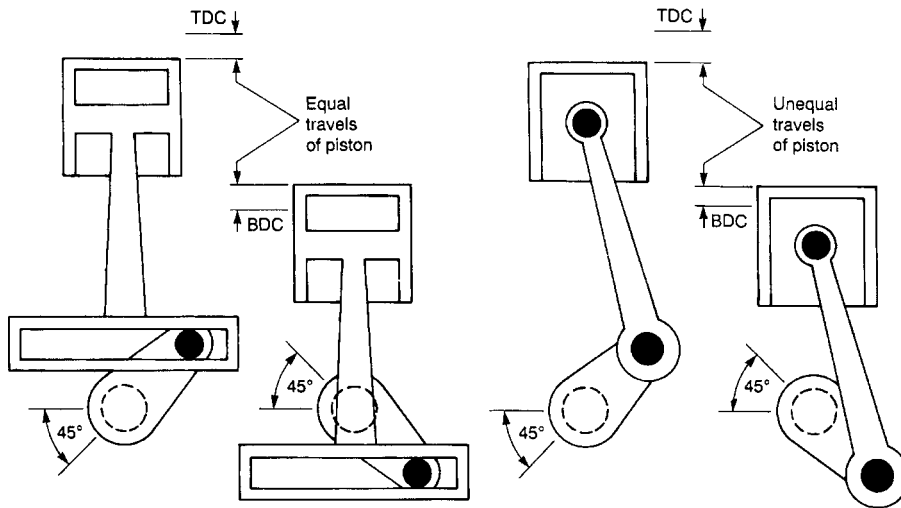


Figure 1.79 Comparison between (a) slotted-bar and (b) hinged connecting-rod and crank mechanisms

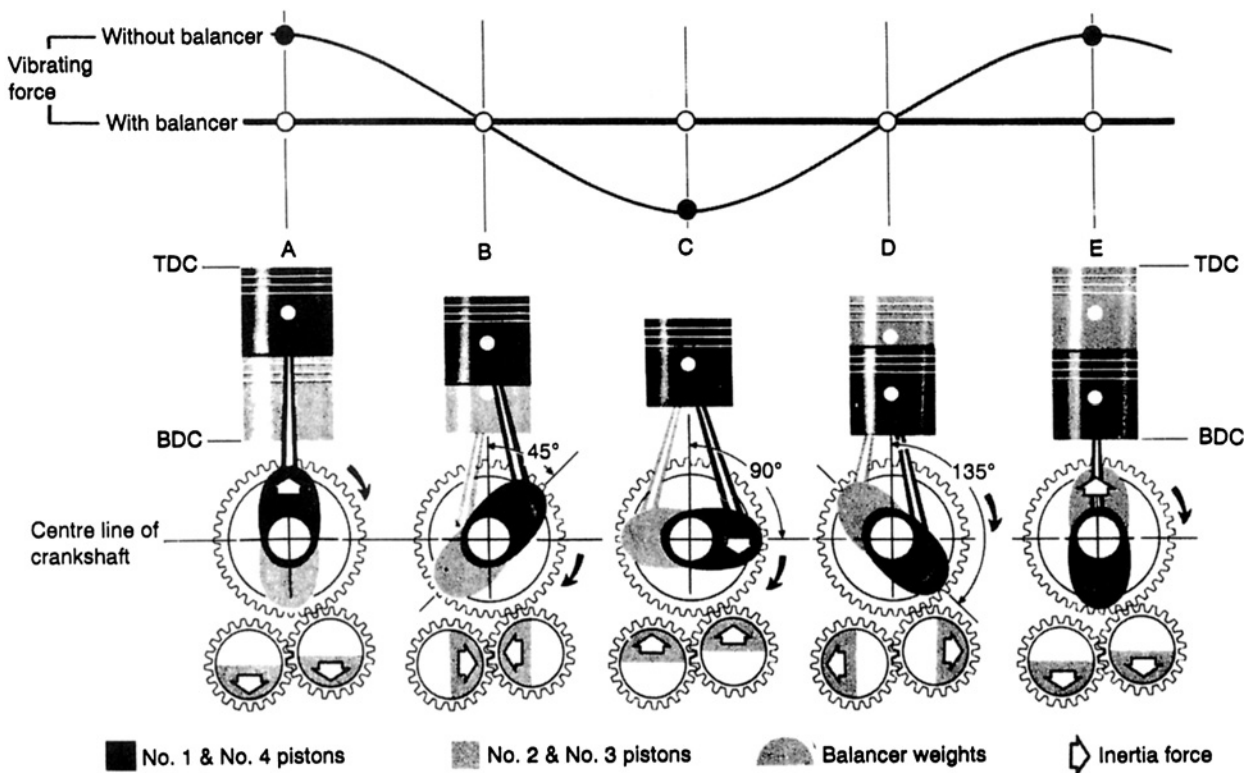


Figure 1.80 Operation of four-cylinder engine harmonic balancer (*International Harvester*)

a non-simple harmonic motion on the piston and is responsible for creating the unbalanced secondary inertia forces with an in-line four-cylinder engine. The purpose of a harmonic balancer is therefore to cancel out these forces by introducing equal and opposite ones.

In Lanchester's original application of the harmonic balancer, two parallel shafts were fitted below and equidistant from the crankshaft axis. These shafts were geared together to run in opposite directions, with one of them being chain driven at twice the crankshaft speed, and they were provided

with bob-weights in or about the plane of symmetry of the engine. The effects of these bob-weights were such that they neutralized each other in respect of their lateral motion, but combined in respect of their vertical motion to give a small harmonic one whose phase was contrary to that created by the error in motion of the pistons. Therefore when the pistons were at their top and bottom dead centres, the bob-weights were always at their lowest position. This sequence of events will be made clear in Figure 1.80, which applies to a heavy-duty diesel engine. In this installation the geared-together

bob-weights are in turn driven from a gear shrunk on to the crankshaft and so positioned that they align with the exact centre of the engine. The balancer gears are matched and timed to each other by suitable markings and also to the crankshaft driving gear. Lubrication for the assembly is provided from the engine main oil gallery.

Despite its ingenuity, few designers of a past era in motor-ing thought it worthwhile to incorporate a harmonic balancer in their four-cylinder engines for passenger cars, their objections being the mainly practical ones of the additional faster working parts and weight and also the theoretical one of creating two otherwise useless lateral forces. It was not until the mid 1970s, when Mitsubishi Motors in Japan introduced their refined version of the harmonic balancer, that interest in its application was revived. Mitsubishi had found that by minimizing engine vibration in this manner, they were able to achieve a reduced level of booming in the body interior and also less intrusive noise at high engine speeds, together with improved durability for the engine auxiliaries and equipment.

The Mitsubishi installation of harmonic balancer, or 'silent shafts' as they call it, is of particular interest because of the way in which the balancer shafts are disposed at different levels on either side of the engine to perform an additional function. Namely, this arrangement provides a vertical separation about which the lateral forces generated by the bob-weight can be utilized to create a resisting couple, which assists in opposing sympathetic rocking vibrations of the engine on its mountings. Both balancer shafts are chain driven at twice engine speed from a sprocket on the nose of the crankshaft. The upper shaft is directly driven from its chain sprocket and therefore rotates in the same direction as the crankshaft, while the lower shaft is indirectly driven from its chain sprocket through a pair of gears, so that its direction of rotation can be opposite to that of the crankshaft (Figure 1.81). Each shaft with its bob-weights positioned in the centre plane of the engine is rigidly supported in front and rear bearings, the latter receiving lubrication under pressure from the former via holes through the centre of the shafts.

1.9 CRANKSHAFT TORSIONAL VIBRATION DAMPERS

The causes of crankshaft vibration

When, say, a heavy and very rugged-looking engine crankshaft is being physically handled in the workshop, the impression given of rigidity is such that the possibility of it being set into a state of vibration might seem unlikely. However, the crankshaft is viewed rather differently by an engine designer, who recognizes that it does in fact behave as an elastic body. This is because in practice the crankshaft cannot be made perfectly rigid, and during engine operation it can be subject to three modes of vibration: torsional, axial and bending.

Torsional vibration

A ready impression of what is meant by torsional vibration may be gained by considering the antics of the simple torsional pendulum used in physics experiments. This piece of apparatus consists essentially of a metal disc suspended in the horizontal plane by a length of wire, the upper end of which is rigidly anchored to a fixed beam (Figure 1.82a). If a disturbing force momentarily turns the disc from its position of rest, then not only will the spring restoring force of the twisted wire return the disc to its original position, but also the inertia of the disc (or natural reluctance to change its state of rotary motion) will cause it to overshoot. This in turn sets up restoring forces in the again twisted wire that cause the process to be repeated in the opposite direction, and so on. The disc will continue to oscillate with diminishing amplitude or rotary movement either side of its original position of rest, until what we can now identify as torsional vibration of the pendulum dies away or decays (of which more will be said later).

If the disturbing force is applied repeatedly, so that the pendulum is excited into a continuous state of torsional vibration, then the amplitude of disc oscillation will depend upon the frequency of application of the disturbing force and the *natural frequency* of vibration of the pendulum. Here it

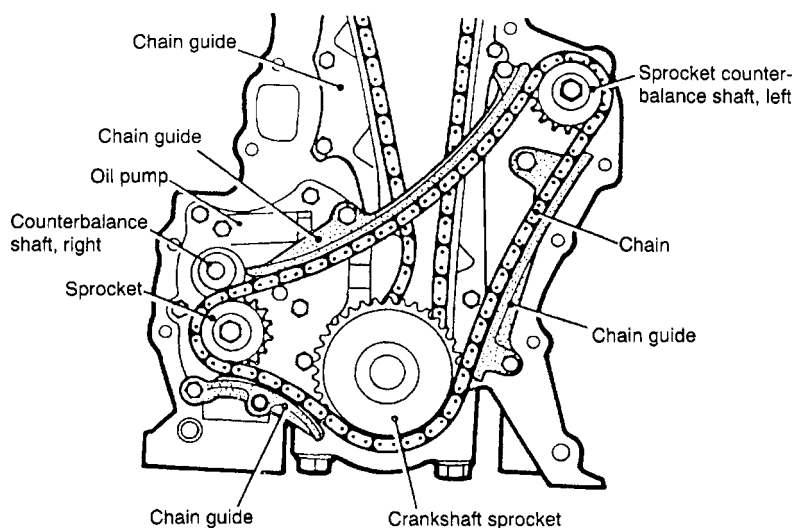


Figure 1.81 Driving arrangements for 'silent shafts' harmonic balancer (Colt-Mitsubishi)

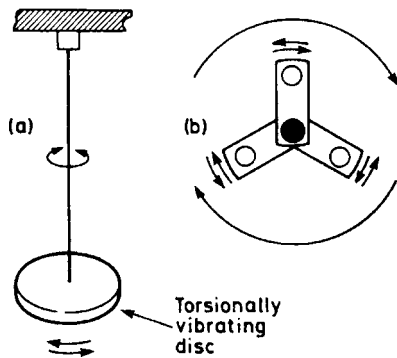


Figure 1.82 Crankshaft torsional vibration: (a) simple torsional pendulum (b) torsional vibration superimposed upon crankshaft rotation

should be appreciated that every body possesses one or more natural frequencies of vibration (the term *frequency* means a certain number of vibrations performed per second) at which it will continue to vibrate when disturbed, until the energy it receives has been dissipated. A few simple experiments conducted with our torsional pendulum would also yield the further useful information that its natural frequency of vibration could be raised by any one or all of several methods. These are to make the wire shorter, to use a thicker wire, and to reduce the weight of the metal disc.

By applying our knowledge of the manner in which the simple torsional pendulum behaves, we are now better able to visualize how torsional vibrations can arise with the much more complex engine crankshaft and how they can be minimized. The disturbing forces applied to the crankshaft are, of course, derived from the pulsating gas and inertia torques acting via each piston, connecting rod and crankthrow combination, as explained in Section 1.4. If the crankshaft were perfectly rigid, then the only effect of these pulsating torques would be to cause some irregularity in its speed of rotation, which could be smoothed out by the action of the flywheel. However, as mentioned earlier, the crankshaft cannot in reality be made perfectly rigid, with the result that the torque pulses are capable of twisting it and therefore of exciting it into a state of torsional vibration. This vibration is superimposed upon the continuous rotation of the shaft (Figure 1.82b).

If the frequency of the disturbing vibrations should coincide with one of the natural frequencies of crankshaft vibration, then a condition known as *resonance* will occur. A danger of resonant vibration is that the energy of the disturbing vibrations may be greater than that lost by the twisting and untwisting of the crankshaft, so that the amplitude of torsional vibration builds up to such a degree that the crankshaft can be over-stressed and eventually suffer a fatigue fracture.

In practice, of course, the design of the crankshaft system is contrived such that its natural frequency of vibration is raised as high as possible. On a similar principle to our simple torsional pendulum, the natural frequency may be raised by making the shaft as short as possible, increasing its diameter and using a lighter flywheel. A resonant or *critical order* vibration would therefore only be expected to occur beyond the normal speed range of the engine, or in other words if the engine is over-revved. However, it may still be necessary in

the interests of both engine smoothness and satisfactory operation of the timing drive to suppress the less critical orders of torsional vibration, which do occur within the normal speed range. For this purpose the crankshaft can be fitted with some form of torsional vibration damper.

Axial and bending vibrations

Axial vibration of a crankshaft is of lesser significance than torsional and bending vibration and is generally regarded as being a by-product of the former. That is, the twisting and untwisting of the crankshaft is accompanied by alternate decreases and restorations in length, which would of course apply to any other body subjected to the same treatment.

Bending vibration of the crankshaft when considered as a beam also results from the pulsating gas and inertia forces imposed upon it via the piston and connecting rod assemblies. As might be expected, this form of vibration is strongly reacted against at the main bearings. In high-compression-ratio engines, the maximum rate of pressure rise during combustion can be such as to cause undesirable vibration of the engine structure and mechanism, especially bending vibration of the crankshaft if it and its bearing supports are not rigid enough.

Types of torsional vibration damper

When describing the vibration of a torsional pendulum, it was implied that in the absence of further disturbing forces the vibration eventually dies out or decays. It does so, of course, because of the natural damping effects of air friction on the surfaces of the oscillating parts and of internal friction in the spring material itself. The internal friction or *damping capacity* of a material is that property which relates to vibration energy being absorbed and subsequently dissipated in the form of heat. Apart from external air and internal material friction there is a further 'apparent' damping effect present in the case of crankshaft operation, this being the energy absorbed and again dissipated as heat by the shearing of the oil films that lubricate the crankshaft, connecting rod and piston bearing surfaces.

Even when the total effect of these natural and apparent damping forces is taken into account, it is by no means sufficient to suppress resonant vibrations of a crankshaft. For this reason it may be necessary to include in the crankshaft assembly a device known as a torsional vibration damper or simply crankshaft damper. The concept of the torsional vibration damper as a means of introducing additional friction to dampen vibration of a crankshaft is quite old; C.H. Bradbury, a one-time leading diesel engine specialist, described it humorously as 'a corrective device for engines designed to run at one speed and sold to run at some other speed'.

Two types of torsional vibration damper may be encountered in petrol engine practice. These are generally classified as the slipper damper that was used in earlier engine designs, and the rubber damper that is in current use.

Construction and operation of the slipper damper

This type of damper is basically a device for separating two masses of different inertia by frictional means, one being the crankshaft and the other usually a pair of small flywheels. The

latter are not keyed to the crankshaft, but are simply spring loaded apart against friction surfaces on the damper hub, which is rigidly attached to the nose of the crankshaft (Figure 1.83a). So although the damper flywheels normally rotate in unison with the crankshaft, it is also possible for them to slip – hence the description slipper damper – relative to the crankshaft, once the friction torque that restrains their angular movement is overcome. Loosely fitting dowel pins are used to prevent relative angular movement between the flywheels themselves, whilst at the same time permitting them axial freedom under the influence of their spring loading.

The concept of the slipper torsional vibration damper is generally credited to Dr F. W. Lanchester, who developed such a device for the Daimler Company just before World War I, although Henry Royce arrived at very much the same solution to the problem also during this period. Early versions of what was once generally referred to as the Lanchester damper tended to be of elaborate construction, since they virtually amounted to a small multiplate clutch assembly with lubricated metal-to-metal rubbing surfaces. Later examples were of much simpler construction, as originally described and in the first instance pioneered by the American Chrysler Corporation in 1925. Non-metallic friction discs were incorporated in these later dampers, their rubbing surfaces being unlubricated.

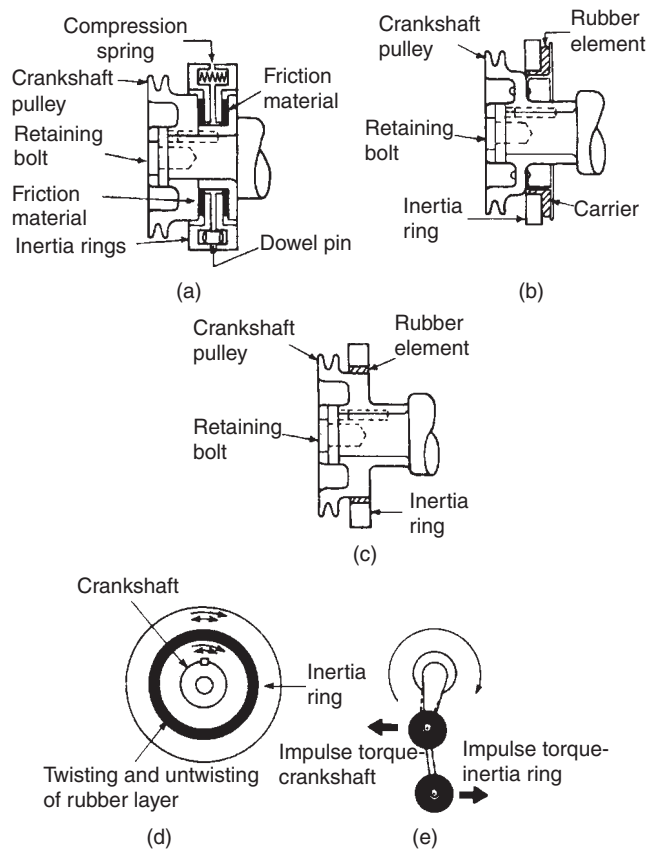


Figure 1.83 Types and action of crankshaft torsional vibration damper: (a) slipper (b) bonded rubber (c) unbonded rubber (d) damping (e) detuning

At the onset of a critical vibration period, the inertia of the damper flywheels is such that they become increasingly reluctant to follow the torsional oscillations superimposed upon the rotating crankshaft, and ultimately slippage occurs between the faces of their webs and the friction discs. That is, the friction torque of the damper that normally restrains the flywheels from angular movement relative to the crankshaft is exceeded, so that the damper is in effect behaving like a slipping clutch. As a result the energy absorbed in overcoming friction at the rubbing surfaces and dissipated in the form of heat is abstracted from the energy stored in the vibrating crankshaft. A serious amplitude of torsional vibration is therefore never allowed to build up. It should be realized, however, that a slipper damper can only really be effective in dealing with the particular frequency of torsional vibration for which its spring loading and hence slipping torque has been predetermined.

Construction and operation of the rubber damper

One of the shortcomings of the slipper damper was that its performance did not always remain consistent over an extended length of service, as a result of deterioration of its friction surfaces. The ensuing engine vibration periods could also be accompanied by noisy operation of the slipper damper, or what was sometimes referred to by service engineers as slipper roar. Put another way, a misbehaving slipper damper can have a worse effect on crankshaft torsional vibration than no damper at all. Recognition of these difficulties led to the development of a much simpler type of torsional vibration damper in which the two masses of different inertia, represented by the crankshaft and in this case a single small flywheel, were separated by both frictional and elastic means through the medium of rubber.

In long-established practice the rubber damper essentially comprises three concentric parts, these being a carrier cum hub assembly that is rigidly attached to the nose of the crankshaft, a ring-shaped flywheel or inertia ring that may be grooved to accept a V-belt, and a layer of rubber which is either bonded to each of these components (Figure 1.83b) or sandwiched between them under precompression, as in later designs (Figure 1.83c). It will be appreciated that there is no other connection between the carrier and the ring apart from that established by the intervening layer of rubber. The advantages of the later non-bonded version are that it is less costly to manufacture and allows a wider choice of rubber specification. Coincidentally, it was also Dr F. W. Lanchester who first patented a rubber type of crankshaft torsional vibration damper in 1928, and again the American Chrysler Corporation who, in the mid 1930s, developed the idea in the general form that we know it today.

Although the construction of the rubber damper is simpler than that of slipper and viscous dampers (Section 2.5), its operation is rather more complex because it acts as both a damper and a detuner. So far as the former duty is concerned, the onset of a critical vibration period causes the inertia ring to behave in a manner similar to the flywheels of a slipper damper, except that in this case the rubber layer is continually being twisted back and forth (Figure 1.83d). Since the rubber possesses a useful amount of internal friction, the

energy it absorbs and rejects as heat is abstracted from the vibrating crankshaft.

So much for its friction damping function. The *detuning* function of a rubber damper is perhaps not so readily visualized until it is appreciated that, unlike the slipper and viscous dampers, its inertia ring forms part of a torsional spring drive system with its own natural frequency of vibration. It is, in fact, analogous to that of the torsional spring drive used in clutch centre plates described later, except that the wind-up action of the circumferential coil springs is replaced by the rubber layer being deflected in torsional shear. By careful tuning of the damper vibration characteristics, its inertia ring can be induced to move in anti-phase with the torsional oscillations of the crankshaft and thus oppose their build-up. As a further simplification, the action of the damper as a detuner may be compared to that of a double-jointed pendulum, where the application of a disturbing force to the upper half can cause both halves to swing in opposite directions; in other words, their masses will move in anti-phase (Figure 1.83e).

Methods of mounting torsional vibration dampers

For maximum effect a torsional vibration damper is always mounted on the front end of the crankshaft (Figure 1.84), since it is this end that suffers the greatest amplitude of vibration as a result of being furthest away from the *nodal point* (position of no vibration) of the crankshaft. The fact that the nodal point lies much closer to the rear end of the crankshaft can be explained by the greater inertia of the nearby flywheel and hence its reluctance to vibrate. If flywheels were mounted at both ends of the crankshaft then the nodal point would lie midway along the length of the shaft. It has sometimes been observed that it is one matter to design a damper of adequate capacity, but quite another to make its hub stay put on the nose of the crankshaft, because any joint transmitting a pulsating torque is inherently trying to fidget itself loose. A mechanically efficient joint is therefore essential for securing the damper and may be achieved by several methods, which most conveniently lend themselves to illustration (Figure 1.85). Split wedge rings and flange fittings are usually associated with heavy-duty diesel engine practice. An interference-fitting hub is sometimes favoured in American passenger car engine design.

1.10 VALVE TRAIN

General background

Chiefly for reasons of accessibility, the inlet and exhaust valves in very early motor vehicles were arranged in two separate rows, one on either side of the cylinders and operated from beneath by similarly positioned camshafts. This long since obsolete arrangement of side valves provided what was known as a *T-head engine* (Figure 1.86a). Its combustion characteristics were later recognized as being poor, and the contrasting hot and cold (exhaust and inlet) sides of the engine could lead to cylinder distortion problems.

The T-head engine was then superseded by the *L-head engine* (Figure 1.86b), in which the inlet and exhaust valves were arranged in a single row on one side of the cylinders and again operated from beneath by a similarly positioned camshaft. Engines with this particular arrangement of side valves underwent considerable development and for many years provided a power unit which was generally cheap to produce. It became obsolete during the mid 1950s because its power output was somewhat limited by space restrictions on the usable size of inlet valves and by difficulties encountered in adequately cooling the exhaust valves.

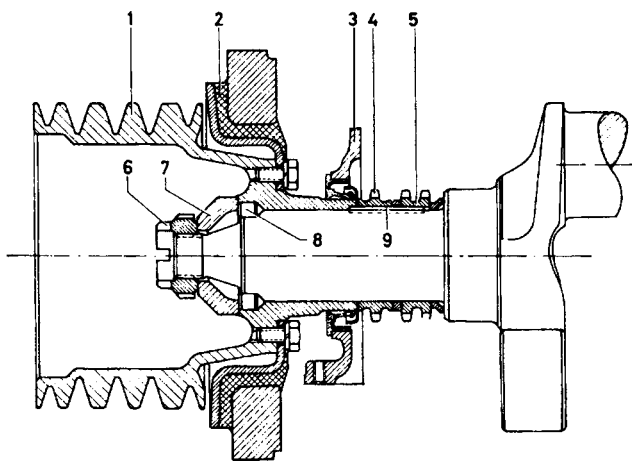


Figure 1.84 Example of attention to detail in rubber type vibration damper installation (Mercedes-Benz)

- 1 V-pulley
- 2 vibration damper
- 3 cover with radial sealing ring
- 4 oil pump sprocket
- 5 camshaft sprocket and distributor idling gear
- 6 clamp nut
- 7 pressure flange
- 8 dowel pin
- 9 Woodruff key

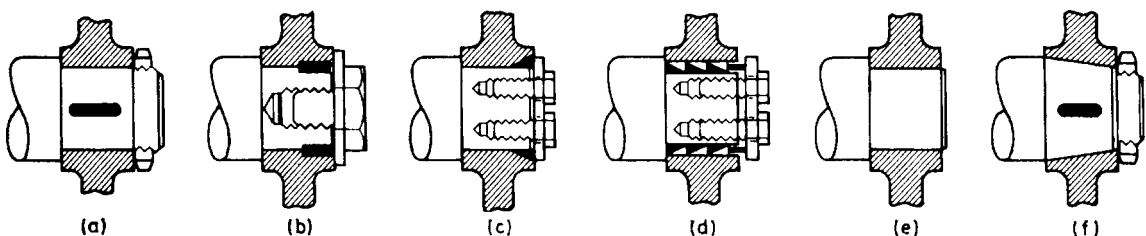


Figure 1.85 Flangeless methods of hub mounting for crankshaft vibration dampers: (a) parallel with key (b) parallel with dowel pins (c) parallel with wedge ring (d) parallel with wedge rings (e) parallel interference (f) taper with key

An early compromise between positioning the valves alongside the cylinders and in the head above them was a combination of these two locations. This resulted in the occasional *F-head engine* being built with overhead inlet and side exhaust valves (Figure 1.86c), both sets of valves being operated from a single camshaft mounted in the crankcase. Each overhead inlet valve was operated by a push-rod-and-rocker system. The main advantage of this type of layout was that larger inlet valves could be used, but being heavier they also placed limitations on maximum allowable engine speed. F-head engines were relatively expensive to produce and have been obsolete since the early 1960s.

While these various developments of the side valve engine had been applied to many touring cars, the designers of racing and sports cars had fairly early recognized that better engine performance would be more readily obtained by placing both the inlet and the exhaust valves over the cylinders, albeit with a certain amount of mechanical complication and less quiet operation. In varying degrees such arrangements allowed for more efficient shapes of combustion chamber and for a less tortuous and therefore faster route to be taken by the ingoing mixture and the outgoing exhaust gases.

Furthermore, the inlet and exhaust valves could either be arranged in a single vertical or near-vertical row, or be separated into two rows and mounted at an included angle to each other. These two arrangements of in-line and inclined valves are thus said to provide *I-head* and *V-head engines*, respectively (Figures 1.86d and e). Their mode of operation can be either directly from a single or a pair of cylinder-head-mounted or *overhead camshafts*, or indirectly through a push-rod-and-rocker system acting upon one, and in some earlier cases two, crankcase-mounted or *side camshafts*.

The present-day requirement for high performance from a medium-power engine has resulted in generally higher maximum crankshaft speeds, typically in the region of 5500–6000 rev/min. This can be explained by recalling the factors governing the power output of an engine since, if neither piston displacement nor mean effective pressure can be further increased, then the only other way to raise maximum power is to permit higher crankshaft speeds. For this reason, it becomes increasingly more important to avoid erratic operation of the valves at high engine speeds. As a result, about 80 per cent of car manufacturers now offer models with

the engine valves operated directly from an overhead camshaft, rather than by the less rigid push-rod-and-rocker system.

Commonly used abbreviations in connection with these valve layouts are as follows:

SV	side valves
IOE	inlet-over-exhaust valves
OHV	overhead valves
SOHC	single overhead camshaft
DOHC	double overhead camshafts

Side camshaft, push-rods and rockers

Camshaft

This serves to open the engine valves positively and to control their closing against the return action of the valve springs. In motor vehicle practice, a one-piece construction is almost invariably used for the shaft and its cams (Figure 1.87). Camshafts are generally produced from hardenable cast iron, which has replaced the case-hardened forged steel material used formerly. The angular spacing of the integral cams is such as to impart the required motion, in correct sequence, to the inlet and exhaust valves in each cylinder. To preserve accuracy of valve motion, the camshaft must be rigid enough to resist deflection under the alternating torsional and bending loads imposed upon it by the valve operating mechanism.

The camshaft journals are supported radially in the crankcase by a series of plain bearings. Since the loading on the camshaft bearings is generally not heavy and the operating speed of the camshaft is only one-half that of the crankshaft, the bearings are sometimes machined direct in the engine structure, especially where this is of aluminium alloy. Otherwise, established practice is to use separate steel-backed bushes that are pressed into machined bores. These bushes are lined with either a white metal alloy or one of the other bearing materials referred to in connection with the big-end

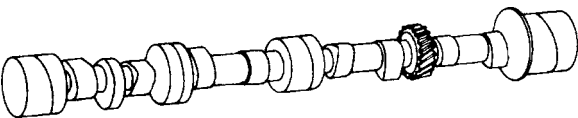


Figure 1.87 Camshaft for a four-cylinder engine

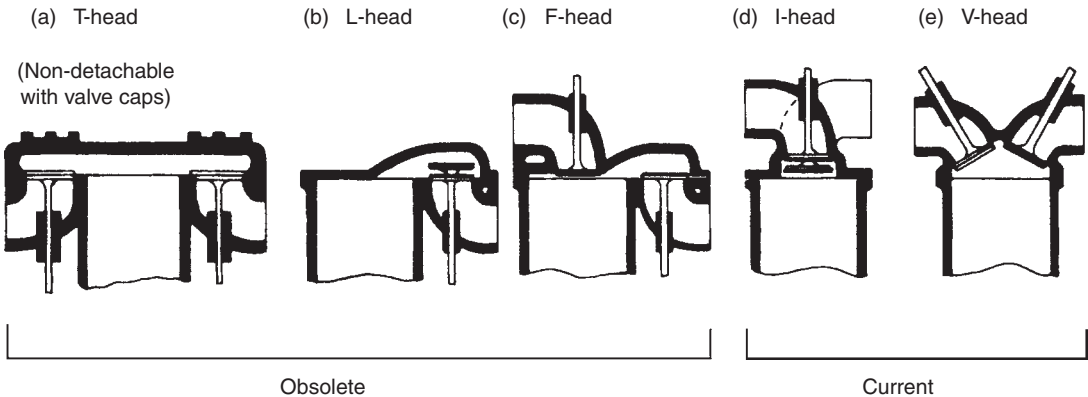


Figure 1.86 Identification of basic valve arrangements

and main bearings (Section 1.7). To enable a camshaft to be inserted endwise through its bushes, the radius of its bearing journals is made slightly larger than the operating radius of its cams. There may also be a progressive, but albeit very small, decrease in radii for successive journals and bushes in the direction of camshaft insertion, so that it does not have to be threaded through close-fitting bearings of all the same size when being inserted (or withdrawn).

Cam followers

To convert the radial motion of the cams into the reciprocating motion necessary for opening and closing the valves, cam followers or tappets must be used (Figure 1.88a). This is because the force exerted by a cam acts perpendicular to its contact surface and therefore does not remain in the direction of follower travel. In other words, whatever mechanism bears directly on the cam is subject to a certain amount of side thrust. Sliding followers are almost invariably used in conjunction with crankcase-mounted camshafts. This type of follower is termed a tappet barrel (Figure 1.88a), and is typically produced from hardenable cast iron. However, since certain combinations of camshaft and tappet materials behave better than others, great care has to be exercised by the engine designer to ensure their compatibility and avoid scuffing. Various surface treatments are also applied to the cams and tappet barrels, in order to assist the running-in process of their highly stressed contacting surfaces.

Push-rods

These are required to transmit the reciprocating motion of the cam followers to the valve rockers (Figure 1.88a). Both ends of the push-rod form part of ball-and-socket joints which accommodate the angular movements of the push-rod arising from the straight-line motion of the tappet barrel on the one hand and the arcuate motion of the valve rocker on the other.

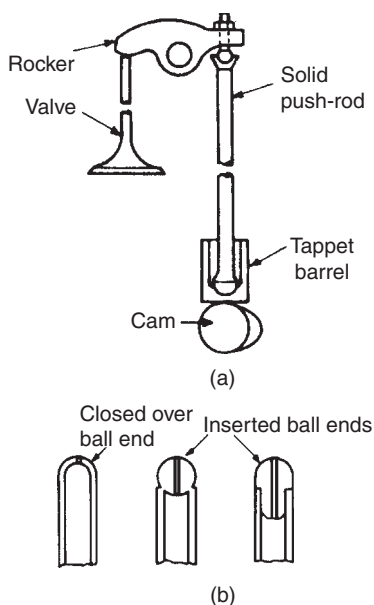


Figure 1.88 (a) Push-rod valve operation (b) types of pushrod

Since the push-rod is part of a valve train that constitutes a vibrating system, it must combine maximum rigidity with minimum weight. Push-rods are generally produced from steel and may be of either solid or tubular construction. In comparing solid and tubular push-rods of the same strength, the latter (Figure 1.88b) usually offer some reduction in reciprocating weight and can also serve as an oil conduit in the valve train lubrication system.

Valve rockers

The function of these is to cause both a reversal and a magnification of the motion imparted by the cam and follower to the valve. The valve rocker, or rocker arm, is a short rigid beam that oscillates about an offset pivot of either the journal bearing or the ball-and-socket type. An advantage of the latter type is that it makes the rocker inherently self-aligning, but it is necessary to introduce some means of restraining lateral movement. In cross-section the depth of the rocker arm greatly exceeds its width, since the bending loads imposed upon it are mainly within the plane of oscillation, with very little side loading.

Valve rockers may be of either solid or hollow construction (Figure 1.100a and b), and may be produced from steel forgings, iron castings, steel pressings, or in some later applications from aluminium alloy. Solid rockers oscillate about a stationary hollow shaft known as the rocker shaft. This is supported by a series of pedestals mounted on the top deck of the cylinder head, each pair of rockers generally being separated from the next pair by one of the pedestals (Figure 1.89). Sideways location of individual rockers is against adjacent rocker shaft pedestals, the rockers in each pair being held apart by means of a compression spring, spring clips or spacer tube. For some designs a separate bronze bushing is pressed into the rocker bearing bore, while in others the rocker is hardened all over with a plain bore bearing directly on the rocker shaft. A curved pad is machined on the end of the rocker where it contacts the valve stem tip, so as to allow the partly

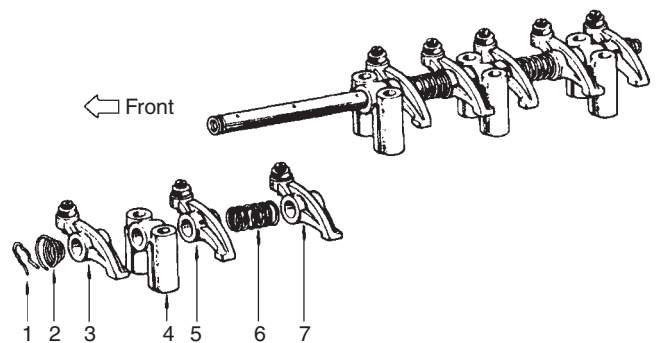


Figure 1.89 Rocker shaft assembly for a four-cylinder engine (Toyota)

- 1 retainer clip
- 2 conical spring
- 3 valve rocker arm, no. 1 exhaust
- 4 rocker shaft pedestal
- 5 valve rocker arm, no. 1 inlet
- 6 compression spring
- 7 valve rocker arm, no. 2 inlet

rolling and partly sliding motion that occurs between them. Although they have been used with conventional rocker shafts, hollow rockers are now usually mounted on individual pivot posts with either hemispherical or part-cylindrical seatings (Figure 1.100b), the latter being currently favoured in American practice. The pivot posts may be pressed and in some cases screwed into bosses on the top deck of the cylinder head.

Overhead camshaft and sliding followers

Camshaft

For a cylinder-head-mounted camshaft, modern practice tends to favour an increase in the number of intermediate bearings to one between each pair of cams. The bearing installations for overhead camshafts are currently similar to those of crankcase-mounted camshafts. An exception is where the bearings are supported in pedestals with detachable caps, which permit the camshaft to be inserted from above the engine. Also, this particular arrangement enables a reduction to be made in the diameter of the journal bearings, since the cams no longer have to be assembled through them (Figure 1.90).

Cam followers

The use of direct-acting sliding followers with a cylinder-head-mounted camshaft demands relatively large-diameter tappets (Figure 1.90). This increase in their base area is dictated by the absence of a multiplying leverage (otherwise provided by rocker arms) in the valve train, so that larger cams have to be used to give the desired amount of valve lift. Since an overhead camshaft is in even closer proximity to the valves than it was the obsolete side valve layout, the tappets must necessarily be both hollow to fit over the valve springs and short in length. For this reason they are aptly termed 'inverted-bucket tappets', although they may sometimes be referred to as Ballot-type tappets, having originally been introduced by the French Ballot company for its straight-eight engine in 1919.

In a recent Toyota development of the inverted bucket tappet, the tappet body is cold forged from aluminium alloy with a sprayed-on iron coating for its sliding surface. A tappet setting steel shim is recessed into its head portion and a stiffening steel disc is caulked beneath its head where contact is made with the valve stem tip (Figure 1.36). This construction promotes better fuel economy by virtue of generally lightening

the valve train, because the valve spring load can be reduced so that less torque is required to drive the camshaft. Furthermore, the valve train is less prone to chatter, since the tappet is lighter in weight and not so hard as steel and is therefore quieter in operation.

Overhead camshaft and pivoting followers

Camshaft

Similar remarks apply as in the previous system regarding the number of installation of the camshaft bearings.

Cam followers

These are known as the pivoting type because they swing between the camshaft and the valve tips and act as either a simple (straight) lever arm (Figure 1.100g and h) or 'bell-crank' (angled) lever (Figure 1.100i).

Modern applications of the former type generally feature a forged steel lever arm, or finger rocker as it is often called, which is self-aligning on a ball-and-socket pivot mounting (Figure 1.91), although journal bearing pivots have been used in the past. To restrain lateral movement of the ball-and-socket-mounted finger rocker, the valve stem tip may be recessed into the curved contact pad on the end of the arm (Figure 1.100g). To reduce valve train friction in some more recent applications, the sliding contact between the cam and finger rocker is replaced by a rolling one, the rocker being slotted in its central portion to embrace a roller mounted on needle bearings, the modern Ford V6 engine being an example of this practice.

The bell-crank type of pivoting rocker is supported on either plain journal or, more recently, needle roller bearings from a conventional rocker shaft. This is mounted above the camshaft, so that the downwards-extending lever arms of the bell-cranks bear on the cams and the outwards-extending arms contact the valve tips (Figure 1.100i).

Comparison of different systems

Side camshaft, push-rods and rockers

The advantages of this method of valve operation are generally that the timing drive to the camshaft is uncomplicated and therefore less expensive to produce, and that tappet clearance

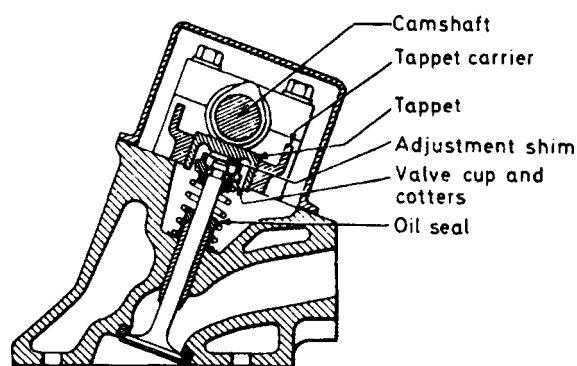


Figure 1.90 Overhead camshaft and sliding followers

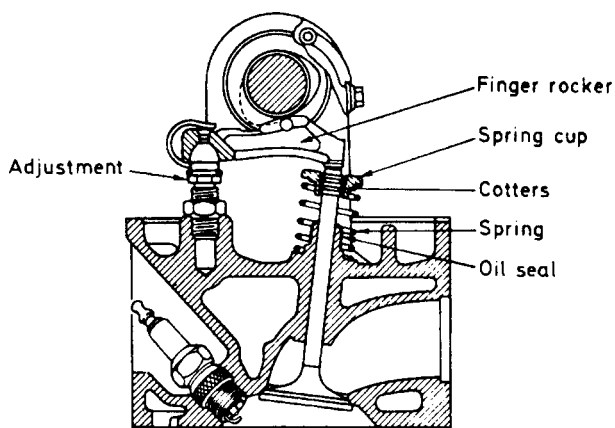


Figure 1.91 Overhead camshaft and pivoting followers

adjustment can be conveniently performed. Disadvantages are concerned with its greater tendency towards vibration, which is due to flexing of the push-rods and rockers and creates false motions of the valves. There may also be the need for frequent tappet clearance adjustment because of the number of points subject to wear.

Overhead camshaft and sliding followers

For this method of valve operation the advantages are chiefly its lower inertia and greater rigidity in the interests of high-speed operation and a reduced requirement for tappet clearance adjustment. Its disadvantages include the need for a more elaborate and therefore expensive timing drive to the camshaft(s); the engine in general tends to be more difficult to service, although tappet clearance adjustment in particular need not be laborious in modern versions, and in any event should seldom be required; and the engine height is increased.

Overhead camshaft and pivoting followers

Here the advantages are that it can provide a further reduction in inertia over that of sliding followers, together with some magnification of the motion imparted by the cam to its valve. It is also simpler to incorporate a ready means of tappet clearance adjustment. Disadvantages are generally that finger rockers do not possess the same degree of rigidity as sliding followers; there must inevitably be some side thrust transmitted to the valves, and effective lubrication can be more difficult to attain for minimum wear between the cams and followers.

Further developments in valve actuating systems

The advent of four-valve combustion chambers for passenger car engines, as discussed in Section 3.3, has necessarily led to some revision of the actuating mechanisms that have traditionally been used with two-valve installations. An interesting comparison between the various mechanisms that may be used, and which were investigated by the Toyota company during the development of their own four-valve engines, is shown in Figure 1.92. It will be noticed that in the second and third examples of SOHC layouts, a single cam and forked

rocker is used to actuate one of the two pairs of valves. The final choice made by Toyota on the basis of valve train rigidity and cost effectiveness was the direct-acting second example of DOHC. This less complex system also minimized friction losses in the valve train, which contributed to improved fuel economy in the engine speed range most used during everyday driving. Furthermore, this particular system had already proven virtually maintenance-free in high-performance engines.

A new approach to camshaft manufacture has been adopted by Ford for their V six-cylinder engine, which was introduced in 1995 and features DOHC for each bank of cylinders. As a weight saving measure and also to reduce inertia loading on the timing drive, each camshaft is of tubular construction with pressed-on sintered metal cam lobes. In the continuing pursuit of engine refinement with multiple valve cylinders an interesting development by BMW has been to counterbalance the eccentricity of the paired cam lobes by offsetting, in the opposite sense, the shaft sections between adjacent cams.

Identifying the parts of a cam profile

At this stage we must consider some of the whys and wherefores of what is probably the most versatile of all machine elements, namely the cam itself. According to their construction, cams used in engineering are usually classified into three types, known as disc, face and cylindrical cams. In the valve train of the automotive engine we are concerned only with the first-mentioned type. To convert rotary motion into a reciprocating movement, the basic parts of a simple disc cam profile comprise the base circle, the flanks and the nose (Figure 1.93). The last two parts constitute the cam lobe or lifting portion of the cam, since they determine the extent to which the follower can be lifted clear of the base circle. The period of valve opening, or valve event, is derived from the angular relationship between those points on the cam lobe where the opening flank begins and the closing flank ends. It therefore includes any interval during which the valve dwells in the fully open position, the governing factor here being the length of the nose arc. Conversely, that portion of cam rotation during which the base circle is operative with respect to the follower determines the valve closed period.

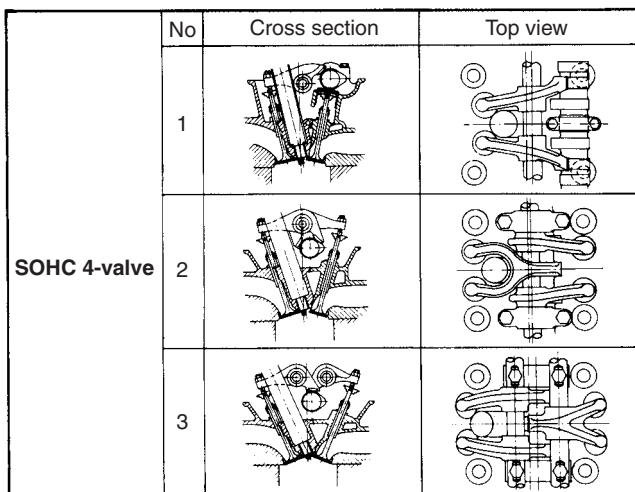


Figure 1.92 Examples of actuating mechanisms for four-valve layouts (*Toyota*) HLA, hydraulic lash adjuster (hydraulic tappet)

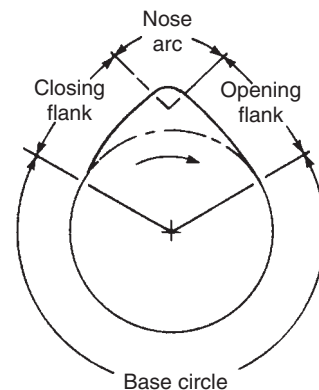


Figure 1.93 Basic parts of a simple cam profile

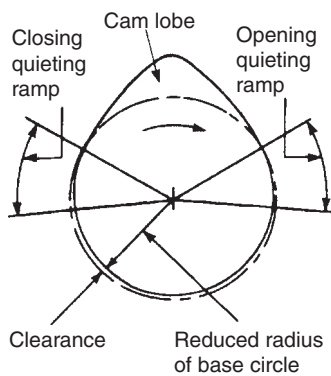


Figure 1.94 Simple cam profile modified by addition of quieting ramps

The cam profile must also take into account the effects of valve or tappet clearance, so as to minimize noise from this source. For this purpose the simple cam profile is modified to include quieting ramps between the base circle and the flanks (Figure 1.94). These quieting ramps take the form of inclines that occupy 15 to 30° of camshaft rotation with a rise equal to the tappet clearance. Geometrically they are blended into the cam profile by reducing the radius of the base circle to correspond with this clearance and then connecting the undercut base circle to the opening and closing points on the flanks by means of the ramps. Dynamically these ramps serve to reduce the velocity at which the valve initially leaves and finally returns to its seating. Impact loading is thus reduced throughout the valve train, which contributes both to quieter operation and to extended valve life. Ramp heights are reduced for engines that feature hydraulic tappets because these virtually maintain a zero clearance throughout the valve train as explained later.

Valve lift and open period

From the description of the basic parts of a cam profile just given, it should be evident that a complete rotation of the cam momentarily lifts its follower by an amount equal to the overall cam height minus the base circle diameter dimension, the amount of lift being known as the cam lift (Figure 1.95). Here we must be careful to distinguish between cam lift and valve lift, since they are not necessarily one and the same thing. This is because the cam lift may be either less than, or equal to, the actual valve lift, depending upon whether or not a multiplying leverage is provided by valve rockers. The various arrangements of valve rockers were described earlier in this section, and it only remains to add that they usually confer a multiplying leverage in the region of 1.5:1. A further point is that valve lift will always be reduced in the presence of tappet clearance, so in the absence of valve rockers it would be more correct to the state that valve lift is nearly equal to cam lift.

The maximum lift of a valve generally approximates to one-quarter of its head diameter, since any greater opening than this gives diminishing returns in gas flow capacity. That is, the restriction to flow in the port and through the valve opening becomes equal when optimum lift occurs.

The period of valve opening or valve event has already been defined as the angular relationship between those points

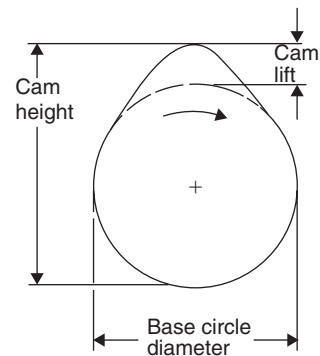


Figure 1.95 Extent of cam lift

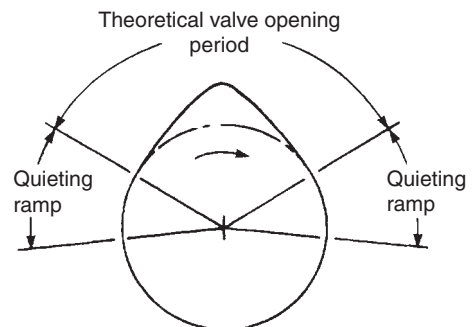


Figure 1.96 Cam valve opening period

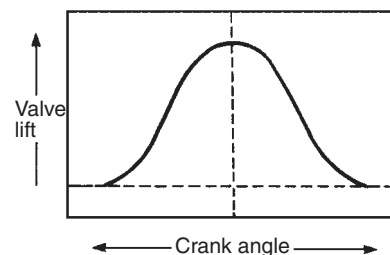


Figure 1.97 Valve opening diagram

on the cam lobe where the opening flank begins and the closing flank ends (Figure 1.96). Whilst this is fine in theory, it is rather more difficult in actual practice to determine the precise points at which the valve begins to open and close. This is because the tappet clearance is usually taken up somewhere near the middle of the quieting ramp when follower lift for each degree of cam rotation is very small indeed. It is for this reason that when verifying the nominal valve timing of an engine, the checking tappet clearance is normally specified to be greater than the running clearance, so as to ensure that valve lift does not occur until the beginning of the cam opening flank when the initial movement of the valve will be more readily detected. Taking matters a step further, if the valve lift for every say 5° of crankshaft rotation is measured and plotted on a graph then we arrive at a *valve opening diagram* (Figure 1.97). The significance of this is that the larger the area under the curve, the more gas will flow through the valve during its open period. A valve opening diagram is therefore of immediate interest to an engine designer and also anyone involved in

preparing or supertuning competition engines. It is of much less interest to an automotive service engineer, however, who is more likely to be concerned with measuring cam height during engine overhaul to check whether it lies within the wear limits laid down by the manufacturer.

Types of cam profile

If each valve could be opened to its full lift instantaneously, dwell in the fully opened position for the desired period of opening and then be closed instantaneously, then clearly the area under the curve of its opening diagram would be at a maximum. Equally clearly, too, these ideal conditions of valve operation are unattainable in practice, since the infinite acceleration and deceleration of the valve train components would subject them to mechanically destructive forces. In the real-life automotive engine, it is therefore necessary to employ a cam profile that will open and close the valves in a more considerate manner. Although the theoretical considerations underlying the design of modern cam profiles are far removed from the realms of automotive service engineering, it may nevertheless be of interest to at least identify and briefly describe the various types of cam profile that have been used past and present, as follows.

Concave cams

A roller follower has to be used in conjunction with this early type of cam profile, where each concave flank is an arc of a circle that is tangent to both the base circle radius and the nose radius (Figure 1.98a). Although this type of profile lifts the follower with constant acceleration and deceleration, the transition from one to the other occurs very suddenly. Another disadvantage lies in the need to use a roller follower, which being heavier in construction creates larger inertia forces than a simple flat-faced one and thus makes the concave cam unsuitable for application to high-speed engines.

Tangent cams

Either a roller or a curve-faced follower is required for this again early type of cam profile, which is very simple in construction since each straight-line flank is a tangent joining the base circle radius with the nose radius (Figure 1.98b). It is sometimes described as a fast cam, because so long as the follower remains on the flank it continues to be accelerated. Then follows a very sudden transition from acceleration to deceleration, which is confined to that period of cam rotation when the follower is being lifted by the nose radius and is consequently very high in value. This in turn demands the use of a much increased valve spring force and generally makes the tangent cam again unsuitable for high-speed operation of an engine.

Convex cams

An essentially flat-faced follower is used in conjunction with this type of cam profile, in which each convex flank is an arc of a circle that is tangent to both the base circle radius and the nose radius (Figure 1.98c). A characteristic of this form of cam is that the follower is lifted for only a short period at high acceleration, this being followed by a much longer period of relatively leisurely deceleration, so that the controlling force exerted by the valve spring need not be too

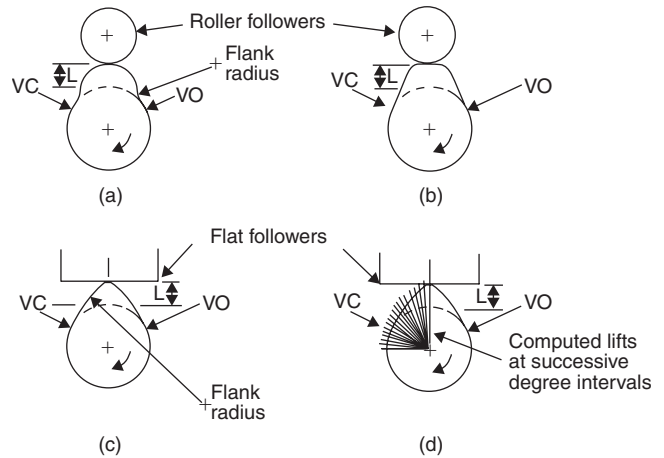


Figure 1.98 Types of cam profile: (a) concave (b) tangent (c) convex (d) non-geometric convex. L lift; VO valve opening; VC valve closing

great. As with the previously described concave and tangent cams, the convex cam still suffers from the disadvantage that the transition from acceleration to deceleration occurs very suddenly. However, the three-arc convex cam was widely used prior to the advent of the modern convex cam of non-geometric type described next.

Non-geometric convex cams

Here again an essentially flat-faced follower is used with cam profiles of this type, where each convex flank is not simply an arc of a circle, but has been tailor-made from a series of computed lift values at successive degree intervals of cam rotation (Figure 1.98d). By this means it becomes possible to make less sudden the transition from acceleration to deceleration during lift of the follower and also, of course, likewise during its return. Furthermore, cam profiles of this type can be mathematically developed to compensate at certain critical operating speeds for the effects on valve lift caused by any lack of rigidity in the valve train components.

Wear considerations related to cam form

The precise alignment required to avoid edge loading or digging in between nominally parallel cam and tappet contact faces is, in practice, very difficult to guarantee. Since edge loading could lead to excessive surface stresses, the contact faces of the tappet barrels may be made very slightly convex and are thus better able to accommodate any such discrepancies in alignment. This particular feature is usually combined with cams that are both tapered across their width and offset axially relative to the axes of the tappet barrels (Figure 1.99). The effect of this geometrical relationship between the cam and tappet contact faces is to induce rotation of the tappet barrel as it reciprocates, thus securing a more satisfactory distribution of wear over its working face. In modern OHC practice only the offsetting of the cams to induce rotation of inverted-bucket tappets is usually necessary, the cam and tappet contact surfaces being ground parallel.

In more recent American engine designs there has been a revival of interest in the roller type of sliding cam follower,

the roller being used at the foot of a hydraulic tappet (this type of tappet is described later). The roller is provided with needle rolling bearings at its pivot and operates against a nodular iron camshaft with modern cam forms. Careful design is necessary to ensure that the roller is of optimum size to minimize contact stress. Also, the roller track may require a crowned profile, so as to compensate for any slight misalignment between the roller and its cam. Apart from conferring greater durability on the valve train, it is claimed by one manufacturer that this arrangement reduces friction by

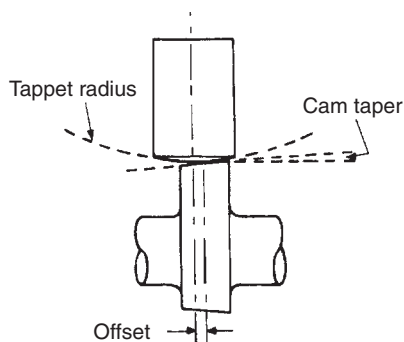


Figure 1.99 Cam and tappet contact geometry

about 8 per cent and is therefore a contributory factor in improving the all-important fuel economy. A roller may similarly be applied to the pivoting rocker of an overhead camshaft installation, as described earlier.

Valve clearance: manual adjustments

The design of an engine must take into account the effects of thermal expansion and contraction of the valve train components, relative to that of the engine structure. Since these effects could result in the valves being held off their seats when they should be firmly closed, a small operating clearance has to be introduced into the reciprocating parts of the valve train. This is generally termed the valve or tappet clearance, and provision for either its manual or automatic adjustment is incorporated in all engines. The latter form of adjustment will be dealt with later.

For engines with push-rod overhead valves and solid rockers, valve clearance is usually adjusted at the rockers by means of a locking nut and hardened screw, the ball end of which registers in the push-rod socket (Figure 1.100a). The clearance may be checked by inserting a feeler gauge blade of appropriate thickness between the rocker pad and the valve stem tip. (A useful practical hint is not to slacken completely the locking nut while turning the adjusting screw.)

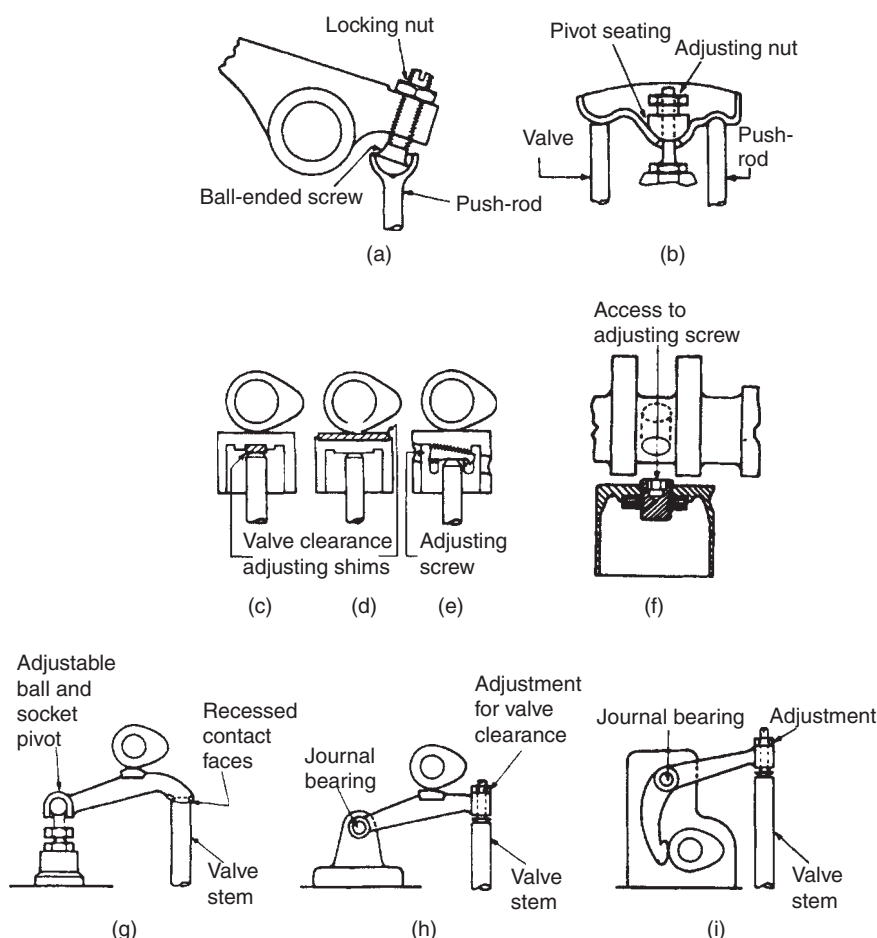


Figure 1.100 Valve clearance adjustments

Where manual adjustment of the valve clearance is employed for push-rod overhead valves and hollow rockers, a screw-threaded portion at the upper end of the pivot post receives a self-locking nut, against the underside of which the pivot seating abuts (Figure 1.100b). Turning the self-locking nut thus allows the valve clearance to be set by virtue of either lowering or raising the rocker pivot. In the case of an overhead camshaft and inverted-bucket tappets, the adjustment can be a time-consuming operation in those installations where valve clearance is set by the selective assembly of graduated-thickness shims, which are inserted between the valve stem tip and the underside of the tappet head (Figure 1.100c). Other less inaccessible means of adjustment have, however, been introduced in recent years for inverted-bucket tappets.

In one now widely used system discs of appropriate thicknesses are located in recesses in the tappet heads and are directly contacted by the cams, so that if a valve is held depressed after being operated by its cam the disc may easily be changed (Figure 1.100d). Another system utilizes the wedging action of a taper-faced adjuster cum locking screw, which is located transversely in the tappet and forms an abutment for the valve tip (Figure 1.100e). Yet another and particularly ingenious system incorporates a self-locking screw in the head of the tappet. Access to this screw, which bears directly against the valve stem tip, is gained through a divided cam and cross-drilling in the camshaft (Figure 1.100f).

For manual adjustment of valve clearance with an overhead camshaft and self-aligning finger rocker, the ball pivot is usually screwed into a boss in the top deck of the cylinder head, so that raising or lowering the ball pin alters the valve clearance accordingly (Figure 1.100g). In the case of journal bearing pivots, a screw-type adjustment for valve clearance is generally provided at the valve tip end of the rocker (Figure 1.100h). The same applies to the bell-crank type of pivoting rocker (Figure 1.100i). An interesting point to note in connection with these various methods of valve clearance adjustment is that those where the screw adjuster remains stationary serve to reduce the inertia of the valve train.

The effects of incorrectly setting tappet clearances

Accurate setting of the valve or tappet clearances in accordance with the manufacturer's specification is of vital importance, otherwise at least the valve timing of the engine can be affected and at worst valve failure may result. In giving incorrect valve timing, should the clearance be set smaller than specified, the valve will tend to open earlier and close later than intended, and vice versa should the clearance be greater.

The effect of setting the tappet clearance much too close, perhaps in a misguided attempt to gain quieter running, can result in the valve not fully closing on to its seating. Apart from loss of cylinder pressure, this will allow the high-temperature combustion gases to blow past the valve face and, since no heat can escape to the valve seat, the valve head temperature becomes so high that rapid burning and destruction of the face occurs. This situation is usually aggravated by charring of the oil film on the valve stem, which causes the valve to stick in its guide. The effect of setting the tappet clearance much too wide is to produce excessively noisy operation and also wear caused by pounding of the valve against its seating. Furthermore, the amount by which the valve is lifted from its seating will be reduced.

Hydraulic tappets

Since their introduction to the American motor industry on the Pierce-Arrow car of 1932 (a make of car that we shall be meeting again later in connection with early power-assisted steering), self-adjusting hydraulically operated tappets are often employed in preference to the simple mechanical type. Although the latter can periodically be manually adjusted to compensate for changes due to wear, they are unable to accommodate the effects of thermal expansion. Therefore a running clearance must be established in the valve train, which not only creates a potential source of noise, but also can be subject to error in its setting. Hence, the reasons for using hydraulic tappets include quietness of valve train operation, constant valve timing and the elimination of valve clearance adjustments in service. These advantages are obtained because tappets of this type automatically maintain zero clearances throughout the valve train, under practically all operating conditions, because their action is such as to compensate for the differences in thermal expansion of the valve train components and the engine structure. They also simplify assembly of the engine and therefore reduce cost of production.

In construction, the hydraulic tappet is essentially a telescopic device that receives a pressurized supply of oil from the engine lubrication system. The engine valve opening load is imposed upon a spring-loaded inner plunger, which has either a ball or a disc type of non-return valve that normally acts to close off the foot of the plunger. An exceedingly fine working clearance is provided between the plunger and the tappet body in which it is free to slide.

Its operating cycle begins each time an engine valve closes, at which point the tappet dwells upon the base circle of the cam and the plunger return spring extends the tappet assembly to absorb any clearances that exist in the valve train. As any extension of the tappet will create a depression beneath the non-return valve of the plunger, the valve opens so that oil under pressure from the engine lubrication system is admitted to the tappet compression chamber (Figure 1.101a). This is formed between the closed end of the barrel and the underside of the plunger. As soon as the cam begins to lift the tappet, the increased pressure on the trapped oil maintains the non-return valve closed, so that the column of oil behaves like a rigid strut to transmit the opening load developed between the cam and the engine valve (Figure 1.101b). There is an intentional slight leakage of oil from the compression chamber, which takes place between the plunger and its operating bore in the tappet barrel. This controlled leakage is known as tappet *leak-down*, and its purpose is to ensure that the engine valve always returns fully to its seating, as once again the tappet returns to the base circle of the cam ready for the next operating cycle.

Although once generally applied to the valve trains of push-rod-operated overhead valve systems, hydraulic tappets are now increasingly used to similar advantage as self-adjusting pivots for the finger rockers in modern overhead camshaft installations. That is, the hydraulic tappet body remains stationary and its plunger provides a pivot mounting for the finger rocker. This arrangement has the advantage of minimizing the inertia of the valve train, since the adjustment mechanism does in effect remain stationary. However, there has been a recent tendency in some multiple valve DOHC engines to incorporate a hydraulic self-adjusting facility in direct-acting

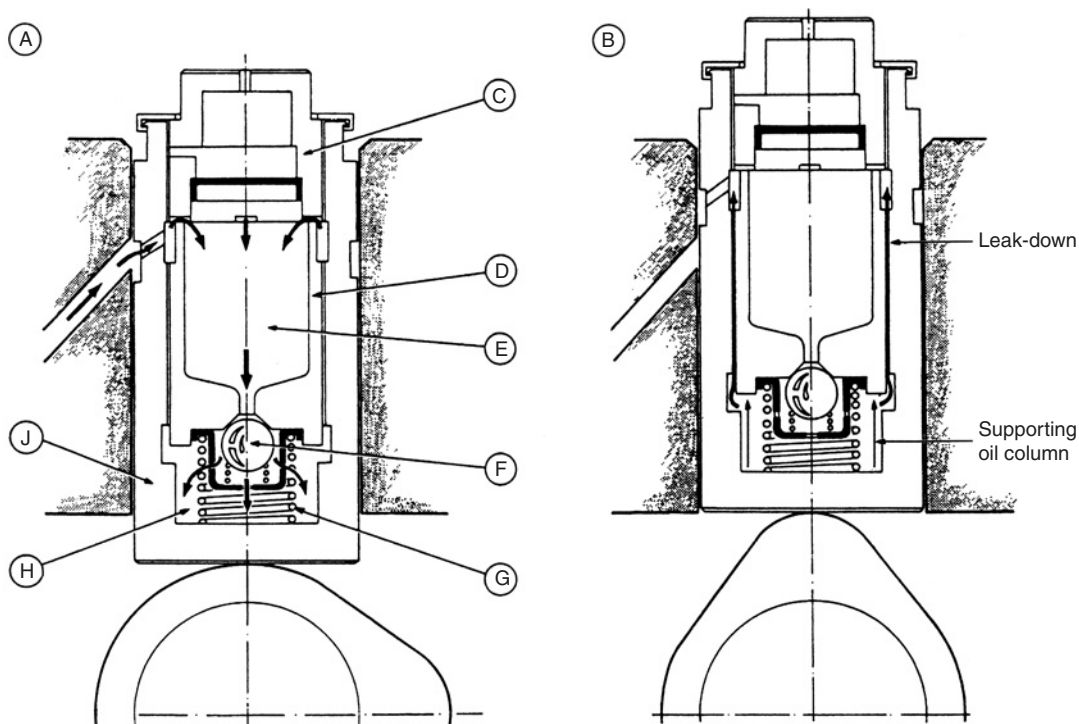


Figure 1.101 Operating principle of hydraulic tappet: (a) tappet absorbing valve train clearances (b) tappet acting as rigid strut to open valve

- | | | |
|------------------|--------------------|--------------------|
| A Valve closed | D Tappet cylinder | G Tappet spring |
| B Valve open | E Feed chamber | H Pressure chamber |
| C Tappet plunger | F Non-return valve | J Tappet body |

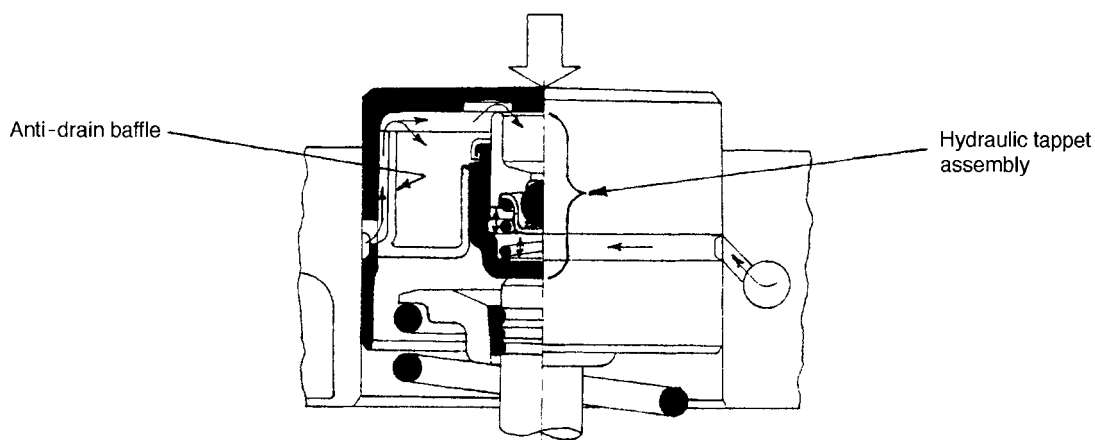


Figure 1.102 Schematic arrangement of a hydraulic bucket tappet showing method of oil feed

sliding type followers, thereby producing a very compact 'hydraulic bucket tappet'. Its operating principle is identical to that already described for other hydraulic tappet applications, but an 'anti-drain' function may also be included. The relevance of the latter is that the nearer to the vertical that the tappet operates, the greater is the chance that air may be induced to enter its compression chamber when significant expansion of the tappet occurs. In other words, following the leak-down of a tappet under the spring load of an engine valve that remained partially open when the engine was switched off and

then gradually closed, thereby leaving the tappet in a compressed state prior to the engine being started again. The purpose of an anti-drain baffle in the tappet (Figure 1.102) is therefore to ensure that, when the tappet expands on starting the engine, the level of oil retained in the tappet is high enough to exclude air from its compression chamber, which otherwise results in a noisy tappet until the air is eliminated. From similar considerations it must also be arranged that oil does not drain away from the hydraulic tappets supply gallery in the cylinder head when the engine is not running.

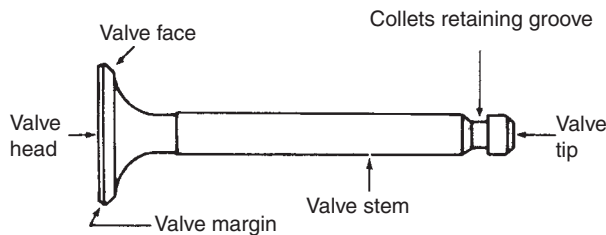


Figure 1.103 Terminology of the poppet valve

Engine valves

It has long been established practice to employ what are termed *poppet valves*, which comprise a disc-shaped head with a conical seating and a stem that acts as a guiding surface (Figure 1.103). Their main advantages over other possible alternative valve forms are as follows:

- 1 They are self-centring as they close on to the cylinder head seating.
- 2 They possess freedom to rotate to a new position.
- 3 It is relatively easy to restore their sealing efficiency in service.

For mechanical strength and to assist gas flow, the valve stem is blended into the head portion to form a neck of fairly generous radius, but a certain amount of relative flexibility between them is generally permitted. Under the influence of cylinder pressure, the valve head may therefore better conform to its seating should there be any distortion present. The valve stem is provided with a few hundredths of a millimetre working clearance in its guide; this is usually increased slightly for the hotter exhaust valves to allow for their greater expansion.

It is usual for the head diameter of the exhaust valve to be made less than that of the inlet valve, since cylinder gases may be more easily evacuated at exhaust pressure than admitted at induction depression – truly, as Spinoza observed, ‘Nature abhors a vacuum’! Another reason for favouring a smaller-diameter exhaust valve is generally to reduce its thermal loading by virtue of the shorter path of heat flow.

The conical facing of the valve makes an angle of either 45° or 30° with the plane of the head. Although the former angle provides a higher seating pressure for a given valve spring load, the latter angle permits a greater gas flow for a given valve lift. Hence in some engines the face angles of the exhaust and the inlet valves may be 45° and 30° respectively. To improve further their seating conditions, the exhaust valves may be installed with a differential face-to-seat angle of up to 1° (Figure 1.104). The effect of this is initially to concentrate the seat bedding towards the larger-diameter and combustion side of the valve face, so that any subsequent dishing of the valve head then tends to centralize the seat bedding.

In practice, the face angle of the valve may therefore either match, or have a slight positive interference, with the seat angle. Also, the actual bedding area should be neither too narrow, which would hinder heat dissipation, nor too wide, which would reduce seating pressure and be less effective in breaking up deposits.

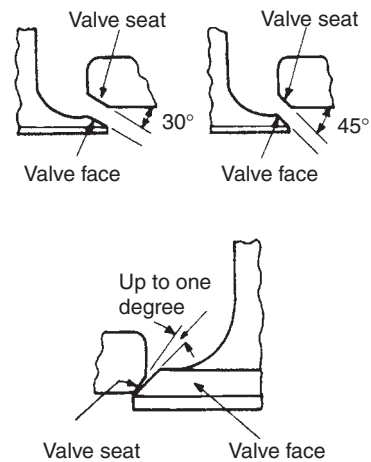


Figure 1.104 Valve seating angles

Valve materials

The valves must endure a particularly arduous existence, since they are subjected to severe mechanical and thermal loading. Exhaust valves in particular have to withstand mean operating temperatures that may approach 900°C , with about 75 per cent of this heat being transferred to the cooling system through the valve seat and the remainder through the valve guide. Seat face burning or ‘guttering’ must therefore be resisted.

Since the thermal loading on the inlet valves is less severe, they are generally produced from low-alloy silicon-chromium steel, which is also known as Silchrome from its American origins dating back to 1926. Its specification combined 3 to 3.5 per cent silicon (which until then had always been looked upon as an unavoidable impurity) with 8 to 9 per cent chromium and 0.4 to 0.5 per cent carbon. For light-duty applications a 1 per cent chromium steel may instead be chosen. A one-piece valve construction is used with these materials, since they can be locally hardened for tip, groove or seat wear resistance, while the stem may be chromium plated for the same reason.

The most widely used material for the exhaust valves is chromium-manganese-nickel steel, such as that designated 21-4N and referred to again later in connection with diesel engine valves (Section 2.6). This high-alloy material has a combination of hot strength and corrosion resistance that meets the requirements of most engines at temperatures up to over 800°C . The durability of the exhaust valves has sometimes been enhanced by the application of an aluminium coating to their heads, this being a General Motors process to form a tough corrosion resistant layer with valve steel and improve heat conductivity. Bimetal two-piece exhaust valves are also now employed, because they are more economical to manufacture. A necessarily expensive high-alloy steel need to be used only for the head portion, where the maximum value of hot strength is required; a less expensive low-alloy steel can be used for the stem, which confers the best guide wearing properties without the aid of chromium plating.

It may be of interest to mention that the joining of two-piece valves is accomplished by a process known as friction welding. This process is performed by the heat liberated

when one of the parts being butted together is rotated against the other stationary part under an axial load, the final weld occurring when rotation is stopped. Friction welding was originally exploited by German engineers to join plastics components during World War II, but was afterwards vigorously developed by Russian engineers to join metal components. In the case of two-piece valves the friction weld in the stem is positioned to remain in the guide during valve travel.

Truly has it been said that the engine valve represents 'a metallurgical triumph over a mechanical monstrosity'!

Internally-cooled exhaust valves

Since the early 1980s the increased thermal loading that has been imposed on high-performance engines, first by turbocharging (Section 9.2) and then more recently by multiple valves (Section 3.3), has led to a modest revival of interest in the 'internally-cooled' exhaust valve. This function is accomplished by providing a hollow interior for the valve which, before its tip is sealed, is partially filled with a material that rapidly transfers heat from its exposed head surface to the stem portion and thence to the valve guide and cylinder head cooling jacket. The purpose of only partially filling (about half) the valve interior is to ensure that the enclosed material is shuttled to and fro by the opening and closing motions of the valve, as though shaking a cocktail drink, which therefore facilitates the transfer of heat from its head to the stem.

This type of exhaust valve was originally developed to meet the arduous duty encountered in aircraft piston engines, the first heat transferring material used being mercury, which rather interestingly was also favoured by Mercedes-Benz for the very powerful supercharged engines of their legendary Grand Prix cars of the 1930s. However, the real breakthrough in the development of the internally-cooled exhaust valve can be attributed to S.D. Heron, a British engineer working in America, who during the same period developed the sodium-cooled valve that posed less of a sealing problem. The advantages of using metallic sodium are several, not least among them being that as a conductor of heat it is over six times better than the valve steel and is also lighter in weight. Since it has a low melting point of 97.5°C it is liquid at valve operating temperatures and therefore when shaken it transfers heat by convection as well as conduction, whilst it has a high boiling point of 880°C that avoids vaporization. The sodium-cooled exhaust valve is usually furnished with a Stellite seat facing alloy (Section 2.6) to complement its high-duty application, so not surprisingly this type of valve construction has only received limited use for automotive engines owing to its greater cost. The Nissan high-performance 4.5 litre V8 engine with four-valve cylinders, which was introduced in 1990, is a modern example where sodium-cooled exhaust valves have been found advantageous in improving knock resistance and increasing torque when the engine is operated under high speed, heavy load, conditions.

Valve springs

As mentioned in Section 1.4, the valve springs are required to ensure that the motion of the valves and their operating mechanism follows faithfully that of the cams. Since this

motion is a constantly changing one, inertia forces are created that may be either positive or negative. The former tend to maintain the component parts of the valve train in contact with one another, whilst the latter act to separate them. Hence, the valve springs serve to counteract the unwanted negative inertia forces. The valve springs must also maintain adequate sealing pressure for the valves during their intended closed period, in which respect they are assisted by the cylinder gas pressure acting upon the valve heads during the compression and power strokes.

In conventional practice, the valve springs consist of wire wound in the form of a helix or coil. The valve closing load is conveyed axially along the spring, which stresses the material principally in torsion. For the valve open and the valve closed conditions, the ratio of spring loads is usually in the region of 2:1. Since the valve spring is compressed between parallel abutments, its end coils are ground flat and square with the spring axis. The coil ends are also diametrically opposed, so as to minimize an inherent tendency towards bowing of the spring as it is being compressed. Because they are subjected to severe service, valve springs are produced from high-duty materials, these generally being either hard-drawn carbon steel or chrome-vanadium steel. To reduce stress concentration in the spring wire and thus make it less liable to fail by fatigue, the valve spring may be shot peened. In this process the wire surface is bombarded at high velocity with metal shot, which induces a residual compressive stress that discourages crack propagation in the material.

Valve spring surging

If the frequency of load application in the valve train should happen to coincide with the natural vibration frequency of the valve spring itself, a condition known as spring 'surge' may develop as a result of resonance effects. This phenomenon occurs as a wave motion, which is generated by the inertia of the individual spring coils, such that successive compressions and extensions travel along the spring from coil to coil, causing them to be unequally loaded. The effect of this surging is to oppose normal valve spring loading, so that true valve motion is disturbed.

To lessen any tendency towards surging within the operating speed range of the engine, the valve springs are designed to have a high natural frequency of vibration. Various additional measures may be taken to minimize surge, such as the use of double springs, mechanical spring dampers, and progressive rate springs. Double springs are arranged concentrically about the valve stem, each having a different natural frequency of vibration. Furthermore, the outer diameter of the inner spring may be made equal to the inner diameter of the outer spring, so that (lubricated) rubbing occurs between them (Figure 1.105a). This rubbing contact promotes a friction damping effect, which suppresses surge by dissipating its energy as heat. In some earlier designs the inner coil spring was made from rectangular section wire, so as to obtain the maximum load capacity in the minimum space (Figure 1.105a). The incidental advantages of using double springs include greater spring stability and less risk of engine damage in the event of a spring breaking. Another method of promoting friction damping is to encircle the valve spring

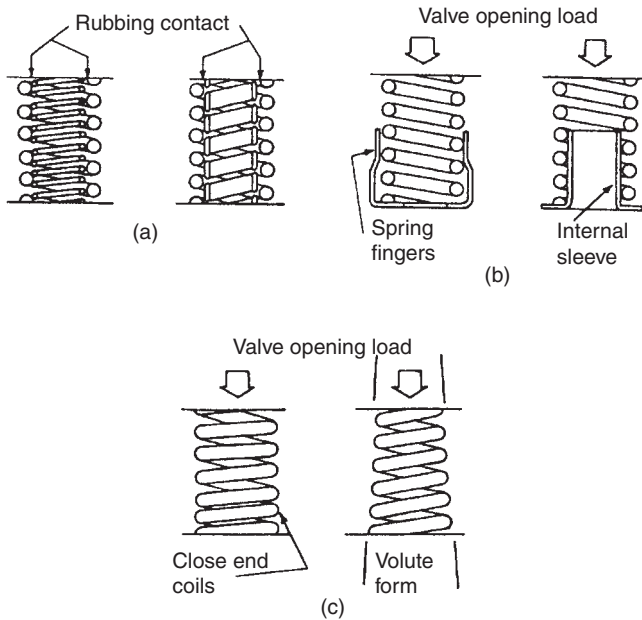


Figure 1.105 Valve spring arrangements to combat surging: (a) double valve spring assemblies (b) valve spring dampers (c) progressive rate valve springs

with spring fingers. These are located from the stationary end of the valve spring and press inwards to make rubbing contact with its centre coils (Figure 1.105b). Alternatively, an internal sleeve may be installed within the spring to achieve a similar effect (Figure 1.105b).

Progressive rate springs are also commonly used to suppress surge. Springs of this type either are close coiled at their stationary end or, less commonly, utilize a volute form with the smallest diameter and hence stiffest coil at the moving end (Figure 1.105c). In both cases, the effect is to vary the number of active coils in the spring during valve lift, which produces changes in its natural frequency of vibration so that resonance is avoided. Space limitations generally preclude the use of surge-free low-inertia springs of the hair-pin and torsion bar types, as alternatives to the helical valve spring in automotive engine practice.

For readers who may be interested in the engine technology of Grand Prix cars, where engine speeds of around 18 000 rev/min are now being achieved, the motion of their valves is controlled not by steel but by pneumatic springs. This concept of using valve stem plungers moving in chambers of compressed air to maintain the valves in contact with their cams, was devised by Bernard Dudot in 1986 for Renault Grand Prix engines.

Methods of valve spring retention

Before leaded petrol was phased out, the different methods of valve spring retention were related to the varying extents that rotation of the exhaust valves was to be encouraged. It was then generally recognized that gradual rotation of the valves during engine operation was beneficial to valve life. This is because it promotes a more uniform temperature distribution around the valve seat, as well as improving heat

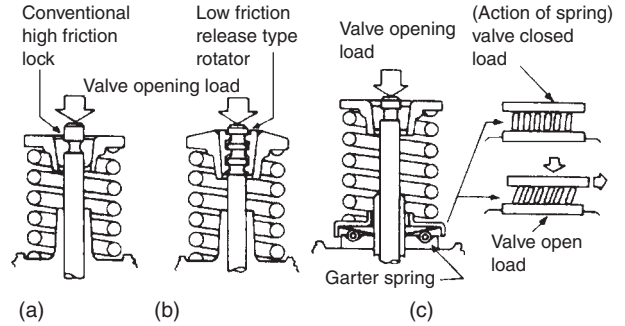


Figure 1.106 Methods of valve spring retention

transfer from the valve by virtue of more effective removal of seat deposits. The valves are generally credited with an inherent tendency towards rotation, which is variously attributed to engine vibration and the slight winding up of the valve springs as they are compressed when the valve is opened. Several devices of American origin may be employed, however, to impart either a non-positive or a positive rotational movement to the valves.

High-friction-lock retention

This was once the conventional method of retaining a valve spring, the moving end of which encircles a spigot washer with a central conical recess. Seating in the latter is a pair of split-type wedge collets which in turn engage a shallow groove machined near the top of the valve stem (Figure 1.106a). The pressure exerted by the valve spring therefore holds the collets tightly in place, yet they can readily be removed (albeit with a little persuasion from a soft-faced mallet in some cases!) by depressing the spring while holding the valve closed. This method of valve spring retention makes no pretence at encouraging rotation of the valves during engine operation.

Release rotators

This method of valve spring retention confers a non-positive rotary movement to the valves by merely permitting them to rotate relatively freely under the influence of engine vibrations. Such action is made possible by dispensing with the radial clamping exerted by the high-friction-lock method of retention and transmitting the spring load to the valve stem through a multiple-thrust collar arrangement, which is provided by abutting collets (Figure 1.106b). This is in contrast to the gap that must necessarily exist between the sides of wedge collets.

Positive rotators

In this method of valve spring retention, valve rotators of the positive type are incorporated as intermittent motion devices. A modern version consists of an enclosed garter spring, which is sandwiched between a thrust washer and the lower valve spring washer, there being the normal high-friction lock against the upper end of the valve stem (Figure 1.106c). During operation the increase in valve spring load that accompanies valve opening causes the coils of the garter spring to lean over. In so doing they act as friction sprags and transmit a torque to the valve spring and its upper retaining washer, which in turn imparts a small rotational movement to the valve itself.

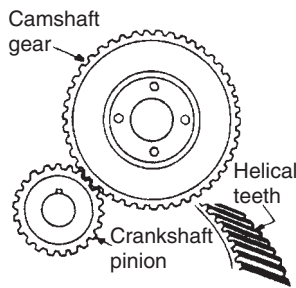


Figure 1.107 Arrangement of gear timing drive

With the requirement to use unleaded petrol, there is less incentive to impart a rotational movement to the valves. In the absence of what was previously a 'lubricating' film of lead oxides and lead sulphides forming on the valve seating, it is now thought that valve rotation could lead to scuffing wear.

1.11 TIMING DRIVE

Methods of driving camshafts

In conventional four-stroke engine practice one or more camshafts, according to valve train requirements and arrangement of the cylinders, are driven from the front end of the crankshaft. Since the axes of the crankshaft and the camshaft(s) are parallel to one another, a positive drive (non-slipping) between them may be afforded by any of the following means of power transmission: gear, chain, and toothed belt.

Gear train

Gear drives were once widely favoured for both crankcase- and cylinder-head-mounted camshafts. Except in the case of competition engines their use is now confined to the former application, helical teeth being generally specified in the interests of smooth and quiet operation (Figure 1.107). These qualities derive from the progressive engagement characteristics of their inclined teeth, one pair remaining in mesh until the following pair are partially engaged (Section 14.1). To reduce further any noise from gear-type drives, a non-ferrous acoustically dead material such as aluminium alloy may be specified for the camshaft gear. A steel crankshaft pinion is retained in this event.

Chain and sprockets

Until the advent of the toothed-belt timing drive for cylinder-head-mounted camshaft(s), the chain and sprockets drive was the most widely used for both crankcase- and cylinder-head-mounted camshafts. The two types of chain drive employed are known as the roller chain and the inverted-tooth chain, each being designed to avoid sliding contact between the teeth of the sprockets and the chain.

Historically, the bush roller chain was invented and patented as a means of power transmission by Hans Renold in 1880. In construction, the side plates of the inner link pairs are rigidly connected by bushings, within which oscillate the bearing pins connecting the side plates of the outer link pairs, and upon which rotate the rollers that engage the sprocket teeth (Figure 1.108). Any wearing of the pivot joints will tend to increase the pitch of the chain, that is the linear distance

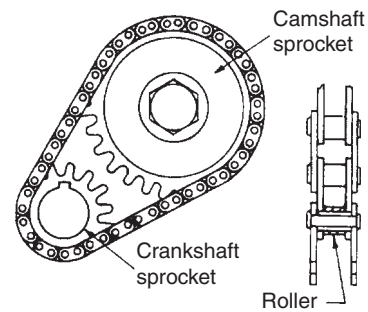


Figure 1.108 Arrangement of roller chain timing drive



Figure 1.109 Inverted-tooth chain timing drive

measured between adjacent roller axes. To accommodate this apparent stretch of the chain, the sprocket teeth are curved in profile so that, as the rollers ride higher on the teeth, contact is still maintained between each roller and tooth included in the arc of chain wrap.

According to the power to be transmitted, roller chains may consist of either a single or two or more strands; the multistrand chains provide a greater drive capacity for a given pitch of chain. In a recent example of a roller chain timing drive for a double-overhead camshaft engine, the sprocket wheels are provided with plastic inserts to support the chain and reduce any tendency to whine.

The inverted tooth chain drive (Figure 1.109) was another Renold invention and, although originally used in Britain until the mid-1920s, it subsequently found widespread application for driving a crankcase-mounted camshaft during the side-valve engine era in America. This type of timing chain comprises a multiplicity of overlapping toothed links, which are connected together by means of either bearing pins and bushes or special rocker pivots. The teeth formed both on the chain links and the sprockets are of tapered profile, so that full-length bedding is established along their contact faces. Any wear occurring in the pivot points of this type of chain is accommodated by the link teeth moving higher up the sprocket teeth. To prevent the inverted-tooth chain from sliding endwise off its sprockets, either the outermost links are extended radially inwards so as to embrace the sides of the sprocket teeth, or central guide links engage a groove in the sprocket. Similar to the roller chain, the drive capacity of the inverted-tooth chain increases in proportion to its width. For both roller and inverted-tooth timing chains the sprockets are usually made from either steel or cast iron. In American practice an inverted-tooth timing chain may be used in conjunction with an aluminium alloy camshaft sprocket that has nylon teeth.

Toothed belt

Originally adopted in 1963 by the Hans Glas Company in Germany as a method of driving an overhead camshaft for a

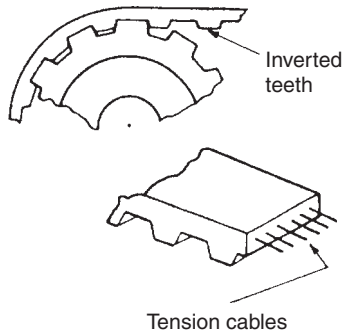


Figure 1.110 Arrangement of toothed-belt timing drive

motor car engine, the inverted-tooth belt was actually invented by an American, Richard Case, of the US Royal Company some 25 years previously and first found application to the synchronous needle and bobbin drive of sewing machines.

To provide a positive timing drive, as opposed to the non-positive friction drive conferred by an ordinary fan V-belt, the belt teeth engage with axial grooves in the peripheries of the crankshaft and the camshaft pulleys (Figure 1.110). Tooth profiles are based on either trapezoidal or semi-circular forms, which permit a smooth rolling motion upon engagement and disengagement of the belt with its pulleys, similar to the action of gearing. The shear strength of the teeth in mesh always exceeds the tensile strength of the belt. In this context the tension loads are transmitted through the neutral axis of the belt by a layer of either fibre-glass cords or steel cables, their strands being spirally wound in alternate directions to stabilize the running of the belt. The main body of the belt is constructed from neoprene synthetic rubber, and to protect the teeth it is faced with a hard-wearing polyamide (nylon) material that also has a low coefficient of friction against the pulley grooves. With some products the intended direction of travel for the timing belt is arrow-marked on its back surface. Axial location of the belt may be effected by a flanged jockey pulley. Lubrication is not, of course, required for a toothed timing belt, which is generally enclosed for protection against the elements and as a safety precaution.

Camshaft drives compared

Apart from quietness of operation, a timing drive must further meet the all-important criteria of low cost, ease of assembly and maintenance, and a long trouble-free life. In purely general terms, the relative merits of the various methods of driving camshafts may be summarized as follows.

Gear drive

This better maintains the accuracy in timing of the camshaft and, especially important in the case of diesel engines, the fuel injection pump. In contrast significant changes in timing can occur owing to apparent stretch in chain drives over an extended period of service, since this particular effect cannot be compensated for by any tensioning device. Gear drives, as always, must be carefully engineered and therefore tend to be expensive, but in high-grade practice they can be made both quieter and longer lived than chain drives. Their application is, however, mainly confined to timing drives for

crankcase-mounted camshafts, since roller chain and toothed-belt drives offer a much simpler layout for driving overhead camshafts.

Chain drive

This can readily accommodate either short or long centre distances between the crankshaft and camshaft(s) and is therefore equally suitable for both crankcase- and cylinder-head-mounted camshafts, although the heavier inverted-tooth chain is less well suited to the latter application. Unlike a helical gear timing drive, no end thrust forces are generated at the sprockets of timing chains, which are also less intolerant of any slight misalignment. On the debit side, a timing chain is subject to the cumulative effects of slight wear occurring at the joints of its many links that results in apparent stretch. For most chain runs it therefore becomes necessary to incorporate either a manual or an automatic adjuster to maintain correct chain tension.

Furthermore, at certain critical engine speeds fluctuations in torque can arise in the rotating elements of the timing drive, which can excite the chain into vibration and require the addition of rubber-faced damper shoes to bear against the 'tight' side of the chain (Figure 1.114 and 1.115).

Toothed belt

As earlier inferred, this form of timing drive is complementary to the modern revival of cylinder-head-mounted camshafts. Its advantages include inherent quietness and smoothness of operation, lightness in weight and no lubrication requirements. Hence, this type of timing drive can be accessibly mounted external to the engine main structure. On the debit side, the greater width of the toothed belt tends to demand more installation space at the front of the engine. Also whereas the gear and the chain types of timing drive are usually expected to last the normal service life of an engine, the toothed belt has to be considered a consumable item, a typical early recommendation being that it should be inspected with a view to replacement after 30 000 miles (48 000 km). Higher engine speeds and the adoption of four-valve layouts have both increased the duty imposed on the toothed belt timing drive, even so the present-day replacement interval has tended to become longer and in some cases doubled to 60 000 miles (96 000 km). Whilst this reflects the improvements that have been made in timing belt construction, it is still essential to observe the particular manufacturer's recommendations in this respect.

There has nevertheless been a trend by some manufacturers to revert to a chain timing drive, because they represent a more compact installation and can offer greater durability with lower maintenance. Also, there can be the additional duty to drive a high-pressure fuel delivery pump where diesel and direct injection petrol engines are used.

Timing belt misbehaviour in service

The mileage interval specified by the vehicle manufacturer for replacing a timing belt is intended to provide a margin of safety, before the belt ultimately fails by fatigue of its constantly flexing tensile members. However, there are other factors that can arise in the operation of a timing belt, which at least can prematurely shorten its useful life and at worst lead

to breakage, with potentially major damage to the engine where piston to valve contact occurs through loss of timing.

As a general guide to some of the problems that may be encountered with timing belts in service, the following is a list of their likely causes:

Undue wear on belt teeth

Belt tension either insufficient or excessive causing incorrect meshing with toothed pulleys

Undue wear between belt teeth

Belt tension excessive causing too deep meshing with toothed pulleys

Belt abraded by rough tips on toothed pulleys

Undue wear on edge of belt

Misalignment between the toothed pulleys

Damaged flange on toothed pulley that provides axial location for belt

Teeth cracking at roots and peeling from belt

Belt tension insufficient to prevent undue bending of the teeth as they ride too high on the toothed pulleys, and can lead to the belt jumping teeth

Group of teeth sheared from belt

Sudden overload imposed on the belt by seizure of a driven auxiliary

Structural deterioration of belt

Oil or diesel fuel contamination of the belt causing swelling, separation and ultimate destruction of the rubber composite material

Back surface crazing of belt

Excessive frictional heat caused by seizing of either an idler pulley or driven auxiliary acting against the back surface of the belt

Belt torn apart

Acute bending of the belt either before or during installation causing a local weakness and tearing between adjacent teeth
Rogue object caught up between the belt and a toothed pulley causing local over-stretching and tearing across a group of teeth.

Since a toothed belt timing drive is normally quiet in operation, any noise that develops can provide an audible warning of distress. A whining noise can indicate excessive tensioning of the belt, while a slapping sound is associated with insufficient tensioning.

Attachment of the camshaft timing wheel

The once traditional method of attachment was to key the timing wheel on a reduced-diameter extension of the camshaft,

endwise retention being effected by a central clamping nut. To prevent relative rotation between the clamped parts a 'Woodruff' key was used, this being close fitting only on the sides of its keyway and self-aligning by virtue of its semicircular form (Figure 1.111a). In modern practice, the function of the Woodruff key is generally replaced by that of an off-centre dowel pin (Figure 1.111b). The dowel pin is simply a plain cylindrical pin that is made close fitting in corresponding holes in the timing wheel and the end face of the camshaft, these two members being spigotted together for radial location. Their endwise clamping is effected by a central bolt.

For heavier-duty applications, a ring of close-fitting bolts may perform both the clamping and the driving functions, radial location being supplied by a spigot fitting for the timing wheel (Figure 1.111c). This particular method of attachment also lends itself to a fine adjustment feature for setting the valve timing, since if two sets of holes are provided in the timing wheel it is then possible to make incremental adjustments of half a tooth pitch to the timing. Alternatively, if there is no requirement for such fine adjustment, the correct positioning of the timing wheel may be guaranteed by offsetting one of its ring of bolt holes to correspond with a similar offset hole in the camshaft end.

The attachment of the crankshaft timing wheel has been dealt with in Section 1.8.

Controlling end float of the camshaft

In arrangements where the oil pump (together with ignition distributor) is still provided with a right-angle drive from the engine camshaft, via a pair of crossed-helical or skew gears, the obliquity of their teeth is such that an end thrust must be reacted against by the camshaft bearing arrangements. Similar considerations apply, of course, where a helical gear timing drive is used. For these reasons it therefore becomes necessary to control any endwise movement of the camshaft.

Axial location for a crankcase-mounted camshaft is commonly effected by the parallel surfaces of a collar thrust bearing, which acts on the timing wheel end of the camshaft. For some applications, a thrust plate bolted to the front wall of the crankcase is provided with a semicircular opening, the edge of which engages with a clearance groove around the camshaft front journal (Figure 1.112a). In other designs, the thrust plate may completely encircle a similar clearance groove that is formed partly by a shoulder machined on the front end of the camshaft, and partly by the inner end face of the timing wheel boss. Less commonly, a true collar thrust

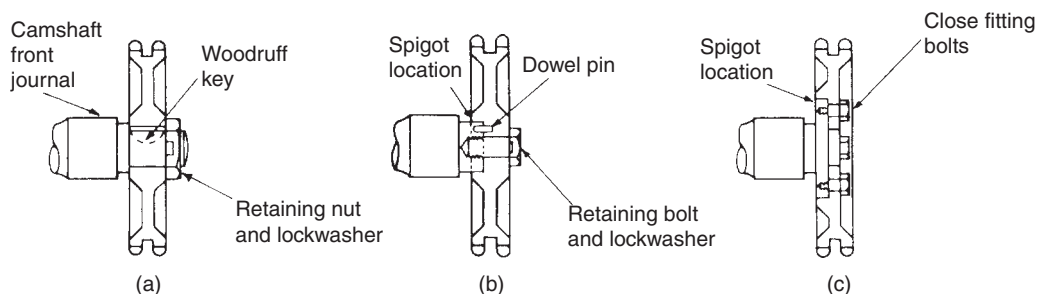


Figure 1.111 Camshaft timing wheel mountings

bearing in the form of a flanged front journal may be employed (Figure 1.112b). The flange registers in an annular clearance groove in the forward part of the bearing housing, which is counter-bored and provided with a closing thrust plate.

Cylinder-head-mounted camshafts are located axially by collar thrust bearings similar to those just described, except where detachable bearing caps are used. Here, the end thrust loads may be reacted against on either side of the front bearing housing by the inner end face of the timing wheel boss on the one hand and by a flange formed immediately behind the camshaft front journal on the other (Figure 1.113a). Where an inverted-tooth belt drive is used for overhead camshafts, axial location may be effected from the rear end of the camshaft, again by methods similar to those already described (Figure 1.113b).

For all these various thrust bearing arrangements the camshaft end float or end play typically falls within the range of 0.15–0.38 mm (0.006–0.015 in).

Timing drive installations

Whereas only a one-stage timing drive is demanded by a crankcase-mounted camshaft for push-rod-operated overhead

valves (Figure 1.114), either one or more stages of chain drive may be needed for cylinder-head-mounted camshafts. For in-line cylinder engines with a single overhead camshaft a one-stage timing drive suffices (Figure 1.115), but where double overhead camshafts are used a two-stage drive is required (Figure 1.116). The latter entails the use of an idler

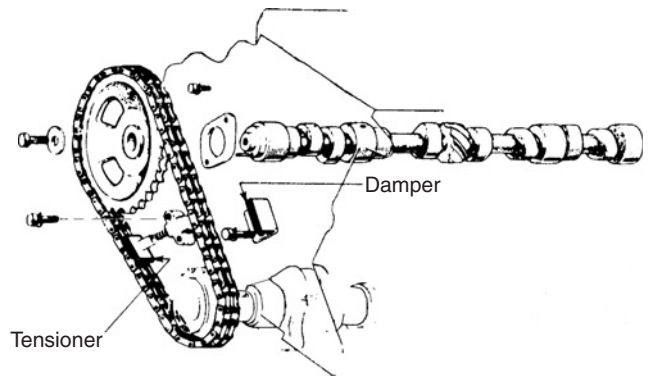


Figure 1.114 Exploded view of a one-stage roller chain timing drive (Toyota)

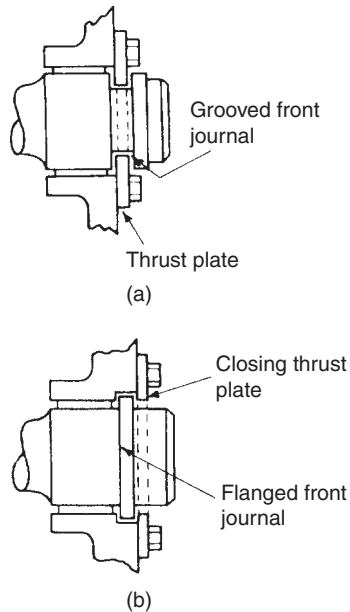


Figure 1.112 Collar thrust bearings for crankcase-mounted camshaft

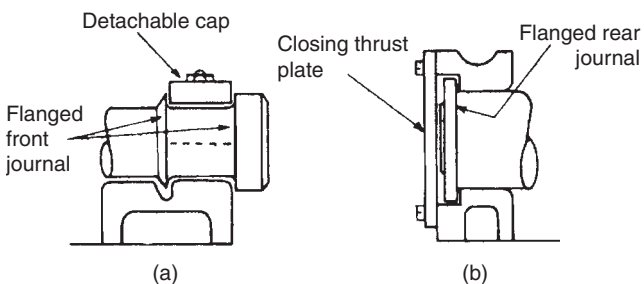


Figure 1.113 Thrust bearings for cylinder-head-mounted camshaft(s)

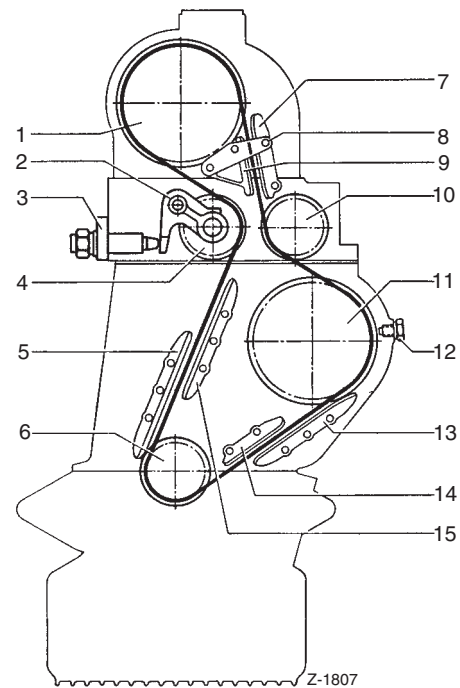


Figure 1.115 A one-stage roller chain driving a single overhead camshaft (Mercedes-Benz)

- | | |
|---|-------------------------|
| 1 camshaft sprocket | 9 chain guide, inside |
| 2 sprocket bearing and tension sprocket | 10 reversing sprocket |
| 3 chain tensioner | 11 intermediate gear |
| 4 tension sprocket | 12 locking screw |
| 5 chain guide, outside | 13 chain guide, outside |
| 6 crankshaft sprocket | 14 chain guide, inside |
| 7 chain guide, outside | 15 chain guide, inside |
| 8 support bracket for chain guide, inside | |

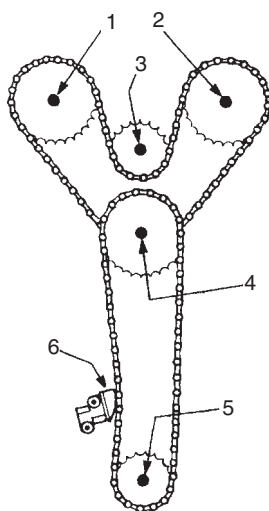


Figure 1.116 A two-stage roller chain driving double overhead camshafts (*Renold*)

- 1,2 camshaft sprockets
- 3 secondary stage manually adjustable idler sprocket
- 4 tandem idler sprockets (one hidden behind the other)
- 5 camshaft sprocket
- 6 primary stage slipper shoe chain tensioner

sprocket assembly comprising two sprockets in tandem, so that the primary stage transmits drive from the crankshaft sprocket to one of the idler sprockets, and the secondary stage is from the other idler sprocket to each camshaft sprocket. This arrangement also enables the 2:1 timing drive reduction ratio to be shared between the primary and secondary stages, thus permitting the use of smaller-diameter sprockets than would otherwise be possible.

A similar arrangement is required for V-formation cylinder engines with double overhead camshafts on each bank, although the installation becomes further complicated in that the secondary stage must be duplicated. The detailed manner in which such a multi-stage timing drive has been accomplished by Nissan engine designers for their modern V six-cylinder engine is best visualized by considering the arrangement in its entirety (Figure 1.117). Where a single overhead camshaft is used on each bank of a V engine, a much simpler one-stage drive is possible.

For in-line cylinder engines utilizing a toothed-belt timing drive, a one-stage drive only is required for both single (Figure 1.118) and double overhead camshaft installations (Figure 1.30). Note also in the latter example that the coolant pump is positively driven by the timing belt. Prior to the advent of direct type of ignition systems that do not need a distributor (Section 10.5), it was established practice for the distributor to be driven in tandem from the rear of an overhead camshaft. This source of drive may now be utilized to power a high-pressure pump, which delivers fuel to the common rail of a modern direct injection petrol engine (Section 6.9). An auxiliary or jack shaft that is mounted in the crankcase in a similar manner to that of a camshaft may be included in the timing drive to operate the oil pump and also, in earlier practice, a mechanical fuel pump (Figure 1.118). Otherwise, the oil pump is driven directly from the front of the crankshaft

(Figure 1.30 and 3.10) and an electrical fuel pump is used. A one-stage toothed belt timing drive may also be used for V engines with a single overhead camshaft on each bank, but separate belts with side-by-side drive pulleys on the crankshaft are required for horizontally opposed cylinder engines (Figure 4.13).

Further developments in timing drive installations

In the case of engines with four-valve combustion chambers where a narrow, rather than a wide, included angle is adopted for the two pairs of valves, the closeness of the two camshafts may preclude the use of a one-stage toothed-belt drive to both. That is, a toothed-belt drive can be taken to one camshaft and some other form of inter-camshaft drive must be added to complete the timing drive.

When Toyota were faced with this particular engine development problem, they found that the use of a chain drive over such a short distance between centres and relatively small sprockets was neither quiet nor reliable enough, because of the torque fluctuations imposed by changes in valve spring load as the valves opened and closed. Instead they chose to use a helical gear drive between the two camshafts (Figure 1.119a and 1.36) and made special provision for ensuring its quiet operation. Since noise can arise from the backlash that results from the tolerance necessary for correct meshing of the teeth in a pair of gears, the driven gear is of divided construction with one part being torsionally spring-loaded against the other to cushion the effect of the backlash (Figure 1.119b and 1.22). Although this method of quietening a pair of gears has been used in earlier timing drive systems, this is the first time it has been applied with such simplicity and to a double overhead camshaft layout. It is referred to by Toyota as their 'inter-camshaft scissors gear drive'.

Timing drive tensioners

Various forms of jockey sprocket, slipper shoe and spring blade tensioners may be arranged to bear against the slack (non-driving) side of roller chain runs. The jockey sprocket type of tensioner is mounted on either a sliding plunger, a swinging arm, or an eccentric bushing. Slipper shoes may similarly be mounted on either a sliding plunger (Figures 1.114 and 1.116) or, less commonly, on a swinging arm (Figure 1.115). Blade-type tensioners generally take the form of rubber-faced spring steel strip, which is so mounted that it bows towards the chain under the influence of an extension spring connecting its ends (Figure 1.120a). A further refinement is for a semirigid blade to be produced from a thermoplastics material, which when hot can deform under the influence of its spring loading and readily adopt the natural contour of the chain run to exert optimum tensioning.

For their automatic operation, all the various types of chain tensioners rely primarily on spring loading (Figure 1.115), which may or may not be adjustable externally. In one version, the spring loading is augmented by hydraulic pressure derived from the engine lubrication system (Figure 1.120b). The oil is admitted behind the spring-loaded plunger of the tensioner mechanism and escapes via a bleed orifice to confer a measure of built-in hydraulic damping action for resisting chain flutter. Chain tensioners of the jockey sprocket and

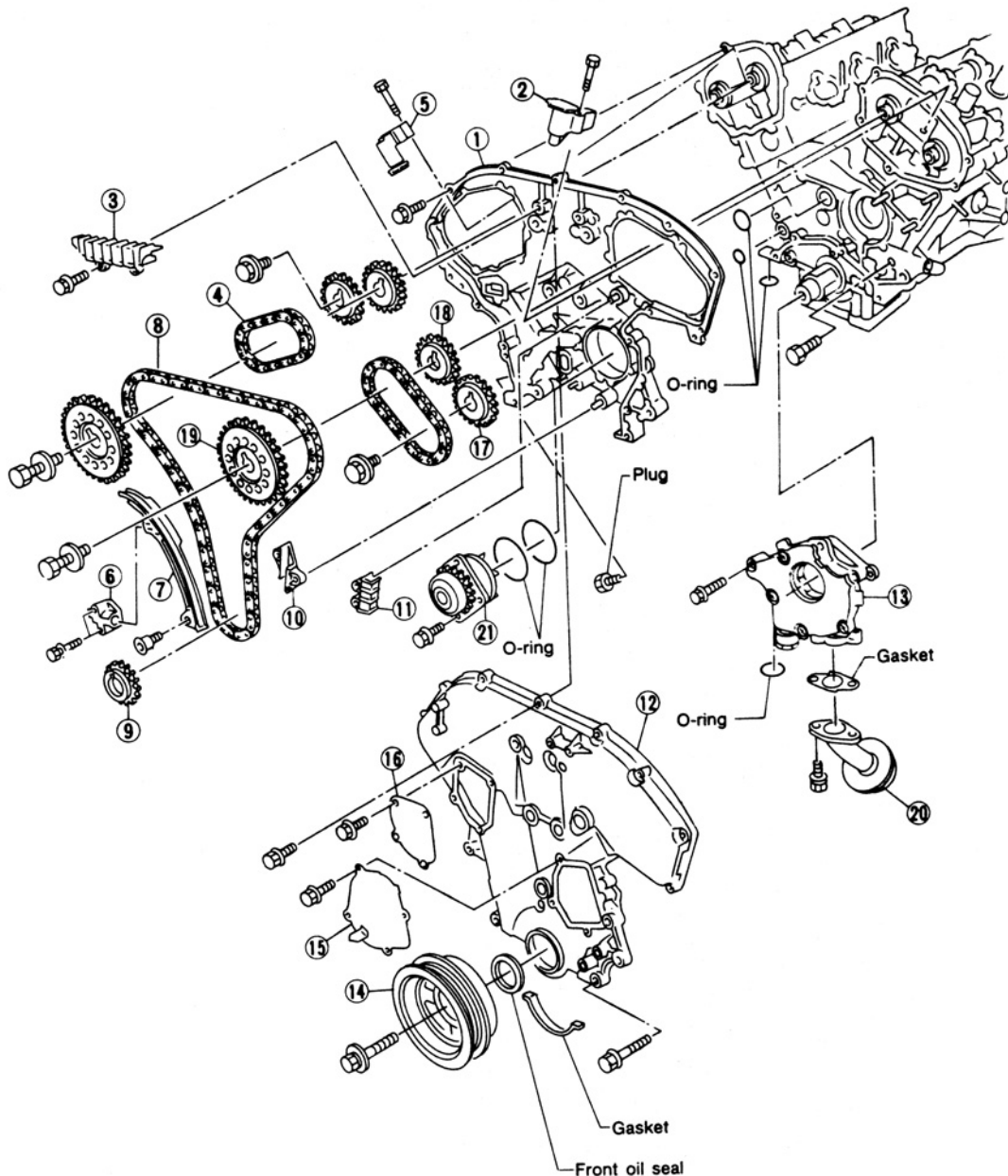


Figure 1.117 A multi-stage roller chain timing drive chain and sprockets for a modern V six-cylinder engine (*Nissan Motor GB*)

- | | | |
|--------------------------------|--------------------------------|------------------------------|
| 1 Rear timing chain case | 8 Timing chain | 15 Water pump cover |
| 2 Left camshaft tensioner | 9 Crankshaft sprocket | 16 Chain tensioner cover |
| 3 Intake chain tensioner guide | 10 Lower chain tensioner guide | 17 Exhaust cam sprocket, 2nd |
| 4 Camshaft chain | 11 Upper chain tensioner guide | 18 Intake cam sprocket, 2nd |
| 5 Right camshaft tensioner | 12 Front timing chain case | 19 Camshaft sprocket, 1st |
| 6 Chain tensioner assembly | 13 Oil pump assembly | 20 Oil strainer |
| 7 Chain slack guide side | 14 Crank pulley | 21 Water pump |

slipper shoe types generally incorporate a non-return mechanism. Its purpose is to prevent any significant loss of adjustment under adverse conditions of chain loading, which may force back the tensioner an undue amount. The non-return or locking feature usually takes some form of pawl and ratchet device (as in a handbrake mechanism), in which the movement of the tensioner in taking up chain slack is followed up

by the pawl engaging successive teeth on the ratchet. Over-tensioning of the chain is avoided by the teeth on the pawl and ratchet assuming a trailing angle, such that with each increment of travel they always allow the chain tensioner a limited return movement against its spring loading.

With toothed-belt timing drives for overhead camshafts, a jockey pulley is usually arranged to guide the back of the

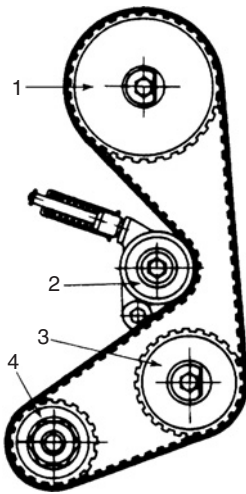
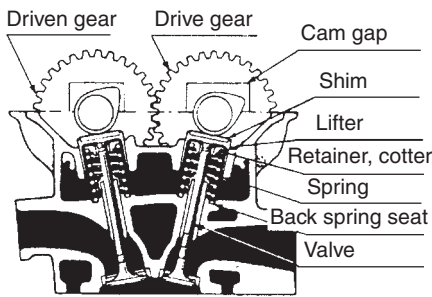
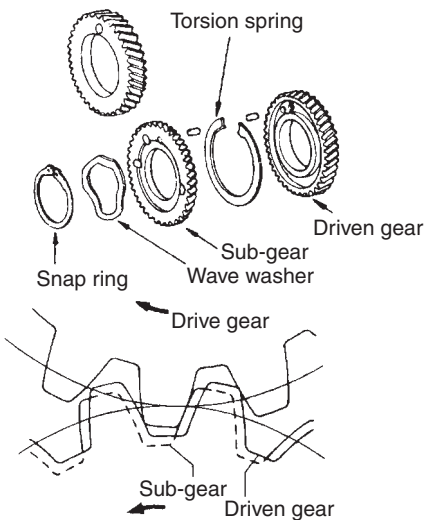


Figure 1.118 A toothed belt driving an overhead camshaft (*Lancia*)

- 1 camshaft toothed pulley
- 2 spring-loaded jockey pulley
- 3 toothed pulley for auxiliary units drive shaft
- 4 crankshaft toothed pulley

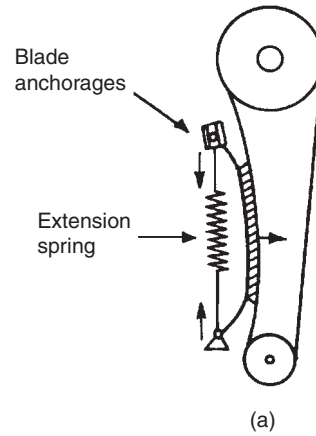


(a)

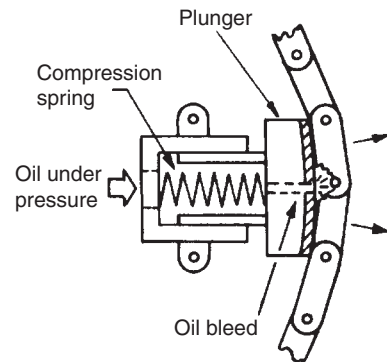


(b)

Figure 1.119 Upper timing drive for narrow-angle four-valve layout: (a) geared-together camshafts (b) scissors gear mechanism (*Toyota*)



(a)



(b)

Figure 1.120 Types of automatic timing chain tensioner: (a) spring loaded blade (b) spring loaded plunger

slack side of the belt run (Figure 1.118). It is either manually tensioned or automatically spring loaded, so as to absorb manufacturing tolerances in the timing drive and to facilitate installation of the belt when the tensioning arrangements are released.

1.12 THE PRINCIPLES OF VALVE TIMING

Basic considerations

So far in our understanding of the four-stroke principle, we have accepted that the opening and closing of the inlet and exhaust valves is timed to coincide exactly with the beginning and ending of the induction and exhaust strokes. For a variable-speed motor vehicle engine such a tidy approach to valve timing would result in very inefficient operation. In actual practice a departure must therefore be made from the basic valve timing implied by the four-stroke principle, this being early acknowledged by the French automotive pioneer Fernand Forest who originally patented the idea of valve timing in 1906. Let us now examine how this departure can be accounted for in relation to the factors involved, which may be listed as follows:

- 1 Inertia effects of the incoming and outgoing cylinder gases
- 2 Elastic nature of the incoming and outgoing cylinder gases

3 Mechanical stresses imposed by rapidly opening and closing the valves.

In physics we learn that all bodies possess inertia, no matter whether we are talking about a cannonball or a column of gas, and that the effect of inertia is to resist a change of motion. The incoming and the outgoing flows of cylinder gases are therefore subject to inertia effects. As a consequence there is an unwelcome reluctance of the gases to commence flowing, followed by a welcome reluctance of them to cease flowing.

Also in physics we discover that the volume of a gas, unlike a liquid, is not fixed and that this property of great compressibility is possessed by all gases. The elastic nature of any burnt cylinder gases remaining in the combustion chamber at the end of the exhaust stroke can, therefore, be such that they expand and impede the flow of incoming air and petrol mixture at the beginning of the induction stroke.

In engineering we appreciate that when a material is acted upon by a force, every particle in the material acts upon every other particle with which it is in contact and sets up what is termed a state of stress. This imposes definite limits as to the rapidity with which a valve can be opened and closed, before breakdown of lubrication and overstressing of the contact surfaces of its cam and tappet occurs. Due allowance must therefore be made for the short time it necessarily takes to open and close each valve during its open period.

How then can the basic valve timing of the four-stroke principle be modified in practice to accommodate the various effects mentioned? It is done by providing for the *lead* (advance time) and *lag* (delay time) of the inlet and exhaust valve periods of opening, or *valve events* as the American industry prefers to call them.

Lead, lag and overlap

The inlet valve is given a lead in opening before the piston reaches top dead centre on the exhaust stroke (Figure 1.121a), so that least resistance is offered to the incoming flow of air and petrol mixture as the piston begins its induction stroke. It is also provided with a lag in closing after the piston reaches bottom dead centre and is beginning the compression stroke (Figure 1.121a), so as to take advantage of the reluctance of the incoming mixture to cease flowing as the piston ends its induction stroke. The maximum amount of air and petrol mixture is therefore induced to enter the cylinder, upon which depends the power developed by the engine.

Similarly, the exhaust valve is given a lead in opening before the piston reaches bottom dead centre on the power stroke (Figure 1.121b), so that the burnt gases are already leaving the cylinder under their own pressure as the piston begins its exhaust stroke. Therefore the engine expends less energy on expelling the exhaust gases than would otherwise be the case. The exhaust valve is also provided with a lag in closing after the piston reaches top dead centre and is beginning the induction stroke (Figure 1.121b). This not only better scavenges the combustion chamber of exhaust gases, but also lowers the cylinder pressure to facilitate flow of the incoming air and petrol mixture.

By modifying the basic valve timing of the four-stroke principle in the manner described, it necessarily follows that the opening of the inlet valve before top dead centre on the

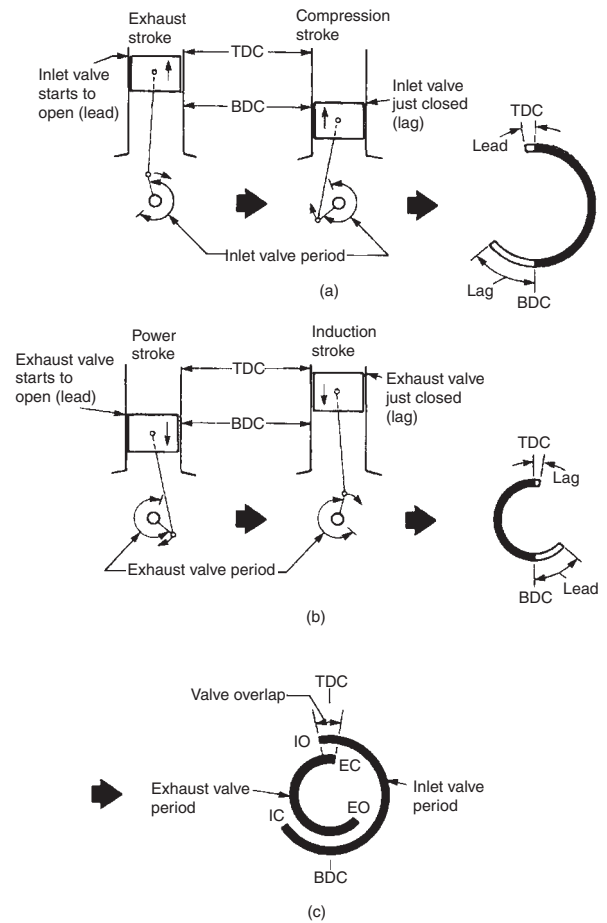


Figure 1.121 The principles of valve timing: (a) exhaust and compression (b) power and induction (c) valve timing diagram

IO inlet valve starts to open
 IC inlet valve just closed
 EO exhaust valve starts to open
 EC exhaust valve just closed

exhaust stroke, and the closing of the exhaust valve after top dead centre on the induction stroke, result in a period when both valves are either partially or fully open. The superimposing of that part of the valve periods where the inlet valve opens before the exhaust valve closes is termed the *valve overlap* (Figure 1.121c).

Valve timing diagrams

For convenience of expression the opening and closing points of the valves have traditionally been shown in the form of a valve timing diagram (Figure 1.121c), although the trend in present-day service manuals is merely to tabulate this data, or simply to quote the number of degrees before top dead centre when the inlet valve begins to open. For tabulating valve timing information the commonly used abbreviations BTDC and ATDC denote before top dead centre and after top dead centre, and refer to the positions of the crankshaft as the piston is respectively advancing towards, and retreating from, the combustion chamber. Similarly, BBDC and ABDC denote before bottom dead centre and

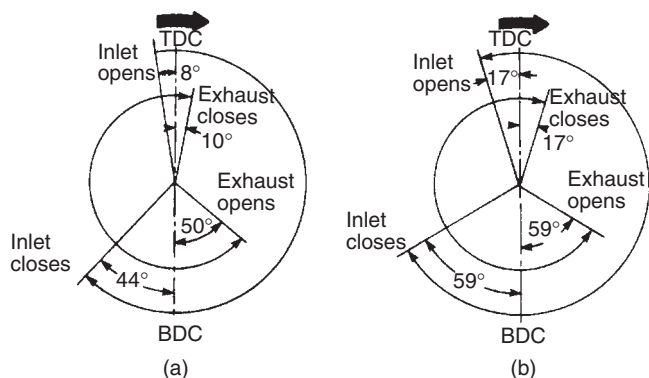


Figure 1.122 Typical valve timing diagrams for four-stroke petrol engines: (a) medium-performance car (b) high-performance car

after bottom dead centre, and correspondingly relate to the opposite sense of piston travel.

An early method of expressing valve timing was simply to relate the opening and closing points of the valves to measured distances of the piston from top dead centre. However, since piston travel is quite small near each end of the stroke for comparatively large angular movements of the crankshaft, this method did not guarantee sufficient accuracy of setting the valve timing in practice, so angular rather than linear values are always quoted.

In everyday engine overhaul the valve timing details are really only a matter of academic interest, since the actual process of setting the valve timing can usually be accomplished simply by ensuring correct alignment of correlation markings on the relevant parts of the timing drive mechanism. However, the need for actually verifying the valve timing values of an engine can arise, for example, in specialist tuning or supertuning of a competition engine where the camshaft(s) has been reground to a specification different from that issued by the original manufacturer.

When comparing the valve timing diagrams for the petrol engine of medium- and high-performance cars (Figures 1.122a and b), it will be noticed that with the latter the valve periods of opening (or events) are extended to greater values of crankshaft angularity. This is especially true in relation to the later closing of the inlet valve, so that at high operating speeds the increased lag allows as much pressure energy as possible to be generated in the cylinder by the incoming air and petrol charge, prior to its further compression by the rising piston. There will also be an increase in the value of valve overlap for the high-performance engine. This means, in effect, that at the top dead centre of the piston both inlet and exhaust valves will be unseated to a greater extent (neither the exhaust valve fully closed nor the inlet valve fully opened) and thereby encourage better breathing of the engine, again at higher operating speeds.

For the medium-performance engine any such increase in the value of valve overlap would be unrealistic, since it could promote a reverse flow of exhaust gases in to the induction system, especially with the increase in valve area of four-valve combustion chambers. This would, of course, tend to occur when manifold depression was high during light and closed-throttle running conditions. The ensuing contamination of the

fresh air and petrol charge by exhaust gases would result at least in poor low-speed performance and unstable idling, and at worst in actual misfiring, if the oxygen content of the contaminated charge became insufficient to support combustion.

Variable valve timing

Although the basic valve timing implied by the four-stroke principle was early modified for better engine performance, as long as the valve timing is fixed with respect to crankshaft rotation it must still remain something of a compromise. More specifically, it would be an advantage if the valve overlap period could be varied, so that its duration could be shortened for engine starting and during low-speed running and lengthened for high-speed running. By varying the valve timing in this manner it should therefore be possible not only to optimize engine performance at all running speeds, but also to improve fuel economy and reduce harmful exhaust emissions. In this context it should be noted that any overlap dilution of the air and fuel mixture with exhaust gas constitutes a simple form of exhaust gas recirculation (EGR) (Section 11.2).

The concept of variable valve timing has long proven an attraction to engine designers, but until recently its practical realization was always considered to be fraught with mechanical complication. Credit is therefore due to the Italian Alfa-Romeo company which, in 1982, was the first to introduce a relatively simple form of variable valve timing on a production car engine. This particular system, known as variable inlet valve timing or simply VIVT, is applicable only to a double overhead camshaft valve train and essentially involves mounting the drive sprocket on the inlet camshaft through an axially movable sleeve or timing piston. This is splined both internally and externally where it engages the camshaft and sprocket respectively, the internal splines being helical and the external ones straight. When the timing piston is moved axially against its return spring, it therefore causes the camshaft to advance rotationally in relation to its sprocket by virtue of the helical spline connection. The timing piston is hydraulically actuated by engine oil pressure, which is admitted via a solenoid valve controlled from the electronic management system for the engine.

A further development of this type of variable valve timing system is the Renold 'Camphase' camshaft angle phase controller. Instead of providing alternative angles of either advance or retard for the inlet camshaft, the purpose of the Renold system is to vary continuously the angular relationship between the inlet camshaft and crankshaft, termed the 'phase angle', over a range of 30° crank angle according to the dictates of the engine management system. Hence the optimum phase angles can be pre-programmed into the electronic control unit and the actual phase angles adjusted to these values. The device is hydraulically actuated by engine oil pressure via solenoid valves, a closed-loop control system being used to monitor the phase angle of the camshaft.

At the heart of the phase angle controller is a linear-to-rotary slidable piston, the bore of which is formed to engage a helical shaft that is mounted as an extension to the camshaft (Figure 1.123). The piston is externally of square form and engages a complementary bore in the phase controller outer body, which carries the camshaft driving sprocket. Therefore

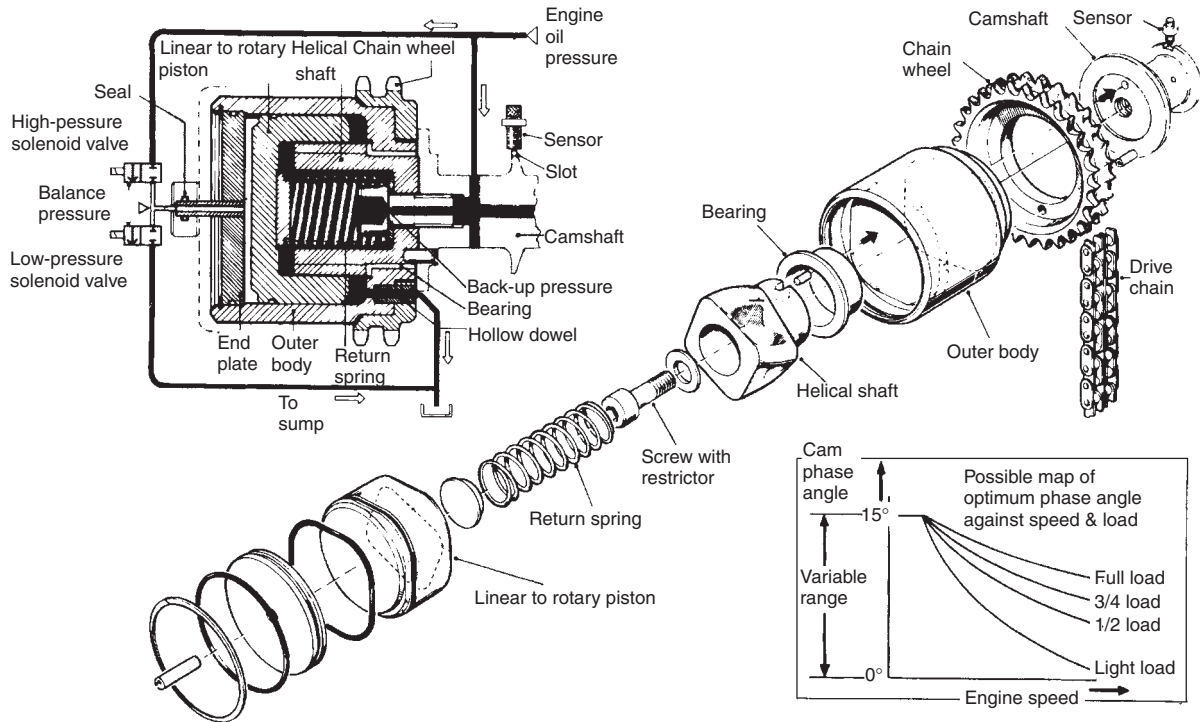


Figure 1.123 Camphase camshaft angle phase controller (*Renold Automotive*)

any enforced endwise movement of the piston over the helical shaft will cause partial rotation not only of the sliding piston, but also of the outer body of the phase controller through which it is sliding. By this means the camshaft can be partially rotated in relation to its driving sprocket.

The piston is actuated hydraulically by engine oil pressure controlled by two solenoid valves. When the high-pressure (HP) valve is opened it admits a balancing oil pressure against the head of the piston, the force on which exceeds that exerted by the return spring and moves the piston endwise to a new position towards the camshaft. Closing the high-pressure (HP) valve and opening the low-pressure (LP) valve releases the oil under pressure to the sump, so that the return spring causes the piston to retreat away from the camshaft. A moderate back-up pressure is maintained within the piston for lubrication and vibration damping purposes. The operation of the phase controller is therefore such that by opening and closing the HP and LP valves, the endwise position of the piston can be adjusted and hence the phase angle between the inlet camshaft and crankshaft varied continuously. Any misbehaviour of the camshaft phase controller in service will be detected and indicated by the engine management system, the device itself possessing a failsafe characteristic that allows continued running of the engine.

Another approach to variable valve timing is that pioneered by the Honda company, which is designated 'VTEC' for variable valve timing and lift electronic control system. Instead of allowing a controlled rotational movement for the camshaft relative to its driving sprocket, it utilizes three cam lobes and three rocker arms for each pair of valves in the four-valve combustion chambers (Figure 1.124). The two outer

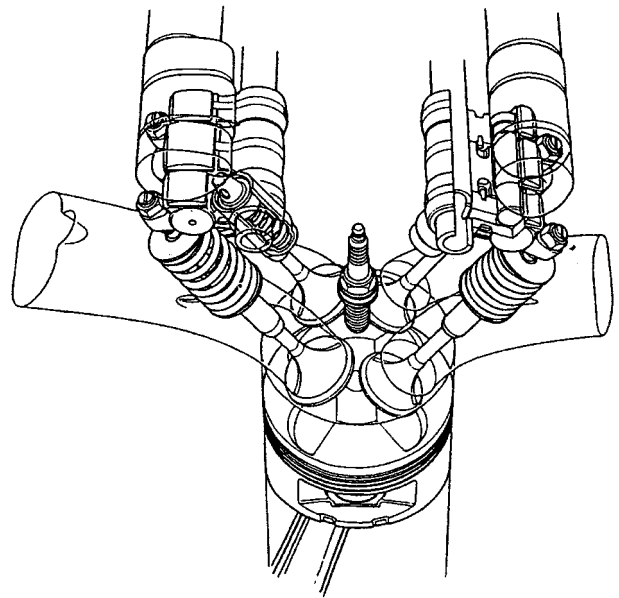


Figure 1.124 Variable Valve Timing and Lift Electronic Control (VTEC) (*Honda*)

cam lobes are active for the low and middle engine speed ranges, while the central cam with an increased lift profile becomes active for higher engine speeds to optimize breathing and power development.

In the lower- and mid-rpm ranges, the two outer rocker arms actuate each pair of valves in the conventional manner,

while the central rocker arm swings freely against a spring-returned plunger. At high rpm, the two outer and central rocker arms are locked together by a laterally engaging hydraulic plunger assembly, which is pressurized from the engine lubrication system via an electronically controlled spool valve. When this valve opens the central cam lobe with its increased lift profile becomes dominant, since its rocker arm compels the outer rocker arms to follow the same motion. The increased lift and longer duration of opening for the valves therefore ensure optimum timing for the gases that flow into and out of the cylinders, which contributes to greater power output.

Port timing for the two-stroke petrol engine

The port timing of a two-stroke petrol engine is determined by considerations similar to those affecting the valve timing of a four-stroke unit. That is, the port timing must take into account the time lapse before the ports are either fully uncovered or fully covered, and also the inertia effects of the incoming and outgoing flows of the crankcase and the cylinder gases. Some compromise is inevitable, however, since the ports are necessarily uncovered and covered by the piston at equal angles on either side of the crankshaft dead centres (Figure 1.125). The port timing diagram of a conventional two-stroke engine is therefore said to be symmetrical.

In order to avoid dilution of the crankcase charge, the exhaust port is uncovered at about 15° of crankshaft angularity in advance of the transfer port, so that the latter remains covered until the cylinder pressure drops below that existing in the crankcase. This is generally referred to as the *blow-down* period in the two-stroke engine. However, as a consequence of symmetrical port timing, this lead in exhaust port opening

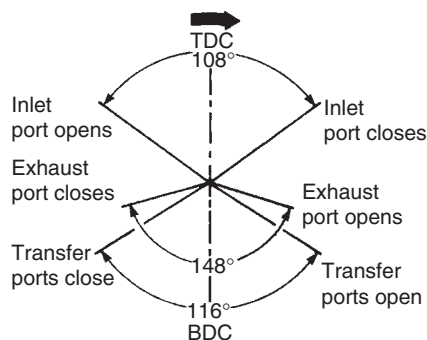


Figure 1.125 Typical port timing diagram for a motor car two-stroke petrol engine

conversely becomes a lag in closing. A part of the scavenge charge transferred from the crankcase therefore tends to short-circuit across the cylinder and be lost through the exhaust port before it closes, which is naturally detrimental to fuel consumption. This condition was especially prone to occur in early two-stroke engine designs that utilized a cross-scavenging effect on the burnt gases leaving the cylinder. A notable improvement in this respect was made possible by the development of a loop-scavenging system and later by the addition of either a reed or a disc valve to control inlet port opening and closing. The operating principles of engines incorporating these features were explained in Section 1.3. Values for the valve, rather than port, timing of an advanced concept supercharged two-stroke petrol engine for a passenger car are included in Figure 1.10, which shows its operating principle.

2 The diesel engine

2.1 SUITABILITY FOR ROAD TRANSPORT

It is a matter of common knowledge that diesel engines have found widespread application in stationary power-generating units and as prime movers for marine propulsion, rail locomotion and especially, by virtue of their more economical operation than the petrol engine, road transport.

The first four-stroke diesel engine to achieve a commercial success in road transport was that introduced by the German firm of MAN in 1924, but it was the intensive research and development carried out on the diesel engine by such British firms as Gardner and AEC during the 1930s and not least by H.R. Ricardo with his 'Comet' swirl combustion chamber (Section 3.5) which probably made the greatest contribution to this major advance in heavy-vehicle technology and its beneficial effect on the economics of road transport operation. Lighter-duty diesel engines later found modest application as power units for passenger cars, following the fuel crisis of the mid-1970s. More recently, the considerably improved driveability of diesel-engined passenger cars in terms of performance and refinement has been such that they now account for about one-third of new cars sold in Europe. This is not difficult to understand because when compared to a petrol-engined car, a diesel-engined version can reduce fuel consumption by about 30 per cent and similarly increase the range between refuelling. By the nature of its construction and operating characteristics, there can also be potential advantages in terms of greater durability, better reliability and the use of a safer non-volatile fuel in the event of an accident. As in the case of modern heavy vehicle diesel engines, the direct-injection combustion chamber used in conjunction with advanced fuel injection systems has pointed the way forward for passenger car diesel engines (Sections 3.5 and 8.11).

Since the mid-1960s the diesel engine for heavy vehicles, and later for passenger cars, has increasingly been developed in turbocharged form. This was originally as a means of improving power output in the medium- to high-speed range, but in more recent years the improved combustion associated with turbocharging has been exploited to produce an environmentally cleaner engine. That is, the turbocharger increases the amount of air delivered to the combustion chamber, so that the injected fuel can be burnt more efficiently. The benefits are a lower fuel consumption for the power developed and, as a consequence, better emission control. Furthermore, the engine can be quieter running because the turbocharger has a silencing effect on the induction and exhaust processes. Similar considerations apply to passenger car diesel engines.

For about the first decade of its life the diesel engine was constructed solely in four-stroke form, following which fairly rapid progress was made in developing two-stroke versions for large, slow-speed marine engines. Notable early examples

were those built by the German firm of Krupp and the Swiss firm of Sulzer, the latter being first in the field in 1905. The two main reasons for this development were first that the two-stroke diesel offered a reduction in weight and installation space, and second, that it was possible to reverse easily its direction of rotation – an important consideration for the marine engineer.

Although the two-stroke diesel engine was later developed for other marine, stationary power-generating and rail locomotion purposes, its application to motor vehicles (at least in the United Kingdom) has been on a relatively limited scale. This can almost certainly be accounted for by its fuel consumption being generally less favourable than that of a corresponding four-stroke diesel engine, a matter of prime consideration in the road transport field. Nevertheless, there have been several high-speed two-stroke diesel engines successfully developed for automotive applications. Notable among these have been the uniflow-scavenged engines introduced by General Motors in America and Fodens Limited in England. Another interesting English design was the Commer TS3 two-stroke diesel engine, which had opposed pistons (in common cylinders) with rocker and link connections to the crankshaft. This layout had long been associated with high scavenging efficiency, albeit at the expense of greater mechanical complication.

2.2 OPERATING PRINCIPLES

The four-stroke diesel engine

The essential difference between the petrol and the diesel engine (Figure 2.1) is that the former relies on *spark ignition* (SI) and the latter on *compression ignition* (CI). More specifically, the combustion process in the diesel engine is initiated by spontaneous ignition of the fuel when it is injected into a highly compressed charge of air, which has reached about 800°C. Diesel engine combustion also tends to occur at constant pressure rather than at constant volume as in a petrol engine. This means that in the diesel engine the combustion pressure continues to rise steadily as the piston retreats and the cylinder volume increases, whereas in the petrol engine the combustion process is so rapid that there is very little movement of the piston while it occurs and hence very little increase in cylinder volume. Strictly speaking, though, neither engine fits exactly into either of these categories.

During its early development this type of engine was described variously as compression ignition, Diesel, oil and heavy oil, but it has long since come to be known generically as the diesel engine (with a small 'd'). This acknowledges the major personal contribution made to its development by Dr Rudolph Diesel (1858–1913), who was born in Paris

of German parents. He became a student of mechanics and later entered the well-known engineering works of Sulzer Brothers in Winterthur, Switzerland. It was in the early 1890s that he developed his theories on what we now know as the diesel engine principle and subsequently took out various patents, including a British one granted in 1892. A few years later his theoretical work was embodied in a working engine of practical form built by the famous firm of MAN at Augsburg.

In fairness, however, it must be added that Diesel's concept of sparkless ignition was actually predated by the pioneering work of an English engineer, Herbert Ackroyd Stuart (1864–1927). In 1890 he patented an engine operating on a similar principle, but which required a vaporizer surface at the end of the cylinder. For starting the engine, the vaporizer required the application of external heat. Hence, the first true

compression ignition engine is generally attributed to Rudolph Diesel.

In the four-stroke diesel cycle the following sequence of events is continuously repeated all the time the engine is running (Figure 2.2):

- 1 The *induction* stroke, during which air only is taken into the combustion chamber and cylinder, as a result of the partial vacuum or depression created by the retreating piston.
- 2 The *compression* stroke, in which the advancing piston compresses the air into the very small volume of the combustion chamber and raises its temperature high enough to ensure self-ignition of the fuel charge. This demands compression pressures considerably in excess of those employed in the petrol engine.
- 3 The *power* stroke, immediately preceding which the fuel charge is injected into the combustion chamber and mixes with the very hot air, and during which the gases of combustion expand and perform useful work on the retreating piston.
- 4 The *exhaust* stroke, during which the products of combustion are purged from the cylinder and combustion chamber by the advancing piston and discharged into the exhaust system.

As in the case of petrol engine, the timing for the opening and closing of the inlet and exhaust valves, and also that for injecting the fuel, departs from the basic four-stroke operating cycle.

Comparison of diesel and petrol engines

The following generalizations may be made on the use of diesel versus petrol engines in commercial vehicles and, more recently, cars:

- 1 The diesel engine has better fuel economy than the petrol engine. This is because its thermal efficiency is 30–36 per cent, compared with the 22–25 per cent of a petrol engine. As indicated in Section 1.3, thermal efficiency is the ratio of useful work performed by the engine to the internal energy it receives from its fuel.
- 2 The diesel engine has generally proved to be more reliable, to need less maintenance and also to have a longer life than an equivalent petrol engine. These advantages derive mainly from its sturdier construction and cooler running characteristics.

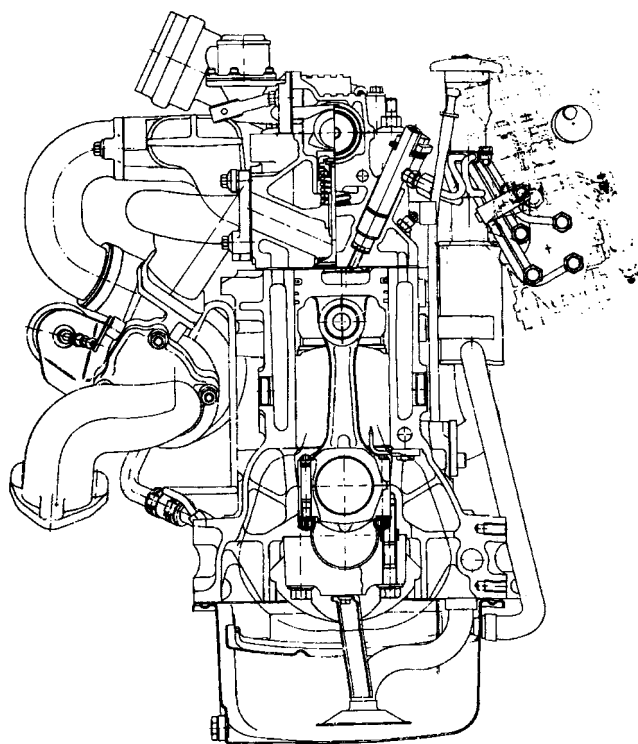


Figure 2.1 Cross-section of a four-stroke turbocharged direct-injection diesel engine (Perkins)

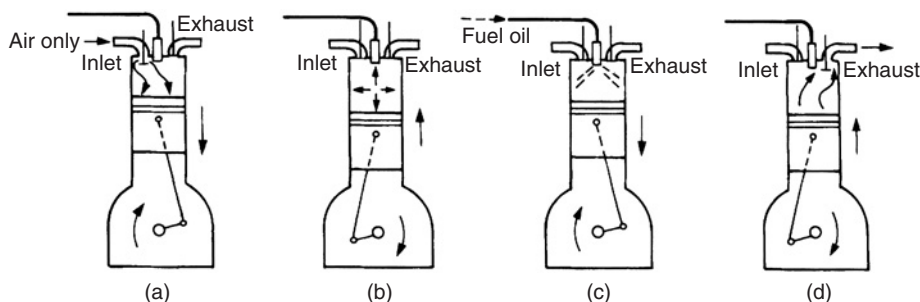


Figure 2.2 The four-stroke diesel engine cycle: (a) induction (b) compression (c) power (d) exhaust

- 3 Although a petrol engine develops its maximum power at higher rotational speeds than an equivalent diesel engine, the latter can provide better pulling power. This is because the maximum turning effort or torque exerted by the crankshaft of the diesel engine is greater and also better maintained over a wider range of engine speeds.
- 4 A disadvantage of the diesel engine is that it tends to be heavy and bulky in relation to its power output. This is explained by the greater operating pressures and loads that have to be catered for in the construction of the diesel engine.
- 5 The noise and vibration level of the diesel engine, especially under idling and low-speed operation, compares unfavourably with the petrol engine. Again, this is chiefly a function of the much higher cylinder pressures in the diesel engine. More recent diesel engine motor cars have nevertheless been praised for their low level of noise at motorway cruising speeds.
- 6 The diesel engine is sometimes criticized for having smoky exhaust of unpleasant odour, although it is the invisible products of combustion in the exhaust gases of a petrol engine that are more harmful to the environment. Avoidance of a smoky exhaust with a diesel engine is largely a question of good driving technique, regular maintenance and proper adjustments.
- 7 A safety consideration is that the fuel oil used in motor vehicle diesel engines is far less dangerously flammable than petrol, thus reducing the fire risk in the event of an accident. For taxation purposes, the fuel oil used in automotive diesel engines is referred to as DERV (an abbreviation of diesel engine road vehicle).
- 8 Finally, the basic cost of the diesel engine, together with its associated fuel injection equipment, is generally higher than that of an equivalent petrol engine.

The two-stroke diesel engine

Since this type of engine may take several practical forms, our attention here will be concentrated on the high-speed version suitable for application to road transport (Figure 2.3).

Apart from the expected difference that air only is introduced into the engine cylinder prior to the injection of fuel oil, another departure from two-stroke petrol engine practice is that, instead of using crankcase compression, a rotary blower is used to charge the cylinder with low-pressure air. This type of blower is sometimes also used for supercharging four-stroke cycle engines (Section 9.2). The distinction that must be made here, however, is that whereas a supercharger is used simply to increase power output of a four-stroke engine, a similar blower is essential for a two-stroke diesel in order that it shall work at all.

Furthermore, a pair of exhaust valves is located in the cylinder head to provide a uniflow system of scavenging. This means that there is no change in direction for the cylinder air stream, which is in contrast with the loop system of scavenging described for the two-stroke petrol engine in Section 1.3. The two-stroke diesel engine is therefore mechanically more complicated.

In the basic two-stroke diesel cycle, the following sequence of events is continuously repeated all the time the engine is

running and while the rotary blower is supplying air to the inlet ports of the cylinder (Figure 2.4):

- 1 The *induction-exhaust* event. Air only is admitted to the cylinder during the period the inlet ports are uncovered by the piston, which occurs towards the last quarter of the power-exhaust stroke and about the first quarter of the induction-compression stroke. During this part of the cycle, the exhaust valves are opened just before the cylinder inlet ports are uncovered and then closed just before the ports are covered again. This sequence of exhaust valve events not only ensures that the exhaust gas pressure falls below that of the scavenging air supply, and thus prevents any return flow of exhaust gases, but also leaves the charge in the cylinder slightly pressurized prior to final compression. Hence, the combination of uncovered inlet ports and open exhaust valves allows air to be blown through the cylinder, which removes the remaining exhaust gases and, by the same token, fills it with a fresh charge of air. Since neither the air nor the exhaust gases change direction in passing through the cylinder, the term uniflow scavenging can justifiably be applied.

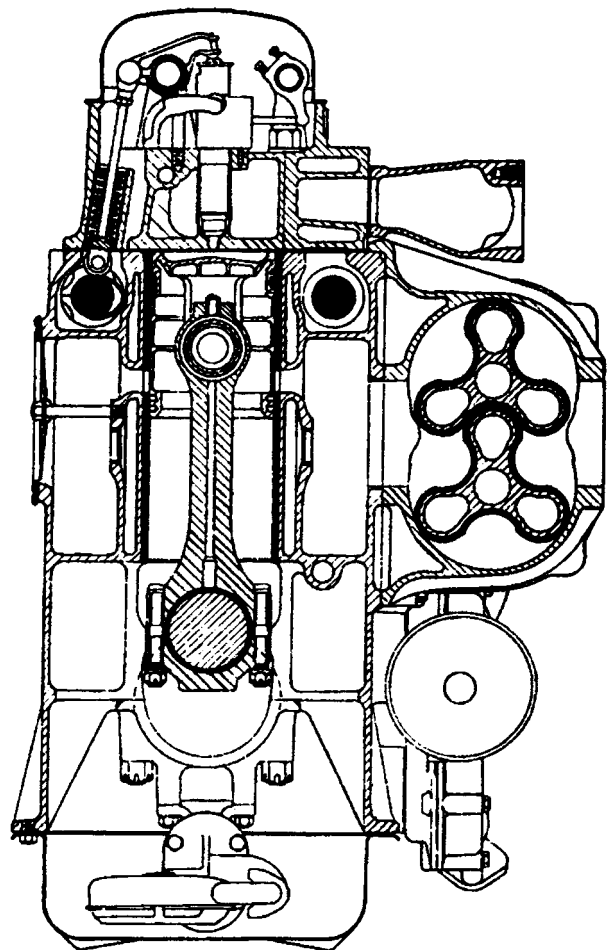


Figure 2.3 Cross-section of a blower-charged two-stroke diesel engine (General Motors)

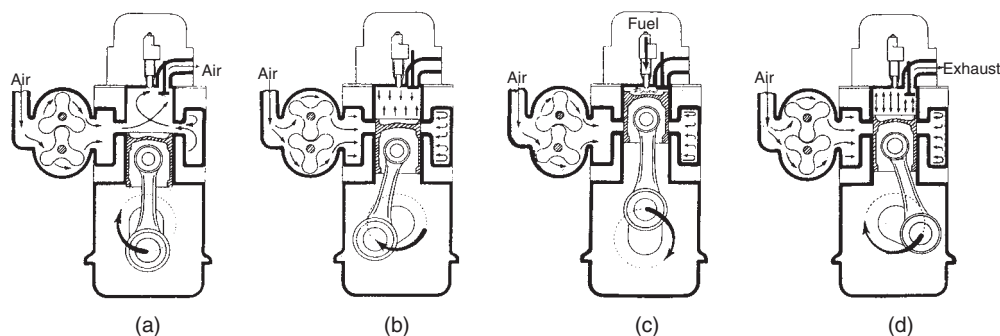


Figure 2.4 The two-stroke diesel engine cycle: (a)/(b) induction-compression (c)/(d) power-exhaust (*General Motors*)

2 The *compression-power* event. The remaining three-quarter portions of the *induction-compression* and *power-exhaust* strokes occur in a very similar manner to that of the four-stroke diesel engine; that is, the advancing piston compresses the air into the lesser volume of the combustion chamber and raises its temperature high enough to ensure self-ignition of the fuel charge. This is injected into the combustion chamber just before the piston begins to retreat on its power-exhaust stroke.

It should be noted that the operating cycle of the two-stroke diesel engine has been described in terms of events rather than strokes in order to assist understanding.

2.3 CYLINDER BLOCK, CRANKCASE AND HEAD

Cylinder block and crankcase

A distinctive feature of the cylinder block and crankcase construction for an automotive diesel engine, as compared to that used in for example marine applications, is that the crankshaft is underslung from the upper part of the crankcase in petrol engine fashion, rather than being supported from below on a bed plate. It will, therefore, be evident that the diesel cylinder block and crankcase (Figure 2.5) is basically similar to that of a petrol engine and likewise must contain the gas loads, albeit of increased intensity, which strive to force apart the cylinder head from the main bearings below. In addition to the appreciable mechanical loading, the diesel cylinder block and crankcase must also be able to withstand severe thermal loading and, in view of current legislation, radiate the least possible noise during engine operation. As in petrol engine practice, the automotive diesel cylinder block is almost invariably cast integral with the crankcase, so as to combine the required rigidity with a reasonably low weight.

Although the thickness of metal sections in the diesel cylinder block and crankcase are unlikely to differ very much from those of a modern high-output petrol engine, since they are largely determined by the requirements of foundry technique, other differences in design may be encountered. For example, a feature that is rarely found in petrol engines but is fairly common in diesel practice, especially with V cylinder arrangements, is the provision of transverse tie bolts as

well as vertical studs for securing the main bearing caps (Figure 2.6). This system serves to reinforce the crankcase and confer the necessary rigidity of mounting for the main bearings, because their supporting walls have to react against the dynamic loads imposed by two inclined and offset cylinders. Furthermore, the vertical main bearing studs may extend upwards through the cylinder block of in-line cylinder engines, to act also as cylinder head studs and thus place the entire engine structure in a beneficial state of compressive stress, thereby countering the opposite effect created by cylinder gas pressure. A rather obvious difference in construction is noticeable at the front end of many diesel engines, where provision must be made for mounting a much larger size of timing cover, since the timing chest usually has to accommodate a more comprehensive gear train, as described later.

Although not constituting a basic difference in design, it should be noted that in long-established diesel engine practice for commercial vehicles either wet (Figure 2.5) or, less commonly, dry cylinder liners are employed in order to extend effective engine life for the several hundred thousand miles demanded by vehicle fleet operators. For endwise location it is usual for both types of liner to have a top flange, which is recessed in the closed deck of the cylinder block. In the case of wet liners a narrow 'fire-ring' spigot to protect the adjacent edge of the cylinder head gasket may be formed around the top surface of the flange and engage freely with a circular groove in the mating face of the cylinder head (Figure 2.7). Also, a bead of special silicone sealant may be specified for application to the underside of the liner top flange on engine build. Once the liner has been driven fully home, a clamp is temporarily applied to the liner thereby squeezing out the excess sealant for removal and then allowing that remaining to dry.

Owing to the greater mechanical loading and increase in noise level encountered with the automotive diesel engine, as compared to the petrol engine, aluminium alloy is rarely used as a cylinder block material but a cast iron of similar composition to that mentioned later for the cylinder head is generally specified and typically possesses a slightly higher tensile strength. For greater resistance to impact loading the main bearing caps may be produced from malleable or nodular cast iron. A strongly constructed aluminium alloy sump may nevertheless be used in conjunction with the cast iron cylinder block.

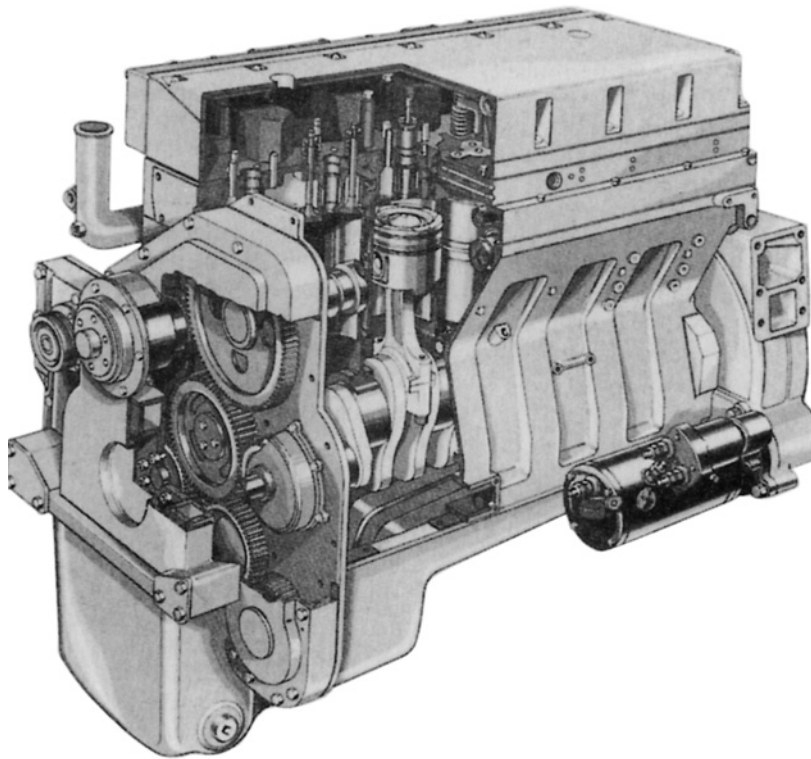


Figure 2.5 The Cummins in-line engine with four valves per cylinder and panelling of the crankcase to reduce noise

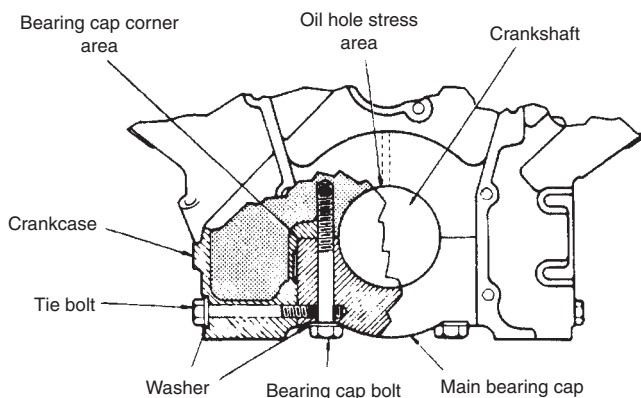


Figure 2.6 Diesel main bearing cap with transverse tie bolts (*International Harvester*)

For passenger car diesel engines increasing use is now being made of a lighter weight cast iron known as 'compact graphite iron', usually abbreviated to CGI. This material offers about a one-third greater tensile strength and an improved fatigue resistance, as compared to grey cast iron. However, it does call for more expensive high-speed machining techniques to be adopted on production. The relevance of CGI to modern diesel engine design is that it either allows a lighter construction that bears comparison with an aluminium alloy, or its greater strength can be utilized to contain higher cylinder pressures in the interests of efficient combustion and hence improved emissions control.

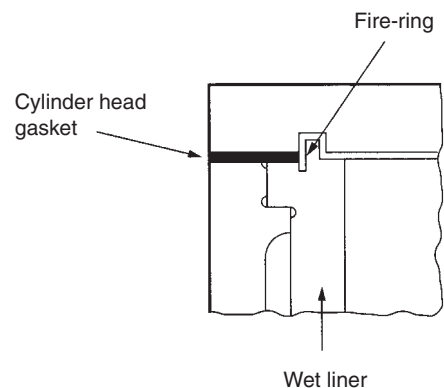


Figure 2.7 Diesel engine wet cylinder liner with fire-ring

Cylinder head

The cylinder head for a diesel engine is generally more complex in construction (Figure 2.8) and has a more severe cooling requirement than its petrol engine counter-part. A one-piece cylinder head for an in-line cylinder engine, and similarly one for each bank of a V cylinder layout, can be regarded as established modern practice; although a two-piece cylinder head (two separate heads) for an in-line cylinder engine may still be used in the case of very large commercial vehicle applications, mainly to ease the problem of handling heavy units during service operations. Individual heads are, of course, employed for the cylinders of air-cooled diesel engines (Figure 5.3).

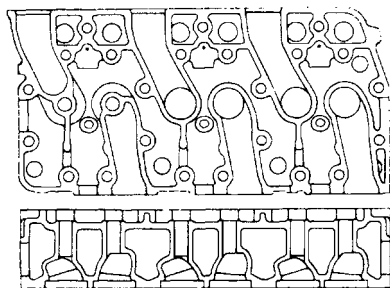


Figure 2.8 Section through a diesel cylinder head casting (Midcyl)

Structurally, provision must be made in the diesel engine cylinder head for incorporating a passage for each fuel injector, either through the medium of an inserted metal tube where direct injection is utilized, or by way of an integrally cast boss to suit indirect injection. The latter system also requires the upper part of the combustion swirl chamber to be formed in the cylinder head, whilst the lower half comprises an inserted machined plug that contains the throat portion connecting the chamber to the cylinder. Since this component acts as a hot-spot in the system, as explained later, it is installed with an annular air gap to minimize heat transfer to the cylinder head metal. From the viewpoint of maintaining cylinder head gasket sealing efficiency, it is essential that no settling should occur at the abutment face of the chamber insert, because any sinkage of its lower face into the sealing surface of the head would locally relieve gasket pressure and result in gas leakage (Figure 2.9). To avoid interference with the piston at top dead centre the valves are typically recessed into the surface of the cylinder head (Figure 2.18), and also they are generally provided with inserted valve guides and seats in diesel practice.

Although in common with the petrol engine the very hot metal on the gas side and the less hot metal on the coolant side of the combustion chamber results in a private tug-of-war, owing to the difference in expansion across the thickness of metal wall, the problem becomes more acute in the diesel cylinder head because of a greater lack of uniformity in heat distribution across that part of the head covering the cylinder. This chiefly arises from there being less latitude in positioning and spacing of the inlet and exhaust valve seats, and similarly the swirl chamber where this is applicable. To avoid fatigue cracking in the thermally stressed regions bridging these components, it is necessary to ensure an adequate flow of coolant through the cylinder head; in some designs, coolant entering the head is directed through channels to impinge first above the combustion chamber roof (Figure 2.10).

The materials used for the cylinder heads of motor car engines have already been described in Section 1.6. Except for some motor car diesel engines and also the individual cylinder heads of air-cooled diesel engines, which utilize an aluminium alloy, the cylinder head material for commercial vehicle diesel engines is a cast iron. A high-strength chromium-iron alloy, such as that developed by Midcyl and known as Chromidium, may typically be specified. This type of material is characterized by an extremely high resistance to wear and corrosion, and careful control of metal composition enhances the pressure

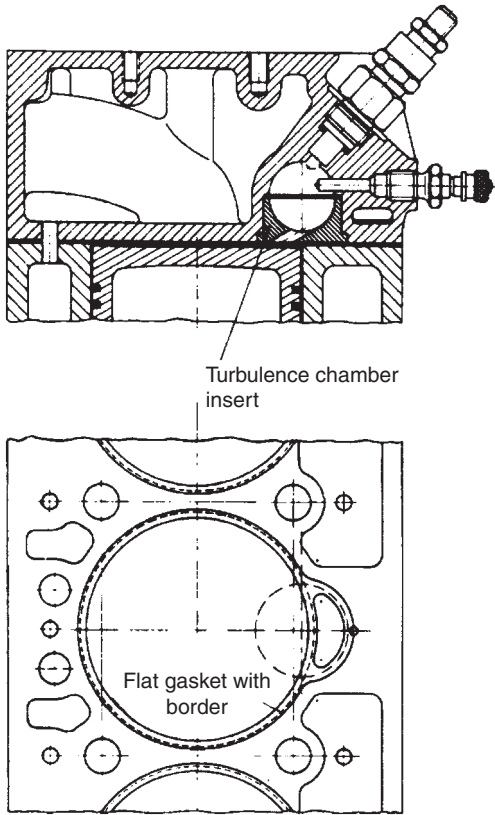


Figure 2.9 Gasket sealing arrangement for a diesel swirl-type combustion chamber (Reinz)

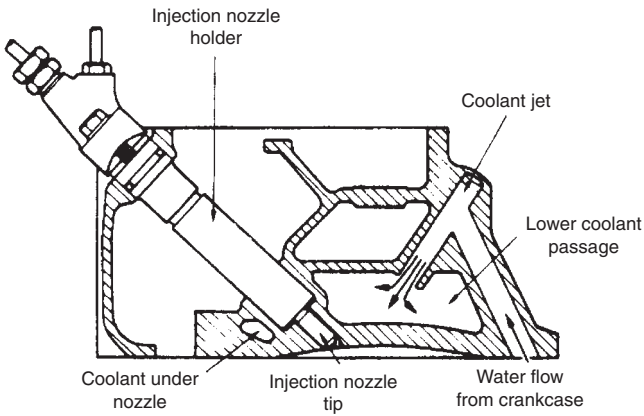


Figure 2.10 Directed coolant flow in a diesel cylinder head (International Harvester)

tightness of the finished casting. The following material specification for Chromidium cylinder head iron is given by way of interest:

Carbon	3.10–3.50%	Manganese	0.50–0.80%
Silicon	1.85–2.40%	Chromium	0.20–0.30%
Sulphur	0.15% max.	Brinell hardness no.	190–240
Phosphorus	0.20% max.		

British Standards BS 1452 Grade 14.

2.4 PISTONS AND CONNECTING RODS

Pistons and rings

A diesel engine piston (Figure 2.11) differs from its petrol engine counterpart in the following basic respects.

Increased sectional thickness of metal

This is concentrated in the upper portion of the piston for better heat dissipation and to afford the maximum possible support for the more heavily loaded gudgeon pin bosses, in which the pin is made fully floating. The shape of the piston crown is, of course, largely determined by the design of the combustion chamber, since with a direct-injection system the latter is to all intents and purposes formed within the piston head. In any event, the crown section of the piston has a generous metal thickness to withstand the higher cylinder gas pressures without distorting and to provide a ready path for heat flow to the rings without cracking. To reduce thermal loading on the top compression ring, the piston top land is of increased depth. Temperatures must generally be kept below about 200°C in the top ring groove, in order to avoid ring sticking and scuffing problems due to carbon deposits and lacquer formation. A second land of increased depth is also provided to contain the higher mechanical loading on the top ring.

Longer in relation to diameter

It may be recalled (See Section 1.4) that pistons assume a trunk form to present a sliding bearing surface against the cylinder wall, which thus reacts against the side thrust arising from the angular motion of the connecting rod. Making the piston skirt relatively longer spreads the greater side thrust forces developed in the diesel engine over an increased surface area. Hence, the unit pressure is reduced so that separation between the piston and the cylinder wall is better maintained by the film of lubricating oil, and wear through direct contact is minimized.

May carry more rings

In long-established practice, the diesel engine piston has generally been equipped with three instead of two compression rings and two rather than one oil control ring. More recently, however, there has been a definite trend to reduce the numbers of both compression and oil control rings to the same combination as that used in modern petrol engine practice. Nevertheless, an unfilled groove may be provided below the gudgeon pin to allow the addition of a second oil control ring in later service. Technically, the number of rings used depends upon the severity of application and the projected design targets for oil consumption, blow-by and friction effects, the latter being related to fuel consumption. Where ring-belt temperatures tend to be unusually high, which can promote ring sticking and blow-by, a traditional remedy in diesel engine practice generally has been to use a top ring of 'keystone' or wedge-section form. It acts on the principle that the slightest movement of the tapered ring in its correspondingly shaped groove will vary its side clearance, therefore discouraging any build-up of deposits. An included angle of 10° for the ring faces is typical. Wedge-section top rings may also be adopted for highly rated passenger car

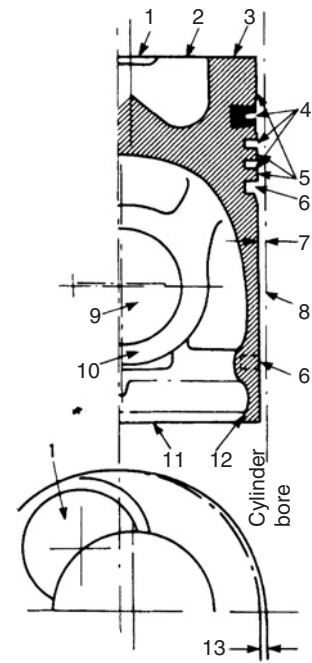


Figure 2.11 Terminology for a diesel engine piston with bonded insert (GKN Pistons)

- | | |
|----------------------------|--------------------|
| 1 valve pockets | 8 skirt |
| 2 combustion chamber | 9 piston pin hole |
| 3 crown | 10 piston pin boss |
| 4 compression ring grooves | 11 open end |
| 5 lands | 12 register |
| 6 oil control ring grooves | 13 ovality |
| 7 minimum clearance | |

diesel engines, although in this case a single-side wedge form may be preferred, so as to combine the advantages of this type of ring with those of a conventional rectangular section ring.

Need for top ring carrier

For naturally aspirated (non-pressure-charged) diesel engines where cylinder pressure and temperature conditions are not too demanding, the piston may be simply of monometal (all aluminium alloy) construction. Otherwise, where a more hostile cylinder environment is expected, the piston is generally of bonded insert construction, since the principal limiting factor of the monometal piston is ring groove wear. The insert is bonded into the aluminium alloy and forms a wear-resistant carrier for the top and sometimes also the second compression ring. It is made from a nickel cast iron alloy with hot expansion characteristics that closely match those of the aluminium alloy used in the main body of the piston. Even so it will be noticed that the insert is of tapered form, which ensures that the piston alloy grips it on cooling and contracting in diameter. An aluminium alloy having a 12 per cent silicon content and usually described as a eutectic aluminium alloy (eutectic signifies an alloy composition that solidifies at one temperature rather than over a range of temperatures, and

ensures good castability) is commonly used both for diesel and petrol engine pistons.

Requirement for oil cooling of piston

The importance of avoiding top ring sticking in the diesel engine piston has earlier been mentioned. Clearly this could become a problem with the ever increasing performance of the turbocharged engine, especially in relation to the current trend of reducing the depth of the piston top land, so that its surrounding dead volume is minimized for more efficient combustion. A significant proportion of the combustion heat to which the piston is exposed flows through the rings to the cylinder cooling jacket, the piston itself being cooled only by contact with the cylinder wall and a constantly renewing thin film of oil, which has originally been pumped from the sump. The reservoir of oil in the sump therefore acts as a medium for transferring excess heat away from the piston. Further use can be made of this particular heat transfer medium by arranging for a standing jet of oil to be directed against the underside of the piston crown. For this purpose a series of strategically sited nozzles receive oil under pressure from the main lubrication system of the engine and spray it up into the pistons (Section 4.4). This also permits closer fitting pistons for quieter running.

Two-piece articulated pistons

Although two-piece pistons with steel crowns and aluminium skirts are not unfamiliar to the marine diesel engineer, it has only been in recent years that pistons of this configuration have been adopted in preference to monometal pistons in some high-output turbocharged diesel engines for heavy vehicles such as the Scania DSC 12 engine (Figure 2.20).

Higher power ratings inevitably result in greater thermal and mechanical loadings being imposed on the cylinder components, which have therefore had to be further developed to

remove any undue limitation on engine performance and reliability. In particular, higher power ratings generally attract increased piston temperatures and these can have serious practical disadvantages in engine operation. Apart from promoting a tendency towards ring sticking and scuffing due to carbon deposits and lacquer formation, at elevated temperatures the hot strength of aluminium alloys becomes significantly less than their cold strength, which could potentially lead to failures in the highly loaded regions of the piston crown boss supports and the gudgeon pin boss bearings.

To overcome the problems that can otherwise appear with a conventional monometal piston, even though it may incorporate a bonded insert ring carrier and benefit from oil cooling, engineers resorted to a two-piece piston construction. This comprises a forged steel crown that carries the piston rings and an aluminium skirt and therefore combines the best of both worlds, because the steel crown better tolerates the higher temperatures and also expands less to maintain a more constant operating clearance for the rings, while the aluminium skirt still provides a light-weight means of guidance for the piston. However, unlike the two-piece piston used in marine diesel engine practice where the crown is typically bolted to the skirt, those designed for heavy vehicle engines are of more compact articulated construction (Figure 2.12). That is, the connection between the crown and skirt is made solely through the gudgeon pin, so that the crown and semidetached skirt are mutually self-aligning with advantage being taken to minimize the running clearance of the cooler operating skirt. The latter may also incorporate an upper oil collecting gallery to shake cooling oil supplied by sprayer nozzles (Section 4.4) up between the crown and its ring carrier, especially since the steel crown has a lower heat conductivity than if it were made from an aluminium alloy. An oil drain from the gallery is provided through return ducts situated remotely from the oil spray. Another feature of construction, again borrowed from marine diesel engine practice, is that the inner faces of the piston crown gudgeon pin bosses are tapered outwards towards their lower ends and



Figure 2.12 A dismantled two-piece articulated piston and connecting rod assembly (Scania)

embrace the matching tapered sides of the connecting rod small-end boss, the purpose of this arrangement being to present an increased bearing area against the gudgeon pin on the firing stroke.

Connecting rods

The connecting rod has sometimes been described as one of the most important components in conventional engine design. Attention to detail in its design has always been of prime importance, and can perhaps be judged by recalling one aspect of the wartime development of the famous Rolls-Royce Merlin aero engine. Here, the alteration from a stepped to a generously radiused shoulder at the junction of the shank to big-end bolt lugs significantly reduced stress concentration in the forked connecting rods and was one of the important modifications that ultimately made possible a safe doubling in power output.

Returning to our comparison of petrol and diesel engine connecting rods, as far as the latter are concerned there is currently no departure from what has long been the established form for this component. Rather the difference lies in its more rugged proportions to withstand the increased compressive loading resulting from the higher combustion pressures in the diesel engine. More specifically, the shank of the connecting rod is often quite noticeably flared out at the junctions respectively with the small-end boss and the big-end bearing housing (Figure 2.13), again to spread the load over these parts with the minimum of stress concentration and to ensure that the thin-wall bearings are rigidly supported so they can retain their roundness.

As in petrol engine practice, the joint between the connecting rod and cap is sometimes made at an angle, so that the big end can accommodate the maximum diameter of crankpin, whilst still allowing the connecting rod and piston assembly to be withdrawn through the top of the cylinder during overhaul. In any event, the lateral location of the cap must be sufficiently positive to avoid mismatch at the joint faces of the big-end bearing, which may entail the use of either serrated or stepped joints (Figure 2.14). With an angled joint, setscrews (instead of bolts and nuts) are normally required to clamp the cap to the rod. For heavy-duty applications two bolts may be used on each side of the big-end bearing assembly (Figure 2.15).

The materials from which connecting rods are forged generally comprise a carbon-manganese steel for passenger car petrol engines and a low-alloy 1 per cent chromium steel for commercial vehicle diesel engines, the latter material being the harder after their heat treatment.

Big-end bearings

Here again, the connecting rod big-end bearings of a diesel engine perform the same function as those in a petrol engine and likewise are more heavily loaded than the main bearings, where the equivalent loading is shared between two adjacent ones. Also, the big-end bearing is subjected to centrifugal loading enforced by the circular path of the lower part of the connecting rod, an effect that is absent in the case of main bearings supporting a fully counterbalanced crankshaft.

However, a basic difference does arise between the loading characteristics of diesel and petrol engine big-end bearings. In transmitting the reciprocating forces from piston to crankshaft and vice versa, it is the upper half-liner in the connecting rod that bears the brunt of the loading in a diesel engine, whereas it is the lower half-liner in the connecting rod cap that is on the receiving end in a petrol engine. The reasons for this contrary behaviour are explained later where the distinction will be made between the predominant sources of loading on diesel and petrol engine bearings. These are cylinder gas pressures and inertia forces of the reciprocating parts respectively, always remembering that the inertia forces vary

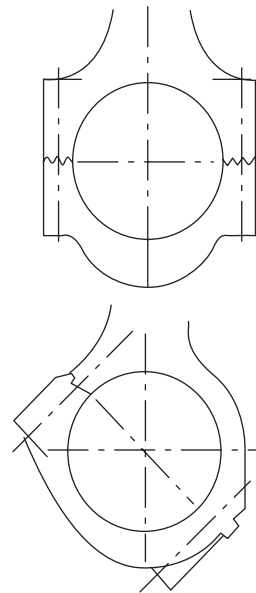


Figure 2.14 Examples of diesel connecting rod cap location (Glacier)

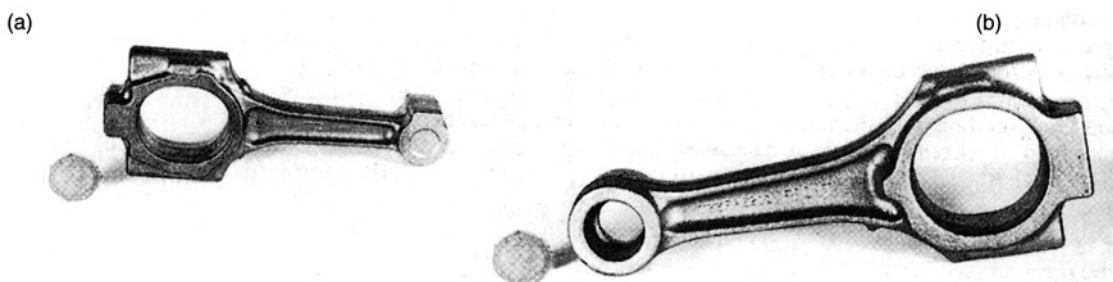


Figure 2.13 Comparison of integral connecting rod cap and forgings for (a) petrol and (b) diesel engines (GKN-SDF)

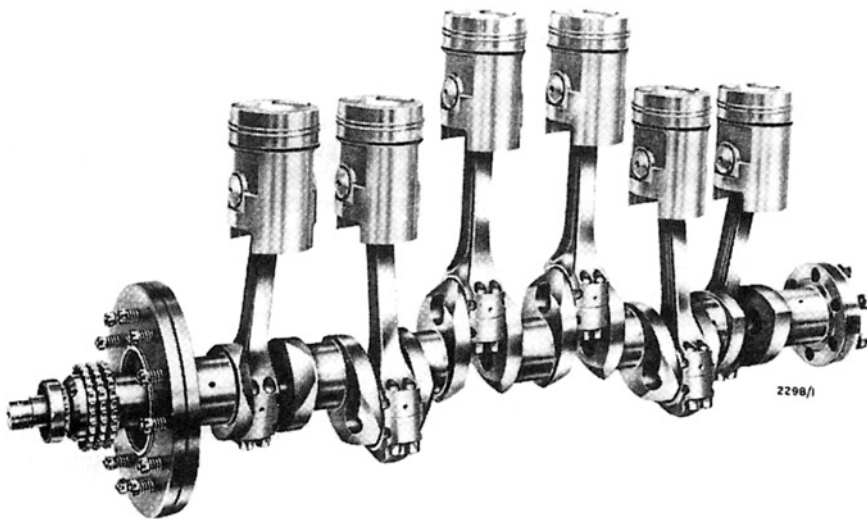


Figure 2.15 Crankshaft assembly of a long-established design of diesel engine (*Gardner*)

as the square of the engine speed and are virtually unopposed by cylinder gas pressure on the exhaust strokes.

Until recent years, the lining material requirements for diesel big-end and main bearings could usually be satisfied by one and the same material. Traditionally this has been either a copper-lead mixture or a tin-aluminium alloy. With the advent of pressure charging, however, the loading on the big-end bearings can rise above the performance capabilities of these materials, since the comparative bearing pressures under maximum-torque running conditions can be in the order of 35 MN/m^2 (5250 lbf/in^2) for naturally aspirated and 50 MN/m^2 (7500 lbf/in^2) for turbocharged diesel engines. To provide the increased load-carrying capacity therefore requires the highest-duty bearing materials available, such as lead-bronze and aluminium-tin-silicon compositions with softer overlays of either lead-tin or lead-indium.

In modern engine production, it is not unusual for a specialist manufacturer to develop and supply what is termed a 'piston module', which comprises the piston, piston rings, connecting rod and bearings.

2.5 CRANKSHAFT ASSEMBLY AND MAIN BEARINGS

Crankshaft

The predominant source of loading on the bearings of a diesel engine and hence on the crankshaft itself derives from high cylinder pressures. Under normal conditions of combustion, even with a naturally aspirated engine, peak cylinder pressures can be up to about 10 MN/m^2 (1500 lbf/in^2), which is therefore in the region of twice that encountered in the petrol engine, whilst for a turbocharged diesel engine the cylinder pressure may be up to 14 MN/m^2 (2100 lbf/in^2).

Clearly then a diesel engine crankshaft (Figure 2.15) must be of robust proportions, and to this end the diameter of the main bearing journals is usually made not less than 0.75 of the cylinder bore dimension and the diameter of the crankpins not less than about 0.65 of this dimension. These values are, of

course, at least on the upper limit for those quoted in Section 1.8 relative to petrol engine crankshafts, and contribute to a substantial and beneficial overlap between main journal and crankpin of typically around 20 to 35 mm (0.8 to 1.4 in). In fact the rigidity of the crankshaft can be such that for shorter versions used in V engines no torsional vibration damper may be needed. Naturally a main bearing has always been used on either side of each crankthrow, as is now the case with most high-compression petrol engines, so that the crankshafts for in-line six-cylinder and V eight-cylinder diesel engines are carried in seven and five main bearings respectively with more recent V six-cylinder versions having four main bearings. The lengths of both main and big-end bearings generally approach 0.30 of the cylinder bore dimension.

When describing the materials and manufacture of petrol engine crankshafts for motor cars, it was mentioned that although the majority of their crankshafts are produced from iron castings of the spheroidal graphite type with a tensile strength of about 63 kg/mm^2 (40 tons/in^2), crankshafts forged from higher-strength low-alloy steels are still used for heavy-duty applications. Included in this category are, of course, commercial vehicle diesel engines. Although there is no hard and fast rule regarding the material preferences of different engine manufacturers, a few examples can be given. The material used for the crankshaft of one medium-capacity, naturally aspirated diesel engine is a 1 per cent chromium steel with a tensile strength of 71 kg/mm^2 (45 tons/in^2), while that for a turbocharged version of the same engine is a 1 per cent chromium-molybdenum steel with an increased tensile strength of 79 kg/mm^2 (50 tons/in^2). The effect of a small addition of molybdenum as an alloying element is to impart a relatively uniform hardness and strength to the material. At the other end of the scale, the material specified for the crankshaft of a large-capacity turbocharged diesel engine can be typically a 3 per cent chromium-molybdenum nitriding steel with a tensile strength of up to 94 kg/mm^2 (60 tons/in^2). Nitriding or nitrogen hardening is a process performed by heating the material to a temperature of 500°C

and passing ammonia gas over it. On contact with the hot steel the ammonia gas breaks down into hydrogen and nitrogen, so that the latter diffuses into the steel to form nitrides at the surface. The benefits conferred by this treatment are an extreme surface hardness coupled with a high fatigue resistance.

Main bearings

Although the crankshaft main bearings of a diesel engine perform the same function as those in a petrol engine and, in established modern practice, are also of the same thin-wall construction, their loading characteristics can be somewhat different. This is because in the diesel engine, the predominant source of loading derives from high cylinder pressures generated at relatively modest operating speeds; whereas in the petrol engine, it is imposed by high inertia forces arising from acceleration and deceleration of the reciprocating parts at much increased operating speeds. Furthermore, the duration of loading for each complete working cycle of the engine is brief for cylinder pressure and protracted for inertia forces. If excessive the former type of loading tends to promote fatigue failures and the latter type wiping failures of a bearing. In the unlikely event of such failures occurring in normal service, they would be characterized by random cracking and superficial melting of the bearing respectively.

It will therefore be recognized that in modern practice both diesel and petrol engines can pose problems of severe bearing loading and heating with their consequent adverse effects on oil film thickness and bearing durability. For this reason an identical choice of high-duty lining materials is often now made to satisfy the main bearing requirements of both types of engine. Materials in current use are copper-lead mixtures and tin-aluminium alloys with typically 70 per cent copper and 20 per cent tin contents by weight respectively, the relative merits of these two classes of bearing material having been compared in Section 1.7. This common requirement for high-duty bearing materials was rarely in evidence some forty years ago, when copper-lead mixtures were confined almost exclusively to diesel engine applications and all but large-capacity petrol engine bearing requirements could still be satisfied by white metal alloys. At that time the tin-aluminium bearing as we know it today had yet to become established and, of course, petrol engine operating speeds were much lower.

Torsional vibration damper

For particularly arduous duty such as may be encountered in the larger sizes of automotive diesel engines, the rubber damper can have certain limitations in respect of its durability, since the high damping forces involved could result in overheating of the rubber element. A viscous damper, so called because the damping forces are generated by the drag of a very heavy or viscous fluid, is of generally more rugged construction and may therefore be the preferred choice for these applications.

This type of damper basically comprises a small fly-wheel that is completely enclosed in a hermetically sealed (airtight) casing, which is rigidly attached to the nose of the crankshaft via a flange fitting against the front pulley (Figure 2.16a). The flywheel ring is not in any way attached to the correspondingly

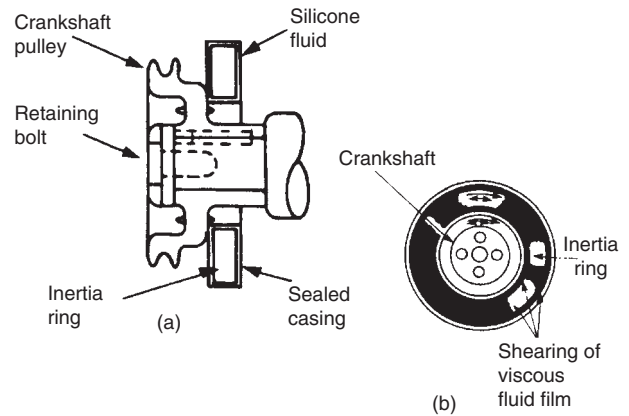


Figure 2.16 Torsional vibration damper: (a) construction (b) action

annular-shaped casing, but is completely separated from its interior walls by an all-enveloping film of silicone fluid, albeit only a few hundredths of a millimetre thick, so that to some extent it can also damp axial vibration of the crankshaft. It is therefore purely the drag of the silicone fluid, which characteristically retains a nearly constant and high viscosity (resistance to flow) over a wide temperature range, that tends to cause the casing and enclosed flywheel to revolve in unison.

In the absence of crankshaft torsional vibration the enclosed flywheel tends to rotate in unison with its casing, because the force required to shear the viscous film of silicone fluid that separates them is quite appreciable. Here it should perhaps be explained that just as one solid body resists the action of sliding when in physical contact with another, so do the multiplicity of molecular layers in a fluid mass resist sliding over one another in the manner of a pack of playing cards pushed endwise. With the onset of a critical vibration period, however, the inertia of the damper flywheel is such that it cannot follow the rapid oscillatory movements of its casing attached to the crankshaft (Figure 2.16b). The relative motion thereby created between the flywheel and its casing results in the molecular layers of the fluid film undergoing a shearing action, which process abstracts energy from the vibrating crankshaft and dissipates it in the form of heat from the damper. An important consideration in the fitting of a torsional vibration damper to the crankshaft of a modern heavy-vehicle diesel engine is that of protecting the comprehensive gear train of its timing and auxiliary drives from vibrational disturbances.

Historically, it was once again Dr F.W. Lanchester who first proposed the idea of a viscous damper to control crankshaft torsional vibrations, and it was the subject of a patent by him as long ago as 1910. However, it was the further research by the Americans G.J. Dashefsky and Capt H.M. Jensen at the New York Naval Shipyard in 1930 that produced the first really practicable viscous damper as we would recognize one today. Rather interestingly these early researchers used furniture glue as the viscous medium, so it was the World War II development of silicone fluids and their application to the viscous damper by the American Houdaille-Hershey Corporation in 1946 that led to finality in design.

2.6 VALVE TRAIN AND TIMING DRIVE

Valve train

The valve train layouts for diesel engines are generally similar to those used with I-head petrol engines. For heavy-duty diesel engines the valves are usually operated through the medium of push-rods and rockers, which receive their motion from a crankcase-mounted camshaft with either tappet barrel or finger rocker followers. In both cases these followers may have a roller contact with the camshaft. In some designs where four smaller valves per cylinder are used and typically those in which the more bulky unit injector system of fuel injection (Section 8.9) would be unable to accommodate two larger valves, each pair of valves is operated from a single overhead rocker that acts upon a vertically sliding cross-head or bridge piece which straddles their stem tips. The cross-head is provided with an adjusting screw beyond the rocker tip, so as to ensure equal operation of both valves (Figure 2.17). In practice, the adjusting screw is slackened sufficiently to allow the other end of the sliding cross-head to be brought into contact with its valve tip under finger pressure. The adjusting screw is then advanced into contact with its own valve tip and held while the locknut is tightened. Valve adjustments proper are carried out in the conventional manner, but with the difference that the clearances are measured between the rocker lever pad and cross-head platform for each pair of valves.

Less commonly, the valves of a heavy-duty diesel engine may be operated from an overhead camshaft, which either attacks them directly via interposed bucket tappets or indirectly through rockers with the object of gaining easier valve clearance adjustment. However, the overhead camshaft has been widely adopted as part of the valve train for light-duty

diesel engines such as used in passenger cars. In late 1996 General Motors Europe introduced a bridge type of four-valve actuation for their direct-injection passenger car diesel engines of 2.0 and 2.2 litre capacities, a point of interest being that the bridge piece sits directly on the hydraulic bucket tappets that surmount the paired valves. By virtue of requiring only one instead of two overhead camshafts, a 30 per cent reduction in valve gear friction is claimed to the benefit of engine fuel consumption.

Comparison of diesel and petrol engine valves

In vehicle diesel engines the poppet valves used are of the same established form as those described for petrol engines, but with more generous metal sections for the heads and with noticeably longer stems (Figure 2.18). The former is provided in the best interests of heat dissipation and mechanical strength and the latter arises from the valves seating in the bottom face of the cylinder head, rather than in the roof of a combustion chamber as for most petrol engines. Although for normally aspirated diesel engines the head diameter of the inlet valve conventionally exceeds that for the exhaust valve, both valves are made equal in diameter in the case of turbocharged versions, where energy from the exhaust gases is utilized to pressure charge the air induction system. To reduce seat wear, the face angle of the inlet valves for turbocharged versions is usually reduced from 45° to 30° , since the inherently lower seating pressure with the latter angle tends to compensate for the higher cylinder pressure. For some direct-injection diesel engines a shroud or masked portion is forged integral with the inlet valve head (Figure 2.19b), the purpose of departing from a plain head (Figure 2.19a) being to promote induction swirl (as explained later). This type of valve does, of course, require

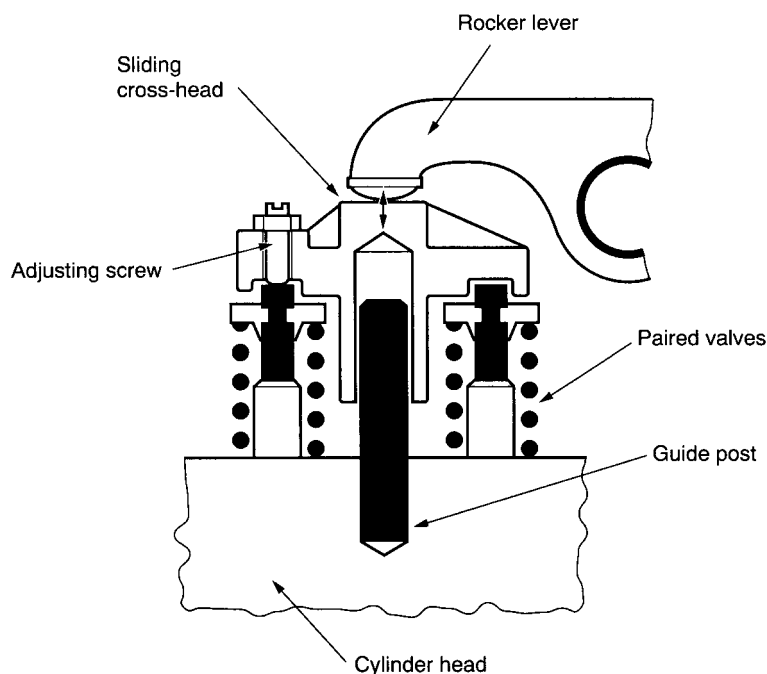


Figure 2.17 Rocker and cross-head valve actuation for a four-valves per cylinder diesel engine

a flat or keyway to be machined on the stem to locate the shroud at the optimum position in the intake port. As in the petrol engine, the limiting factor in the durability of the diesel engine valve train is usually the exhaust valve.

As compared with petrol engines, the materials used for the inlet and the exhaust valves of diesel engines must cater for a much wider range of application and power outputs, as summarized below.

Diesel engine inlet valves

For moderate-duty applications silicon-chromium steel is again commonly used, although for indirect-injection engines where the valves tend to be hotter running a chromium-manganese-nickel steel is often demanded. The latter material has long been known by its original designation 21-4N, and is a high-alloy stainless steel of the austenitic group with a nominally 21 per cent chromium content. Steels in this group are the most resistant to corrosion of any of the stainless steels. For large-capacity turbocharged diesel engines a variety of materials may be used. These range from 1.5 per cent nickel-chromium-molybdenum steel to austenitic chromium-nickel steel, according to running temperature and mechanical loading. Stellite hard facings may be employed to resist

wear on the seats of some inlet valves and will be described later in connection with exhaust valves (Figure 2.19c). The significance of their use for inlet valves is that with turbocharging the valve seats can run very dry, because the pressure at the inlet ports inhibits any oil leakage down the valve guides.

Diesel engine exhaust valves

For moderate-duty applications, the two most widely used exhaust valve materials are the high-alloy stainless ones, chromium-manganese-nickel (21-4N) and chromium-manganese-nickel-niobium (21-42) steels. Niobium alloys have a great resistance to heat and oxidation, a point of interest being that niobium metal itself has the very high melting point of 2415°C. The stems of valves made from these materials are usually chromium plated for durability. As in petrol engine applications, the use of a bimetal two-piece exhaust valve construction is becoming more common (Figure 2.19d).

In the case of large-capacity turbocharged diesel engines there are several high-alloy materials in use. These include, apart from chromium-manganese-nickel and chromium-nickel austenitic steels, a material known as Nimonic 80A. This is one of the Nimonic series of nickel-chromium (rather than chromium-nickel) alloys having a basically 80 per cent nickel and 20 per cent chromium composition, which were developed during World War II to withstand the most severe conditions of stress and temperature encountered in aircraft gas turbine engines. Stellite seat facing alloys are generally employed for high-duty applications and contain a high proportion of cobalt. This material has a particular relevance to diesel valves, since it can resist corrosion from the combustion of the sulphur content in fuel oils.

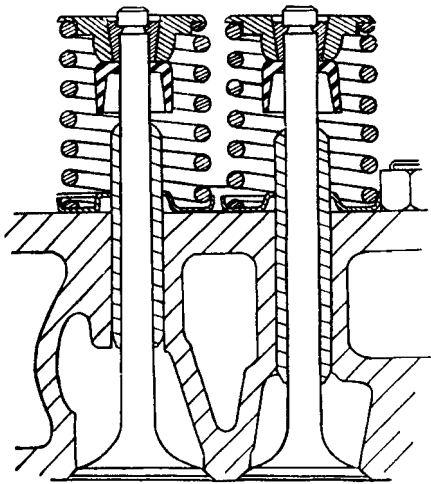


Figure 2.18 Arrangement of diesel engine valves (*Perkins*)

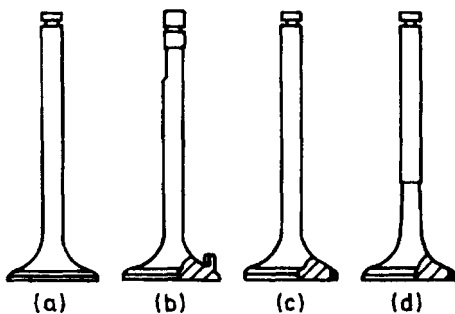


Figure 2.19 Types of diesel engine inlet and exhaust valves: (a) plain inlet (b) shrouded inlet (c) hard-faced exhaust (d) hard-faced bimetal exhaust

Timing drive

A once typical front mounted timing drive for a diesel engine comprises four gears: one each for the camshaft and fuel injection pump drive, a crankshaft pinion, and an idler gear that engages the other three (Figure 5.3). All four gears have teeth of helical form and are produced from steel for this type of heavy-duty application. Alternatively, a triangulated layout of double-row or triple-row roller chain has been used for the timing drive, the length of the chain being kept to a minimum so that the effects of wear have the least disturbance on the timing of the fuel injection pump. In more recent practice, there has been a need to ensure greater precision of operation for the valve train and unit injectors, which has resulted in the camshaft for in-line cylinder engines being mounted either high in the cylinder block, or on the cylinder head itself, and in both cases this inevitably entails using an extended (Figure 2.5) and sometimes a compound (Figure 2.20) train of gears. Furthermore, the gear train may be mounted at the rear of the engine with the input gear being sandwiched between the crankshaft flange and the flywheel (Figure 2.20). The purpose of this reversal in arrangement of the timing drive will be understood by recalling that when a crankshaft suffers a period of torsional vibration, its maximum amplitude will be greatest at the front end (Section 1.9). Therefore to reduce vibration, wear and now importantly noise in the timing drive, it is clearly advantageous to mount

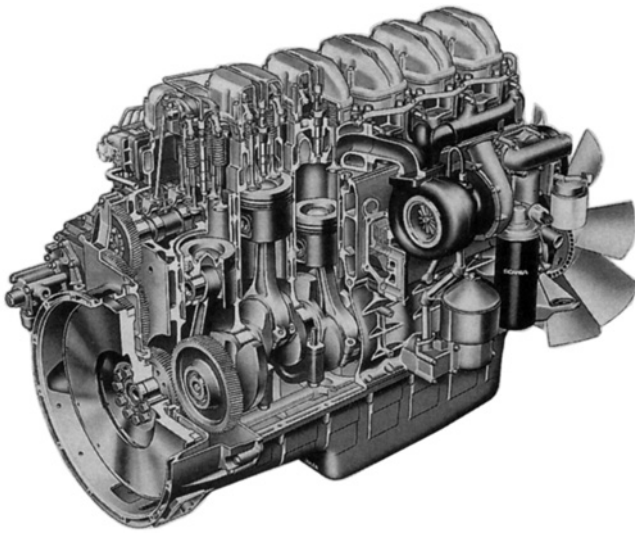


Figure 2.20 The 12-litre Scania DSC 12 in-line engine with four valves per cylinder and rear mounted timing gear (Scania)

the gear train at the opposite end of the crankshaft where the torsional vibration is much less. An ideal but hardly practicable course of action would be to site the timing gear train at the nodal point (position of no vibration) of the crankshaft, but this unhelpfully lies somewhere between the rear crankthrows. Another advantage of a rear mounted timing drive is that it can provide a suitably geared power take-off facility (Section 14.5).

As a matter of historical interest, it is worth mentioning that some designers of passenger car engines had early recognized the technical merit of a rear mounted timing drive, notable examples that may be familiar to classic car enthusiasts being those found in the in-line six-cylinder engines of the high-grade AC and Alvis cars of a past era. Other designers objected to this arrangement on the grounds of its inaccessibility in service, but such criticism proved to be largely unfounded because the less highly stressed timing drive seldom required attention.

In the case of light-duty diesel engines used in passenger cars, a toothed belt is now generally used for the timing drive to the overhead camshaft and also includes the fuel injection pump, which is mounted at about mid-height on the engine.

Valve timing for the four-stroke diesel engine

As in the case of the four-stroke petrol engine, the timing for the opening and closing of the inlet and the exhaust valves departs from the basic four-stroke operating cycle. The valve timing values may similarly be presented in the form of a diagram (Figure 2.21). There is a greater likelihood that actual verification of these values may be called for, at least in respect of inlet valve opening and exhaust valve closing points, when reassembling a diesel engine after overhaul. The practical implication of this is that since the piston approaches as closely as possible to the cylinder head, then any deviation from the valve timing specified by the manufacturer can result in the very real danger of the valves fouling the piston

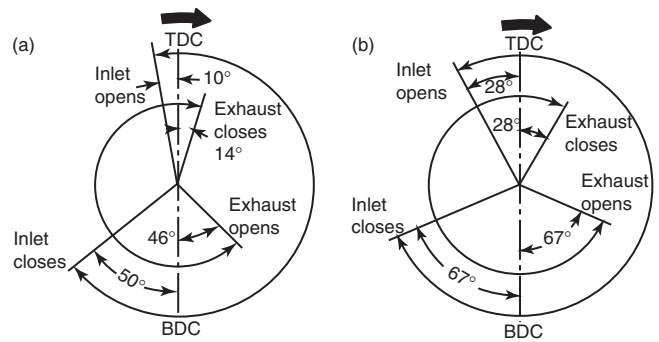


Figure 2.21 Typical valve timing diagrams for four-stroke diesel engines: (a) commercial vehicle, medium speed (b) passenger car, high speed

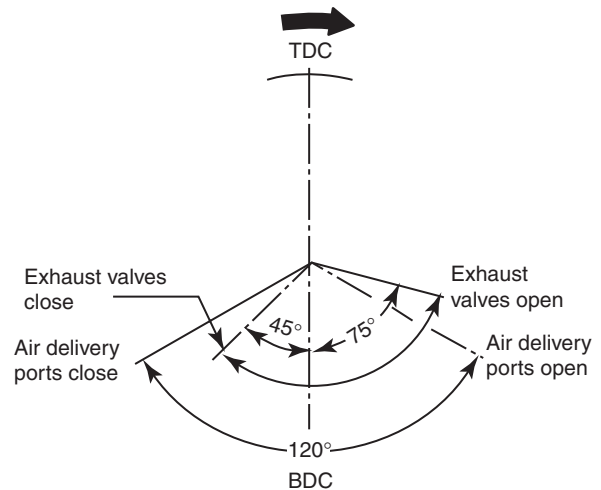


Figure 2.22 Typical port and valve timing diagram for a commercial vehicle two-stroke diesel engine

when the engine is started. It is also for this same reason that diesel engine manufacturers warn that valve clearance adjustments should never be attempted with the engine idling, because severe damage could result from inserting a feeler gauge between the valve and its rocker under these conditions.

A further critical factor lies in the choice of value for the inlet valve lag, so that the all-important pressure energy generated during the compression stroke is always at a maximum. More specifically, the inlet valve must neither close too late, which could be detrimental to easy starting of the engine, nor close too early, which could detract from maximum performance. The value of the exhaust valve lead is generally less critical with respect to diesel engine performance, because of the lower operating speeds. For turbocharged diesel engines a valve timing with an increased degree of overlap is generally specified so that, with the earlier opening of the inlet and the later closing of the exhaust valves, the pressurized charge of air can be utilized for more efficient scavenging of the exhaust gases.

As compared with medium-speed diesel engine practice in commercial vehicles (Figure 2.21a), the valve timing

diagrams for high-speed engines used in modern diesel-powered passenger cars (Figure 2.21b) usually reveal an increase in duration of both inlet and exhaust valve events. This is, of course, in the interests of providing similar performance characteristics to those that would be expected from a comparable petrol engined car.

Port and valve timing for the two-stroke diesel engine

With reference to the typical port and valve timing diagram for this type of engine (Figure 2.22), the main point to be noticed in comparison with that for a two-stroke petrol engine is that

the exhaust event need no longer be symmetrical. This is, of course, made possible by the use of mechanically operated poppet exhaust valves. A sufficiently early opening of the exhaust valves can therefore be provided, so as to initiate thorough scavenging of the exhaust gases just prior to the air inlet ports being uncovered by the piston on the power-exhaust stroke, without incurring a too late closing of the exhaust valves. Since the latter action would reduce the amount of pressure energy generated during the induction-compression stroke, the exhaust valves are usually timed to close just before the air inlet ports are covered again by the piston.

3 Combustion chambers and processes

3.1 BASIC LAYOUTS OF COMBUSTION CHAMBERS

General background

The general arrangement of the combustion chamber in respect of its geometric shape, disposition of the valves, siting of either the sparking plug or the fuel injector, and cooling provision is always regarded by an automotive engineer as a critical factor in the design of spark-ignition (petrol) and compression-ignition (diesel) engines. An enormous amount of both theoretical and practical research has been devoted to this aspect of engine design, especially since Sir Harry Ricardo in England began his pioneering investigations into the combustion process in the early 1920s. The aim then was to prevent knocking in the engine cylinders, or detonation as it was originally known. Some fifty years later, the research priority shifted to the reduction of atmospheric pollutants emitted from the engine and to improving its fuel economy. In modern research computer simulation programs are widely used to predict and refine the combustion process. For our purposes, therefore, we can only attempt to summarize the basic essentials, past and present, as they are generally recognized today.

Geometric shape

To obtain the highest possible thermal efficiency from an engine it is necessary to minimize heat loss to the combustion chamber walls. For this reason, the geometric shape of the combustion chamber must always be such that the ratio of its total surface area to its volume is kept as low as possible. This factor together with the absence of any recessed areas in the chamber is of particular importance in current petrol engines, since it also contributes to a reduction in harmful hydrocarbon (HC) exhaust gas emissions (Section 11.1), by ensuring more complete combustion of the air and fuel charge.

For both petrol and diesel engines the geometric shape of the combustion chamber is contrived, wherever possible, to promote a controlled degree of charge agitation or *compression turbulence* to the cylinder gases. The reasons for promoting this turbulence are not, however, the same for both types of engine. In the case of the petrol engine, the turbulence is associated with propagating the chemical reaction in the air and fuel charge, whereas for the diesel engine a relatively shorter period of turbulence is required for rapid mixing of the air and injected fuel just prior to combustion, the inherent excess of air reducing harmful exhaust gas emissions (Section 3.5). On the other hand too much turbulence is to be avoided, because it can increase heat loss to the combustion chamber walls. Where compression turbulence is difficult to promote in what otherwise may be efficient

designs of combustion chamber, it then becomes necessary to obtain a similar effect by promoting a swirling motion to the incoming cylinder charge (either air and fuel or air only), this being known as *induction swirl*.

Disposition of valves

It may be recalled that it has long been established practice to mount both the inlet and exhaust valves over each cylinder in either I- or V-head manner. The latter arrangement in particular is noted for its 'easy breathing' quality in petrol engines. For diesel engines it would, in any event, be physically impossible to obtain a sufficiently high compression ratio without the compact clearance volume associated with the I-head.

In the interests of achieving a high volumetric efficiency, a single inlet valve must be made of adequate size, typically about 45 per cent of the cylinder bore diameter. It should also be provided with ample clearance around the head, so that the incoming cylinder charge does not impinge on the combustion chamber or cylinder walls. The open areas of both the inlet and exhaust valves must, of course, be such as to give adequate flow capacity for the incoming and outgoing cylinder gases. As earlier noted, there is nothing to be gained by lifting a poppet valve from its seating by an amount exceeding one-quarter of its head diameter. A 30° instead of a 45° angle is sometimes used for the inlet valve seating, because it permits a greater gas flow for a given valve lift.

Siting of sparking plug or fuel injector

The siting of the sparking plug electrodes in the combustion chamber of the petrol engine was early recognized by H.R. Ricardo as being one of the important factors controlling pinking or detonation, this being an abnormal condition of combustion described later. To achieve what Ricardo termed the highest useful compression ratio (usually abbreviated to HUCR) and be free from detonation, the distance of flame travel within the combustion chamber must not be unduly long in any one direction. This ensures that there is least time available for a detonating pressure wave suddenly to build up ahead of the flame front. A short flame travel is beneficial for rapid and efficient combustion with least emission of unburned hydrocarbons.

A further compromise usually sought in siting the sparking plug is that preferably its electrodes should initiate combustion in the hottest and therefore exhaust valve region of the combustion chamber, so that the flame front travels from the hottest to the coolest parts. On the other hand the plug electrodes also need to be cooled and scoured of any residual exhaust products by the incoming fresh cylinder charge.

Any reader who takes an interest in vintage cars is likely to know that a 1920s 3 litre Bentley, for example, has two

sparkling plugs serving each cylinder with one on either side. In fact it was not uncommon practice during that era for designers of high-grade passenger cars to employ twin sparking plugs, especially those associated with firms that also manufactured aircraft piston engines. For the latter it early became conventional practice to use two sparking plugs per cylinder, each served by its separate magneto ignition system. This duplication was partly for reasons of reliability and partly to promote more rapid combustion by initiating two flame fronts from opposite sides of the combustion chamber.

It is therefore of interest that the idea of using twin sparking plugs was revived in the late 1980s by Alfa-Romeo and applied to their long-acclaimed four-cylinder in-line engine. In this case the main advantage sought was the ability to ignite leaner mixtures, especially at low engine speeds, so that both fuel consumption and harmful exhaust emissions could be reduced without sacrificing the smooth running qualities of the engine. These gains may be attributed to less delayed combustion, which results at best from the ignitable charge having greater exposure to the two separate points of ignition and at worst from the increased likelihood that it will be in proximity to at least one of the points of ignition. A further refinement of the twin sparking plugs concept is for them not to be fired simultaneously, the purpose being to reduce combustion noise by avoiding an excessive initial pressure rise as the air and fuel charge is ignited.

With regard to the siting of the fuel injector nozzle in diesel engines, this depends upon the type of combustion system used, as described later. In general terms, a centrally disposed nozzle is required for direct-injection engines with open combustion chambers, whereas for indirect-injection engines an offset nozzle spraying into a combustion antechamber of either the precombustion or swirl type is employed.

Cooling provision

If a high volumetric efficiency is to be obtained, it is essential that the exhaust valve in the combustion chamber be maintained as cool as possible. Otherwise, the heat radiated from it will reduce the density of the incoming charge, either air and fuel or air only, so that a lesser quantity enters the cylinder for compression. A no less important reason for maintaining adequate cooling of the exhaust valve is that of avoiding detonation. Special provision must therefore be made to direct a vigorous flow of coolant around the high-temperature surfaces of the cylinder head combustion chamber in the region of the exhaust valve.

3.2 COMBUSTION IN THE PETROL ENGINE

Basic considerations

The combustion process is concerned with the chemical reaction that takes place between a hydrocarbon fuel – petrol – and oxygen from the air, and which continues until a point of chemical equilibrium is attained. For the moment, we can say that this simply means that the hydrocarbon fuel is oxidized or, more specifically, burned (since the latter implies very rapid oxidation) to carbon dioxide and water, which are oxides of carbon and hydrogen respectively. It is, of course, the intense heat that accompanies this chemical reaction

which causes the cylinder gases to expand and pressurize the cylinder, such that the piston is driven away from the combustion chamber on its power stroke.

It will be recalled from Section 1.3 that the sequence of events in the four-stroke petrol engine was defined as induction-compression-power-exhaust, but in some popular types of publication the power stroke may be described as the explosion stroke. The latter may therefore give the impression that the operation of such an engine is one ‘Of dire combustion and confused events’, to borrow an expression from Shakespeare. With a real engine, however, the energy conversion taking place in the combustion chamber is, or at least should be, somewhat better ordered. In this context, we must first recognize that the combustion process can take two basic forms, which in physical chemistry may be classified as follows:

Spontaneous combustion This has no external cause and occurs purely as a result of a chemical reaction that is simultaneously set up throughout a mixture of combustible gases, and which is accelerated by the evolution of heat until completion.

Progressive combustion This does have an external cause which, in our case, is an electric spark from the ignition system, and occurs as a *flame front* that radiates from a combustible nucleus developed around the spark. As combustion progresses, the unburnt and the burnt portions of the mixture are separated by a *reaction zone* in the flame front, the flame cooling and slowing down as it approaches the combustion chamber walls, finally to become extinguished (Figure 3.1).

Normal and abnormal combustion

In further considering the combustion process that takes place in the petrol engine and which is normally of a progressive

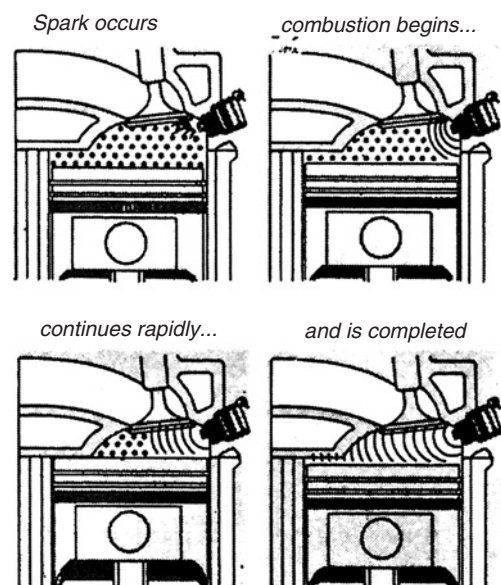


Figure 3.1 Visualizing normal combustion in the petrol engine (Champion)

character, we must always remember that we are talking about a phenomenon occupying only a few milliseconds. Paradoxically, it can also be said that probably more time has been devoted by scientific researchers to investigating the combustion process and related combustion chamber design than to any other aspect of automotive engine operation. The reasons for this are basically threefold, the first two being historical in character and the last being of more recent origin: detonation, roughness and pollution.

Detonation

For high-efficiency engines operating at high speeds, it will be evident that a very fast rate of burning of the combustible charge is required. As with any mixture of gases, however, there must be a limit to which the rate of burning may be increased, beyond which the combustion process undergoes a sudden and dramatic change that results in detonation. In normal combustion, the burning progresses in a steady and uniform flame front that travels across the chamber at about 20–40 m/s (65–130 ft/s), this being known as the *combustion speed*. With detonation the flame front similarly advances across the chamber, but about ten times faster, and before it can complete its journey the unburnt gases ahead of it – or end gases as they are usually termed – become heated to such an extent by the overall pressure rise in the chamber that their self-ignition temperature is exceeded (Figure 3.2). When this occurs a condition of spontaneous combustion is created, resulting in a high-pressure detonation wave travelling through the chamber at a speed comparable to that of sound. Quite understandably, this produces a shock loading on the engine structure and components sufficient to generate a clearly audible knocking or pinking noise. If this abnormal combustion is allowed to persist, it can cause serious damage to the engine, such as blowing a hole in the piston crown (Figure 3.3).

Since the early 1920s the control of detonation has been improved in several ways, chiefly by experiment with different

combustion chamber shapes, more efficient cooling arrangements for the engine and the production of more knock-resistant fuels. The knock resistance of a petrol is defined in terms of its ‘octane rating’. A higher octane rating signifies a greater resistance to engine knocking. There are two different test procedures for determining the octane rating, these being the Research Method giving a Research Octane Number (RON) and the Motor Method giving a Motor Octane Number (MON). Basically, the former rating indicates the resistance to engine knocking during acceleration, whilst the latter rating similarly relates to high-speed running. The octane number of a petrol is established by matching its knock resistance in a test engine against a reference fuel. This comprises a mixture of high knock resistant iso-octane and low knock resistant n-heptane. The actual octane number of the petrol then corresponds to the percentage by volume of iso-octane in the reference fuel, which exhibits the same knock-proneness as the petrol in the same engine running under similar controlled conditions.

In modern practice, combustion knock is often monitored by a knock control system, which comprises a vibration sensor acting in conjunction with the electronically controlled ignition of an engine management system. By relating an adverse signal from the knock sensor to a positional signal from the crankshaft sensor, it becomes possible to retard the ignition timing for any particular cylinder until the knock disappears. A knock sensor is mounted on the cylinder block between pairs of cylinders, where vibration is best amplified by the less stiff walls of the coolant jacket. It therefore converts mechanical vibration into electrical signals and is tuned to detect vibrations typically in the 6–8 kHz (6000–8000 cycles/s) frequency range. A knock control system therefore makes a further contribution to more stable ignition timing and its beneficial effects on engine operation and emission control.

Roughness

Detonation and roughness have sometimes been described as the twin evils of the automotive petrol engine. The reason for this is that whereas better control over detonation allows the use of higher compression ratios to improve thermal efficiency, as described in Section 1.3, it can also result in the maximum rate of pressure rise during combustion being

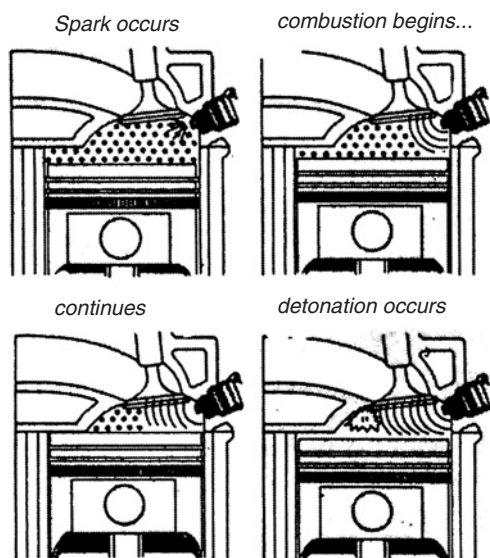


Figure 3.2 Visualizing detonation in the petrol engine (Champion)

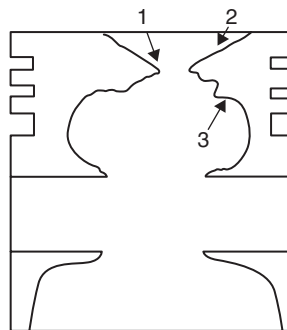


Figure 3.3 Typical piston damage from severe detonation: 1 sharp ragged edges blown out below; 2 occasional metal pitting; 3 possible cracks on underside (Champion)

such as to cause undesirable vibration of the engine structure and mechanism, especially bending vibration of the crankshaft. Although this problem of rough operation originates from the combustion process, its practical solution usually lies in providing greater rigidity for the engine parts.

Pollution

As a simplification it was earlier mentioned that during the combustion process the hydrocarbon fuel is burned to carbon dioxide and water. Unfortunately this rather tidy and relatively harmless state of affairs does not actually exist in practice. Evidence of this truth lies not only in the unpleasant odour of the exhaust gases, but also in the common knowledge that they contain poisonous carbon monoxide. It is the presence in fuel of compounds other than hydrocarbons, and also the fact that the combustion process may only be partially completed, which account for the harmful exhaust emissions. More recently a further concern to environmentalists has been the emission of carbon dioxide (CO_2) which, although non-toxic as earlier implied, is an unwanted contributor to global warming. That is, the heat absorbed from the sun cannot leave the atmosphere in the presence of a buildup of carbon dioxide.

The general topic of harmful exhaust gas emissions, their legislated regulation and control is later dealt with in Section 11, and the role played by the geometric shape of the combustion chamber has already been mentioned in Section 3.1.

3.3 PETROL ENGINE COMBUSTION CHAMBERS

The six basic types of combustion chamber employed, past and present, in automotive petrol engines may be classified as follows:

Hemispherical	Bowl-in-piston
Lozenge	Split-level
Wedge	Pentroof

Hemispherical chamber

The hemispherical form of combustion chamber with inclined overhead valves has long been regarded as the classic type for high-output engines (Figure 3.4), especially when used in conjunction with double overhead camshafts after the

style introduced as long ago as 1912 by the talented Swiss designer Ernest Henry (1885–1950). Nowadays the hemispherical chamber is also recognised as offering a beneficial reduction in harmful exhaust emissions.

A high volumetric efficiency or ‘easy breathing’ is afforded by this type of chamber, because not only may large and well-cooled valves be accommodated, but also the radial disposition of the valves ensures that their open areas are not masked at all by the chamber walls. Moreover, the sparking plug is usually sited very close to the central axis of the chamber, so that flame travel distances are minimized and practically equalized for good detonation control (Figure 3.5a).

Since the hemispherical chamber lacks a squish area, charge agitation is normally effected by induction swirl (both these aspects of charge agitation will be described later). A dome-head piston is sometimes required to reduce the combustion chamber clearance volume and obtain a high compression ratio, in which case the dome may be offset slightly to promote compression turbulence. As a consequence of departing from a true hemispherical shape, however, the ratio of total surface area to volume of the combustion space is

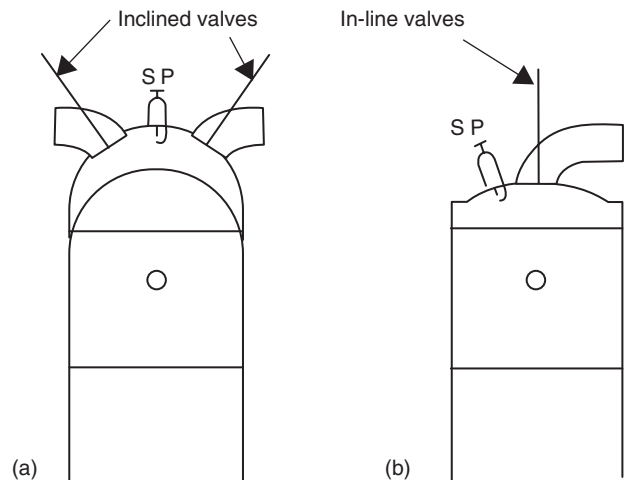


Figure 3.5 Basic layouts of combustion chambers: (a) hemispherical (b) segmental

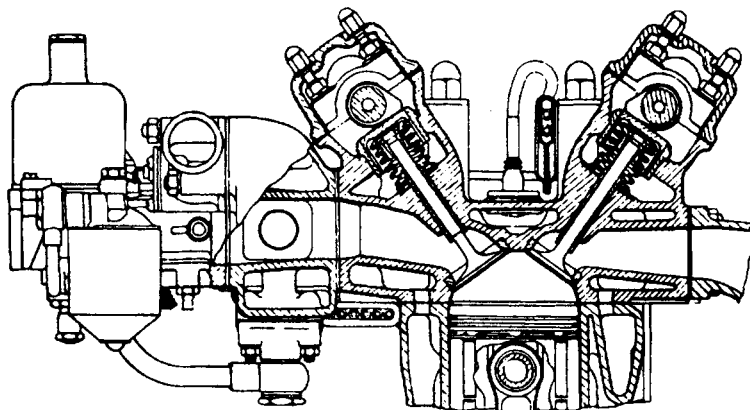


Figure 3.4 The classic Jaguar version of a hemispherical head

increased. This in turn reduces thermal efficiency, owing to the greater heat loss from the combustible charge.

More recently, the basic hemispherical chamber has undergone further changes, such that in some designs it more approaches a segmental form. That is, it assumes a comparatively shallow part-spherical shape with in-line rather than inclined valves (Figure 3.5b). Again, the object is to obtain a favourable ratio of total surface area to volume, thereby improving thermal efficiency and reducing the quantity of incompletely burned products of combustion. Hence, better control is maintained over harmful exhaust emissions.

Lozenge chamber

The lozenge or inverted-bathtub form of combustion chamber is now little used, but it long represented the simplest type for overhead valve engines of moderate output (Figure 3.6a). This type of chamber was evolved in the early 1920s from the plain cylindrical form, as a result of increased compression ratios demanding reduced clearance volumes. The valves are mounted in-line, and in plan view the chamber was originally semicircular at each end with straight sides, its greatest width being such that it overlaps the cylinder bore, but in cross-section it is appreciably narrower than this. To improve volumetric efficiency, which otherwise tends to be a limiting factor with this type of chamber, the valves may be slightly inclined to the cylinder axis and the chamber end walls may be ‘undercut’, where they tend to encroach upon the open areas of the valves (Figure 3.6b).

The chamber side walls may also slope away from the roof, so that the sparking plug entering through one side at a fairly steep angle can be brought closer to the valves. This improves combustion efficiency, in which connection the sparking plug is usually biased slightly towards the exhaust valve. Hence the combustion process is initiated in the hottest, and completed in the coolest, regions of the chamber for good detonation control. A certain amount of compression turbulence is provided by a *squish area*, formed where the flat top of the piston overlaps the combustion chamber opening and approaches very closely the underside of the cylinder head. The squish area, with its high ratio of surface area to volume, also acts as a *quench area*, in as much that it serves to cool the remote parts of the charge or end gases. For this reason the lozenge type of combustion chamber may also be offset laterally from the cylinder axis in the direction of the sparking plug.

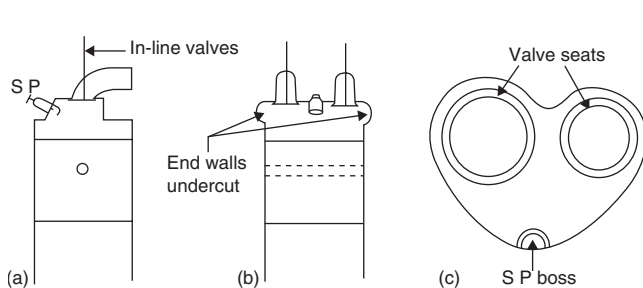


Figure 3.6 Lozenge combustion chamber: (a)/(b) basic layout (c) plan view of Weslake version

A later development of this form of combustion chamber, which was introduced by the British Motor Corporation (BMC) in 1952 for their ‘A Series’ engine and subsequently adopted for the trend-setting Austin and Morris Mini cars, saw it further modified and made heart-shaped in plan view. The apex portion then directs the incoming cylinder charge towards the sparking plug and also shields it from heat being radiated by the exhaust valve. This version of lozenge combustion chamber is known as the Weslake type (Figure 3.6c), so named after H. Weslake, an English researcher into gas flow through cylinder heads who invented it in the late 1940s.

Wedge chamber

The wedge or ramp form of combustion chamber for overhead valve engines is a comparatively later development. It first appeared in the late 1930s on some American engines, since when it has been widely used for both moderate- and high-output engines. This type of chamber is generally D-shaped in plan view, the circular part of which follows the cylinder opening. In cross-section, this circular portion tapers uniformly away from where the sparking plug is situated in its side wall and terminates at the approach to a very narrow squish area (Figure 3.7). The latter is completed by the top of the piston where it approaches very closely to the underside of the cylinder head. The valves are generally mounted in-line and also inclined to correspond with the angled chamber roof, thus facilitating the flow of gases to and from the cylinder for good volumetric efficiency. Distribution of the clearance volume is therefore such that the greater portion of the combustible charge is concentrated around the sparking plug, so that fairly short and nearly equal flame travels are achieved for smooth combustion. Furthermore, the sloping roof of this part of the chamber reduces its ratio of surface area to volume, thereby improving thermal efficiency as a consequence of less heat loss. The squish area generally covers 25 per cent or more of the piston head surface area and, as in other combustion chamber designs, it also acts as a quench area for good detonation control.

However, in light of current requirements to reduce harmful exhaust emissions, without incurring unacceptable penalties in fuel consumption and power output, the greater ratio of total surface area to volume of the combustion space tends to put the wedge chamber at a disadvantage compared with the hemispherical type.

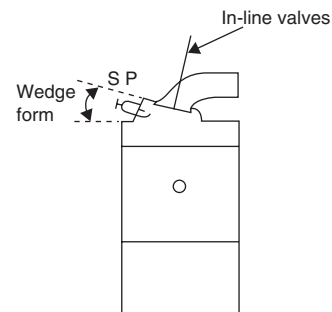


Figure 3.7 Basic layout of wedge combustion chamber

Bowl-in-piston chamber

The bowl-in-piston or piston cavity form of combustion chamber is of more recent origin than the wedge type. Its limited application to petrol engines dates from the early 1960s, following earlier investigations by S.D. Heron in America. In this design the combustion chamber proper takes the form of a cylindrical recess located in the piston head (Figure 3.8a). As compared with other types of chamber utilizing in-line valves, the open areas of the valves are less masked, although they are in themselves limited by the size of valves that can be circumscribed by the cylinder bore (Figure 3.8b). The annular rim of the piston is generally relieved about the gudgeon pin axis, so that a safe working or bumping clearance for the valves is maintained against the piston during their overlap period. A high thermal efficiency is usually claimed for this type of combustion chamber, since its flat roof contributes to a low ratio of total surface area to volume. Furthermore, the geometric symmetry of the bowl-in-piston chamber affords minimum and near-equal flame travels, the sparking plug being positioned as close as is practicable to the central axis of the chamber. Charge agitation is initiated by induction swirl and completed by compression turbulence. The former is derived from the usual angled porting, whilst the latter is promoted by the flat-topped rim of the piston approaching very close to the underside of the cylinder head and thus providing an annular squish area. Again, as in other combustion chamber designs, the squish area also serves as a quench area.

One disadvantage of this type of combustion chamber is that it tends to make for a heavier piston construction. Another, in the context of reducing harmful exhaust emissions, is that the hotter running piston tends to increase the production of unwanted nitrogen oxides.

Split-level chamber

Of further interest is the split-level form of combustion chamber. This is also known as the May fireball, so named

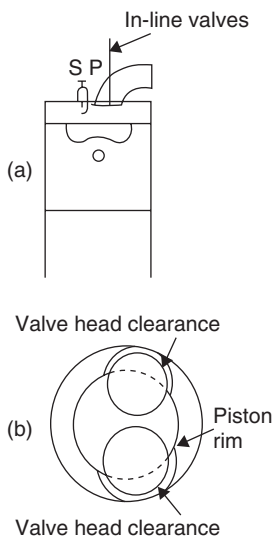


Figure 3.8 Basic layout of bowl-in-piston combustion chamber

after Michael May, the Swiss researcher into combustion chamber design, who first developed the concept in the mid 1970s. This type of combustion chamber was used in the renowned Jaguar V12 engine.

The essential features of the split-level combustion chamber are a lower-level and therefore shallow collecting zone beneath the inlet valve, and an upper level formed by a deeply recessed exhaust valve that constitutes the combustion zone of the chamber (Figure 3.9). A swirl-promoting guide channel is provided between the collecting and combustion zones with a tangential approach to the latter. The combustion chamber is completed, of course, by the flat top of the piston and the sparking plug is sited in the wall of the upper-level combustion zone.

As the piston closely approaches the combustion chamber towards the end of its compression stroke, that portion of the air and fuel charge residing in the collecting zone is forcibly ejected, via the tangential guide channel, into the combustion zone of the chamber, wherein a rapid swirling motion is generated in the vicinity of the sparking plug. This action makes possible a fast and efficient burning of air and fuel cylinder charges that are leaner than normal, which in turn allows an increase in engine compression ratio without incurring detonation. Hence, the advantages generally attributed to the split-level type of combustion chamber are those of reduced fuel consumption and increased thermal efficiency. Similarly, there is a reduction in harmful exhaust emissions, as would be expected with any new design of combustion chamber if it is to prove acceptable. However, in the context of present-day engine design that generally favours inclined overhead valves, the split-level chamber was necessarily limited to in-line vertical or near vertical overhead valves.

Pentroof chamber

In engine terminology, the designation 'multiple valves' signifies that each cylinder has more than one inlet and one exhaust valve. Since the early 1980s there has been a trend in combustion chamber design towards providing two inlet and two exhaust valves for each cylinder (Figure 3.10), instead of the customary one of each. Four-valve combustion chambers were pioneered by the earlier mentioned Ernest Henry and were actually race proven even before World War I, but thereafter were mainly confined in application to racing car and a few high-performance sports car engines such as for the vintage Bentley cars of the 1920s. The reason for their

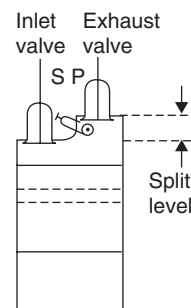


Figure 3.9 Basic layout of split-level combustion chamber

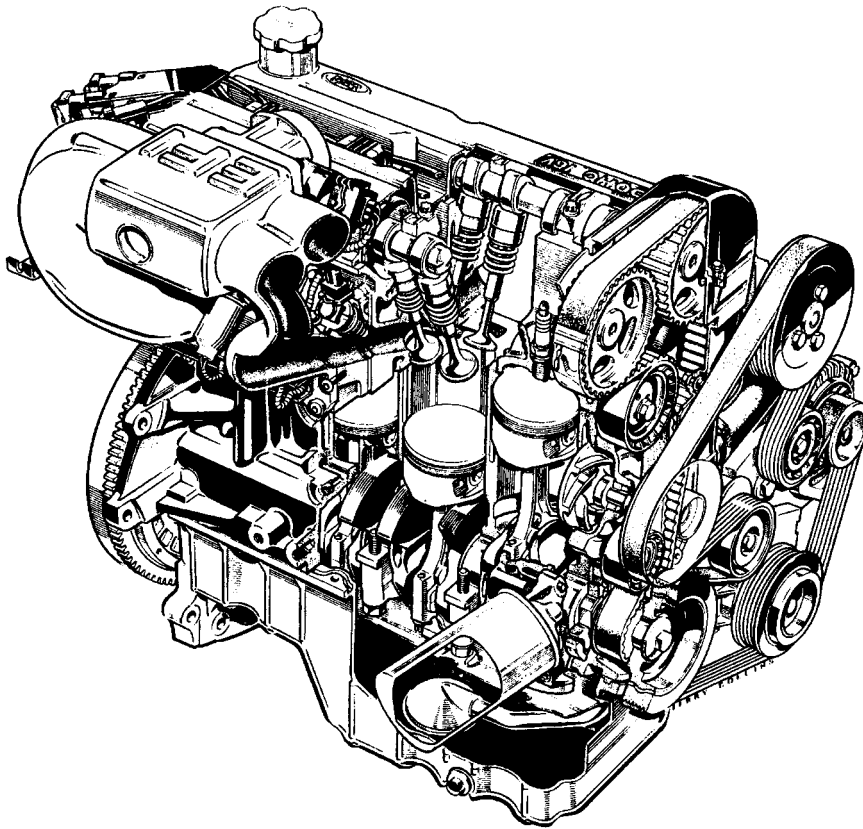


Figure 3.10 A modern four-valve cylinder engine with pentroof combustion chambers and double overhead camshafts (*Ford*)

use was to maximize power output whilst avoiding destructively high valve temperatures.

What then is the reason for now using four-valve combustion chambers in the engines of many volume-produced cars, especially considering the additional complication and cost incurred with such an arrangement? The answer to this question has to be more comprehensive and lies in the requirements of the modern petrol engine given in Section 1.1, namely to provide an optimum performance with high fuel economy and low pollution.

The advantages and disadvantages of the four-valve combustion chamber, which is aptly named the pentroof type, may be summarized as follows:

- 1 The combined area of the four largest valves that can be accommodated in a given combustion chamber always exceeds that of the two largest which may be used. Doubling the number of valves therefore increases the area through which the incoming mixture and outgoing exhaust gases can flow. As a result the engine can breathe more freely at low valve lifts in the interests of better performance, especially with regard to improving mid-range torque characteristics and hence drive-ability of the car (Section 1.1).
- 2 The opening and closing motion of the valves is easier to control at high engine speeds, because each valve is smaller and therefore lighter in weight. This can permit an increase in engine operating speeds, which again contributes to

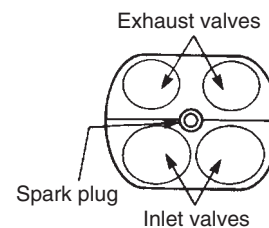


Figure 3.11 Plan view of four-valve combustion chamber

better performance. To reduce the bending moments in the camshafts of four-valve configurations, their bearings may be located between the valves of a pair as in Cosworth designs rather than between each pair of valves as in conventional practice.

- 3 The provision of two inlet valves creates greater turbulence for the mixture flowing into the cylinder, so that better mixing leads to more efficient burning of lean mixtures and hence contributes to better fuel economy.
- 4 The provision of two exhaust valves minimizes the intensity of hot-spots in the combustion chamber, thereby avoiding too high combustion temperatures which can promote the formation of polluting nitrogen oxides in the exhaust gases.
- 5 The four-square arrangement of the valves allows the spark plug to be centrally positioned in the combustion chamber (Figure 3.11), so that flame travels can be both

equalized and minimized for good detonation control. An increase in compression ratio may then be possible to provide better performance and fuel economy.

- 6 The four valves generally and the two exhaust valves particularly should run cooler, since not only are they individually smaller in size with shorter heat flow paths, but also their seatings are better spaced for more uniform cooling. This reduction in valve operating temperatures should therefore make for greater durability. Even so, the higher combustion temperatures encountered have necessitated the use of sodium-cooled exhaust valves by some manufacturers (Section 1.10).
- 7 The modern four-valve arrangement inherently demands a V-head with its greater mechanical complexity and increased friction losses. Apart from doubling the number of valve components, there is almost invariably a requirement for twin overhead camshafts to operate them together with their associated additional journal bearings, whereas with a more ordinary two-valve I-head engine only one camshaft is necessary. Similarly, the extra demands placed on the engine lubrication system calls for an increased capacity oil pump.
- 8 The engine is more costly to manufacture owing to the increased number of valve components used, together with the additional machining required on the cylinder head for the extra valve seats, valve guides and camshaft bearings.
- 9 The engine may prove more costly to service, but this will to some extent depend on whether greater durability is realized in practice with the four-valve arrangement, such that optimum engine performance can be maintained for extended mileages.

An interesting comparison between two combustion chambers of the pentroof type with four valves, which have been developed by Toyota, is shown in Figure 3.12. The sports type is of classic layout as would be expected for a responsive high-performance engine where maximum power is developed at high revolutions. The high-efficiency type is intended to produce better torque in the low to medium speed range together with more economical operation in everyday town driving. With the latter type it will be noticed that in addition to the short flame travel distance derived from the centrally disposed sparking plug, a narrow included angle has been adopted between the pairs of inlet and exhaust valves to allow the design of a very compact combustion chamber. Furthermore, the larger squish areas at the outer edge of the cylinder are conducive to quicker combustion.

Finally, it should be mentioned that it is by no means unusual to encounter two inlet and two exhaust valves in each cylinder of commercial vehicle diesel engines (Figure 2.5), and even three of each in larger-capacity diesel engines used for rail traction purposes. It may also be of interest to recall that, in the mid-1930s, Rolls-Royce developed a form of pentroof or ramp type of combustion chamber to improve turbulence in their four-valve aero engine cylinder heads. Curiously though, the encouraging results achieved on a single-cylinder test engine did not successfully translate to the war-time Merlin engine, which had a more conventional type of combustion chamber.

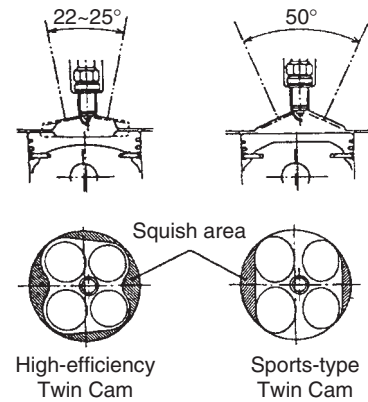


Figure 3.12 Comparison of four-valve combustion chambers with narrow- and wide-angle valves (*Toyota*)

Three- and five-valve combustion chambers

The less common present-day use of three-valve combustion chambers is again a revival of much earlier design practice, which sought to improve engine performance. In modern applications there are the further aims of better controlling exhaust emissions and reducing fuel consumption.

Historically, engine designers originally preferred an arrangement of two exhaust valves and one inlet valve, based on the supposition that more resistance was offered to the exhaust gases leaving the cylinder than there was to the fresh mixture entering it. However, it was Ettore Bugatti the designer of the once famous sports and racing cars that bore his name who, in the early 1920s, introduced the converse arrangement of two inlet valves and one exhaust valve. The two smaller inlet valves and passages serve to increase the velocity of the incoming charge for more efficient filling of the cylinder. Modern versions of the three-valve combustion chamber continue to follow this practice, and it is used, for example, by Mercedes-Benz for their V-cylinder engines (Figure 3.13), including the engine specially developed for their prestigious Maybach car introduced in 2002.

As compared to the more widely used four-valve combustion chambers, the advantages generally claimed for modern three-valve layouts are as follows:

- 1 Dispensing with one exhaust valve reduces heat loss from the engine cylinder, which not only promotes a rapid warm-up, but also the higher exhaust temperatures enable a catalytic converter to reach its 'light-off' temperature more quickly and reduce exhaust emissions (Section 11.2).
- 2 The extra space created in a three-valve combustion chamber readily allows the use of twin sparking plugs. These can provide a shorter flame travel and less delayed combustion, so that both fuel consumption and exhaust emissions can be reduced (Section 3.1).
- 3 For a three-valve combustion chamber it is only necessary to employ a single overhead camshaft to operate both the inlet valves and the exhaust valve, thereby reducing frictional losses in the valve train and hence improving fuel economy.

- 4 A three-valve combustion chamber and its associated less complex valve train necessarily contribute to a saving in engine weight, which further serves to reduce fuel consumption.

Five-valve combustion chambers with three inlet and two exhaust valves have typically been limited in application to turbocharged engines of ultra high-performance cars, such as certain models of Audi and Ferrari. Although paradoxically the Mitsubishi company was the first to mass produce a small capacity, three-cylinder engine with a five-valve arrangement for a Japanese minicar in 1989.

As compared to a four-valve combustion chamber, the main advantage claimed for the five-valve arrangement is that the increase in total area of the valves allows the maximum benefit from turbocharging to be realized. Induction swirl may be increased by arranging for the outer inlet valves to open slightly earlier than the one offset between them, thereby ensuring more rapid combustion, for better performance and lower emissions.

Altering the compression ratio

The ordinary motorist usually associates the term *engine tuning* simply with the various items of preventive maintenance that seek to restore, as far as possible, the economical running qualities of the engine originally intended by the manufacturer. To the dedicated sporting motorist, however, engine tuning can involve dimensional modifications being made that depart from the original mechanical specification of the engine, in order to increase power output. One such modification is to raise the compression ratio as high as possible without incurring detonation. For obvious reasons, this type of work is normally undertaken by specialist engine tuning establishments.

It may be recalled (Section 1.3) that the compression ratio was defined as the swept volume plus the clearance volume divided by the clearance volume. The only practicable approach to raising the compression ratio of an existing engine is therefore to reduce the clearance volume, which may be accomplished in several ways:

- 1 Fitting a thinner cylinder head gasket.
- 2 Skimming metal from the cylinder head face.
- 3 Installing raised-dome high-compression pistons.
- 4 Skimming metal from the base of detachable cylinder barrels.

The third method is generally preferred, because it does not alter the vertical distance between the cylinder head and the crankcase, thereby preserving the original operating geometry of the valve train. Otherwise, the removal of metal from say the cylinder head face may require that either steel shims be added beneath the rocker pedestals, or the push-rods be shortened in engines so equipped.

Variable compression engines

Altering the compression ratio of an engine while it is actually running, that is creating a 'variable compression' engine, has in the past been associated solely with single-cylinder test engines used in research laboratories for assessing combustion characteristics and the knock rating of fuels. The earlier

mentioned Sir Harry Ricardo pioneered this type of research engine, which took the form of raising and lowering the complete cylinder and head assembly in relation to the crankshaft. In construction the lower part of the cylinder jacket was screw threaded so that the cylinder could be slidably positioned in the crankcase by means of a rotatable ring nut, thereby varying the compression ratio.

It is therefore of interest to record that Saab Automobiles have experimentally applied variable compression principles to a supercharged multi-cylinder engine. In this case it allows the compression ratio to be optimized by the engine management system, based on inputs that relate to engine speed, load and fuel quality. Structurally, one side of the integrated five-cylinder block and head is hinged from within the crankcase, the opposite side being linked to a computer controlled hydraulic actuator system. By this means the compression ratio of the engine can be varied between 8:1 and 14:1 by tilting the block and head in relation to the crankcase and hence the crankshaft. This therefore has the effect of altering the clearance volume remaining above the pistons at TDC. An oil-tight enclosure between the hinged upper and lower fixed parts of the engine is formed by a special rubber bellows seal.

3.4 COMBUSTION IN THE DIESEL ENGINE

Basic considerations

The distinguishing feature of the diesel engine is that its combustion process is initiated by spontaneous ignition of the fuel when it is injected into an agitated, highly compressed and therefore exceedingly hot charge of air. Here we are thinking in terms of pressures and temperatures in the region of 5 MN/m^2 (750 lbf/in^2) and 800°C respectively, the temperature rising much faster towards the end of the compression stroke than it does at the beginning. Since the compressed cylinder air attains a temperature that is higher than the self-ignition temperature of the fuel, this being 350 to 450°C and governed by a number of complex factors of which not the least is cylinder pressure, a condition of spontaneous combustion is created.

With the diesel operating cycle it thus becomes possible to ignite a spray of finely atomized, but non-volatile, fuel without the aid of an electric spark. It does, however, mean that the compression ratios for diesel engines must be made appreciably higher than those adopted for petrol engines, and they generally lie between 16:1 and 22:1. If compression ratios as high as these were attempted in petrol engines, then the highly volatile charge of air and fuel admitted during the induction stroke would detonate and burn uncontrollably. Fortunately, no such limitation applies to the diesel engine, because it breathes only 'pure' air. The actual compression ratio chosen for a diesel engine is influenced by combustion chamber design and fuel quality, but in any event it is generally made no higher than is necessary for reliable cold starting.

Although it is a topic that has received much attention by researchers, the precise nature of the combustion process in diesel engines is generally recognized as being complex and as yet not completely understood, since it involves a number of interrelated physical and chemical changes. In simple

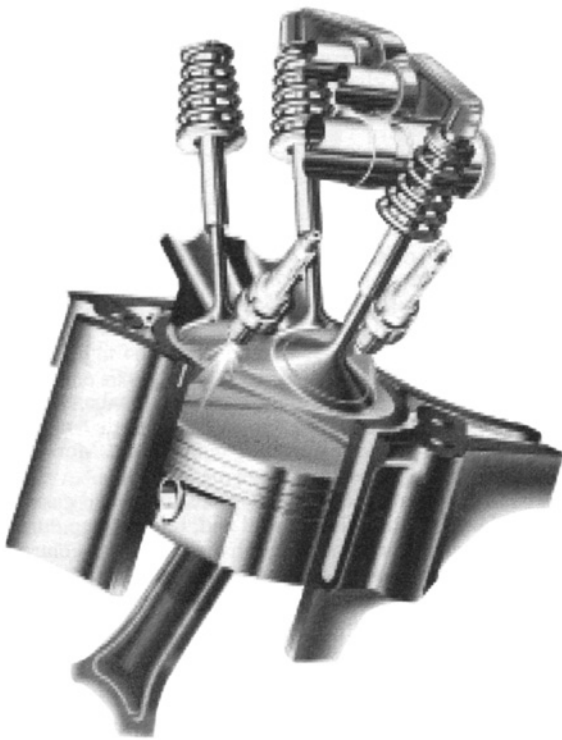


Figure 3.13 Three-valve combustion chamber with twin Sparking plugs (Mercedes-Benz)

terms, however, it would appear that each minute droplet of fuel injected immediately absorbs heat from the high-temperature cylinder air and begins to vaporize on reaching its boiling point. Then as long as the cylinder air temperature appreciably exceeds the self-ignition temperature of the injected fuel, the vapour thus formed will spontaneously ignite and each droplet will burn uniformly from the surface. The combustion of the air and fuel does, of course, raise further the temperature and hence pressure in the cylinder, as a result of which the piston is driven away from the combustion chamber on its power stroke.

Phases of combustion in the diesel engine

An overall impression of the combustion process that takes place in the diesel engine may be obtained by considering it as occurring in three phases, a concept that was originally proposed by H.R. Ricardo in this country over 60 years ago. The three phases may be summarized as follows:

First phase This is known as the delay period, and represents the time between the start of fuel injection and the onset of combustion. The delay period is also sometimes referred to as ignition lag.

Second phase A rapid combustion period, as it is termed, constitutes this phase, corresponding to when combustion is initiated and spreads at a rapid rate through the air and fuel that have come into contact during the delay period.

Third phase This final phase is designated the controlled combustion period, and is characterized by the injected fuel

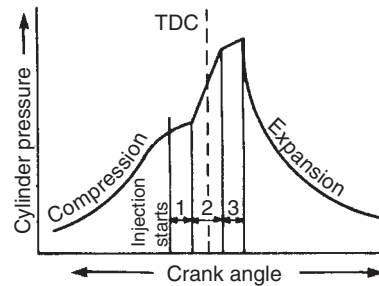


Figure 3.14 The three phases of combustion in a diesel engine

burning almost as soon as it enters the high-temperature combustion space.

In practice, too rapid a combustion period (second phase), with its accompanying high rate of pressure rise, can result in objectionable diesel knock, the sound of which is not unlike detonation in a petrol engine but of much greater intensity. To limit the pressure rise during the second phase of combustion, the delay period preceding it is kept as short as possible, which among other factors depends upon good ignition quality of the fuel. Efficient completion of the third phase of combustion is then achieved by providing for rapid mixing of the injected fuel and the cylinder air.

The three phases of diesel engine combustion can be readily visualized by defining their transition points on a graph as in Figure 3.14, where cylinder pressure has been plotted against crank angle.

The ignition quality of a diesel fuel relates to its ability to minimize the ignition lag during the delay period. It is expressed as a Cetane Number (CN), the higher the number the greater is the tendency for the fuel to ignite and reduce ignition lag. Somewhat similar in principle to determining the octane rating of a petrol, the cetane number of a diesel fuel is established by matching its readiness to ignite against a reference fuel in a test engine. In this case the reference fuel comprises a mixture of fast-burning n-hexadecane and slow-burning methyl naphthalene.

3.5 DIESEL ENGINE COMBUSTION CHAMBERS

Direct-injection combustion chambers

With direct-injection diesel engines the fuel is sprayed through a multihole injector nozzle into an open type of combustion chamber. This is formed partly by a bowl in the piston head and partly by the flat underside of the cylinder head. The basic idea of the bowl-shaped cavity in the piston head is to promote agitation of the cylinder air charge towards the end of the compression stroke, so that combustion is initiated by a process of induction swirl followed by compression turbulence (Section 3.6). Considered in cross-section, the internal shape of the bowl is generally such that it gives a ready impression of conforming to the conical spray pattern of the injector nozzle, which is mounted above in the cylinder head. The raised rim of the piston head prevents the fuel spray from reaching the cooler cylinder walls.

Historically, an important advance in the design of the open type of combustion chamber was made in 1934 when the Swiss Saurer Company introduced their toroidal piston chamber to promote a spiral movement of the air. Various detail refinements have been made to the shape of the bowl since then and these mainly account for the differences in chamber design from one manufacturer to another (Figure 3.15).

Although the combustion process in the diesel engine poses a less severe problem with respect to harmful exhaust emissions than does a petrol engine, since there is a reduction in carbon monoxide by virtue of the excess air involved in the process, increasingly stringent international legislation has demanded that continuing attention be given to reducing exhaust pollutant limits for diesel engines. One area of conflict lies between the requirement for raising combustion temperatures to reduce black smoke or carbon particulates and lowering them to reduce nitrogen oxides. However, the latter can be reduced by retarding the fuel injection timing and, to compensate for the otherwise loss of engine performance, 'square' shaped piston bowls have been developed for direct-injection combustion chambers. These can shorten the ignition delay period by enhancing swirl and squish effects. An example of this type of combustion chamber is that introduced by Perkins and known as the Quadram, where the toroidal recess in the piston crown assumes a four-lobed form. The pockets so formed at each corner of the square bowl then serve to agitate further the swirling vortex of air that becomes trapped in its central cavity and into which the fuel is sprayed. Another advantage associated with the square combustion chamber is that it confers a smoother rise in cylinder pressure for quieter combustion and lower mechanical stresses. A contrasting approach to accommodating retarded injection timing can take the form of a much wider and shallower piston bowl, which is used in conjunction with less air swirl but a greater number of finer fuel sprays injected at higher pressure. Hence there is an increased tendency for the injected fuel to seek out the air rather than vice versa, which again shortens the ignition delay period while avoiding the excessive temperatures associated with too rapid combustion.

These two differing approaches to making the combustion process of direct-injection diesel engines more environmentally friendly, have resulted in what may now be classified as 'high swirl toroidal' and 'low swirl quiescent' forms of combustion chamber (Figures 3.16a and b). In general terms, the large

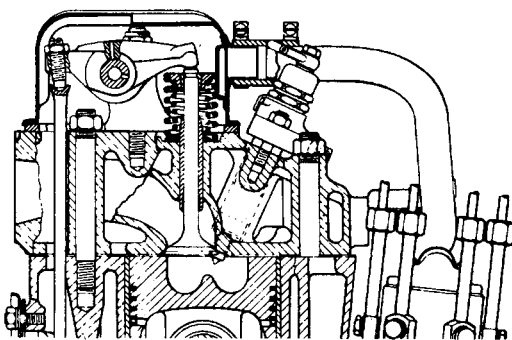


Figure 3.15 A practical application of the direct-injection open chamber

capacity diesel engines used in heavy vehicles require less air swirl than the smaller capacity engines for passenger car application. This difference can be accounted for by the combustion chambers of the latter engines being more restricted in size, together with an increase in engine operating speeds.

Advantages and disadvantages of direct injection

The direct-injection combustion chamber offers the advantages of easy starting and high thermal efficiency with correspondingly low fuel consumption. These advantages derive from a compact combustion space that reduces heat loss and also a method of charge agitation that incurs minimal power loss.

A disadvantage is that the combustion process is not very rapid, owing to the rather long distances the fuel sprays have to penetrate the cylinder air charge. It is also for this reason that what were once thought to be very high injection pressures in the region of 80 MN/m^2 ($12\,000 \text{ lbf/in}^2$), but which are now often more than double this value, are required to provide a sufficiently 'hard spray', as the fuel injection engineer terms it. These higher injection pressures relate to the increasingly energetic fuel sprays demanded by modern low swirl quiescent combustion chambers, wherein a greater number of finer fuel sprays must be injected with considerable rapidity, typically from an eight-hole injector nozzle with hole sizes of only 0.22 mm (0.009 in). Furthermore, the centrally disposed injection nozzle tends to impose physical restrictions on the sizes of the inlet and exhaust valves that can be accommodated, which is to the detriment of volumetric efficiency.

However, the advantages of the direct-injection diesel engine are such that it continues to be used over an ever widening range of commercial vehicles, both light and heavy, and is acknowledged as pointing the way ahead for future development of the diesel engine in road transport. Although its disadvantages initially made it less suitable for diesel engined passenger cars, it is nevertheless becoming established practice for their modern counterparts to use direct injection systems of the common rail type (Section 8.11)

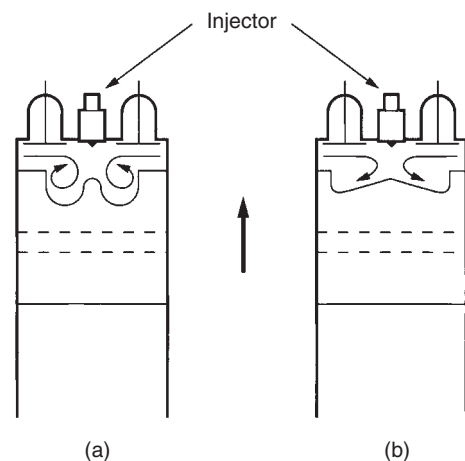


Figure 3.16 Basic layout of direct-injection diesel combustion chambers (a) high swirl toroidal (b) low swirl quiescent

together with turbocharging, thereby taking advantage of the better fuel economy that can be obtained as compared with indirect injection.

Indirect-injection combustion chambers

For indirect-injection diesel engines the fuel is sprayed from an essentially single-hole injector nozzle into a divided type of combustion chamber, which is formed partly in the cylinder head and partly by a saucer-shaped (as opposed to a bowl-shaped) depression in the piston head. Divided combustion chambers actually fall into two quite distinct categories: precombustion, and swirl.

Precombustion chamber

This represents the earliest form of divided combustion chamber, and its originator was Dr Prosper L'Orange of the German Benz Company who patented the idea in 1909. Typically pear-shaped in appearance, the precombustion antechamber is approached from the cylinder through a somewhat restricted passage, or burner, as it is appropriately termed (Figure 3.17a). A narrow cone of fuel is sprayed down towards the burner by the injector nozzle. It should be noted that the whole of the fuel is sprayed into the antechamber. The manner in which the combustion process is initiated in this particular form of divided chamber is based on what is known as *combustion swirl*, which is described later.

Swirl combustion chamber

This form of divided chamber was pioneered by H.R. Ricardo (mentioned earlier), who during the 1930s developed several versions of what have long been known as Comet swirl combustion chambers. The swirl chamber differs from the precombustion type in two clearly identifiable respects. First, the antechamber is of basically spherical form with a relatively unobstructed and tangential passage or throat leading from the cylinder (Figure 3.17b). Second, although the injector

nozzle sprays its narrow cone of fuel wholly into the antechamber, it does so in a direction away from the approach passage. Another interesting feature is that the lower portion of the antechamber containing the throat is separately inserted into the cylinder head and retained with minimal metal-to-metal contact to limit heat transfer to the cooling system (Figure 2.9). It thus forms a hot-spot over which air flows during the compression stroke, thereby raising the temperature of the air to a level higher than that to be expected from compression alone and beneficially shortening the combustion delay period. A process known as *compression swirl* and described later is responsible for initiating combustion in this type of divided chamber.

Advantages and disadvantages of indirect injection

It was early found by P. L'Orange (introduced in the previous section) that the indirect-injection diesel engine ran more smoothly, and less noisily than its direct-injection counterpart. Indeed, it is these self-same operating characteristics that account for the indirect-injection system in either precombustion (Figure 3.18a) or, more commonly, swirl chamber form (Figure 3.18b) also finding early acceptance in diesel engined motor cars. These characteristics may be explained in simple terms by the combustion process being initially confined within an antechamber, so that the rate of pressure rise in the cylinder is reduced. From this it also follows that the engine structure is not so highly stressed, so it can therefore be made less heavy than is possible with direct-injection.

In order to produce sufficient heat for self-ignition of the fuel, an indirect-injection engine requires a higher compression ratio, usually in the region of 20:1, mainly to compensate for the greater opportunities of heat loss from the cylinder charge. Even so the indirect-injection diesel engine can be more difficult to start from cold. For this reason the antechamber of such systems is generally provided with an electrically heated glow-plug, so that when switched on its incandescent electrode instantly ignites any fuel impinging upon it (Figure 8.48).

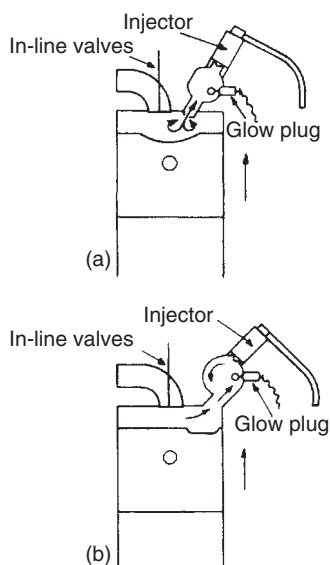


Figure 3.17 Basic layout of indirect-injection diesel combustion chamber: (a) precombustion (b) swirl

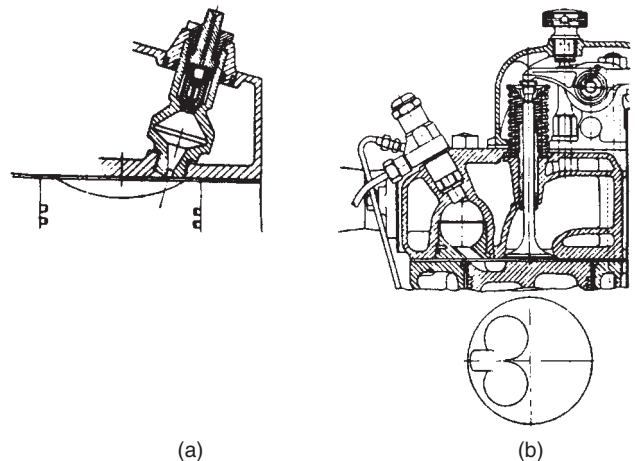


Figure 3.18 Practical applications of indirect-injection divided chambers: (a) precombustion chamber of the Caterpillar Tractor Co. (b) Ricardo Comet Mk III swirl-type combustion chamber

The use of such high compression ratios in the indirect-injection diesel engine naturally demands a heavy-duty battery and starter motor installation.

3.6 CYLINDER CHARGE AGITATION

Basic requirements

The need for a controlled degree of agitation for the cylinder charge of either air and fuel in the petrol engine, or air only in the diesel engine, was referred to in connection with the geometric shape of combustion chambers (Section 3.1). It now remains to examine how compression turbulence and induction swirl may be generated in both petrol and diesel engines, and also compression swirl and combustion swirl in diesel engines only (although compression swirl does occur in the more recent split-level type of combustion chamber for petrol engines).

Compression turbulence

In petrol engines this is promoted by arranging for the piston crown to be brought into very close proximity with a limited portion of the combustion chamber roof. By this means a squish area is formed, which again is a term that was first applied by H.R. Ricardo. The effect of this squish area is such that, as the piston approaches the end of its compression stroke, the air and fuel charge trapped in the diminishing space is forcibly ejected across the combustion chamber. In so doing it enters the enlarged and, in some cases, streamlined portion of the chamber, which imparts a turbulent motion to the charge (Figure 3.19a).

Compression turbulence in direct-injection diesel engines is similar in principle, except that the squish area is actually formed by the annular rim of the piston bowl, so that air is

ejected radially inwards to the centre of the piston as it approaches the end of the compression stroke (Figure 3.19b). By virtue of the toroidal form of the piston bowl, the air that is forced to enter therein is set into turbulent motion and thereby assists the injected spray of fuel to seek out its oxygen supply.

Induction swirl

For both petrol and direct-injection diesel engines, induction swirl may be generated simply by arranging for the incoming charge to enter the cylinder in a tangential direction. An angled bias may therefore be adopted for the intake port in the cylinder head, so that the entering charge is deflected by the cylinder walls (Figure 3.20a). As a consequence of the angular motion imparted to the charge, a beneficial vortex (swirling) action is set up on the induction stroke and continues during the compression stroke. A characteristic known as the 'swirl ratio' is equal to the swirl speed divided by the engine speed with both being expressed in rev/min. Another method of promoting induction swirl for direct-injection diesel engines is to mask one side of the inlet valve, so that the incoming air is provided with a similar directional control (Figure 3.20b). An additional mechanical complication is introduced with this particular method, since the valve stem must be keyed against rotation. In the now established four-valve pentroof form of combustion chamber for petrol engines (Figure 3.22a and b), research into the burning of leaner mixtures by the Gaydon Technology company in the mid-1980's showed that on the induction stroke a beneficial swirling action can be created across the cylinder, this being termed barrel swirl or 'tumble' to distinguish it from the axial swirl described earlier (Figures 3.21a and b). A tumble motion is an important factor in the induction and compression events of the Mitsubishi Gasoline Direct Injection GDI engine (Section 6.9).

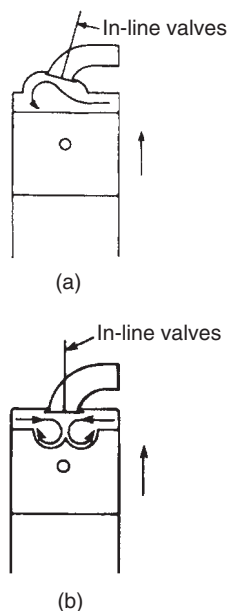


Figure 3.19 Compression turbulence in (a) petrol only (b) mainly diesel engines

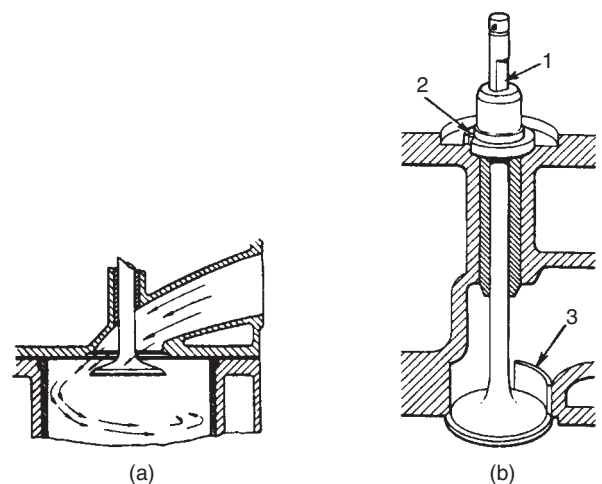


Figure 3.20 Induction swirl in petrol and diesel engines: (a) directed inlet port for petrol and diesel (b) shrouded inlet valve for diesel only.

- 1 flatted stem to prevent valve rotation against thimble key
- 2 thimble locating peg
- 3 shroud

The concept of induction swirl actually dates back to before World War I, when Dr K. Hesselmann of the Swedish Atlas Company first developed it for submarine diesel engines.

Compression swirl

This effect is present with indirect-injection diesel engines where a swirl combustion chamber is used. A characteristic

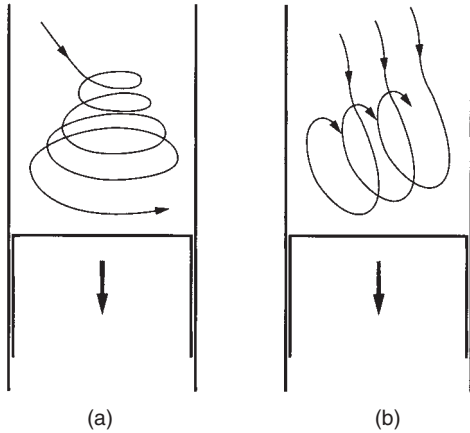


Figure 3.21 Identifying induction swirl patterns (a) axial (b) barrel

of this form of chamber is that during the compression stroke about three-quarters of the cylinder air is forced to enter it, via a communicating passage or throat (Figure 3.17b). The latter is relatively unobstructed in its tangential approach to the swirl chamber, which is of spherical form to promote rapid swirling of the air past the fuel sprayed from the injector nozzle. Hence, the oxygen in the air can seek out its fuel supply, which is of course the converse situation to that found with compression turbulence in direct-injection diesel engines.

Combustion swirl

Again applicable to indirect-injection diesel engines, this effect is present with the precombustion chamber type, which receives only about one-third of the cylinder air during the compression stroke (Figure 3.17a). As mentioned earlier, the precombustion chamber is approached through a relatively restricted passage, down towards which the fuel sprayed from the injector nozzle is directed. Ignition of the fuel in the precombustion chamber causes a sudden increase in pressure therein, which results in partially burned fuel and gas being forcibly ejected into the cylinder clearance space and setting up a rapid swirling motion for completion of the combustion process.

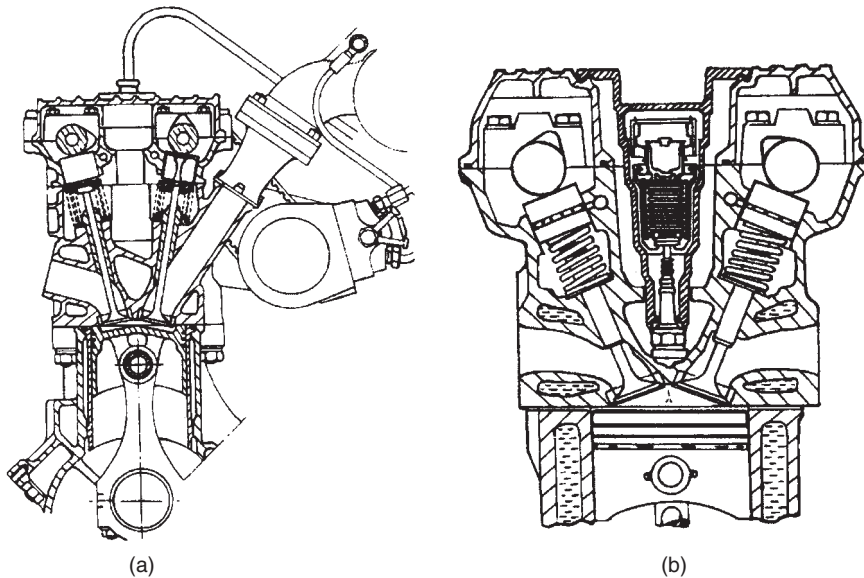


Figure 3.22 Arrangements of four-valve porting to induce (a) barrel swirl (Renault) and (b) axial swirl (Saab)

4 Engine lubrication

4.1 FRICTION AND WEAR

Basic considerations

The main factors that can reduce the performance and life expectancy of the engine and other parts of the motor vehicle are the closely related phenomena of friction and wear.

Friction can be described as the resistance that a material develops against sliding contact. When an apparently smooth surface is viewed through a powerful microscope, innumerable tiny projections and depressions are seen; these are termed surface *asperities*. They may be likened to a mountain chain, so that when two surfaces are rubbed together their action is analogous to that of two mountain chains sliding across one another peak-to-peak (Figure 4.1a).

Without lubrication this friction between the rubbing surfaces results in the removal or transfer of solid material from one to the other and constitutes what is known as wear (Figure 4.1b).

Forms of wear

Several different forms of wear can arise. They may be summarized as follows:

Abrasion occurs whenever hard foreign particles are present between the rubbing surfaces. This type of wear was once a contributory factor in the excessive wearing of engine cylinder walls and piston rings, until it was better controlled by the efficient use of air and oil filters.

Adhesion arises whenever the rubbing surfaces are subjected to extreme contact pressure and the frictional heat generated between them causes momentary welding together of surface

asperities. This type of wear occurs when the tiny welds between opposing asperities are broken by continued sliding of the surfaces.

Corrosion refers to the deterioration of the rubbing surfaces as a result of chemical attack. This type of wear was once the major factor in the excessive wearing of cylinder walls and piston rings, until the advent of chromium plating for the latter.

Fretting is confined to rubbing surfaces where the motion between them is almost imperceptible and typically arises from vibration. It causes deterioration of the rubbing surfaces by the formation of corrosive debris.

Fatigue takes the form of cracking that is initiated just below the contacting surfaces. It may be associated with both rubbing and rolling surfaces that are subjected to repeated deformations. The service life of a rolling bearing can be ultimately limited by this form of wear.

4.2 THE LUBRICATION PROCESS

Reducing friction and wear

The object of engine lubrication is to reduce friction, heating and wear of the working parts or, more precisely, wherever interacting surfaces are in relative motion. It is, of course, accomplished by introducing a film of lubricating oil between the various bearing surfaces to keep them apart, thereby providing a condition of fluid friction rather than dry friction. For effective lubrication the shear strength of this film must be lower than that of the rubbing surfaces it separates. In other words, metal-to-metal contact is avoided by the film of lubricant shearing within itself, as its layers of molecules slide over one another in the manner of a pack of playing cards being pushed endwise (Figure 4.2).

Forms of lubrication

Lubrication is a relatively well-ordered process which is recognized as existing in several forms. These are summarized in the following sections.

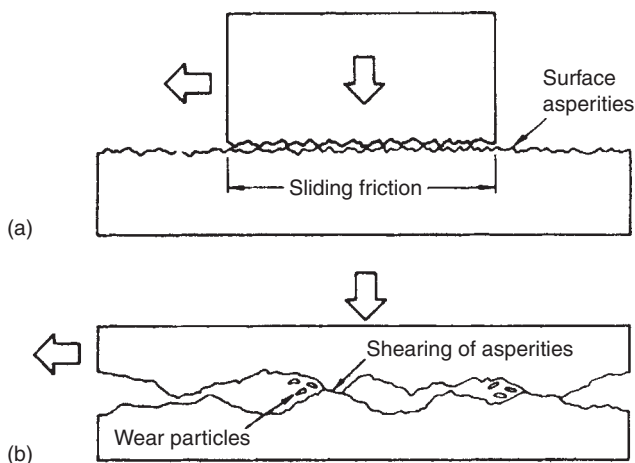


Figure 4.1 Sliding friction causing wear

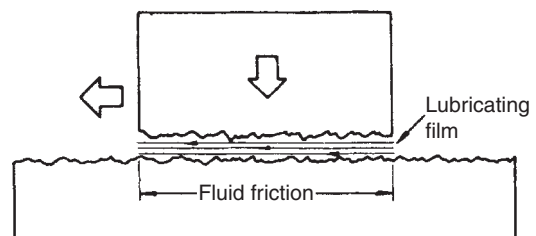


Figure 4.2 Lubrication preventing wear

Hydrodynamic lubrication

This occurs when the clearance condition between the load-carrying surfaces of a bearing favour the formation of a wedge-like film of oil (Figure 4.3). The moving surface of the bearing then acts as its own oil pump to crowd more oil into the converging clearance space occupied by the oil wedge, finally to establish a relatively thick film of oil throughout the entire bearing clearance. This process can occur not only in journal bearings with cylindrical surfaces, but also in slider bearings with plane surfaces. It should be realized that an appreciable oil film pressure can be generated in the engine bearings by hydrodynamic lubrication, which may be more than fifty times the oil supply pressure, or what is commonly termed the engine oil pressure. To maintain a condition of hydrodynamic lubrication it is, of course, essential that a sufficient quantity of oil is always supplied to the bearings by the engine lubrication system.

Elastohydrodynamic lubrication

This represents a modified form of hydrodynamic lubrication. It occurs under conditions of extremely high pressure between the moving surfaces, such as can arise between cams and followers, gear teeth and in rolling bearings where little more than a line or a point contact may exist to support the working load. The viscosity or resistance to flow of the oil trapped between such small interacting surfaces may then increase to such an extent that the oil film momentarily becomes more rigid than the metal surfaces themselves, which then elastically deform to allow the formation of an extended oil film and thus provide lubrication (Figure 4.4).

Boundary lubrication

This exists in a bearing when either the speed of the moving surface is too low for hydrodynamic lubrication, the load acting

is too high, or there is an insufficient supply of oil. It therefore relies essentially upon the adhesion of oil molecules to the load-carrying surfaces, so that they are separated by a relatively thin film of lubricant (Figure 4.5). There is, however, always the danger that this thin film may be penetrated by surface asperities, the resulting metal-to-metal contact then leading to wear unless the oil has been fortified by suitable additives.

Hydrostatic lubrication

This can occur in the form of a squeeze film of oil in a bearing if the load momentarily reverses in direction and the speed of the moving surface is very low. Under these conditions the load-carrying surfaces are initially separated by a relatively thick film of oil, which resists being squeezed out from the diminishing bearing clearance during the short interval that the load reverses in direction (Figure 4.6). A cushioning film of oil can therefore be maintained and then restored to full thickness before the next load reversal occurs.

Mixed-film lubrication

A familiar example of a change from one state of lubrication to another occurs in the engine main and big-end plain journal bearings, which are subjected to a complex system of loading. These bearings experience partly hydrodynamic and partly hydrostatic lubrication during normal running, but develop boundary lubrication when starting and stopping (Figure 4.7).

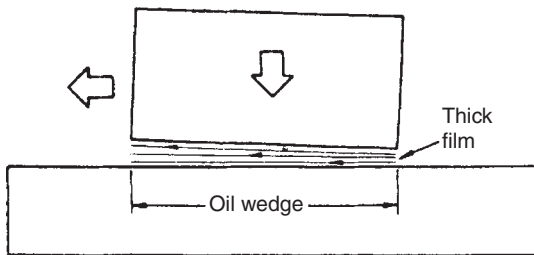


Figure 4.3 Hydrodynamic lubrication

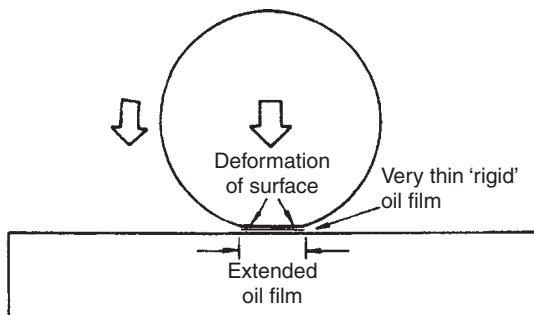


Figure 4.4 Elastohydrodynamic lubrication

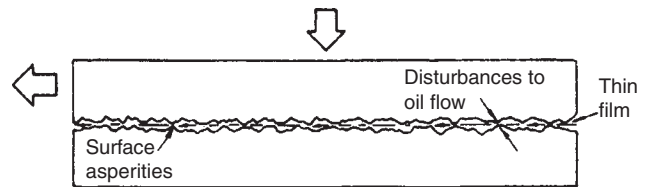


Figure 4.5 Boundary lubrication

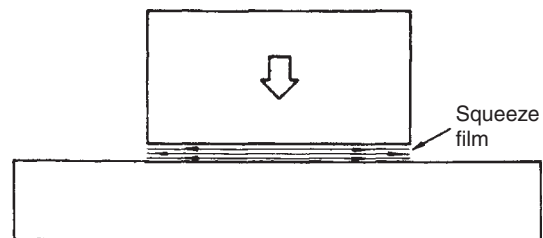


Figure 4.6 Hydrostatic lubrication

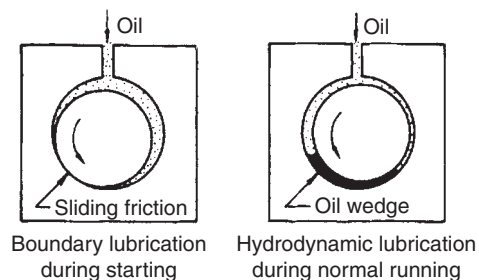


Figure 4.7 Mixed-film lubrication in journal bearing

During the latter stages of operation the shaft journal attempts to climb the side of the bearing wall, either prior to the generation of an oil wedge or after its collapse, so it is under these conditions of starting and stopping that frictional wear of engine bearings is most likely to occur. The transition from one form of lubrication to another is generally referred to as a state of mixed-film lubrication.

Oil film thickness in engine bearings

From our simple explanation of hydrodynamic lubrication, it might appear that so long as a film of oil exists between the bearing surfaces, it does not greatly matter how thick it is. In reality, however, the minimum oil film thickness must be such as to accommodate less than ideal operating conditions, which may exist in a bearing. These can include minute deflections under load, slight distortions arising from thermal effects and insufficient smoothness of the bearing journal. In the latter case, it must be recognized that the highest load capacity of the bearing is developed when the thickness of the oil film is least.

Although the prediction of oil film thickness in a hypothetical bearing, based on the theory of hydrodynamic lubrication proposed by Osborne Reynolds in 1886 (Beauchamp Tower had discovered the practice a few years before), has long been a mathematician's delight, the automotive service engineer is more likely to be curious about the variations in oil film thickness that occur in a real-life engine bearing. By using advanced technology instrumentation, the Castrol oil company can in fact measure the variations in oil film thickness in a big-end bearing with the engine running (Figure 4.8). Castrol refer to this as their BOFT (bearing oil film thickness) test, which forms part of their extensive research programme

to demonstrate the ability of an engine oil to provide effective protection and smooth running power output.

4.3 ENGINE LUBRICATING OILS

Mineral oils

Lubricants can be produced from animal, vegetable and mineral oils; it is the last of these, refined from petroleum and treated with additives, that are suitable for the lubrication of motor vehicle engines. Petroleum is generally thought to have been formed by the decomposition of tiny marine organisms that died, sank to the bottom of the sea and became mixed with sediment. Over many millions of years the accumulating layers of sediment subjected the trapped organic matter to immense pressure and heat, which converted its softer parts into droplets of crude oil mixed with natural gas.

Crude oils are essentially hydrocarbons – compounds whose molecules comprise hydrogen and carbon atoms. The basic types of hydrocarbon found in crude oils are classified as paraffins, naphthenes and aromatics. Oils of predominantly paraffinic type are preferred for the production of automobile engine oils, since they are best suited to the extremes of operating temperature involved. Conventional engine oils therefore consist largely of mixtures of hydrocarbons, but also require treatment with certain carefully chosen chemicals that are known as 'additives'.

Synthetic oils

Although the so-called synthetic oils that do not have a lubricant base of mineral oil form have been developed since the 1920s – indeed, a significant proportion of the lubricants

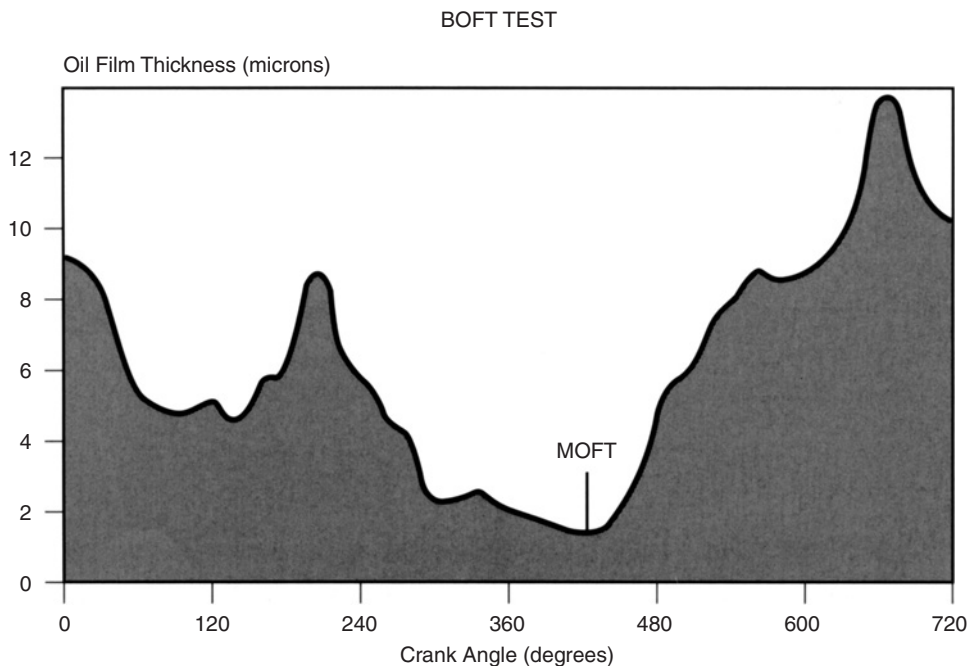


Figure 4.8 Oil film thickness in a big-end bearing of a running engine (Castrol). MOFT – Minimum oil film thickness

used by the German armed forces during World War II were of synthetic origin – it has only been in recent years that oils of this type have become available, chiefly to meet the more severe demands of high-performance vehicle engines, albeit at a higher cost than conventional mineral oils. They are generally classified as either ‘synthetic-based’ or ‘full-synthetic’ oils, the former containing at least 10 per cent synthetic base stock. A chemical compound that has been found suitable for this purpose after processing is polyalpha-olefin (PAO), which may be classed as a synthetic hydrocarbon base stock. It is produced by polymerization (the combination of smaller molecules to form large ones) of ethane gas, to formulate an engine oil with particularly favourable viscosity-temperature characteristics. Such a product is sometimes described by the oil technologist as synthetically engineered polyalpha-olefin.

It is now generally recognized that some synthetic oils will out-perform mineral oils in low-temperature applications and that most also possess better volatility characteristics than mineral oils with equivalent viscosities. In the latter respect this means that thinner oils may be used without necessarily incurring a higher consumption. Other perceived benefits generally depend upon additive composition, as is the case with mineral oils.

Functions and properties of engine oil

The importance of using the correct type and grade of engine oil will be better appreciated if first we consider the essential functions it must perform. These may be summarized as follows:

Lubrication The *viscosity* of the oil is its single most important property, since it represents the internal friction in the film of oil and is caused by the resistance to shearing motion of the oil molecules. More simply, it may be regarded as the resistance to flow of an oil, and it is this property that makes it possible to separate the bearing surfaces in relative motion. To cater for the more difficult conditions of lubrication, such as may arise between the highly stressed cams and tappets, extreme pressure (EP) additives, can be incorporated in the oil. These improve its oiliness by forming a chemical film between the rubbing surfaces.

Entrainment Under less than perfect conditions of lubrication, the flow of oil through the bearings of an engine serves to capture and remove abrasive particles, which otherwise would remain and increase friction and wear.

Sealing To assist in sealing against the leakage or blow-by of the cylinder gases, which is known to cause camshaft wear in some modern overhead camshaft engines, the engine oil must be physically capable of filling the minute leakage paths and surface irregularities of the cylinder bores, pistons and rings. It must therefore possess adequate viscosity stability because of the high-temperature conditions to which it is exposed.

Cooling The conditions under which the oil functions in a cooling capacity generally tend to promote oxidation. This is because the oil is exposed not only to heating – the temperature in the top compression ring groove of the piston may reach 250°C, while even that in the crankcase is typically around 125°C – and agitation, but also to the effects of contact with

metals and contaminants. Therefore, to avoid the formation of gummy deposits and piston ring sticking, engine oils are treated with anti-oxidant additives.

Cleaning The lubricating oil must also contribute to the internal cleanliness of the engine, and it is therefore treated with so-called detergent additives. These can be more accurately described as dispersants because they retain insoluble contaminants in fine suspension, which are removed when the engine oil is changed. Similarly, the corrosive action of acidic products that can contaminate the oil may be neutralized by treating it with anti-corrosion additives. A phenomenon of more recent origin is that known as ‘black sludge’, which manifests itself as a thick tar-like deposit that forms rippling layers in the cooler areas of the engine, especially inside the valve cover, but can flake off to cause obstructions in the lubrication system and at worst can lead to serious engine damage. It first began to appear during the late 1980s in engines designed to minimize harmful exhaust emissions, and has been attributed to a combination of factors that include engine design, operating conditions, fuel quality, engine oil quality and oil change intervals.

Compatibility The exhaust system catalytic converter must be protected against phosphorus induced damage, so to keep the catalyst working cleaner for longer the phosphorus content of the oil must be minimized.

Environmental To reduce the risk of harmful exhaust emissions and also to make for easier and safer disposal and recycling, the amount of chlorine in the oil formulation must also be minimized.

Engine oil additives

Additives are usually chemical compounds of non-petroleum origin, and normally only very small amounts are needed to achieve the desired effect. Their purpose is to confer a service life of acceptable economic length, either by improving a property already possessed by an oil, or by introducing a new characteristic. The different types of engine oil additive and their reason for use, as concisely listed by the Caltex Petroleum Corporation, are as follows:

Viscosity index improvers Reduce viscosity change with temperature; permit reduced fuel consumption; maintain low oil consumption; allow easy cold starting.

Detergents, dispersants Keep sludge, carbon and other deposit-forming material suspended in the oil for removal from the engine with drains.

Alkaline compounds Neutralize acids; prevent corrosion from acid attack.

Antiwear, friction modifiers Form protective films on engine parts; reduce wear, prevent galling and seizing; reduce fuel consumption.

Oxidation inhibitors Prevent or control oxidation of oil, formation of varnish, sludge and corrosive organic compounds; limit viscosity increase that occurs as oil mileage increases.

Rust inhibitors Prevent rust on metal surfaces by forming protective surface films or by neutralizing acids.

Pour point depressants Lower ‘freezing’ point of oil, assuring free flow at low temperatures.

Antifoam agents Reduced foam in the crankcase.

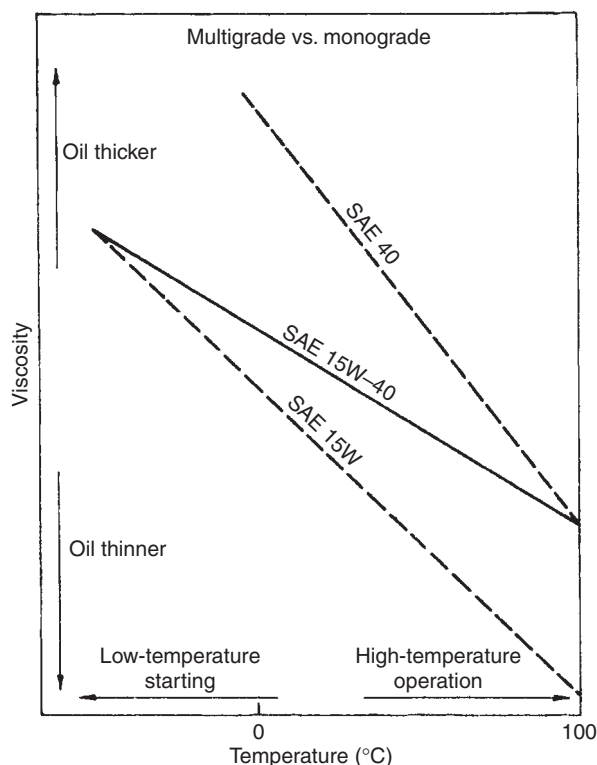


Figure 4.9 Engine oil viscosity change with temperature

Classification of engine oils by viscosity

In the mid 1920s the Society of Automotive Engineers (SAE) in America recommended an arbitrary classification system for engine oils in terms of viscosity only. This system, which has been revised from time to time, has long since been internationally recognized. It established a numerical relationship, defined by SAE numbers, between the viscosities of oils, and specifies minimum and maximum viscosities at stated temperature for an oil to meet a particular grade. In practice this means that the higher the grade number, the higher the viscosity of the oil. On this basis there are now eleven SAE viscosity grades for engine oils, these being SAE 20, 30, 40, 50 and 60 grades, which are intended for normal temperature conditions, while suffix W grades in the range SAE 0W, 5W, 10W, 15W, 20W and 25W are intended primarily for low-temperature conditions.

In common with other liquids, an increase in the temperature of engine oil will cause a decrease in its viscosity (Figure 4.9). As this effect could be quite pronounced with engine oils, which are required to function over a very wide temperature range, they are treated with the earlier mentioned chemical additives known as viscosity index (VI) improvers. The term *viscosity index* expresses the degree of variation of viscosity with temperature, so the higher its value the less will be the change in viscosity of the oil with temperature.

The development of special viscosity index improvers resulted in the introduction of multigrade oils in the early 1950s. These oils are formulated to meet the viscosity requirements of more than one SAE grade and are therefore classified in double SAE numbers, such as 15W-40. The advantages

originally claimed for multigrade oils included easy starting of the engine from cold with minimum oil consumption when hot.

With regard to cold starting, there is a minimum cranking speed below which an engine will not start; this speed is higher for a diesel engine than it is for a petrol engine. A manufacturer therefore determines what is termed the viscosity requirement of an engine; this is the highest value of the sump oil viscosity above which the cranking speed does not exceed the minimum starting speed. Such tests are, of course, related to the cold starting temperatures likely to be encountered in those parts of the world where the car is sold.

Classification of engine oils by performance

In 1947 the American Petroleum Institute (API) introduced a service classification system based on the types and amount of additives used in an engine oil. This was followed in 1952 by a second classification system based on the severity of engine operation. Then in 1970 the API in conjunction with the Society of Automotive Engineers (SAE) and the American Society for Testing and Materials (ASTM) established what are known as 'API engine service categories' to define the overall capability of an engine oil to protect against wear, deposits and self-deterioration. The ability of an engine oil to perform satisfactorily in one or more categories identified by the API is determined by engine tests developed by the ASTM, and the resulting grading system is called 'SAE J183 engine oil performance and engine service classification'. This system can be updated as the need for new categories arises. For petrol engines the existing API service categories and oil descriptions are as follows:

- | | |
|----|---|
| SA | non-additive |
| SB | anti-oxidant, anti-scuff but non-detergent |
| SC | protection against high- and low-temperature deposits, wear, rust and corrosion (1964-67) |
| SD | generally higher levels of protection than SC (1968-70) |
| SE | high-temperature protection more demanding than SD (1971-72) |
| SF | improved anti-wear and high-temperature protection again increased (1980-88) |
| SG | provides improved control of engine deposits such as black sludge, oil oxidation and engine wear relative to oil developed for previous categories; these oils also give enhanced protection against rust and corrosion (adopted 1989) |
| SH | technically similar to above category, but the validity testing of an oil formulation must now be conducted in accordance with the Chemical Manufacturers Association (CMA) code of practice, which excludes self-certifying a product (adopted 1994) |
| SJ | describes an engine oil that is subject to more tests than the previous category, and suitable for passenger cars and light commercial vehicles operating under vehicle manufacturers' recommended maintenance procedures (adopted 1996) |
| SL | similarly describes an engine oil for use in service typical of present and earlier passenger cars, sports utility vehicles and light commercial vehicles. It provides a further advance in tested performance |

levels, including improvements relating to oil consumption, engine cleanliness and oil change intervals (adopted 2001)

And for diesel engines:

- CA light duty
 - CB moderate duty
 - CC moderate to severe duty
 - CD severe duty
 - CE extra heavy duty for engines manufactured since 1983
 - CF introduced in 1994 and intended for service typical of indirect-injection engines.
 - CF-4 provides improved control of oil consumption and piston cleanliness (adopted 1990)
 - CG-4 describes an engine oil suitable for severe-duty diesel engine service and designed to meet US 1994 exhaust emission standards
 - CH-4 superior in performance to service categories API CF-4 and API CG-4, and suitable for engines designed to meet US 1998 exhaust emission standards
 - CI-4 provides a further advance in severe-duty performance levels, especially in relation to engine durability where exhaust gas recirculation is used to meet US 2004 emission limits (adopted 2002)
- (Suffix '4' denotes four-stroke diesel engine categories)

The distinction between rusting and corrosion is that *rusting* occurs from the presence of moisture on those ferrous parts not in continuous contact with the lubricating oil, while *corrosion* arises from the oil becoming contaminated with the acidic products of combustion and attacking various structural and bearing metals.

Since the early 1960s it has been recognized that European requirements for engine oil performance can in certain respects differ from those specified primarily for American operating conditions. Therefore the Committee of Common Market (Automobile) Constructors (CCMC) incorporated additional tests to define the minimum quality for lubricants acceptable to its members. In 1996 the CCMC was superseded by the Association of European Automobile Manufacturers, abbreviated from the French to ACEA. As a result a new classification system for oil performance standards was introduced, which defines the minimum quality level for service fill oils. The system is similar to that of the American Petroleum Institute or API, as it is based on a schedule of physical, chemical and engine tests. A significant difference between the earlier CCMC and current ACEA test sequences is that in the latter case all performance claims must be supported by data generated under the European Engine Lubricant Quality Management System (EELQMS). The ACEA categories or sequences are prefixed by the letters A, B or E to denote petrol, light duty and heavy duty diesel engines respectively, followed by numbers that currently range from 1 to 7 to rate ascending oil performance, and suffixed by the year of issue. Taking a few examples: ACEA A3-02, ACEA B4-02 and ACEA E5-02 category oils relate to recent petrol-engined passenger cars, diesel-engined passenger cars and diesel-engined commercial vehicles. To meet the more demanding 'Euro-4' exhaust emission limits for heavy duty diesel engines (Section 11.3), ACEA categories E6

and E7 have now been introduced; the former to reduce ash accumulation and make the oil compatible with exhaust after-treatment systems, and the latter to minimize abrasive wear of the engine parts including the valve train.

In briefly summarizing the classification and requirements for engine oils, it has to be stressed that a service engineer must always consult the engine manufacturer's recommendations, to ensure that the correct grade and type of oil is used at oil change intervals.

Oil sump capacity and oil level

The amount of oil carried in the sump necessarily represents a compromise, since too great a capacity would hinder warm-up of the engine, while too small a capacity would hasten its contamination. In practice, sump oil capacities generally range from about 3.5 to 7.0 litres (6 to 12 pints) for passenger car engines and in the region of four times these quantities for heavy-vehicle diesel engines. A simple dipstick is all that has traditionally been provided to indicate the minimum and maximum oil levels on most vehicles, although in the past an oil level indicator gauge on the instrument panel was electrically connected to a float unit fitted in the sump on some high-grade passenger cars. In a modern electronically controlled version of an instrument panel low oil level warning system, actuation is by two thermal sensors housed in the sump. One sensor is heated electrically to a temperature higher than that of the oil, while the other sensor continues to register the actual temperature of the oil. When the former sensor is allowed to cool down to the temperature of the latter, a certain time must elapse that will be influenced by the extent of its exposure above a less than required oil level. The actual level of oil in the sump can then be electronically computed based on the signals received from the two sensors.

Too high or too low an oil level will have damaging effects, as indicated in the following sections.

Oil level too high

If the rotating crankshaft approaches too close to the surface of the oil, then the windage it creates can produce foaming of the oil, so that air enters the engine lubrication system and causes inconsistent oil pressure. Should the big ends of the connecting rods actually dip into the oil, then both its temperature and consumption will increase. The latter is because the amount of oil splashed on to the cylinder walls could exceed the controlling ability of the piston rings, so that some of it enters the combustion chambers and is burnt.

As an interesting aside, it was once reported that in oil consumption tests on a fleet of Paris taxis, the average oil consumption was greater for those vehicles where the operator added oil more often and nearer to the dipstick maximum mark, than for those where the operator added oil less often and below the maximum mark.

Oil level too low

This is potentially a more dangerous state of affairs, since if the oil level falls completely below the intake of the oil pump, then obviously no oil will reach the engine working parts, with disastrous results. Even if the oil level has not fallen seriously below the maximum mark, engine oil pressure can still become

erratic with any surging of the oil, caused by the changing motion of the vehicle. Furthermore, not only will a reduced amount of oil in circulation be less able to prevent engine deposits, but also the rate at which the oil deteriorates will be increased owing to its reaching higher temperatures.

Oil deterioration and changing

There are two main causes of engine oil deterioration in service: these are oxidation and contamination, as already mentioned. Suffice it here to summarize matters as follows:

Oxidation

Occurs in the hottest parts of the engine
Thickens the oil and makes cold starting difficult
Reduces oil flow until the engine has warmed up
Produces lacquer deposits that can obstruct fine clearances
Corrodes the internal surfaces of the engine.

Contaminants

Are formed of soot and water from combustion
Are composed of lead compounds from fuel additives
Include abrasive particles admitted through air intake
Tend to clog the oil filtering arrangements
Form lacquers independent of those from oxidation.

The reason for periodically changing the engine oil is therefore not because it simply wears out, but as a result of the oxidation and contamination accompanying the depletion of some (not all) of its additives. These are, of course, gradually destroyed in the process of doing their intended job, which in modern oils they do exceedingly well.

Either the depletion of additives, or the actual engine condition, forms the basis upon which the vehicle manufacturer will specify the oil and filter change interval in service, which may also depend upon the annual mileage and whether the vehicle is subjected, for example, to a fair amount of stop-start driving. Also, a longer than average recommended interval between oil changes may apply only if a high-quality synthetic oil is used. For passenger cars the current practice is typically to change the oil in naturally aspirated petrol engines at intervals ranging from 6000 to 12 000 miles (10 000 to 20 000 km) and for turbocharged versions from 3000 to 6000 miles (5000 to 10 000 km). With passenger car diesel engines the intervals are typically in the range from 5000 to 7500 miles (8000 to 12 000 km). In the case of turbocharged diesel engines for heavy vehicles the intervals typically range from 18 000 to 36 000 miles (30 000 to 60 000 km). It is, of course, always necessary to establish the recommended oil change intervals from the vehicle manufacturer's own service schedule.

Since frequent and prolonged contact with used engine oils may cause serious skin disorders, the Health & Safety Executive have issued precautionary guidance in their publication HS(G)67 titled 'Health & Safety in MOTOR VEHICLE REPAIR'.

Excessive oil consumption

Service complaints of excessive oil consumption may be considered under the following general headings.

Oil leakage

External oil leakage may arise at various locations around the engine unit. It is usually caused by a defective sealing gasket, a discontinuity in a vulcanized sealant joint, an ill-fitting blanking plug, a faulty connection in external pipework, or worn oil seals.

Oil burning

This can be attributed to excess oil reaching the combustion chambers. Its possible sources include worn inlet valve guides and stem seals, combined wear in the cylinders, pistons and rings, or increased oil fling-off from worn bearings.

Oil vaporization

This is concentrated in the high temperature regions of the piston rings and valve guides. It is therefore important to use a high quality oil that maintains its thermal stability at elevated temperatures.

The very fine production clearances built into modern engines, together with the closely controlled formulation of their oils, have greatly contributed to a reduction in complaints of this nature, as compared to earlier service experience.

4.4 ENGINE LUBRICATION SYSTEMS

Wet sump system

A splash type of lubrication system was used for the engines of early motor vehicles. Lubrication of the working parts was initiated simply by allowing the connecting rod big ends to dip into a trough of oil in the crankcase each time they passed through their BDC positions. To improve the lubrication of their bearings, the big ends were often provided with oil scoops (Figure 4.10). Surplus oil that was flung upwards into the crankcase was then collected in galleries from whence it gravitated to the main bearings, camshaft bearings and timing drive. The cylinder walls were similarly lubricated by the oil that was splashed on to them.

Following the purely splash system of engine lubrication came the combined pressure-and-splash or semi-forced system. The essential features of this system were that an engine-driven oil pump delivered oil only to the main bearings; the big-end bearings and other working parts were lubricated in the manner of the purely splash system. An attraction

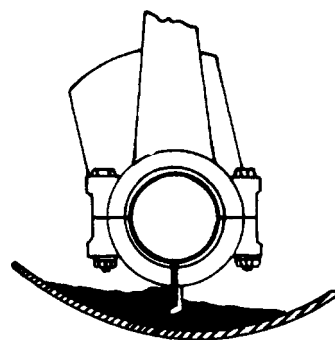


Figure 4.10 Early splash system of engine lubrication

of this system was that no oilway had to be drilled in the crankshaft, as necessary with the fully forced system described below.

The eventual necessity to provide the later high-speed engine with a fully forced lubrication system, arose from the ever-increasing loads and speeds at which the bearings were expected to operate. This meant that the bearings had to be not only lubricated, but also cooled, by the oil circulating through them. We therefore arrive at the long-established fully forced lubrication system, in which oil under pressure is supplied to the crankshaft main bearings, connecting rod big-end bearings and camshaft bearings. Strictly speaking, a fully-forced lubrication system originally included the small-end bearing bushes, which were supplied with oil under pressure from the big-end bearings through drilled connecting rods. The first fully forced lubrication system for motor vehicle engines is generally attributed, like so many features of automobile practice, to the early work of Dr F. Lanchester and was found in the

cars that once bore his name. Even so, the basic idea of forced-feed lubrication to all bearings was applied as long ago as 1890 in the Bellis and Morcom steam engine.

With the fully forced oil circulation system, a major contribution to cooling of the bearings is therefore made by passing a large quantity of oil through them, in excess of the amount needed for lubrication alone.

Oil circuit in the wet sump system

This consists of a network of oilways embodied in both the structure and the mechanism of the engine, the extent and complexity of this network being mainly influenced by the number and arrangement of the engine cylinders and their valve train (Figures 4.11, 4.13 and 4.14a). A more convenient representation of an engine lubrication system may be made with the aid of a block diagram (Figures 4.12 and 4.14b). Extensive use was once made of separate pipes and connecting

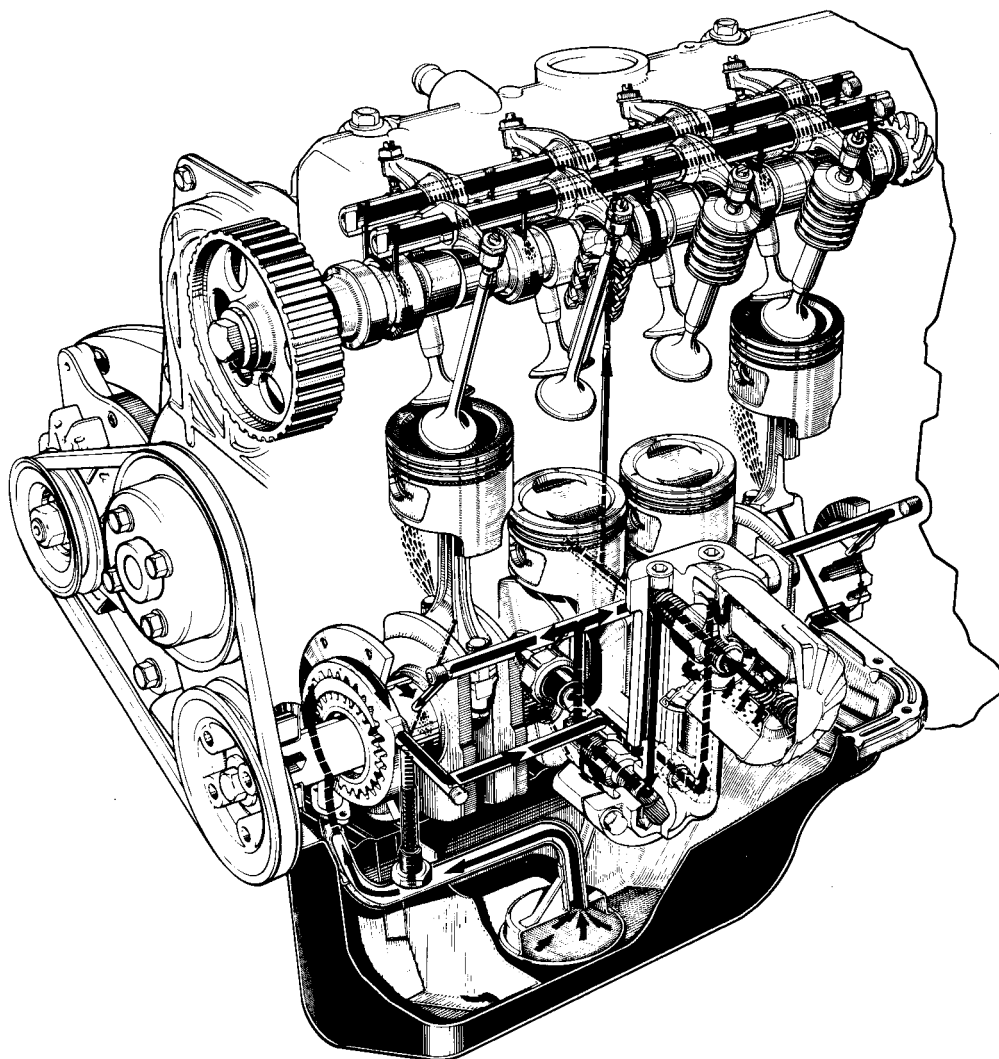


Figure 4.11 A comprehensive forced lubrication system for an in-line cylinder engine (Suzuki)

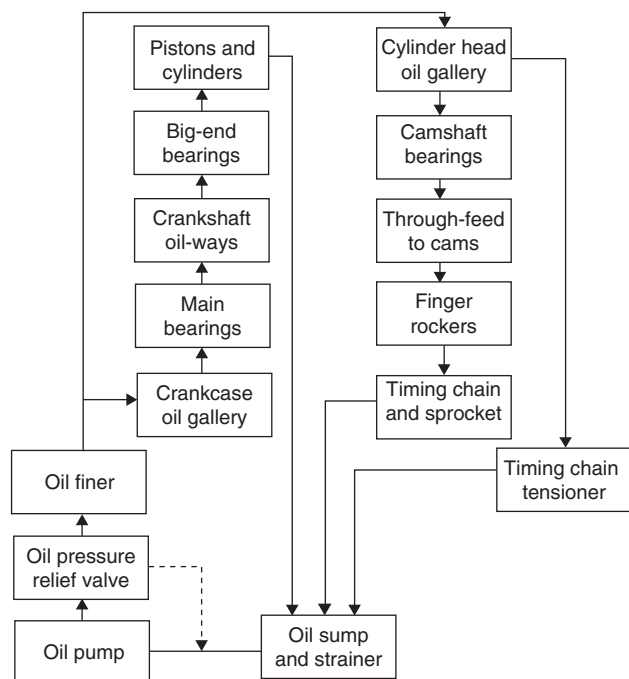


Figure 4.12 Block lubrication diagram for a typical in-line cylinder engine with single overhead camshaft

T-pieces to convey the oil to the various working parts, but since fracturing and loosening of such piping was always a possible source of unreliability, drilled pressure conduits have long since been preferred.

In operation, the engine-driven pump delivers oil from the sump and upwards through a duct in its body, whence it passes into a full-flow filter mounted on the outside of the crankcase. From the filter the oil is directed to either a single or a pair of main oil galleries which extend practically the full length of the crankcase at about the level of the cylinder skirts. A single main gallery is incorporated in the crankcase side wall of inline cylinder engines, while a pair of centrally disposed galleries may be required in the crankcase of some V and horizontally-opposed cylinder engines. The purpose of the main gallery is to distribute the oil to the principal working parts of the engine.

Of major importance is the oil supply to the crankshaft main bearings and their dependent connecting rod bearings. Oil leaving the engine main supply gallery is therefore normally routed directly to the main bearings through communicating ducts in the crankcase webs and main bearing housings. From these ducts the oil passes through an inlet hole in the upper half-liner of each main bearing. At this point, the oil flows into a circumferential groove formed in the actual bearing surface. A proportion of the oil leaving the groove spreads laterally to lubricate and cool the main bearing, whilst the remainder is transferred to perform a similar duty in an

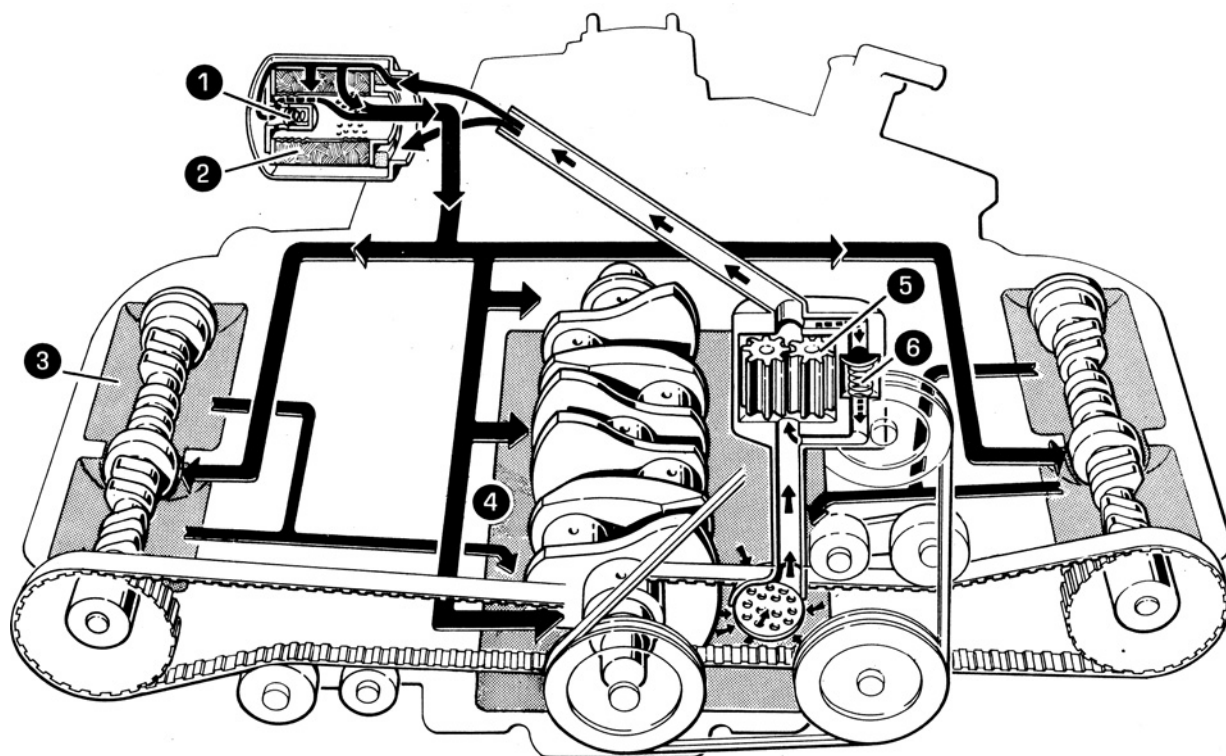


Figure 4.13 Fully forced lubrication system for horizontally opposed four cylinder engine with single overhead camshafts (*Alfa Romeo*)

- | | |
|----------------------|-------------------------|
| 1 By-pass valve | 4 Oil sump |
| 2 Oil filter element | 5 Gear pump |
| 3 Camshaft housing | 6 Pressure relief valve |

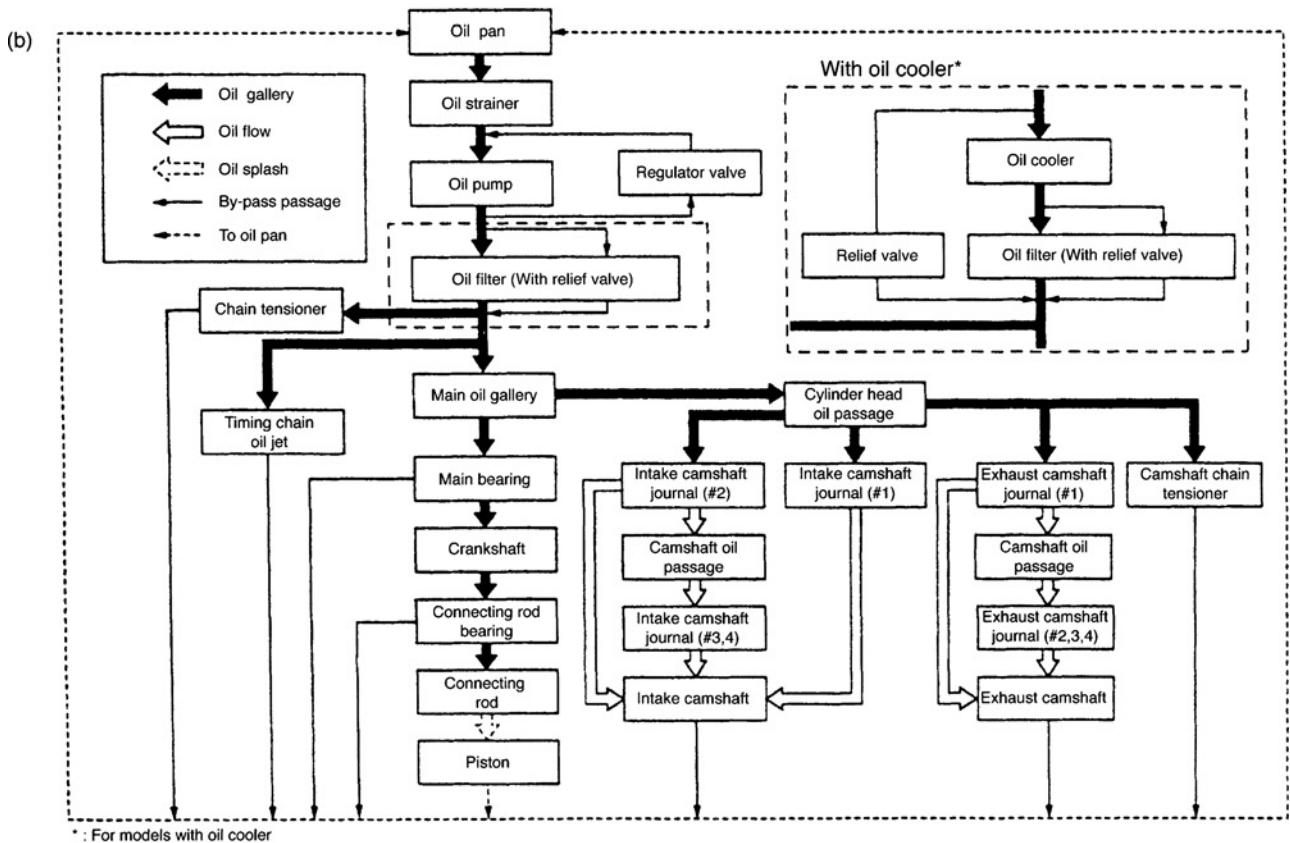
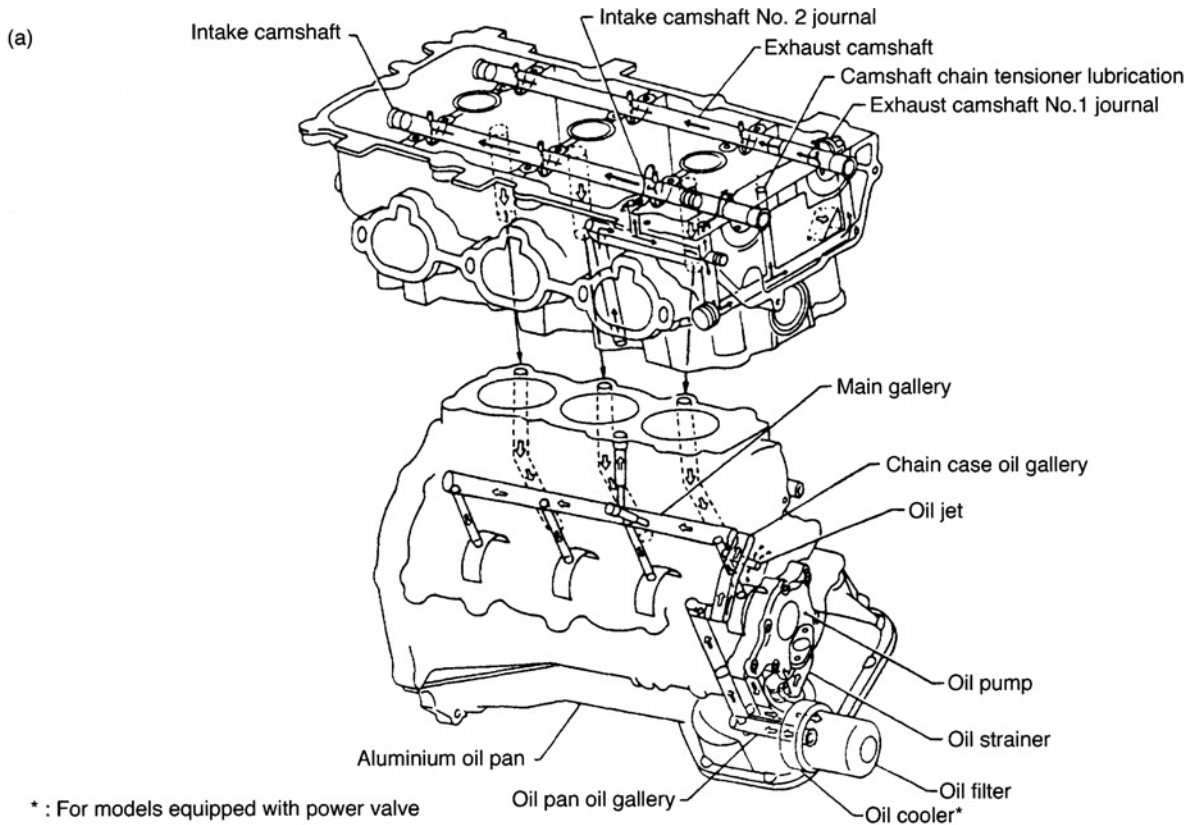


Figure 4.14 Fully forced lubrication system for modern V6 cylinder engine with double overhead camshafts: (a) schematic (b) block diagram (Nissan)

adjacent connecting rod big-end bearing. The latter action is accomplished by means of an angular drilling which registers with the supply groove and passes through the appropriate main journal, web and crankpin (Figure 4.15).

Oil grooving in a bearing liner is always a necessary evil. For some high-performance engines a part-circumferential oil supply groove may be employed to improve the load-carrying capacity of the bearings. In this case, the lower half-liner of each bearing is made grooveless, which in turn may require the main journals to be cross-drilled (Figure 4.15b). Hence, either end of this drilling is always in communication with the grooved upper half-liner, thus ensuring an uninterrupted supply of oil to the big-end bearing with each full turn of the crankshaft.

The connecting rod small-end bearing is more difficult to lubricate effectively, because of its oscillating rather than rotating motion. To assist lubrication, the upper bearing surface of a bushed small-end may be grooved to form a reservoir for collecting oil entering through a drilling in the rod eye (Figure 4.16). With the wider use of the semi-floating gudgeon pin the means of its lubrication assumed more importance, because it became necessary for an oil film to be quickly formed and maintained during cold starting when the gudgeon pin is tight in the piston bosses. Various detail provisions have been made for directing oil into the bores of these bosses, which include broached axial slots and holes drilled at certain angles. The latter may communicate with drain holes from the oil control ring groove (Figure 4.16), or with an oil scoop cast into the four corners of the piston skirt.

For lubrication of gear-type timing drives a jet of oil is usually sprayed directly at the in-to-mesh point of their teeth

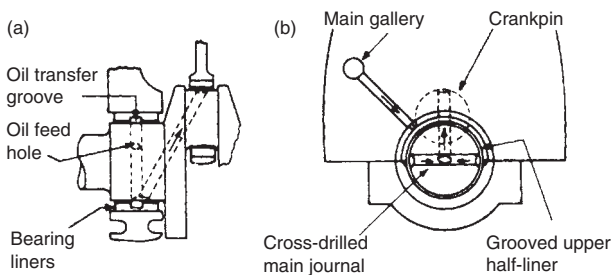


Figure 4.15 Oil supply to the crankshaft bearings

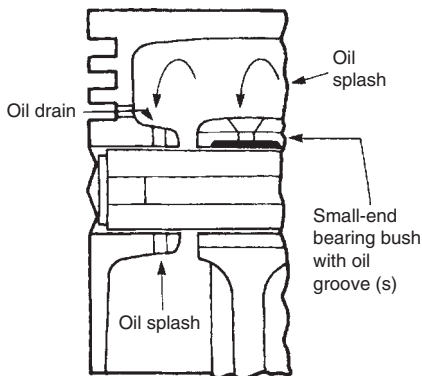


Figure 4.16 Lubrication of small-end bearings

(Figure 4.17). An out-of-mesh spray point may however be chosen (this being an interesting feature of Rolls-Royce V8 engine design), so as to gain the maximum cooling effect during high-speed operation of the gears. This policy is likewise followed by some designers of high-speed industrial gears, where power loss and over-heating could arise from heavy churning of the oil if sprayed at the immediate approach zone to the meshing teeth. The actual amount of oil required to form effective lubricating films between gear teeth is in practice relatively small.

In the case of heavy-vehicle diesel engines an extended network of oilways is incorporated in the crankcase front cover to serve the timing and auxiliary gear drives and their bearings.

With a crankcase-mounted camshaft, driven by either a roller or an inverted-tooth chain, an oil jet is typically arranged to discharge on to the crankshaft sprocket at its meshing point with the chain (Figure 4.18). Where the roller type of chain is used to drive a cylinder-head-mounted camshaft, it is typically lubricated by the controlled leakage of oil from a hydraulically assisted tensioner, as described in Section 1.11.

The manner in which the camshaft bearings receive their lubricating oil depends largely upon whether a crankcase- or a cylinder-head-mounted camshaft is employed. It is usual for the bearings of the former to receive their oil supply via corresponding main bearings. Ducts are incorporated in the crankcase webs and bearing housings to establish communication between the oil supply grooves in the upper half-liners of the main bearings and the camshaft bearing bushes (Figure 4.19a). With a V engine the camshaft bearings may receive their oil supply through cross-ducts leading directly off the main gallery (Figure 4.19b).

Oil for the bearings of a cylinder-head-mounted camshaft may be supplied either directly through ducts extending upwards from the main gallery (Figure 4.20a) or indirectly from a subsidiary gallery in the cylinder head, or similarly by

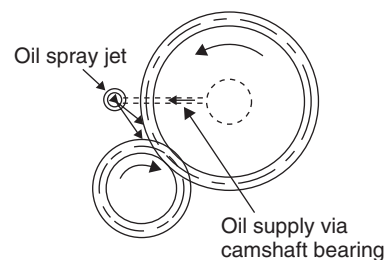


Figure 4.17 Lubrication of gear timing drives

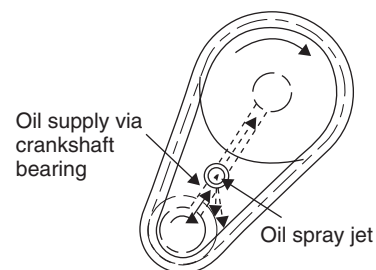


Figure 4.18 Lubrication of chain for crankcase-mounted camshaft

an end-to-end feed from within the camshaft itself (Figure 4.20b). For the now widely used double overhead camshafts, separate oil passages are generally provided to feed the bearings for each camshaft (Figure 4.14). It is usual for the oil supply to overhead camshaft bearings to be restricted, so as to avoid an increase in oil consumption through flooding of the valve mechanism. A further consideration relates to the use of hydraulic tappets in some double overhead camshaft engines, since it must be arranged that oil does not drain away from their supply galleries in the cylinder head when the engine is not running (Section 1.10). To prevent this happening an upper oil retaining valve may be incorporated in the system, which is also provided with a vent hole to facilitate the release of any trapped air when the engine is restarted.

Proper lubrication of the highly stressed cam and tappet contact faces is vitally important. For a crankcase-mounted

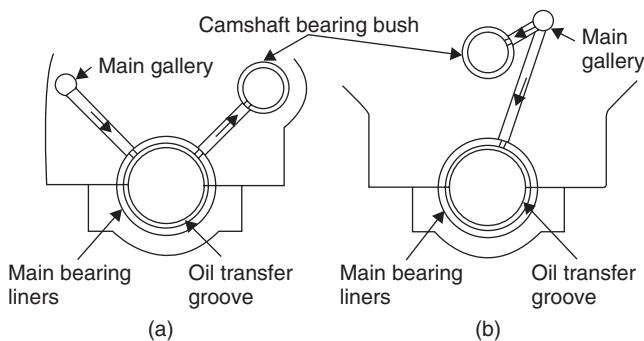


Figure 4.19 Typical oil supply arrangements for crankcase-mounted camshafts

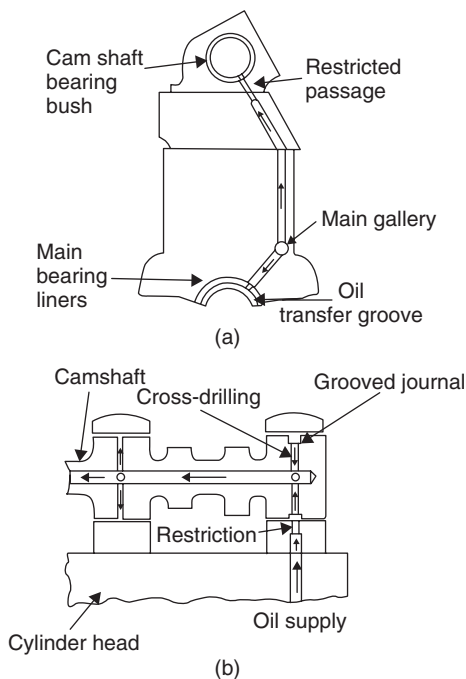


Figure 4.20 Typical oil supply arrangements for cylinder-head-mounted camshafts

camshaft, the cams and tappets may be lubricated partly from the oil being flung off from the connecting rod big-end bearings, and partly by the oil draining down the push-rods from the overhead valve rocker mechanism (Figure 4.21a). Alternatively, the cams and tappets may be provided with an oil bath instead of a splash system of lubrication. The cams then dip into a crankcase trough situated beneath the camshaft (Figure 4.21b). Oil escaping from the camshaft bearings constantly replenishes this trough, its level being controlled by a series of overflow holes, the surplus oil draining back to the sump.

For engines with cylinder-head-mounted camshaft(s), the cams and tappets may similarly be oil bath lubricated, the trough being formed either in a detachable camshaft and tappet carrier, or directly in the top deck of the cylinder head. In other designs, a splash system of lubrication is employed for the cams and tappets. Here the subsidiary oil gallery supplying the camshaft bearings may incorporate squirt holes, so that oil is sprayed on to the cams (Figure 4.22a). This function may also be performed by an external gallery pipe, which extends alongside the camshaft and receives a restricted supply of oil from the main gallery. Where an end-to-end oil feed is adopted for overhead camshaft bearings, the cams and tappets are usually lubricated through additional holes drilled radially inwards from the base circle of each cam to the interior passage of the shaft (Figure 4.22b).

Lubrication of the cylinder walls and piston skirts may be effected either wholly, or partly, by oil being flung off from the big-end bearings of the connecting rods. The cylinders in modern engines of high bore-to-stroke ratio tend to receive more oil from this source, since they are both larger in diameter and closer to the crankshaft.

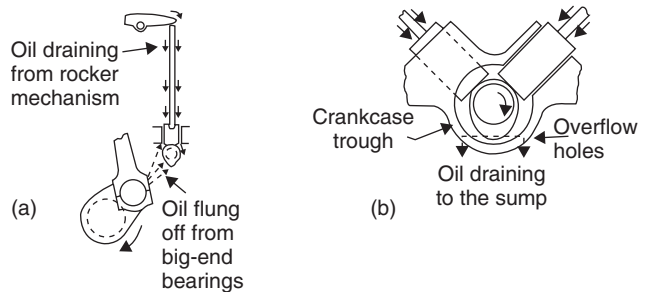


Figure 4.21 Typical lubrication arrangements for crankcase-mounted cams and tappets

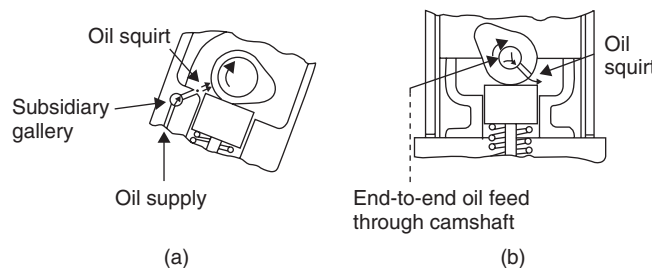


Figure 4.22 Typical lubrication arrangements for cylinder-head-mounted cams and tappets

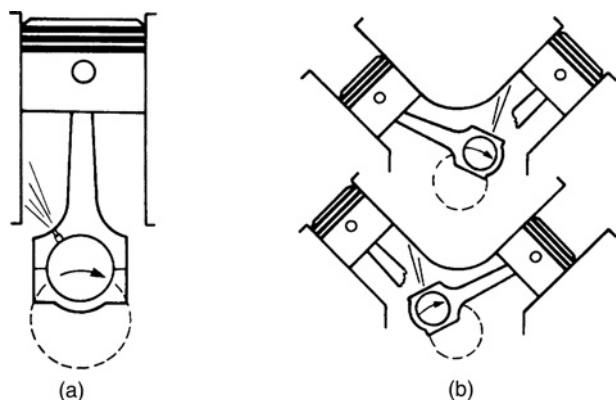


Figure 4.23 Oil spray lubrication of the cylinder walls

In some engines provision is made for the cylinder wall to receive additional lubrication by a fine oil spray, which is directed from a squirt hole drilled in the shoulder of the connecting rod and big-end bearing. The oil spray is usually, but not invariably, directed at the major thrust face of the cylinder (Figure 4.23a). With a V engine it is normal for the oil to be sprayed from the connecting rod of the corresponding cylinder in the opposite bank (Figure 4.23b), so the oil first reaches the upper sides of the inclined cylinders and then gravitates to their lower sides.

Reference has earlier been made to the oil cooling of pistons (Section 2.4), which can lower critical piston temperatures by up to about 30°C. For heavy-vehicle diesel engines it is usually arranged for the separately cooled and then filtered oil leaving the filter head to be divided into two flows; one flow is directed to the main gallery for lubricating the moving parts of the engine and the turbocharger, and the other to an auxiliary gallery that supplies oil to the piston cooling nozzles. A second oil pressure control valve is located in the base of the filter head, which limits the flow of cooling oil to the nozzles according to the pump supply pressure, so that normal oil pressure is not jeopardized at idling and low engine speeds. The oil pump is generally of large capacity where piston oil cooling is employed. On a practical note, it is important to ensure correct rotational alignment of the sprayer nozzles when they are inserted into the crankcase wall and also that their O-ring seals are not twisted in the grooves. For lighter-duty diesel engines such as used in passenger cars, individual nozzle blocks may be bolted to the underside of the cylinder block and directly beneath their oil supply gallery. Each nozzle block has a spring-loaded ball valve at its entrance, so that oil cooling to the piston is automatically cut off during engine idling periods (Figure 4.24).

Dry sump system

The sump was described in Section 1.6 as a reservoir to store the oil that is required by the engine lubrication system. Such an arrangement is further known as a wet sump, as opposed to the less common dry sump. The basic difference between them is the provision of a separate oil tank and the addition of a crankcase scavenge oil pump for the dry sump system. In

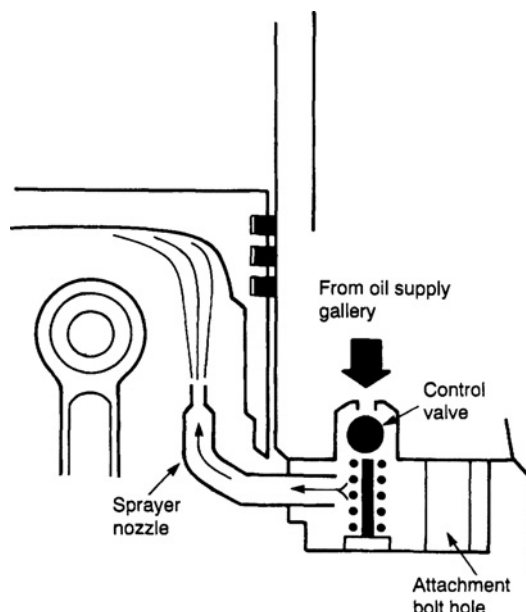


Figure 4.24 Schematic arrangement of oil spray cooling for pistons

other respects the engine lubrication arrangements are similar for both systems. The various types of vehicle in which the dry sump system of engine lubrication may be found, together with its related advantages, may be summarized as follows:

Racing cars The principal advantage here is that even should the oil level in the separate tank become low, it is impossible for surging of the oil content (caused by the rapidly changing motion of the vehicle) to uncover the intake of the oil pump and thereby interrupt the oil supply to the engine bearings. Furthermore, the bulk oil supply is removed from contact with the hot blow-by gases in the crankcase, so it is less subject to oxidation and contamination. Another consideration is that the absence of a bulky sump allows a lower installation height for the engine, which thus contributes to the lowest possible centre of gravity for the vehicle in the interests of safe handling.

Rally cars Dry sump lubrication affords the opportunity of carrying, in a more compact boot-mounted tank, an appreciably larger quantity of oil than would be practicable in the conventional sump. Hence, there is less tendency towards overheating of the greater amount of oil in circulation. Other advantages are similar to those for racing cars, except that the absence of a bulky sump is welcomed for the increased ground clearance it confers beneath the engine over rough roads.

Other vehicles and tractors In these cases, the main benefit to be derived from using a dry sump system is to maintain normal lubrication of the engine when the vehicle is negotiating very steep slopes, such as may be encountered in off-the-road operating conditions.

Operation of the dry sump system

With this type of lubrication system the oil pressure and scavenge pumps are engine driven, either separately at different speeds – the scavenge pump at a higher speed for the reason

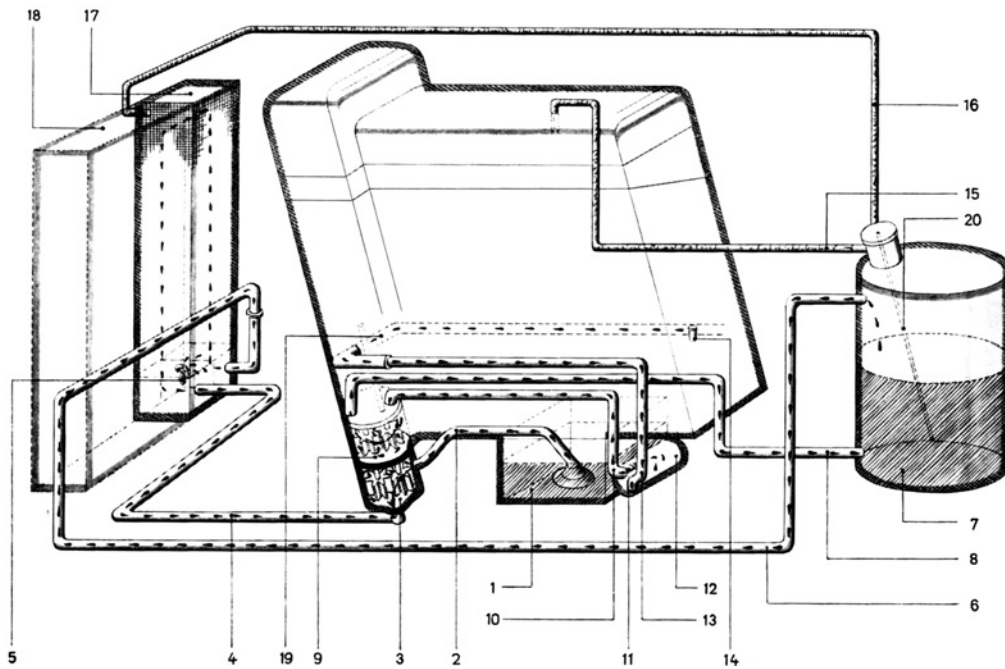


Figure 4.25 Diagrammatic layout of a dry sump lubrication system (Mercedes-Benz).

- | | | |
|---|--|---|
| 1 oil pan | 8 oil line from oil tank to pressure pump | 15 vent line from oil tank to cylinder head cover |
| 2 suction pipe | 9 pressure pump | 16 vent line from oil cooler to oil tank |
| 3 return pump | 10 oil line from pressure pump to oil filter | 17 oil cooler |
| 4 oil line from return pump to oil cooler | 11 oil relief valve in oil filter | 18 radiator |
| 5 by pass valve in oil cooler | 12 oil filter element | 19 main oil duct |
| 6 oil line from oil cooler to oil tank | 13 oil line from oil filter to engine | 20 oil dip stick |
| 7 oil tank | 14 pressure retaining valve in oil main duct | |

mentioned later – or together in tandem. If they are driven in tandem the pressure pump is sealed from the scavenge pump below it. In operation, the pressure pump receives oil from the separately mounted tank and distributes it to the working parts of the engine in a manner similar to that of the wet sump lubrication system.

The oil returning to the base of the crankcase is then collected by the scavenge or return pump and transferred back to the tank. An important consideration here is that the pumping capacity of the scavenge pump must be appreciably greater than that of the pressure pump. The reason for this is that the inevitable presence of entrained air in the returned oil increases its volume. In other words, the scavenge pump must be capable of handling the combined volume of oil and air.

The actual manner in which a dry sump lubrication system is plumbed-in to both engine and vehicle varies, of course, according to the type of application. Since this is a topic that more readily lends itself to illustration than detailed description, a schematic drawing of a dry sump lubrication system once used in a German high-performance car shows a fairly typical installation (Figure 4.25). It will be noted that in this particular case a tandem oil pressure and scavenge pump was employed (Figure 4.26) and also that an oil cooler was included in the scavenge return to the oil tank. Engine oil coolers receive further mention later.

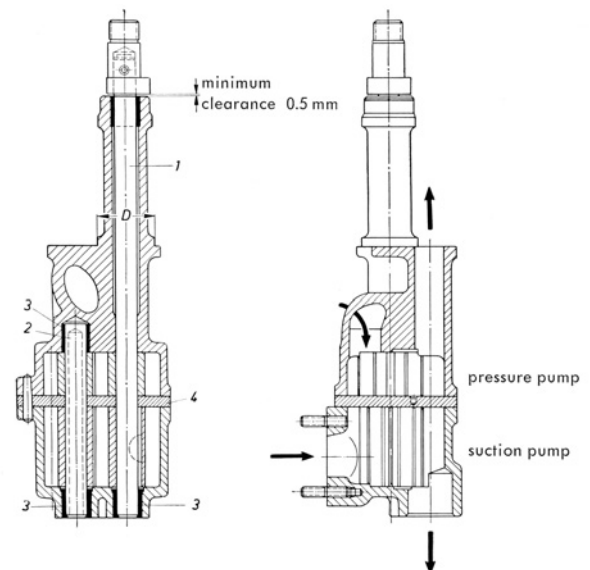


Figure 4.26 Tandem oil pressure and scavenge pump for dry sump system (Mercedes-Benz)

- 1 drive shaft
- 2 shaft
- 3 bearing bushing
- 4 intermediate plate

4.5 OIL PUMPS AND PRESSURE RELIEF VALVES

Developing oil pressure in the fully forced system

In modern practice, engine-driven positive displacement pumps of the rotary type are used to deliver oil and develop pressure in the fully forced lubrication system (Figure 4.27a and c). The term *positive displacement* is used in hydraulic engineering to describe any pump that displaces a definite volume of fluid with each revolution of its drive shaft, regardless of pressure conditions in the circuit it supplies. In other words, the pressure developed by the oil pump depends entirely upon the resistance offered to the flow of oil in the rest of the lubrication circuit. The principal advantage of using this type of pump is that it is compact in design. Because it is not self-priming, the oil pump with its inlet arrangements is usually bolted to a boss on the lower face of the crankcase, so that it is partially submerged in the oil sump but supported independently of it.

External gear pump

This comprises simply an elongated housing and end cover, which enclose a pair of either spur or, for quieter operation, helical gears. A bearing bore in the housing receives the pump drive shaft, rigidly secured to which is the driving gear of the pair. These parts may be either keyed together or machined as an integral pair. A mounting for the driven or follower gear is provided by a spindle supported from the pump housing. The gears generally have a coarse pitch tooth form, so that the least width of gear is required to pump a given amount of oil.

In operation, the oil entering the pump through the inlet port becomes trapped between the teeth of the contra-rotating

gears and the surrounding wall of the pumping chamber. The oil is thus carried around the periphery of the gears (not between them) and then discharged through an opposite outlet port (Figure 4.28a). At this point the action of the intermeshing gear teeth prevents the oil from returning to the inlet side of the pump.

Internal gear pump

This houses a smaller driving gear mounted eccentrically within a larger driven gear. The externally toothed driving gear is rigidly secured to the pump drive shaft, while the internally toothed driven gear runs directly in the cylindrical cavity of the pumping chamber. Since the tooth form of these gears more closely resembles that of a lobe, the driving and driven gears are generally termed the inner and outer rotors of the pump respectively. The number of lobes on the inner rotor is one less than on the outer and so determines the displacement of the pump.

In operation, the oil enters the pump through a kidney-shaped inlet port, as the lobes on the rotors are moving out of mesh. Continued turning of the rotors serves to transfer the oil to a similar kidney-shaped port on the outlet side of the pump, the oil being discharged as the lobes of the rotors move into mesh (Figure 4.28b). The advantages usually claimed for this type of pump are that it is still more compact than the external variety and, because there are fewer teeth in mesh for each revolution, it can be quieter running.

More recently, there have been several examples of crescent-type internal gear pumps, encircling and driven directly from the nose of the crankshaft (Figure 4.27c). The advantages offered include simplicity of installation with particular

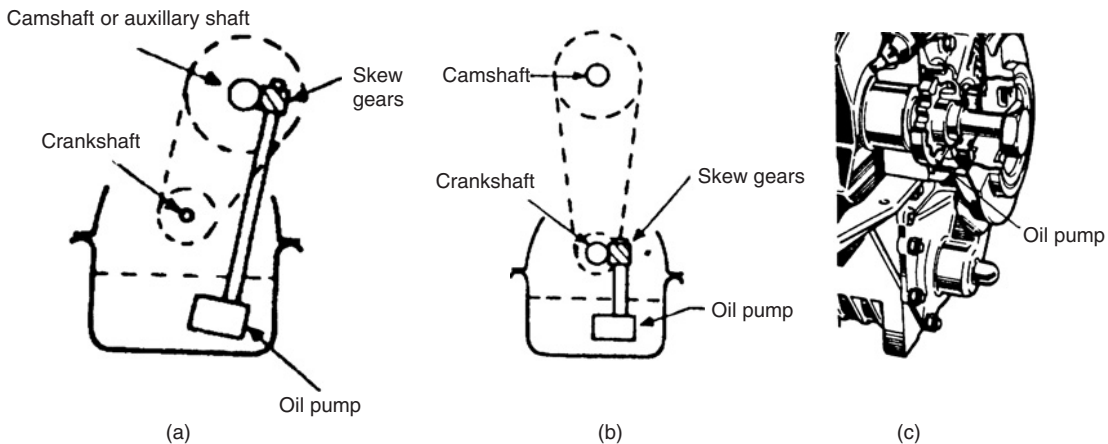


Figure 4.27 Typical pump drives: (a) from side camshaft (b) from crankshaft (c) encircling crankshaft

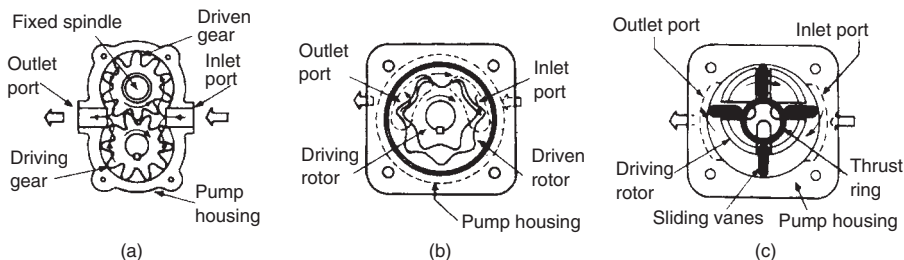


Figure 4.28 Types of engine oil pump: (a) external gear (b) internal gear (c) sliding vane

reference to the overhead camshaft engines and maximum oil pressure can be obtained at half the corresponding speeds of other engines where their oil pumps are camshaft driven, so that substantial oil pressure is maintained during idling.

Sliding vane pump

This takes the form of a driven rotor that is eccentrically mounted within a circular housing. The rotor is slotted and equipped with vanes equally spaced around its periphery, these vanes being free to slide radially within the slots. A

pair of centring thrust rings maintains the vanes in close clearance with their track in the housing.

In operation the vanes are, of course, pressed outwards against the track by their centrifugal action. The oil inlet port is sited where the vanes begin to move away from a point of minimum volume between the eccentric rotor and its housing. After the vanes have passed the point of maximum volume and are again approaching a point of minimum volume, the oil is discharged through an outlet port (Figure 4.28c). This type of pump has the characteristic of giving a continuous rather than a pulsating oil flow, the latter usually being associated with gear-type pumps.

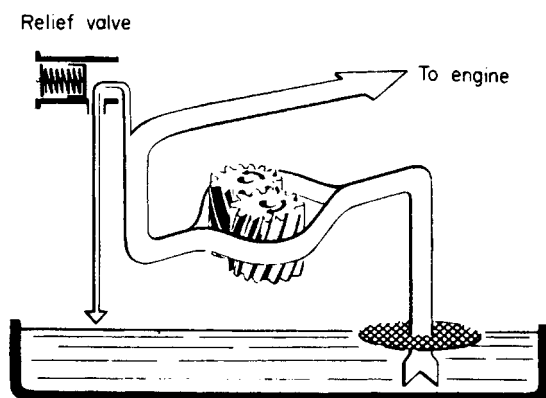


Figure 4.29 Basic action of the engine oil pump and relief valve (Alfa-Romeo)

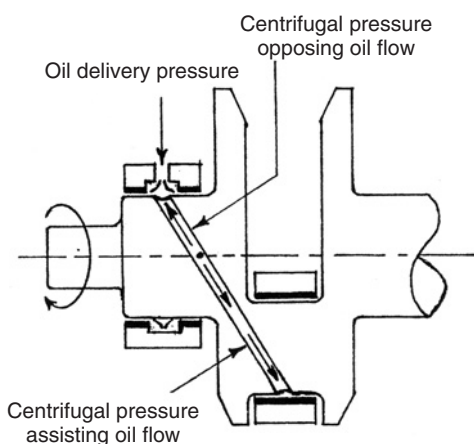


Figure 4.30 Oil flow conditions in a crankshaft

Pressure relief valves

The purpose of the pressure relief valve is to act both as a pressure regulator and as a safety device in the engine lubrication system (Figure 4.29).

A means of pressure regulation is required, since at higher engine speeds the increase in the quantity of oil delivered by the pump is not matched by a corresponding increase in the amount of oil that is able to flow through the bearings. This is because of centrifugal action on the oil already in the ducts of the crankshaft main journals, which imposes an increasing resistance to further oil flowing into them *en route* to the connecting rod big-end bearings. (Figure 4.30).

Under low-temperature starting conditions any increase in oil viscosity or reduction in bearing oil film clearances, or both, imposes an appreciable resistance to the oil flowing in the engine lubrication system. The pressure relief valve must, therefore, act as a safety device to prevent the oil pressure building up to a dangerous level. Otherwise, at worst the pump driving mechanism could be overloaded to its detriment, or the oil filter gasket could be damaged, and at least the excessive pressure drop across the oil filter would cause its own internal pressure relief valve to open and bypass unfiltered oil to the engine bearings.

Types of pressure relief valve

The pressure relief valve consists of either a ball, a poppet or a plate valve that is spring loaded on to an orifice seating and opened by oil pressure.

A disadvantage of the simple ball valve (Figure 4.31a), which has complete freedom to turn in any direction, is that random bedding marks on its working surface can form leakage paths across the valve seating. Leakage arising from this source can be avoided with a guided poppet valve, because this can turn only about its lift axis, so that the bedding condition between the valve and its seating is better preserved.

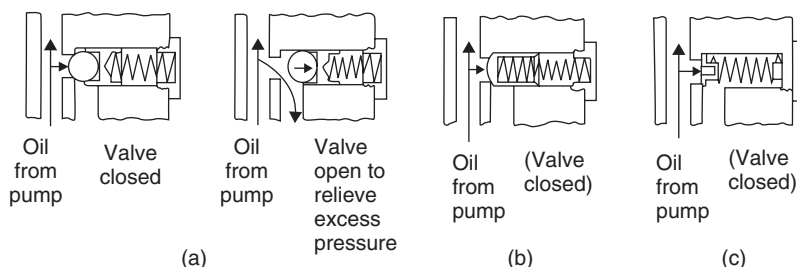


Figure 4.31 Pressure relief valves for engine lubrication systems: (a) ball type (b) poppet type (c) plate type

The poppet valve takes the form of a plunger with either a conical or a hemispherical sealing surface (Figure 4.31b). To minimize friction of the valve sliding in its bore, the plunger guide surface may incorporate annular grooves for oil retention.

A plate-type valve possesses the least inertia and is comprised simply of a thin steel disc (Figure 4.31c). Its edge is usually relieved in several places to prevent cross-binding of the valve in its guide bore.

Pressure relief valve setting

The actual setting of the pressure relief valve is generally related to the rate of oil flow required for adequate cooling of the bearings. In practice, it is typically set to open when the delivery pressure exceeds 280 kN/m^2 (40 lbf/in^2), although higher maximum opening pressures of up to about twice this value may be specified for heavy-vehicle diesel engines. The maximum delivery pressure does not, as explained earlier, bear any relationship to the pressures generated in the bearing oil films.

Oil pressure indicator

In present-day practice it is usual for an oil pressure indicator, as distinct from gauge, to be featured in the comprehensive instrumentation of a high-grade motor car. An oil pressure indicator is electrically operated and comprises two units known as the transmitter and the indicator (Figure 4.32). The body of the former communicates with the engine lubrication system and encloses a flexible diaphragm. One side of this diaphragm is exposed to engine oil pressure, while the other side attacks a flexible bracket carrying one of a pair of sprung-together electrical contacts. The mating contact is carried by a superimposed bimetallic strip, which is also encircled by a heater winding. Within the dashboard indicator unit, the needle derives its arc of movement from another bimetallic strip similarly encircled by a heater winding, both transmitter and indicator windings being connected in series.

When the ignition is switched on to start the engine, it completes an electrical circuit through the indicator and the earthed transmitter units. The current flowing in the transmitter winding heats the bimetallic strip, thereby causing it to bend and open the contacts. Since this action immediately interrupts the supply of current, the strip will cool and the contacts will close again to restore the circuit. In reality, of course, the process is

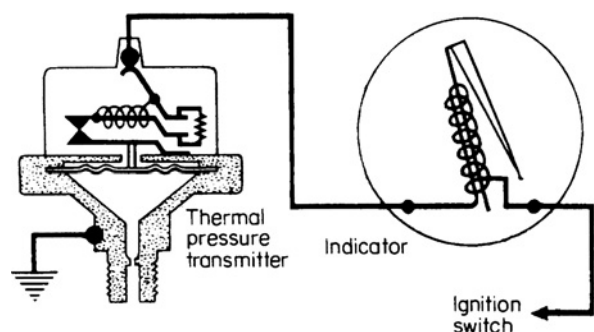


Figure 4.32 Mechanism and circuit of an electric oil pressure indicator (Smiths Industries)

being continually repeated such that the bimetallic strip oscillates to provide pulses of electric current. However, the effect of rising oil pressure on the transmitter diaphragm is to press the contacts more firmly together, so that their rate of oscillation is reduced. In other words, the current pulses sensed by the indicator unit will be longer in duration, thus creating a greater heating effect on its bimetallic strip. The bending of the latter then occurs in such a direction as to cause the indicator needle to move over its scale and record the prevailing oil pressure.

Another form of electrically operated oil pressure indicator similarly utilizes a transmitter diaphragm, but this acts instead against a piezoelectric crystal. Since a physical relationship can be established between the mechanical pressure on the crystallographic material and the voltage generated, a signal of the prevailing oil pressure can again be derived.

Oil warning light

It has long since become established practice to include an oil warning light on the instrument panel of vehicles, so that in the event of the engine oil supply pressure falling below a safe working minimum an illuminated and therefore positive warning is given to the driver.

Electrically the oil warning light is connected in series between the ignition switch and an oil pressure switch (Figure 4.33), the latter being engine mounted. When the ignition is switched on as a prelude to starting, the oil pressure switch contacts are closed to complete the circuit and the oil warning light will be illuminated. After the engine has been started and oil pressure is building up, the warning light will be extinguished by the action of the oil pressure switch opening its contacts and breaking the circuit.

The mechanism of an oil pressure switch typically comprises a reinforced rubber diaphragm that is spring loaded in opposition to the engine oil supply pressure acting on its inner face. Interposed between its outer face and the spring is a moving electrical contact of inverted top-hat form, the rim of which can seat upon a similarly formed stationary contact held within the switch body (Figure 4.34). The effective operating pressure of such a switch is generally somewhere in the range of 35 to 70 kN/m^2 (5 to 10 lbf/in^2).

Since under normal running conditions pressures of this order are exceeded several times by the engine oil supply pressure, the deflection of the switch diaphragm against its spring

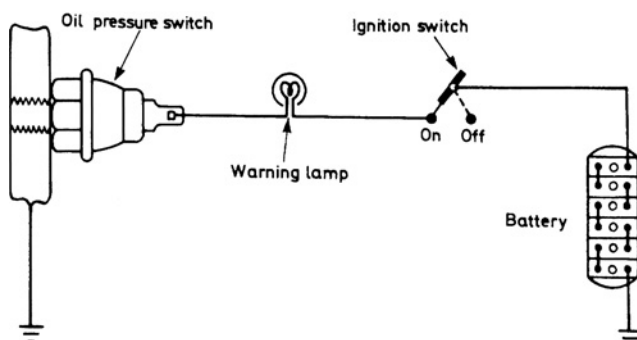


Figure 4.33 Electric circuit for an oil warning light

loading is sufficient to hold apart the switch contacts, so that the oil warning light is no longer illuminated. Conversely, if the engine oil supply pressure falls below the effective operating pressure of the switch, then the contacts close to complete the electrical circuit and the warning light is illuminated.

4.6 OIL FILTRATION AND COOLING

Engine oil filtration

Apart from corrosive wear arising from the acidic products of combustion, the other major form of engine wear is that caused by the presence of abrasive particles in the lubricating oil. This is especially true of those particles of a certain critical size that can span the oil film in the working clearances of the engine parts. Although manufacturers take immense care to build 'clean' engines, traces of casting sand and machining

swarf are not unknown sources of abrasive particles. The running-in process can also be responsible for generating wear particles. During normal running any soot particles contained in the cylinder blow-by gases and hardened by high temperatures can enter the crankcase oil as abrasive particles.

Since the engine lubrication system can never be any better than its oil, it is therefore necessary to include some means of filtering the oil circulating in the system, so as to remove the abrasive particles before they reach the sensitive parts of the engine.

Full-flow filtration

An absorbent filter of the type described later may be arranged to arrest abrasive particles from either a small proportion, or practically all, of the oil being delivered by the pump to the principal working parts of the engine. Modern practice favours the full-flow system in which all the oil is filtered continuously, except that discharged from the main pressure relief valve (Figure 4.35a). Exceptionally this valve may be situated on the outlet side of the filter for better protection from contaminants, in which case the filter takes the full delivery from the pump. The full-flow filter is additionally equipped with its own internal relief valve. Its purpose is to bypass an emergency supply of unfiltered oil to the engine bearings, should the filter element become choked through neglect.

Partial-flow filtration

In this system, which can now be regarded as obsolete, an absorbent type of filter is arranged in the engine lubrication circuit such that 5–10 per cent of the oil delivered by the pump is diverted through it and returned to the sump (Figure 4.35b). Hence, the supply of oil to the bearings is not directly filtered or, in other words, only a small proportion of the circulating oil is filtered. To maintain an adequate supply of oil to the engine bearings the flow through the filter is further limited by a restricting orifice. This is usually situated on the outlet side of the filter, so that it is less likely to become obstructed. An internal relief valve is not required for a partial-flow filter, since if the element becomes choked the oil that would otherwise pass through it simply continues with the main oil flow to the bearings.

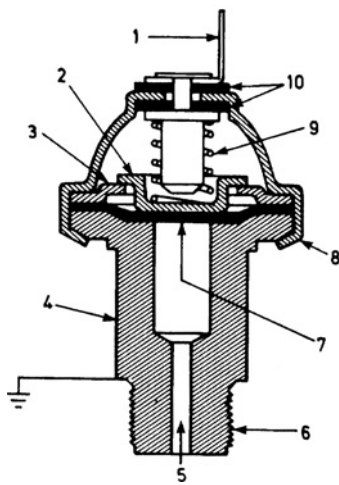
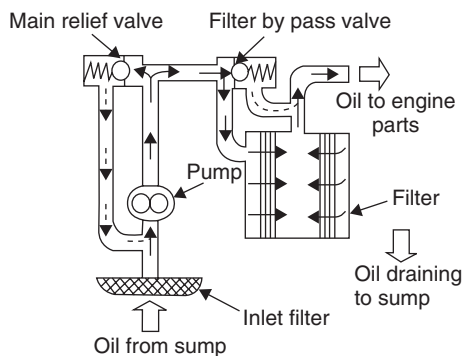
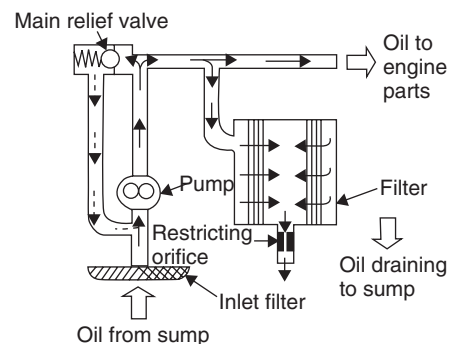


Figure 4.34 Cross-sectional view of an oil pressure switch

- | | |
|-----------------------|------------------------|
| 1 electrical terminal | 6 screw-thread fitting |
| 2 moving contact | 7 diaphragm |
| 3 stationary contact | 8 metal casing |
| 4 metal switch body | 9 compression spring |
| 5 oil pressure | 10 insulation washers |



(a)



(b)

Figure 4.35 Comparison of (a) full-flow and (b) partial-flow systems of oil filtration

It is possible to employ a very fine degree of filtration with the partial-flow system, on account of the slower rate at which oil is passing through the filter. However, the widespread adoption of the full-flow system in which practically all the oil delivered by the pump is subject to immediate filtration, seems to confirm that it performs the filtration process with more certainty. Indeed, it is now recognized in the field of hydraulic engineering that it is statistically impossible to clean the whole fluid content of a system by partial-flow filtration.

Types of oil filter

Various methods of oil filtration may be encountered in the lubrication system of the motor vehicle engine, and they can be conveniently summarized as follows:

Surface filtration For this a coarse type of filter is used, which simply comprises a wire gauze screen such as is fitted over the inlet to the oil pump. Its filtering action is in the manner of a sieve and it is, of course, cleanable upon removal of the sump.

Depth filtration This implies a fine type of filter that once typically comprised a specially processed paper element, but more recently a synthetic fibre material may be used. The replaceable element can be accommodated within either a metal canister made detachable from the engine crankcase, or a housing that is integrated with the crankcase. Because of its lower structural rigidity, the element may be provided with a supporting cylindrical screen. It also generally assumes a pleated form to present the largest surface area of filtering medium for a given size of element. A depth filtering action is obtained as contaminants are caught, and retained, as the oil passes through from the outside to the inside of the porous material.

In modern practice, the pore size of the element may be controlled to arrest particles down to within the region of $5\text{ }\mu\text{m}$ in size. (A micrometre (μm) is equal to $1/1000$ of a millimetre. For purpose of comparison, an ordinary grain of table salt measures $100\text{ }\mu\text{m}$.) As deposits accumulate, the absorbent filter offers increasing resistance to the passage of oil and must therefore be replaced at appropriate servicing intervals, when the oil is changed. Current (full-flow) depth filters are generally sealed and have a spin-on screw fitting, so that service replacement of the element itself is no longer possible, the entire filter being fully disposable.

Another feature that may be found in the modern full-flow filter is the positioning of the element relief valve at the inlet end (Figure 4.36a). The purpose of this is to ensure that any oil bypassed by the relief valve does not pass over contaminated surfaces within the filter casing (Figure 4.36b).

Centrifugal filtration The centrifugal type of filter acts by subjecting the oil entering it to a high centrifugal force. In so doing it separates the heavier contaminants from the lighter oil, depositing them in sludge traps formed in the perimeter of a rotating filter bowl. The latter may be mounted on the nose of the crankshaft and is of large capacity.

Magnetic filtration In practical application this generally comprises a plug that replaces the ordinary sump drain plug and incorporates a powerful permanent magnet. It is thus capable of attracting and retaining ferrous metal particles only, although other contaminants may also be retained by agglomeration (collecting in a mass).

Engine oil cooling

The oil circulating in the lubrication system receives heat that has been rejected from both the normally hot and the friction surfaces of the engine. It is usually recognized that 3.5 to 4.5 per cent of the amount of heat that may be obtained from combustion of the fuel received by water-cooled engine cylinders is rejected to the oil in this manner. This can mean that under severe operating conditions, such as may be encountered for example with turbocharged and also in air-cooled engines where cylinder temperatures are always higher, the temperature reached by the bulk of the oil may exceed 100°C . Since oxidation is one of the main causes of deterioration in engine oil, clearly there is justification in such cases to include an oil cooler in the lubrication system with the object of maintaining the oil temperature at a more normal 80 to 90°C .

Types of oil cooler

Oil coolers may utilize either the passage of air (Figure 4.37) or the flow of water (Figure 4.38) through the engine cooling system proper as the heat exchange agent. Where the latter is water we should not place too much emphasis on the word

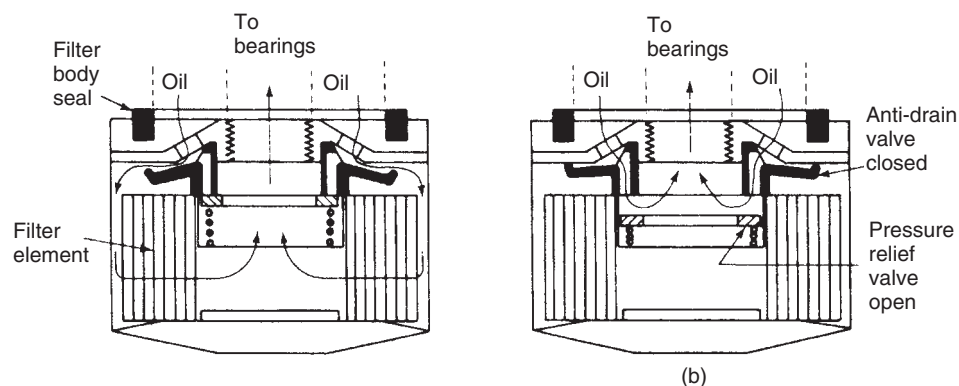


Figure 4.36 A modern full-flow filter of the spin-on type: (a) normal filtering action (b) filter choked and oil bypassed over non-contaminated surfaces

'cooler', since in this type of application an oil cooler does in fact perform two functions:

- 1 Brings the oil rapidly to its most efficient working temperature, because as long as the temperature of the oil is below that of the water the oil will be heated.
- 2 Maintains the oil at its most efficient working temperature under all operating conditions, because as soon as the temperature of the oil exceeds that of the water the oil will be cooled.

Where air is used as the heat exchange agent, necessarily so of course in the case of air-cooled engines, the oil cooler acts predominantly as its description suggests. However, it is then usually required to include an oil cooler bypass valve in the system, which allows the oil to heat up more rapidly from cold by initially restricting its circulation to the engine only. An oil cooler of this type is separately mounted from the engine and comprises a number of finned tubes that are plumbed into the lubrication system. These tubes are exposed to a cool stream of air by virtue of their being mounted across and forward of the engine radiator (Figure 4.39). With an air-cooled engine the oil cooler is mounted directly on the engine and adjacent to the outlet ducting of the cooling fan.

In the case of water being used as the heat exchange agent, the oil cooler is again mounted directly on the engine. It comprises an encased bundle of finned tubes through which the water is circulated from the engine cooling system and over which the oil is pumped in the opposite direction to confer a beneficial counter-flow, suitable piped connections being provided in each case (Figure 4.40).

For a high-performance passenger car oil coolers of both types may be used in series. In a recent Audi installation an

oil-to-water cooler is mounted directly between the engine crankcase and oil filter, and is supplemented by an oil-to-air cooler sited as usual beneath the engine radiator. A thermostat control in the oil-to-water cooler limits the amount of oil flowing to the oil-to-air cooler while the engine is idling with the car stationary. When the temperature of the oil exceeds a pre-determined limit as power output rises with car performance, the thermostat opens to admit a full flow of oil to the oil-to-air cooler so that both oil coolers are brought into action.

4.7 OIL RETENTION AND CRANKCASE VENTILATION

Static oil seals

These are required because to permit assembly of the working parts into the engine, its main structure must necessarily be fitted with detachable oil-tight covers. Although the mating faces of these parts are produced flat and parallel, in practice there are

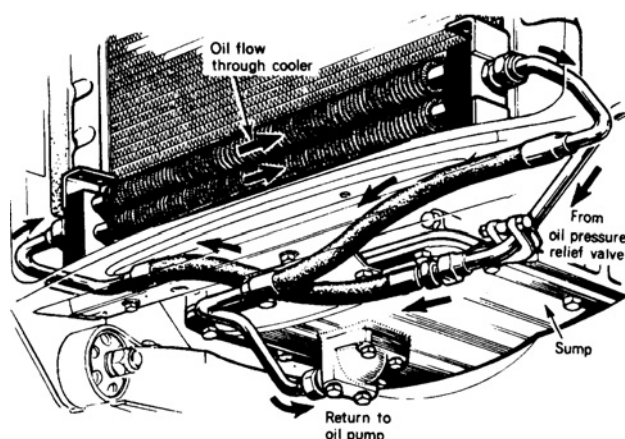


Figure 4.39 Installation of an air-cooled type of engine oil cooler (*Jaguar*)

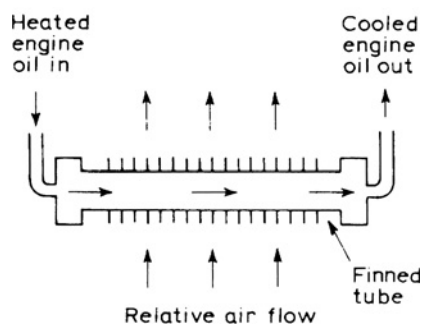


Figure 4.37 Basic principle of oil-to-air cooler

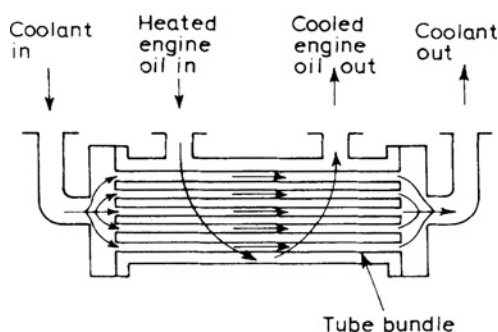


Figure 4.38 Basic principle of oil-to-water cooler

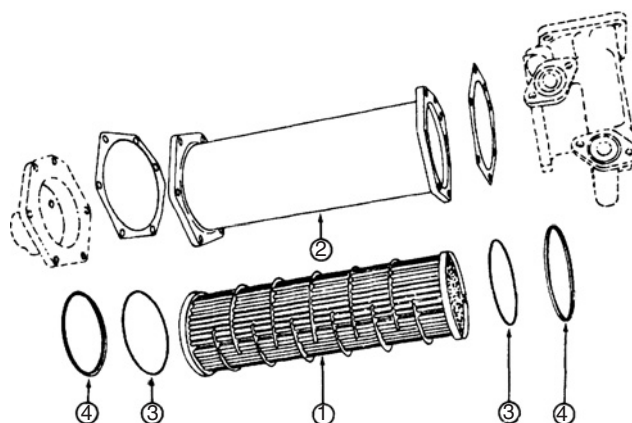


Figure 4.40 Exploded view of a water-cooled type of engine oil cooler (*Cummins*)

- | | |
|-----------|------------|
| 1 element | 3 O-ring |
| 2 housing | 4 retainer |

always minute surface irregularities present. If then a metal-to-metal closure was relied upon, these surface irregularities would offer potential areas of oil leakage. Suitably shaped static or gasket seals are therefore inserted between the various joint faces to prevent oil leakage. The gaskets are manufactured from fibre composition, cork with rubber binders and synthetic rubber materials, so that when compressed by bolt loading they tend to fill any leakage paths.

Typical static sealing applications are: the timing cover, for which a fibre composition gasket may be used; the oil sump, which usually requires a cork-based gasket to accommodate the relative expansion and contraction between itself and the crankcase; and the valve cover, where a synthetic rubber seal is generally appropriate in allowing effective sealing for minimum screw tightness (Figure 4.41).

A modern, less costly alternative to these pre-cut gaskets are room temperature vulcanizing (RTV) gaskets or formed-in-place gaskets, which are applied as a silicone sealing bead of toothpaste consistency. Re-tightening of the joint may be required a few minutes after the sealing compound has been allowed to settle.

Dynamic oil seals

Seals are fitted where each end of the crankshaft emerges from the engine structure. Their purpose is to prevent leakage of oil from the adjacent lubricated areas and also to protect the engine mechanism against the ingress of foreign matter.

The sealing devices used for this purpose fall into two categories:

Clearance seals Although no actual rubbing contact exists between the shaft and seals of this type, they nevertheless

impose a hydraulic resistance against oil leakage by virtue of their screw return pumping action.

Contact seals Their object is to provide a controlled rubbing pressure against the shaft, thereby creating a positive sealing action.

These seals are often used in conjunction with a supplementary oil flinger ring which rotates with the crankshaft. Its action is such that centrifugal force tends to throw off any excess amounts of oil creeping along the shaft towards the main seals. The oil flinger rotates within an annular grooved housing, this in turn being provided with a drain passage for returning oil to the sump.

Contact seals are generally required in modern engines with positive systems of crankcase ventilation, so that advantage can be taken of the latter to maintain a negative pressure in the crankcase and therefore discourage oil leaks.

Sealing the rear end of the crankshaft

Here the sealing requirements are particularly exacting, since the seal has to deal with the relatively large flow of oil from the adjacent rear main bearing. In many constructions the diameter of the crankshaft is substantially increased behind the rear main bearing to form an integral oil flinger, which is followed by a slightly reduced diameter sealing portion.

For clearance seals, either an oil return thread or a band of helical serrations is usually provided on the sealing surface of the shaft (Figure 4.42a and b), which runs within a close-clearance plain bore housing.

With the simple contact seal a flexible packing material is used, the rubbing portion of the shaft being left as a plain journal (Figure 4.42c). The sealing strip is pressed into upper and lower grooves machined in both the rear wall of the crankcase and an extension of the rear main bearing cap. In modern practice, the radial lip type of contact seal made from synthetic rubber has come into widespread use (Figure 4.42d). This type of seal is provided with a spring-loaded flexible sealing lip, which maintains a light rubbing contact with the shaft sealing surface. The sealing surfaces do, in fact, run with a very thin film of oil between them, the positive sealing action being generally attributed to the surface tension effect of the oil film at the exit side of the seal. A press fit must be provided for the seal in its housing and the wiping lip is positioned facing the lubricated bearing.

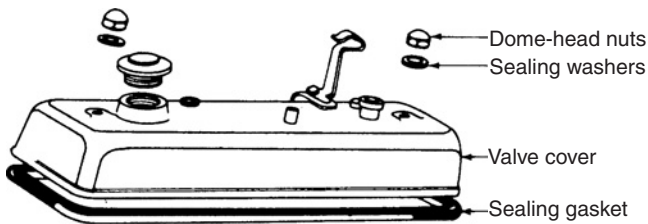


Figure 4.41 A typical static sealing application (Toyota)

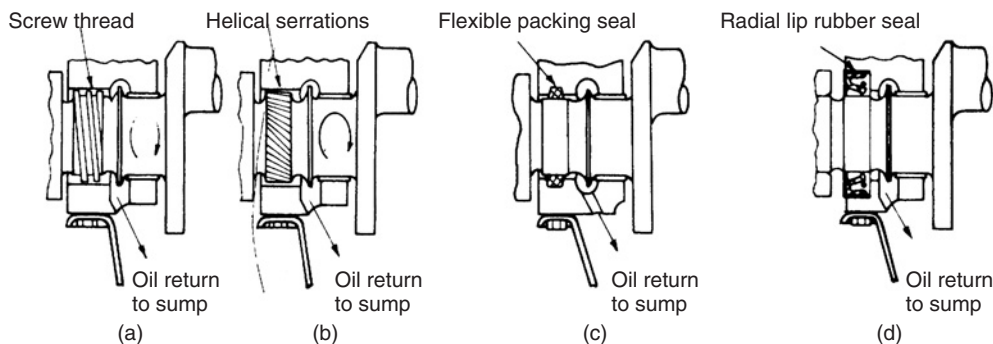


Figure 4.42 Sealing the rear end of the crankshaft

Sealing the front end of the crankshaft

The sealing requirements here are less severe, because the seal is shielded from the oil flowing from the front main bearing by the timing drive sandwiched between them, although this does not apply where an external toothed-belt timing drive is used.

The front end sealing surface is usually furnished by an inner extension of the crankshaft pulley boss. This is left as a plain journal when used in conjunction with either a threaded oil return bush (Figure 4.43a) or a radial lip seal (Figure 4.43b) inserted in the timing cover. Alternatively, an oil return thread may be provided on the pulley boss, which then works within a close-clearance plain bore in the timing cover. Where an oil flinger is also used, it is separately attached to the crankshaft behind the pulley boss.

Valve stems seals

The purpose of these seals is to control the passage of oil between the stems and the guides of overhead valves, especially the oil spill from valve rocker lubrication (Figure 4.44). Any excessive leakage of oil into the cylinders from this source can account for at least half the total oil consumption of a worn engine. If the leakage is bad enough it can produce blue exhaust smoke and hence cause air pollution by emission of hydrocarbons. Detonation or pinking can also arise from

carbon build-up in the combustion chambers, and power output can suffer as a result of carbon build-up on the valve stems restricting their free movement.

Types of valve stem seal

It is fairly common practice to use either a simple oil deflector, or a positive seal, or both. The oil deflector or umbrella seal comprises an inverted synthetic rubber cup, which grips the stem immediately below the valve spring retaining collar. Its depth of side is such that the valve guide entrance is shielded from oil mist when the valve is open, but is exposed to it when the valve is closed (Figure 4.45a)

The positive type of seal again takes the form of an inverted rubber cup, but it is smaller in diameter and secured to the upper end of the valve guide where it seals directly against the valve stem moving through it (Figure 4.45b)

The need for crankcase ventilation

In the similar sense that we open the windows of a room in order to ventilate its interior or, in other words, promote a circulation of fresh air to replace that which has become stale, so too does this basic need arise in the case of a vehicle engine, albeit with additional considerations.

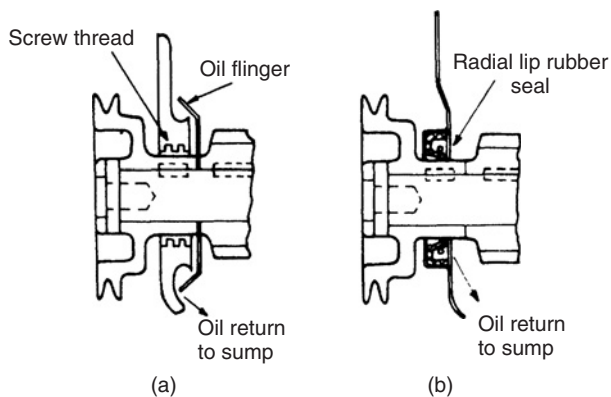


Figure 4.43 Sealing the front end of the crankshaft

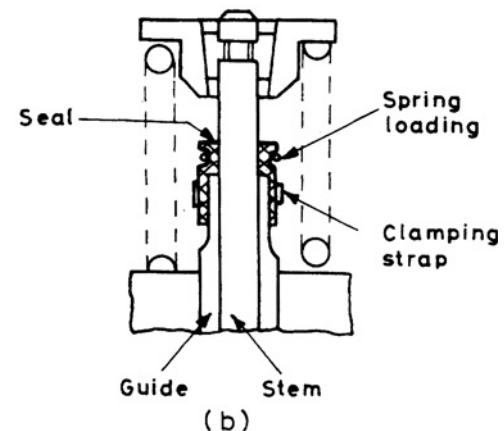
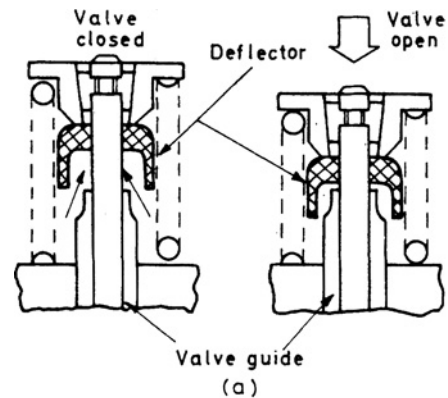


Figure 4.45 Valve stem seals: (a) deflector type (b) positive type

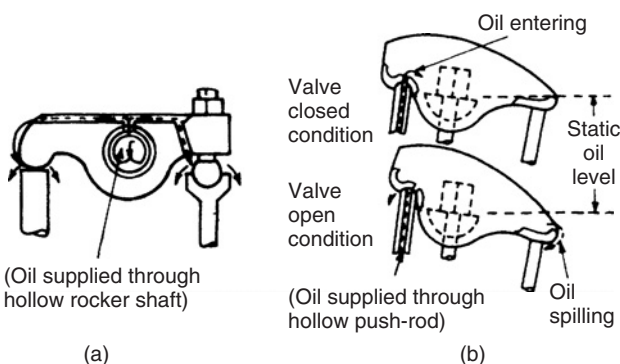


Figure 4.44 Typical methods of lubricating valve rockers

Adequate ventilation of the engine interior may, in fact, be considered as complementary to the satisfactory functioning of the lubrication system for two reasons:

- 1 The crankcase must be purged of harmful cylinder blow-by gases, since their composition is such as to cause both contamination of the oil and corrosion of the working parts.
- 2 The crankcase must be relieved of any build-up of internal pressure, because this could cause oil to leak past the crankshaft sealing arrangements.

Methods of crankcase ventilation

Ventilation of the engine crankcase was originally induced by what we would now term an open system, but from consideration of reducing atmospheric pollution the later closed system has long since become established practice. It is nevertheless useful to recall first the operating principle of an open system.

Open system

Here fresh air at underbonnet pressure entered at the highest point in the system, usually through a combined oil filler and

breather cap. It was induced to circulate through the engine interior by a breather pipe, the lower bevelled end of which protruded into the air stream under the car (Figure 4.46a). Purging of the crankcase gases therefore depended upon the motion of the car through the air. The air circulation within the crankcase was also assisted by the rotation or windage of the crankshaft. A basic limitation of this simple system was that ventilation could be inadequate when it was most needed, namely at low car speeds and during idling.

Closed system

Since the gases escaping from an engine with open crankcase ventilation contribute to air pollution, modern practice demands a positive closed system of ventilation, so that pollution from cylinder blow-by gases becomes negligible. Positive crankcase ventilation (usually abbreviated by PCV) was originally conceived by General Motors in America, where it was recognized in the early 1960s that cylinder blow-by gases could account for perhaps 20 per cent of the harmful emissions from a vehicle engine and its fuel system.

In early forms of closed system crankcase ventilation, the depression existing at the carburettor air cleaner was relied upon to maintain an air flow through the engine interior. For this purpose the breather outlet was simply connected by tubing to the air cleaner, through which the crankcase gases were circulated and subsequently burned in the engine cylinders (Figure 4.46b). A baffle plate flame trap was usually included in the system, so as to avoid a crankcase explosion in the event of an engine backfire.

In modern positive systems of closed ventilation (Figure 4.47) the greater depression existing within the intake manifold is utilized and controlled by either a variable- or a fixed-orifice regulator valve, known as a PCV valve, which is sited upstream from a crankcase oil separator. A variable-orifice regulator valve is interposed between the breather outlet tubing and the intake manifold riser. It consists essentially of a spring-loaded plunger, which opens in response to the intake

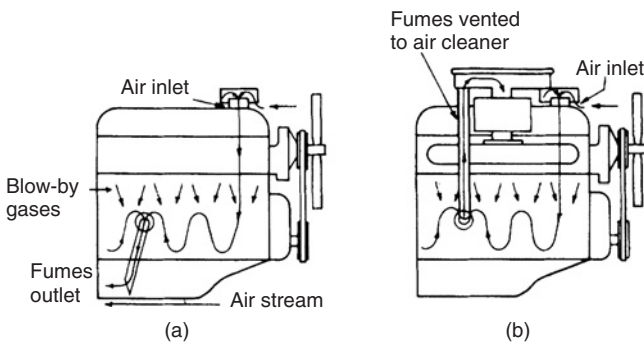


Figure 4.46 Comparison of (a) open and (b) closed systems of crankcase ventilation

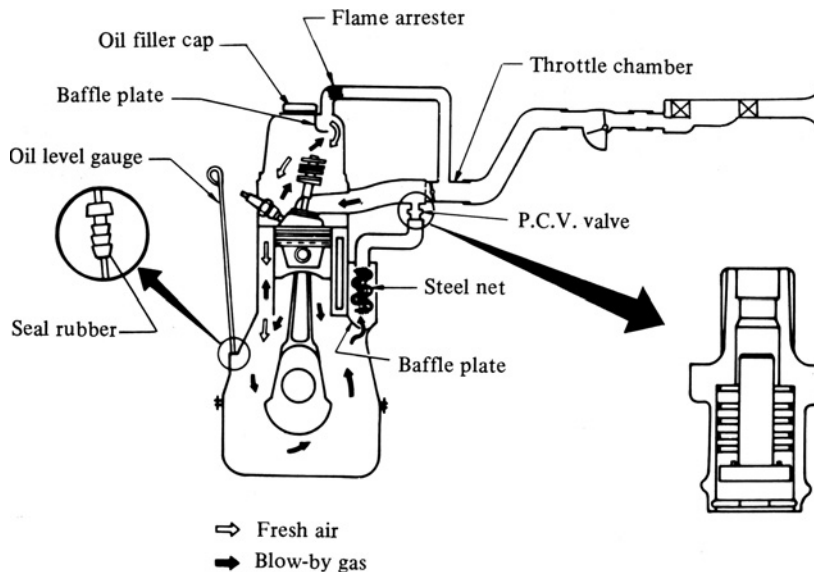


Figure 4.47 A positive closed system of crankcase ventilation (Nissan)

manifold depression acting upon it (Figure 4.48a). When the engine is not running the valve is completely closed. Under normal running the valve opens sufficiently to permit crankcase gases to enter the intake manifold with little restriction. At high manifold depressions during idling and overrun, further travel of the valve plunger against its spring loading closes it in the opposite direction, and the gas flow to the intake manifold is then restricted by an internal bleed orifice in the plunger. This particular feature thus ensures that the carburation process is not upset under these conditions. The valve also acts as a safety device in the event of an engine backfire, since any pressure created in the intake manifold automatically closes the valve completely, thereby isolating the engine interior.

With the other type of regulator valve, a fixed-orifice connection to the intake manifold riser is used in conjunction with a check valve connection to the clean side of the air cleaner (Figure 4.48b). At high manifold depressions the pressure difference across the check valve is such as to hold it against its seating, so that the crankcase gases can only pass through the fixed orifice and into the intake manifold. As engine speed rises, however, the reducing manifold depression behind the check valve and the increasing air cleaner depression ahead of it cause the valve to open from its seating. Hence, the crankcase gases may then enter both the intake manifold and the air cleaner to be subsequently burned in the engine cylinders.

Although PCV valves are designed to minimize the chances of blockage, hence the rattling sound made by their loose-fitting valves when shaken, it is nevertheless important that a PCV valve and its connecting hoses do not become clogged with deposits in extended service. If this does occur the crankcase would become congested with blow-by gases, thus hastening contamination of the engine oil. Also, an

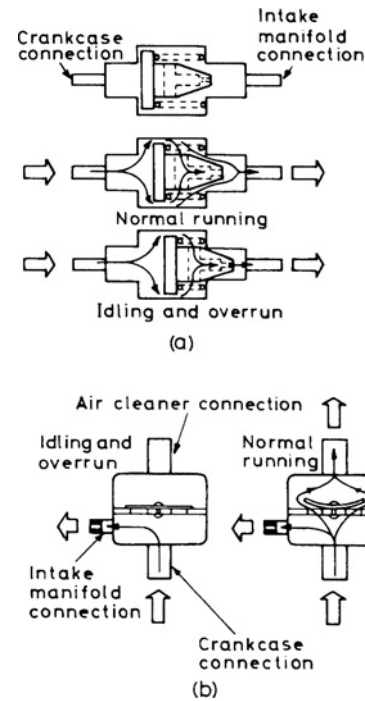


Figure 4.48 Comparison of (a) variable- and (b) fixed-orifice regulator valves

excessive build-up of pressure in the crankcase can jeopardize oil retention to the detriment of oil consumption. It is therefore essential to check for correct functioning of the PCV system at the manufacturer's recommended service intervals, typically every 30 000 miles (48 000 km) or 3 years.

5 Engine cooling, vehicle heating and air conditioning

5.1 HEAT TRANSFER AND COOLING MEDIA

The need for engine cooling

From the viewpoint of converting heat into mechanical energy, it follows that if each piston accomplished its power stroke starting at the temperature of combustion, which for spark-ignition is in the region of 2500°C , such an engine would in theory be highly efficient. To exploit this in practice, however, would impose unacceptably high temperatures on the materials of the working cylinder. In descending order the melting points of cast iron cylinders, steel valves and aluminium pistons are about 1800°C , 1350°C and 900°C respectively, hence all these temperatures would be exceeded in the absence of any cooling.

The behaviour of metals at elevated temperatures also differs from that at normal temperatures. Apart from losing some of their strength they can exhibit a condition known as *creep*, in which the metal deforms slowly and continuously at a constant stress. Excessively high temperatures would in any event cause breakdown of the cylinder lubricating oil films, resulting in the overheated working parts losing their running clearances and seizing up.

For these reasons the engine must be provided with a system of cooling, so that it can be maintained at its most efficient practicable operating temperature. This generally means that the temperature of the cylinder walls should not exceed about 250°C , whereas the actual temperature of the cylinder gases during combustion may, as previously noted, reach many times this figure.

Conversely, there is no merit in operating the engine too cool since this would reduce thermal efficiency and therefore increase fuel consumption. It would also increase oil dilution and hasten corrosion wear of the engine.

It is perhaps worth recalling an observation made many years ago by D.R. Pye, a one-time leading engine specialist, that 'One may go so far as to say that if an engine is not within a narrow margin of overheating at some point or other it is not working at its maximum capacity; or, expressed otherwise, that a change of condition which reduces the cylinder temperatures at critical points under given conditions of running, will raise the limit of power output of which the cylinder is capable.'

Methods of heat transfer

It is a common experience that heat always flows from a hotter to a cooler substance, and a physical principle that the greater the temperature difference between them the more rapidly is the heat transferred, until both substances attain the same temperature. That is, the loss of heat from one substance is equal to the gain in heat by the other. The process

of heat transfer can take place in three ways, all of which are encountered in the engine cooling system, and they may be summarized as follows:

Conduction occurs most readily when heat energy is supplied to a solid substance, through which it is transmitted by direct contact or collision of molecules in the substance itself. The resulting molecular vibration increases as the temperature of the substance is raised and decreases when the temperature is lowered. An *insulator* is a substance that is a relatively poor conductor of heat and therefore acts as a thermal barrier.

Convection occurs when heat energy is supplied to a liquid or a gas and produces density changes in it. These promote *natural convection* currents, because the lower-density warmer matter rises to replace the higher-density cooler matter which descends. The process becomes one of *forced convection* with enhanced heat transfer when the circulation of the liquid is assisted by a pump, or that of the air by a fan.

Radiation occurs when heat energy is transferred by wave motion, there being neither contact of the substances nor molecular vibration within them. Radiant heat energy is emitted by all substances and may be either absorbed or reflected by others. Their ability to emit, absorb or reflect heat depends upon the colour and nature of the surfaces concerned, black rough ones being best for absorption and light polished ones best for reflection.

In practice, motor vehicle engines are designed either for indirect cooling by air through the medium of water or, less commonly, for direct cooling by air. Expressed in everyday language, these two systems are known simply as water cooling and air cooling respectively. Each system possesses certain advantages over the other.

Advantages of water cooling

- 1 Cooling is more uniform because heat is transferred with greater rapidity from the engine metal surfaces to water than it is to air. This can be explained by the specific heat of water being higher than that of air, so that if equal masses of water and air each receive the same quantity of heat, the water will experience the least rise in temperature. Consequently there will be a more rapid transfer of heat as a result of the higher mean temperature differences between the metal surfaces and the water in contact with them.
- 2 Cooling is more constant because the time taken for the water to rise through a given temperature range is longer than that for the same mass of air. This property of the

water possessing a greater *thermal inertia* than the air can be advantageous in maintaining a more nearly constant operating temperature – that is, when the rate of heat emission equals the rate at which heat is generated in the engine.

- 3 Interior heating for the vehicle is improved because outside air may be directed through a heat exchanger matrix, which can be conveniently heated by circulating a portion of the engine coolant through it.

Advantages of air cooling

- 1 Warm-up of the engine is more rapid because of the lower specific heat of the air cooling medium; heat is less readily transferred from the engine metal to the air being circulated around it, as a result of their lower mean temperature difference.
- 2 The system is inherently more reliable because air cooling is immune to either freezing or boiling of the coolant around the cylinder heads and cylinders, and to the loss of coolant. It is also free from any build-up of corrosive products that can restrict coolant passages.
- 3 Less maintenance is required in service because there is neither the requirement to check the cooling medium for level and condition, nor the need to inspect rubber connecting hoses for signs of leakage.

Certain criticisms that are customarily levelled at air cooling should be treated with some reserve: these include less reliability for heavy-duty application, greater power consumption and noisier operation. In the long-established and highly regarded range of heavy-duty Deutz air-cooled diesel engines, produced by the German firm of Klöckner-Humboldt-Deutz AG, the fully integrated air-cooling system has been found to have a substantially lower energy consumption than any water-cooled unit of equivalent output. Independently conducted scientific tests have also determined that these engines are among the quietest power units of equivalent output. Air cooling has also long been successfully used on the compact V eight-cylinder engine used in the Czechoslovakian Tatra car. It is perhaps of interest to note that when tested in the 1980s by a British motoring journalist, this car was praised for its 'remarkable silence'.

5.2 ENGINE AIR-COOLING SYSTEM

Circulation of cooling air

With air cooling, the engine structure is directly cooled by inducing air to flow over its high-temperature surfaces. These are finned to present a greater cooling surface area to the air (Figure 5.1), which in non-motor-cycle applications is forced to circulate over them by means of a powerful fan (Figures 5.2a and b). The car engine structure is almost entirely enclosed by sheet metal ducting, which incorporates a system of baffles. A similar arrangement is used with the diesel engine (Figure 5.3).

These baffles ensure that the through flow of air is properly directed over the cooling surfaces of the cylinders and cylinder heads. To maintain uniform temperatures, the air is forced to circulate around the entire circumference of each cylinder and its cylinder head, the direction of flow being along

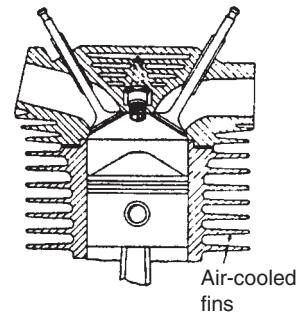


Figure 5.1 An air-cooled cylinder and head

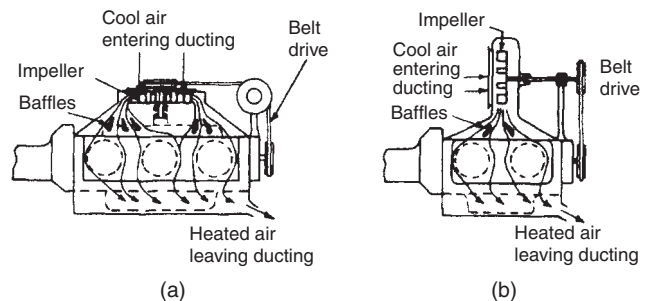


Figure 5.2 Air-cooling systems with (a) vertical (b) horizontal bearing axis for impeller

the cooling fins. These are greatest in number, consistent with providing a sufficient area of flow, on the high-temperature surfaces of the cylinder head in the region of the exhaust valve. The complete system forms what is known as a *plenum chamber* in which the internal air pressure is higher than that of the atmosphere.

Finally, the heated air is discharged from the plenum chamber to the atmosphere, or redirected to heat the car interior.

Air-cooling fan assembly

The forced circulation of air around the engine is generally provided by a centrifugal fan or impeller, which rotates in a spiral-shaped housing. This type of fan is capable of overcoming the appreciable resistance offered to the air as it flows around the ducted and finned cylinders and cylinder heads. The fan is driven by a V-belt and pulley system from the engine.

During operation the air enters at the eye of the impeller via an inlet housing, and flows between each pair of blades. Centrifugal force thus acts upon the enclosed air, which is then discharged under pressure in a radial direction (Figure 5.4). To increase fan efficiency the impeller blades may be curved backwards instead of being straight. The fan housing is also fitted with an inlet ring to minimize any recirculation of air at the point of entry.

Air-cooling throttle valve

This regulates the quantity of air entering the cooling fan, in accordance with engine cooling requirements. It usually takes the form of a movable throttle ring, which acts as a baffle surface at the entrance to the inlet housing (Figure 5.4).

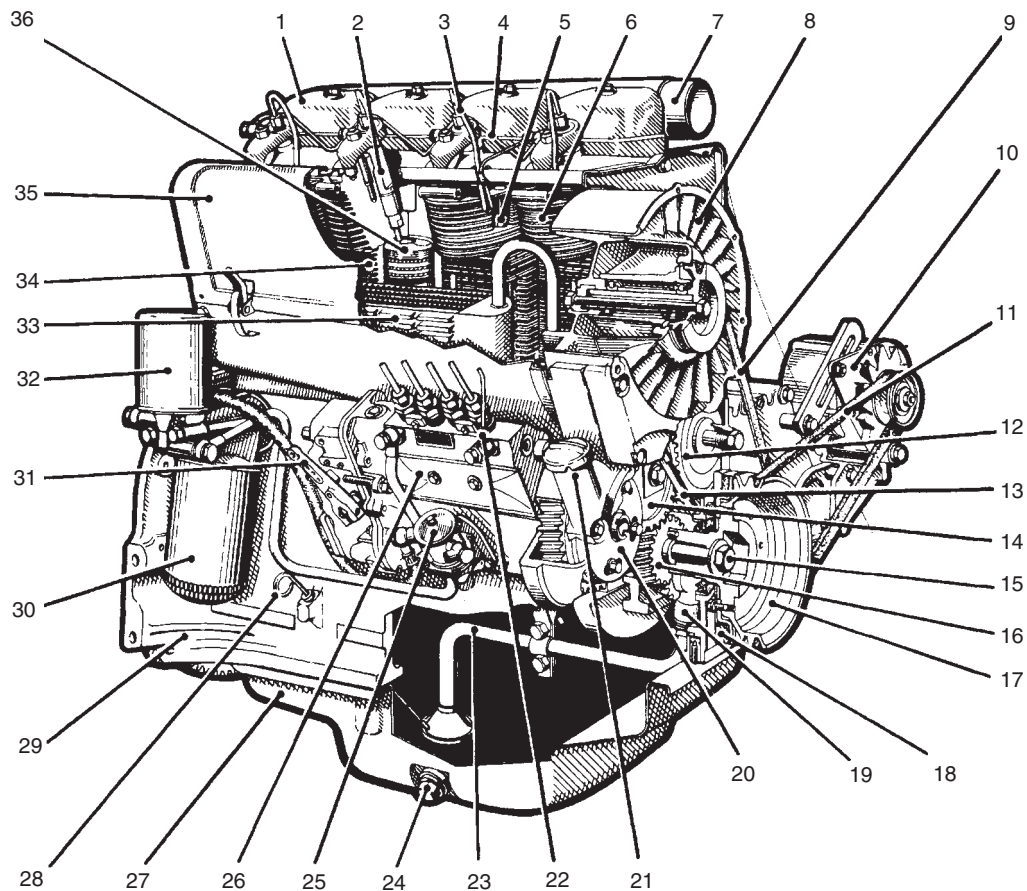


Figure 5.3 Cut-away view of modern air-cooled diesel engine (*Deutz*)

- | | | |
|---|---|--|
| 1 rocker chamber cover | 14 idler gear (driving injection pump and camshaft) | 26 Bosch in-line injection pump with mechanical centrifugal governor |
| 2 injector | 15 anti-fatigue bolt (securing V-belt pulley to crankshaft) | 27 oil sump (Sheet metal or cast iron) |
| 3 injection line to no. 3 cylinder | 16 crankshaft gear | 28 oil dipstick |
| 4 back-leakage line | 17 V-belt pulley | 29 crankcase (cast iron) |
| 5 cylinder head anti-fatigue bolt (four bolts securing each cylinder head with cylinder to crankcase) | 18 vibration damper | 30 oil filter |
| 6 cylinder head (light alloy) | 19 oil pump | 31 speed control lever |
| 7 air intake manifold | 20 injection pump drive gear with advance/retard unit | 32 fuel filter |
| 8 cooling blower (V-belt driven) | 21 oil filler neck | 33 integral oil cooler |
| 9 cooling blower V-belt | 22 overflow line | 34 finned cylinder (grey cast iron), separately removable |
| 10 generator (dynamo or alternator) | 23 oil suction pipe | 35 removable air cowling |
| 11 generator V-belt | 24 oil drain plug | 36 piston |
| 12 camshaft gear | 25 fuel feed pump | |
| 13 oil gallery | | |

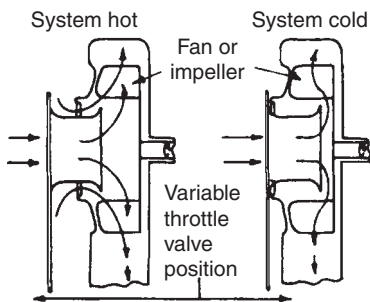


Figure 5.4 Air-cooling fan and throttle value

At normal running temperatures, the throttle ring is retracted from the eye of the impeller to allow free access for the incoming air. When the engine is cold, the throttle ring is advanced towards the eye of the impeller, thus reducing the quantity of air circulated around the engine.

Air-cooling thermostat

Automatic operation of the cooling fan throttle valve is effected by a thermostat, which therefore serves to regulate the rate of engine cooling. It is linked mechanically through a simple leverage system to the throttle valve, so that adequate

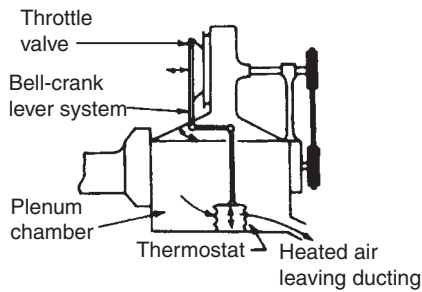


Figure 5.5 Air-cooling thermostat

movement is transmitted to the ring within the available working stroke of the unit. The thermostat is generally installed in the hot air duct leading from the lower part of the plenum chamber (Figure 5.5).

5.3 ENGINE WATER-COOLING SYSTEM

Circulation of cooling water

A very simple form of engine water-cooling system, known as natural or *thermosyphonic* circulation (Figure 5.6), was used in many early motor vehicles up to the late 1930s and provides a useful introduction to the topic. This type of water-cooling system consisted of the following components:

Water jacket This was formed partly in the cylinder block and partly in the cylinder head and surrounded the cylinder walls, combustion chambers and valve ports. It was provided with a lower inlet and an upper outlet connection, the latter sometimes being called the header pipe.

Radiator The purpose of the so-called radiator is to dissipate the engine heat rejected to the coolant. This it does mainly by conduction and convection, rather than by radiation. It originally consisted of an upper header tank that received the heated coolant from the engine; a matrix or core that served to disperse the down flow of coolant into fine streams and also provided for a through flow of cooling air; and a lower collector tank from which coolant was returned to the engine jacket.

Flexible connections Rubber hoses with clips were used to connect the cooling jacket header pipe to the inlet pipe of the radiator header tank and the outlet pipe of the radiator collector tank to the inlet elbow of the cooling jacket.

Other fittings The radiator header tank was fitted with a filler cap and an overflow pipe and the collector tank with a drain tap. Suitable mounting lugs were also provided at the top and bottom of the radiator.

Radiator fan In many, but not all, early thermosyphonic cooling systems a fan was mounted behind the radiator and driven by belt and pulleys from either the crankshaft or the camshaft. Its purpose was to assist the through flow of air normally resulting from the forward motion of the vehicle and also create an air flow when the vehicle was stationary with the engine idling.

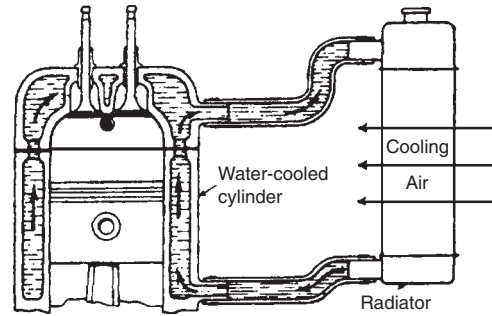


Figure 5.6 The thermosyphonic system of water cooling an engine

Operating principle of the thermosyphonic system

Upon starting the engine from cold, the following sequence of events occurs:

- 1 Cold water in the jacket begins to absorb surplus heat from the engine and expands slightly.
- 2 Since the heated water becomes less dense than the cold water in the jacket, it rises to the header pipe and then flows upwards into the header tank of the radiator.
- 3 This heated water then begins to be forced down in fine streams through the radiator matrix, as a result of more water being heated in the jacket and rising behind it.
- 4 The heated water passing through the radiator is then cooled by the throughflow of cooling air, so that it becomes denser again and readily sinks to the collector tank of the radiator.
- 5 Cooled water is then raised from the collector tank to the inlet of the jacket as a result of the pressure difference created by other cooled water sinking in the radiator and heated water rising in the jacket again.

The rate at which water circulates in the simple thermosyphonic cooling system is in proportion not to engine speed but to the heat output or load placed on the engine. As the engine load increases the circulation becomes more rapid, mainly because the formation of small steam bubbles have the effect of further reducing the density of the coolant being heated.

Purely as a matter of historical interest, perhaps the best remembered application of thermosyphonic cooling was to the legendary Model T Ford car, which remained in production for nearly twenty years up to 1927. Of great simplicity in construction, this was the first car to be mass produced in the true sense of the word. It was also the one about which Henry Ford made his famous remark that any customer could have a car painted any colour that he wanted so long as it was black.

Advantages and disadvantages of thermosyphonic cooling

- 1 It is a simple arrangement of low cost.
- 2 Warm-up is fairly quick.
- 3 Circulation is proportional only to engine load.

- 4 Circulation is not positive.
- 5 Large water passages are required.
- 6 The radiator header tank must have a high location relative to the jacket.

Forced-circulation cooling system

It has long been established practice to include a coolant (or 'water') pump in the engine cooling system to provide what is termed forced circulation. We must now attempt to justify in a little more detail the additional complication of a pump in the cooling system.

With forced circulation it is possible to direct the principal cooling effect to the high-temperature surfaces in the cylinder head, whilst maintaining the cylinder block at acceptable working temperatures by natural circulation. If a relatively high rate of coolant flow is promoted around the exhaust valve seats, exhaust ports and sparking plug bosses, then any tendency towards localized overheating is minimized. From considerations of engine durability it must be appreciated that both the mechanical strength and the corrosion resistance of valve materials decrease with increased operating temperatures. A sufficiently high coolant pressure must also be maintained in these critical cooling regions, so as to avoid the formation of steam pockets that would restrict circulation and impede heat transfer.

The increased rate of coolant flow provided by forced circulation not only improves cooling efficiency, but also permits a reduction in the quantity of coolant required. Hence, the coolant passages in the cylinder block and head may be made correspondingly smaller in size, thus reducing both the bulk and the weight of the coolant jacket. Furthermore, the radiator can be reduced in size, because a higher mean temperature can be maintained between the faster flowing coolant and the air passing through its matrix. This in turn makes for greater convenience of installation, especially where the requirements of modern styling demand a low bonnet line, and also in designs where the radiator is sited alongside a transversely mounted engine.

Advantages and disadvantages of forced-circulation cooling

- 1 It is more complicated and costly arrangement.
- 2 Warm-up is quick as long as a thermostat is used in the system.
- 3 Circulation is proportional to both engine load and speed, the former as long as a thermostat is used in the system.
- 4 Circulation is positive.
- 5 Smaller water passages and radiator are permissible.
- 6 Radiator header tank may be lower with respect to the jacket.

Forced-circulation flow patterns

In practical application the forced-circulation cooling system consists of numerous communicating passages formed within the engine coolant jacket, through which the coolant is circulated around the cylinder walls and combustion chamber regions. To minimize thermally induced stresses in these parts, it is usual for the lower-temperature coolant entering the jacket to be initially circulated over the cooler metal surfaces of the engine. The coolant flow is therefore directed around the

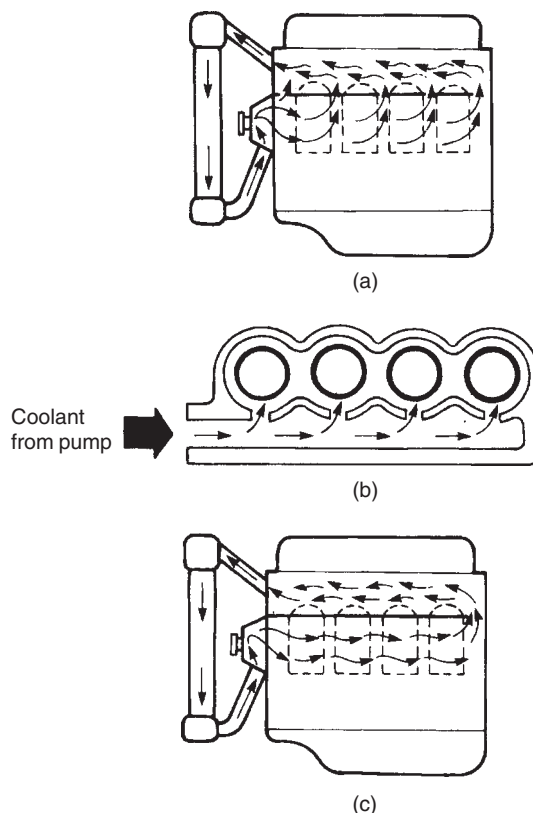


Figure 5.7 Comparison of (a) parallel flow (b) independent parallel flow (c) series flow of coolant between cylinder block and head

cylinder block and then upwards into the cylinder head. In the case of V and horizontally opposed engines, the pump delivers coolant simultaneously to both banks of cylinders. To maintain uniform temperatures throughout the engine, the flow pattern provided between the cylinder block and head can take three forms: parallel, series and reverse.

Parallel flow

In this widely used pattern each cylinder receives a small proportion of the coolant entering at the front end of the cylinder block. After flowing around the 'separately' served cylinders, the coolant then passes upwards through ducts to the corresponding cylinder head regions and thence to the outlet (Figure 5.7a).

More recently there has been a trend to ensure that the separate cooling of the cylinders is done more positively, by directly routing the flow or coolant to each cylinder (Figure 5.7b). This provides for more uniform cooling of the cylinders to reduce any tendency towards detonation and is sometimes known as independent cylinder cooling. The coolant may be arranged to enter each cylinder chamber in a tangential direction, which causes it to swirl around the cylinder barrels as it travels upwards into the cylinder head. This swirling motion of the coolant discourages the accumulation of sludge and rust, since these would greatly reduce cooling efficiency. Mention has earlier been made of directed coolant flow in the cylinder head.

In a variation of this arrangement a dual parallel flow pattern is used, where the coolant from the pump is directed into

the front ends of both the cylinder block and head. The parallel flows then continue along the length of their jackets before combining and returning to the radiator, via a rear mounted thermostat. This arrangement is intended to provide improved cooling for the cylinder head, since about 70 per cent of the heat rejected by the engine to the coolant originates from this source.

Series flow

With this less commonly used pattern, all the coolant entering at the front end of the cylinder block circulates around all the cylinders in turn. It then passes upwards through ducts at the rear end of the cylinder block, whence it flows back along the entire length of the cylinder head to the outlet (Figure 5.7c). The advantages of this particular arrangement are that the cylinder head gasket is required to function only as a gas seal and the absence of coolant passages between the main portion of the cylinder block and the head makes for greater rigidity of these components.

Reverse flow

In the early 1990s General Motors in America, contrary to earlier established principles of engine cooling, introduced a reverse flow system in which the coolant is directed first into the cylinder head and then down into the block. The advantages claimed for this system are cooler cylinder head temperatures and warmer cylinder block temperatures, which respectively raise the detonation limit monitored by the knock control system (Section 3.2) and lower the frictional drag of the piston rings (Section 1.7), thereby reducing the fuel consumption. To minimize thermally induced stresses that could otherwise result from the lower temperature coolant being directed first to the much hotter surfaces of the cylinder head, the thermostatic control is arranged to act on the inlet rather than the outlet side of the coolant pump.

It should be appreciated that the cooling systems developed for modern passenger cars must not only dissipate the greater amounts of heat generated by multi-valve cylinders and turbocharger installations, but also be accommodated in the limited space available in the increasingly crowded engine bay.

Location of the coolant pump

It has long been established practice for the coolant pump to be located on the front face of the cylinder block (Figure 3.10). Until the more recent advent of the radiator mounted and electrically driven fan, it was usual for both the pump and the fan to share a common belt-driven spindle. An advantage of this particular location is that the pump output can be conveniently directed through an internal ducted passage, which not only distributes the coolant flow as evenly as possible among the cylinders, but also directs a vigorous flow upwards to the high temperature regions of the cylinder head.

Coolant pump construction

A non-positive pump of the centrifugal type is invariably used to circulate the coolant, its advantages for this application being as follows:

- 1 It characteristically provides a continuous rather than a pulsating flow.

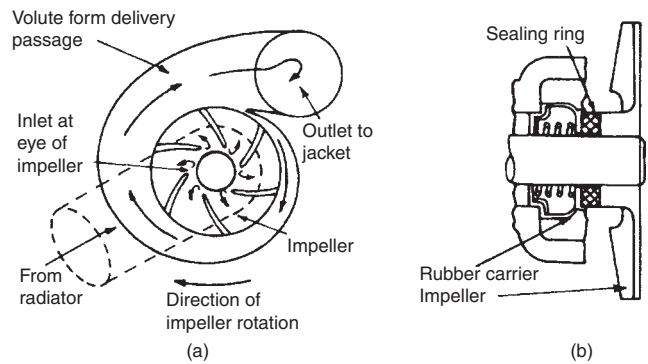


Figure 5.8 Centrifugal coolant pump: (a) action (b) sealing

- 2 Being non-positive (unlike a gear-type oil pump) it neither builds up an excessive pressure, nor overloads its driving mechanism, if resistance to coolant flow is increased anywhere in the system.
- 3 It continues to provide a limited passage for thermosyphonic circulation of the coolant in the event of failure of its driving mechanism.

An engine-driven pump of this type circulates the coolant by means of either a metal or a plastics impeller, which revolves in a spiral chamber (Figure 5.8a). To provide a uniform cross-sectional area for the spaces between the impeller vanes, their height is gradually reduced from inner to outer edges. Further the vanes may be curved backwards instead of being straight, since this reduces turbulence as the coolant enters and leaves the impeller. Also, the spiral chamber may be of volute form to obtain a delivery passage of progressively increasing cross-sectional area. This feature serves to reduce the velocity of the discharging coolant, so that its delivery pressure is correspondingly increased.

To prevent leakage of coolant and to exclude air from the system, the now long-established face seal is used (Figure 5.8b). Mechanical seals of this type originally consisted of a stationary counterface of either cast iron or stainless steel, spring loaded against which was a sealing ring of hard carbon material revolving with the impeller. The sealing ring was bonded to a synthetic rubber carrier to prevent coolant leakage along the impeller spindle. In a modern inverted version of this sealing method, a semi-cartridge construction is employed to support and guide the spring loaded sealing ring or primary seal ring as it is sometimes termed, which is then integrated with a synthetic rubber sealing bellows to become the stationary member, the bellows readily allowing the sealing ring freedom of axial movement to compensate for any misalignment, or wearing, against the counterface or mating ring that now rotates with the impeller (Figure 5.9). A further development of the cartridge type of construction is the 'unitized' assembly, which is handled as a one-piece item on build. It differs only in that a pair of lipped telescopic members are used to support and guide both the stationary sealing ring and its rotating mating ring, the inner supporting member for the latter being made an interference fit on the impeller shaft.

The materials used for the pump seal counterface or mating ring are generally harder than that for the primary seal ring.

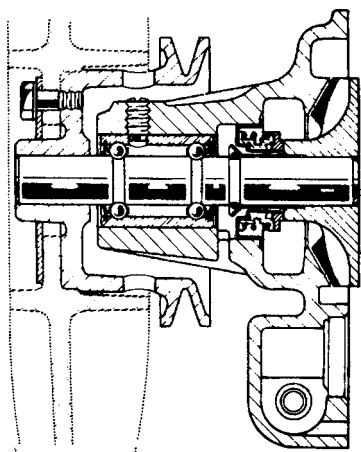


Figure 5.9 Cross-section through a modern coolant pump with a face seal (Alfa-Romeo)

They must be resistant to corrosion and currently may be produced from either sintered or cast metals, or an aluminium oxide ceramic material, and with sprayed coatings also sometimes being used. For the primary seal ring a carbon graphite material, which may be impregnated with a metallic content, continues to offer an acceptable service life with the required resistance to wear that derives from its inherent self-lubricating property, good thermal conductivity and resistance to physical degradation when exposed to the hot coolant. It nevertheless requires sympathetic handling on build, because it is a material that can be easily damaged.

To compensate for high coolant pressure tending to oppose that exerted by the spring on the seal, the sealing ring assembly is also now designed so that the pressure at the eye of the impeller has a limited access to the nonsealing face of the ring. Hence an increase in coolant pressure then assists the spring loading applied to the sealing ring instead of opposing it, thereby reducing the possibility of leakage across the sealing faces. In this context, it should be appreciated that an extremely thin film of coolant must exist between the rubbing faces of the seal to act as a lubricant. A small drain hole is provided in the coolant pump housing between the seal and spindle bearing, so that the latter is protected against any coolant weeps.

Coolant pump operation

In operation, it must always be arranged for the coolant level to be above that of the pump inlet, since this type of pump is not self-priming. The coolant enters the pump near the drive spindle and is admitted to the eye of the revolving impeller, following which it becomes trapped in the spaces between the vanes and pumping chamber wall. A whirling motion is thus imparted to the coolant, which is accelerated to a high velocity and discharged under centrifugal action in the delivery passage. Since the delivery pressure is higher than that existing at the eye of the impeller, a continuous flow of coolant is maintained through the pump. Coolant delivery is not quite proportional to pump speed, since some of the coolant will recirculate to the inlet side via the internal working clearances of the pump. A maximum flow rate of 2.5 to 3.0 litres per

minute for every kilowatt of power (3.3 to 3.9 pints per minute for every horsepower) developed by the engine is typical for motor car petrol engine practice. These flow rates are generally lower at 1.75 to 2.25 litres per minute for every kilowatt of power (2.3 to 3.0 pints per minute for every horsepower) developed in the case of diesel engines, because they reject less heat to the coolant. A relatively greater rate of flow is required for indirect- as compared with direct-injection diesel engines, owing to the more severe heating of their cylinder head valve bridge portions.

Coolant pump misbehaviour in service

Modern coolant pumps are usually maintenance free, and in the event of their unsatisfactory operation, causing engine overheating, it is rare for a manufacturer to advise any other course of action than fitting a service replacement unit. The pump may develop such faults as the following:

Leakage

Abrasive particles originating from metal erosion carried around by the circulating coolant and scoring the seal contact faces.

Worn spindle bearing allowing misalignment to occur between the seal contact faces.

Inefficiency

Metal erosion gradually destroying the form of the impeller vanes.

Excessive end float of the impeller spindle in its bearing increasing the working clearance between the impeller vanes and the pumping chamber.

Apparent inefficiency of the pump caused by a worn or excessively slack driving belt.

Inoperative

Mechanical derangement such as a cracked impeller loosening on its spindle or the pulley adaptor doing likewise owing to loss of interference fit.

Failure of the pump driving belt.

Noisy

Wear in the impeller shaft bearing or seal assembly; also air ingress via the latter.

Cooling fan

The purpose of the fan is to maintain an adequate air flow through the radiator matrix, especially at low vehicle speeds and under engine idling conditions with the vehicle stationary. In other words, the fan serves to speed up the natural draught through the radiator or, alternatively, to create a draught where none would otherwise exist. At high vehicle speeds, the ram effect created by the rapid motion of the car through the air usually promotes sufficient air circulation through the radiator.

A multibladed propeller fan is used to promote an axial flow of air through the radiator of a water-cooling system, its blades normally being irregularly spaced to reduce noise level as fan speed rises. Conventionally, the fan is of the puller type; that is, it is mounted between the radiator and engine so that air is pulled through the radiator as a result of the depression created on the inlet side of the fan. Less commonly, a pusher fan with

its blades set at an opposite angle may be required, typically to push the air through the radiator of a rear-engined bus.

Fixed-drive fan

Although now mainly confined to commercial vehicle engines, the once conventional engine-driven fan is generally operated in tandem with the coolant pump, both sharing a common drive spindle that is rotated by a V-belt and pulley system from the crankshaft. The relative speed of the fan ranges from slightly less to rather more than that of the engine, depending on the particular speed ratio chosen for its pulley drive arrangements. Where space limitations sometimes demand a very low positioned radiator, such as in the case of a forward-control heavy vehicle, the fan may be directly mounted on the crankshaft nose (Figure 5.27).

The fan assembly has often comprised a one-piece steel pressing clamped to the face of the coolant pump drive pulley and spigot located on the end of its spindle. In modern practice the fan may be produced from either an aluminium alloy casting or a plastics moulding with a metal boss insert, to provide a lighter, more easily balanced construction with an improved blade form for greater efficiency. The fan is made as large as practicable, not only for efficient utilization of the available radiator matrix area, but also because the volume of air it displaces increases as the cube of its diameter; thus (in an exaggerated sense), a fan of double the diameter would displace eight times as much air.

Variable-drive fan

Although a fixed-drive fan can be designed to provide an adequate air flow through the radiator under arduous operating conditions, such as during prolonged traffic crawling, its continued operation under more normal running conditions can bring several disadvantages. First, the noise level rises. Second, the engine power absorbed in driving the fan will be greater, especially when it is considered that this increases as the cube of fan speed. There may also be a tendency to over-cooling; the latter is to be avoided because engine life, lubricating efficiency and vehicle heater performance can suffer as a result. To minimize these disadvantages an automatic control is sometimes employed to confer a variable-speed operation on the fan.

The variable-speed fan may be of either the torque-limiting or the temperature-sensitive type, their common objective being to ensure that the fan is not driven at a speed in excess of that required to maintain an optimum cooling system temperature. Both types of fan embody a viscous coupling, which permits relative motion or slip to occur between the driving and the driven members. (It may be recalled here that we have previously encountered this type of coupling in the viscous torsional vibration damper, as fitted to some diesel engine crankshafts.) A driving disc is mounted directly from the coolant pump pulley and revolves within a sealed coupling chamber to which the fan is attached (Figure 5.10a). The coupling chamber is freely mounted on the coolant pump spindle and partially filled with a highly stable silicone fluid, which characteristically retains a nearly constant viscosity with varying operating temperatures.

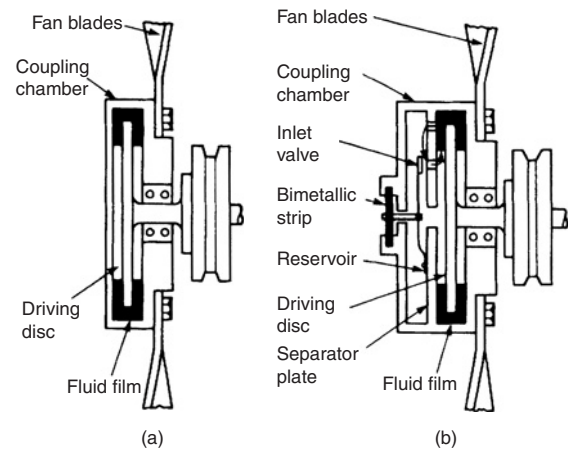


Figure 5.10 Variable-drive fans (a) torque limiting and (b) temperature sensitive

During operation of the simpler *torque-limiting fan* drive the driving disc initially revolves in the stationary mass of fluid and sets up a viscous drag, so that the fluid begins to circulate within the coupling chamber and owing to centrifugal action, moves outwards to fill the narrow gaps between the driving faces of the coupling members. Since these faces are not only arranged to be in close proximity to one another, but also may have mating grooves, torque is transmitted between them by the viscous drag of the silicone fluid, and hence a drive to the fan is established. This drive is sustained with increasing engine speed, until the resisting torque imposed upon the fan by its air flow rises to an extent that overcomes the viscous drag of the silicone fluid and allows the coupling to slip. Maximum fan speed is therefore governed by the ultimate slipping action of the coupling, the latter being predetermined by the viscosity of the silicone fluid and the degree to which the coupling is filled.

The more comprehensive *temperature-sensitive fan* drive differs from the torque-limiting type in that provision is made for automatically varying the degree to which the active part of the coupling is filled. This overcomes the disadvantage that the fan has to operate within a fixed speed range, regardless of cooling system demands. Otherwise, the air flow generated by the fan may be inadequate when vehicle speed is low and engine speed is high, while at low engine speed the uninterrupted drive to the fan may prolong the engine warm-up period.

In construction the coupling chamber of the temperature-sensitive fan is divided into two by a separator plate, thereby forming a working chamber for the driving disc and a reservoir (Figure 5.10b). The fluid in the working chamber is provided with a permanent escape route from an outlet port near the periphery of the separator plate, so that by natural inclination the coupling does not 'want' to drive the fan. However, the working chamber is also provided with an inlet port that is less distant from the centre of the driving disc. This port can be opened and closed by a thermostatically actuated valve, which moves in response to the flexing of a bimetallic strip mounted on the front of the coupling and therefore exposed to the air that has been heated in its passage through the radiator airways. By sensing the coolant temperature in this manner,

the bimetallic strip can be arranged to open the inlet port to the working chamber when the engine coolant becomes too hot. Fluid being expelled to the reservoir can then be continuously readmitted to the working chamber to restore viscous drag against the driving disc and therefore bring the fan into operation.

Off-on drive fan

This type of fan drive is sometimes used on the diesel engines of heavy vehicles to assist quick warm-up of the coolant and maintain a constant engine temperature. It basically comprises a small, spring-loaded, single- or multiplate friction clutch that is built into the pulley drive for the fan. While the coolant remains below a predetermined temperature, a thermostatic control valve located in the radiator top hose directs air from the vehicle's compressed air supply to the fan hub, which overcomes the spring loading on the clutch and disengages the fan from its pulley drive. The fan will now idle until the coolant temperature rises above a predetermined level, at which point the air supplied under pressure to the fan clutch is cut off by the control valve and the spring engages the friction elements to drive the fan for normal cooling.

V-belt drives

It has long been established practice to employ either a single, or a matched pair, of endless V-belts for transmitting drive from the engine crankshaft to the coolant pump and where applicable the cooling fan. The drive also includes various other engine-driven auxiliaries such as the alternator, which usually incorporates in its mounting an adjustment facility for initial tensioning of the belt(s). Driving power is transmitted by the friction forces arising from the wedging action of the V-belt in its pulley grooves. The advantages of employing this type of belt drive include minimum space requirements, quietness of operation at relatively high speeds, and the ability to absorb shock loads.

In cross-section the V-belt may be divided into a neutral load-carrying zone, a lower compression zone and an upper tension zone (Figure 5.11a). The load-carrying members comprise a layer of high-tensile cords of a textile material, this typically being polyester, which thus locate the neutral axis of the belt. These cords are embedded in a synthetic rubber core that both protects and supports them. The core consists of a lower supporting cushion of relatively stiff rubber and an upper layer of more resilient rubber. This particular construction enables the tensile loading in the cords to be transmitted as groove side-wall pressure in the compression zone of the belt, whilst also being able to accommodate the repeated flexing of the belt in the tension zone as it bends around the pulleys. A wear-resistant covering of rubberized fabric material completes the build-up of the V-belt.

Since it first appeared on an American commercial vehicle engine in the late 1970s, the multiribbed V-belt has become increasingly widely used (Figure 5.11b). The advantages associated with this type of belt construction are longer service life, better maintenance of tension, higher transmission efficiency and greater flexibility, which allows the use of smaller-diameter pulleys.

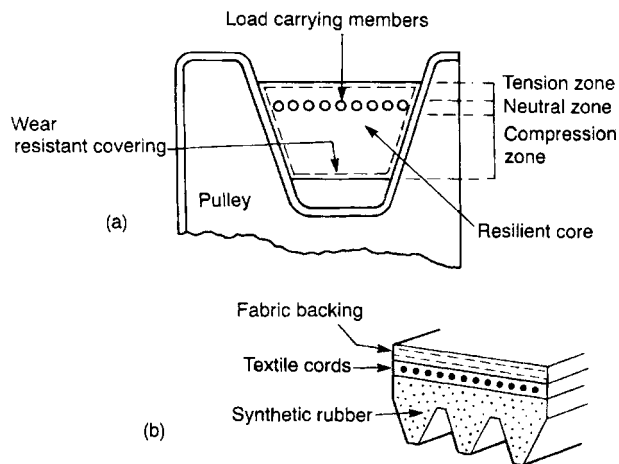


Figure 5.11 V-belt constructions: (a) single (b) multiribbed

Electrically driven fan

The separate electrically driven fan is now widely used in passenger car engine cooling systems. This can be directly mounted from the radiator and switched in and out of operation in accordance with cooling system demands. The switching is thermostatically controlled in response to changes in engine coolant temperature, a sensor being installed for the purpose in the lower half of the radiator. Normally the fan will continue running for a few minutes after the engine has been switched off, this period sometimes being referred to as 'post ventilation'. For greater flexibility of control a two-speed fan and for some high-powered engines twin fans may be used, the latter being individually controlled according to cooling demand. Fuel economy with an electrically driven fan should therefore be improved because the engine better maintains an optimum running temperature, the fan only operates above a preset temperature so that warm-up is quicker, and the fan consumes less energy. The intermittent operation of the electrically-driven fan also makes for generally less noise, although when brought into action at lower engine speeds it may actually be noisier than an engine-driven fan if it has a fixed higher operating speed. An interesting feature of the twin-fan installation for the cooling system of the 2000 Cadillac Northstar V8 engine is that their blades have been acoustically developed with a special curvature, so that operation of the fans is not heard by the car occupants. Not the least advantage of an electrically driven fan is its greater convenience of mounting position; this, of course, accounts for its wide-spread use with modern transversely mounted engines (Figure 5.12).

Fan cowl

A sheet metal or plastic cowl is often made an integral part of the radiator and cooling fan installation. The cowl is funnel shaped to surround the rear face of the radiator matrix and encircle the tips of the fan blades. Its purpose is to contribute to effective operation of the fan by preventing heated air from recirculating through the radiator matrix, which could otherwise lead to engine overheating.

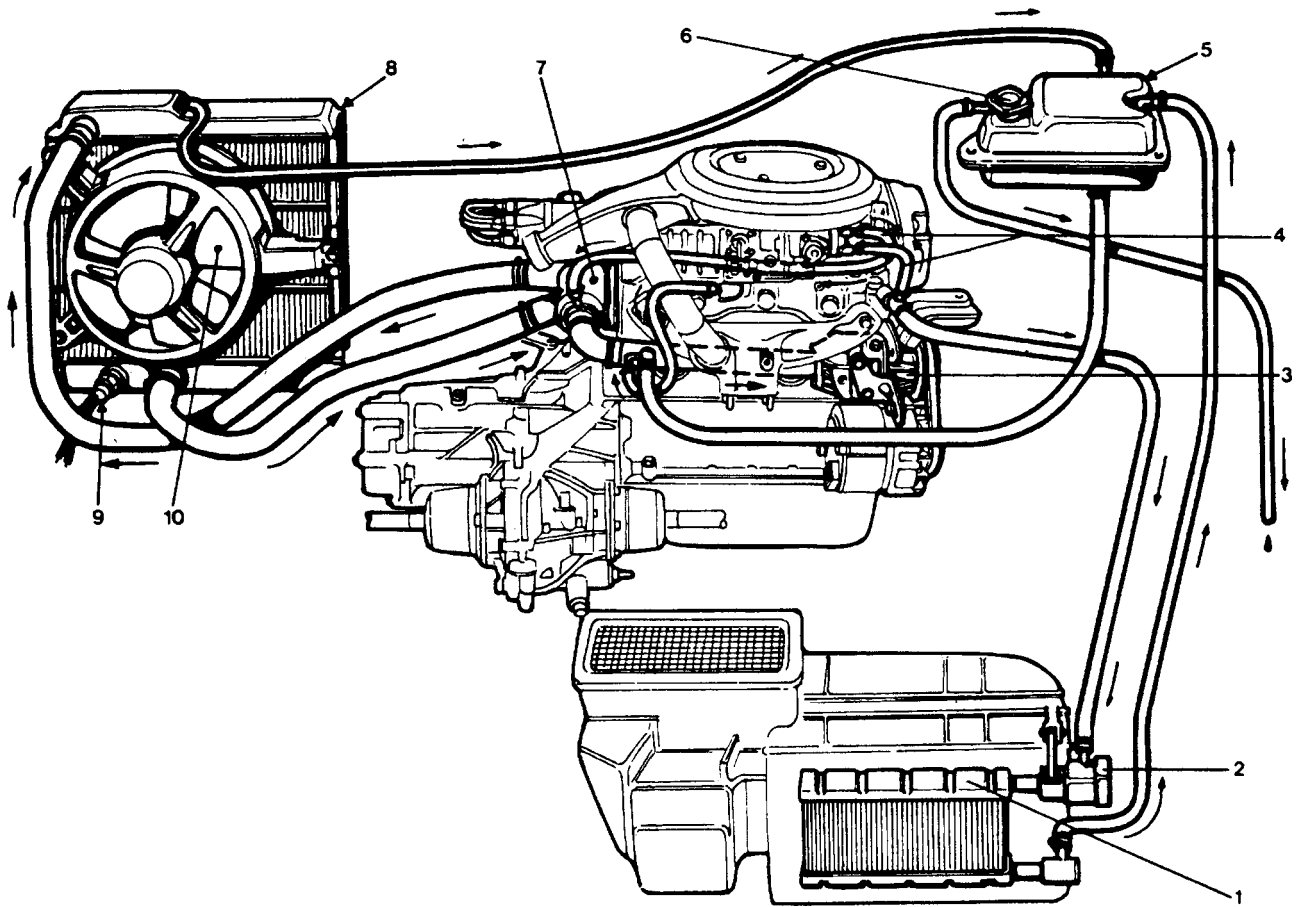


Figure 5.12 Electrically driven fan installation (*Fiat-Lancia*)

- | | |
|------------------------------------|-----------------------------------|
| 1 heater core | 6 cap |
| 2 heater core cock | 7 blending thermostat |
| 3 water pump | 8 cooling radiator |
| 4 semi-automatic choke water hoses | 9 electrofan control thermoswitch |
| 5 filling tank | 10 electrofan |

Radiator construction

The purpose of the so-called radiator is to dissipate the engine heat rejected to the coolant. This it does mainly by conduction and forced convection rather than by radiation. A cooling effect from the radiator is achieved by arranging for the entering heated coolant to be dispersed into fine streams through its matrix or core. Relatively small quantities of coolant are therefore brought into contact with large surface areas of metal. Even greater surface areas of metal must be exposed to the through flow of cooling air, since there is appreciably greater resistance offered to the transfer of heat from the metal surfaces to the air, than from the coolant to the metal surfaces. For this reason the matrix waterways are provided with secondary heat transfer surfaces in the form of metal fins. They are generally shaped to promote turbulence in their airways, which further assists the process of heat transfer between the matrix metal surfaces and the air flowing over them.

Types of radiator matrix

In early practice the construction of the radiator matrix resembled that of a honeycomb. It consisted basically of a

stack of air tubes, the ends of which were bulged and sealed together, so that continuous waterways were formed between them (Figure 5.13a). Although this form of construction lent itself to a variety of pleasing radiator shapes now much admired by historic car enthusiasts, in practice its cooling efficiency was limited by the absence of secondary heat transfer surfaces. The honeycomb radiator was therefore eventually displaced by other forms of matrix construction of which there are three basic types, as described in the following sections.

Cellular or film

With this type of matrix the airways and secondary heat transfer surfaces are formed by a series of zigzag ribbons of copper (Figure 5.13b). Each row of airways is bounded along its sides by a corrugated copper strip. The inward facing peaks of the corrugations act as spacers for the air fins, whilst the outward-facing peaks are relieved across part of their width to form waterway channels. Therefore by interlocking successive rows of complete airways any desired number of continuous waterways may be built up, through which the coolant flows in thin films. This type of matrix construction was once widely

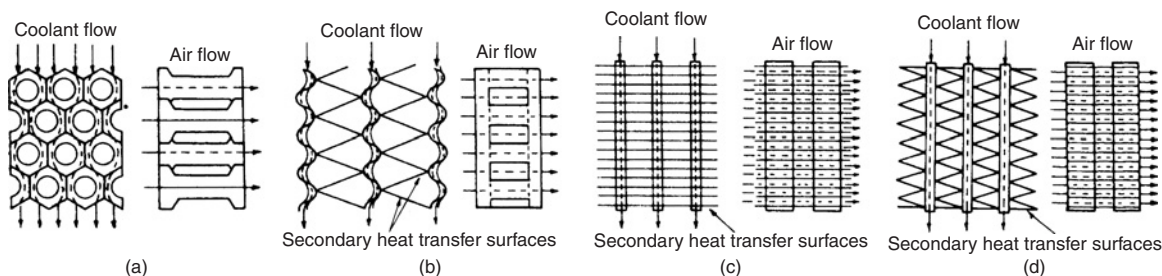


Figure 5.13 Types of radiator matrix: (a) honeycomb (b) cellular or film (c) tube and flat fin (d) tube and corrugated fin

used and generally provided a high rate of heat dissipation for a minimum weight of radiator.

Tube and flat fin

For this type of matrix, also known as the tube and plate type, a series of either in-line or staggered coolant tubes are employed. These in turn are assembled through a stack of continuous air fins, which are spaced fairly closely together and constitute the secondary heat transfer surfaces (Figure 5.13c). The coolant tubes are commonly produced from thin brass strip and assume a flattened oval section, whilst thinner copper strip is used for the air fins. Louvres may be pierced in the latter, so as to promote turbulence in the airways that they form with the coolant tubes. Greater structural strength is usually claimed for this type of matrix, as compared with the cellular or film type, this being an important requirement with the advent of pressurized cooling systems in the late 1940s.

Tube and corrugated fin

Also known as the tube and serpentine fin or the pack type this matrix construction is widely used in modern practice. It comprises alternate rows of waterways and airways, which are provided by brass tubes and zigzag ribbons of copper respectively (Figure 5.13d). Again, the coolant tubes are of flattened oval section and the air fins may be louvred to promote turbulence. There may be two or more rows of tubes, but in more recent practice the requirement for lighter and smaller radiators has led to the development of single-row versions. These have a finer pitch (less separation) for the corrugated fins to maintain an equivalent heat exchange capacity, since the heat radiated from the fins is considerably greater than that from the tubes in this type of radiator. This type of matrix represents a compromise between the cellular or film and the tube and flat fin constructions, in that it generally combines the more efficient heat transfer of the former with the greater structural strength of the latter. An increase in the size of the matrix is generally required for the cooling system of a turbocharged engine, owing to the greater amount of heat that has to be dissipated under maximum torque conditions.

For all types of radiator construction so far described, the sealing joints for the matrix and its header and collector tanks are solder bonded. Since a soldered joint has relatively little strength and both structural integrity and pressure tightness are prerequisites of radiator durability, the necessary strength must be provided by mechanical interlocking

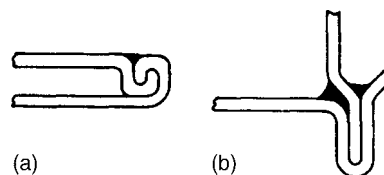


Figure 5.14 Mechanical interlocking of soldered radiator joints: (a) tubes (b) tank

of the parts with the solder connections being used just as a seal (Figure 5.14).

Aluminium and plastics radiators

The increasing use of aluminium alloy in automotive radiator construction, at least as original equipment, is by no means of recent origin. Bolted-on header and collector tanks in cast aluminium were early used in commercial vehicle radiators, and in 1960 the Harrison Radiator Division of General Motors introduced an all-aluminium radiator for a popular American sports car. The mid 1970s saw the introduction of an aluminium and plastics radiator developed by SOFICA in France and now adopted by several German and French car manufacturers. A feature of its construction is that a purely mechanical interlocking process, used in conjunction with synthetic rubber sealing gaskets, is employed for the aluminium alloy fin and tube joints in the matrix and also for joining the matrix and the nylon plastics header and collector tanks.

The advantages claimed for the aluminium and plastics radiator construction are generally those of a saving in weight, lower material costs with no painting requirement, resistance to corrosion and, if required, complex shapes. A possible disadvantage is that in the event of leakage it generally requires service replacement rather than repair.

For all-aluminium radiator construction the methods of joining used, apart from solder bonding, involve either forcible expansion of the tube ends where they enter the header and collector tank plates to provide a simple interference fit, or the application of modern adhesive bonding techniques at these junctures. Both methods lend themselves especially to automated production and the build consistency that it confers. However, the production techniques for copper/brass radiators have also been improved. These have allowed the greater

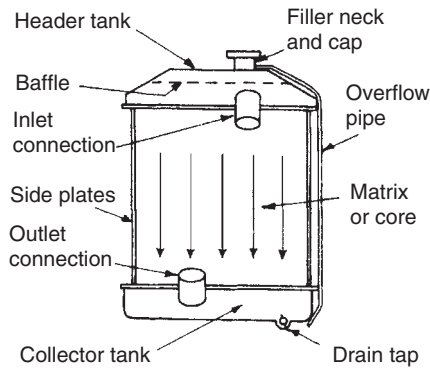


Figure 5.15 Down-flow radiator assembly

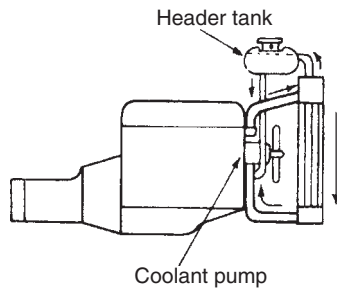


Figure 5.16 Down-flow radiator with separate header tank

strength of the brass tubing to be exploited, especially in reducing tube wall thickness for better heat transfer.

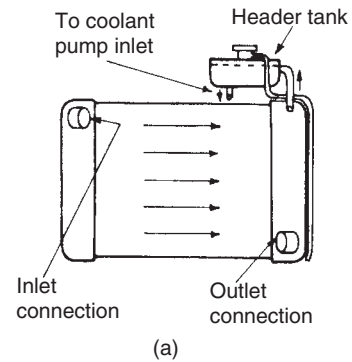
Down-flow and cross-flow radiators

The heated coolant entering the radiator may be dispersed through the matrix in either a down-flow or a cross-flow direction. The differences in radiator installation are described in the following sections.

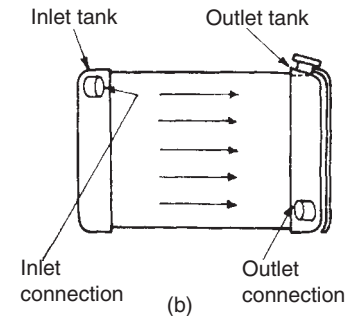
Down-flow radiator

In this long-established form of radiator construction the matrix is attached to an upper header tank and a lower collector tank (Figure 5.15). The header tank performs several functions because, apart from receiving the heated coolant from the engine, it accommodates expansion of the coolant and provides a reserve against any loss of coolant. As its name would suggest, the purpose of the collector tank is to return to the engine the coolant from which heat has been abstracted.

In modern practice, the header tank proper may be mounted separately from the radiator. An advantage of this is to ensure that the radiator is always maintained 'solid' with coolant (Figures 5.12, 5.16 and 5.27). This reduces any mixing of air with the coolant entering the radiator, since aeration of the coolant has an adverse effect on heat transfer efficiency. For the same reason, radiators with an integral header tank may incorporate a submerged perforated baffle that is horizontally positioned just above the inlet pipe orifice, so that the circulating coolant is effectively separated from the air space in the upper part of the header tank. An incidental advantage of using a separate header tank, where its installation is such that



(a)



(b)

Figure 5.17 Cross-flow radiators (a) with (b) without separate header tank

coolant is returned to the pump inlet, is that any tendency towards cavitation in the pump is reduced by virtue of the increase in pressure head at the pump inlet. *Cavitation* may occur in a pump where a restricted inlet passage promotes the formation of bubbles in the circulating coolant. Upon collapse these bubbles containing vapour and gas implode at extremely high pressures and cause erosion damage. A filler neck and cap, together with an overflow pipe connection, is provided on either the integral or the separate header tank.

Cross-flow radiator

In order best to utilize the available air flow through modern radiator grilles, the matrix itself needs to be low and wide. This requirement led to the development in the mid 1960s, notably by General Motors in America, of the cross-flow radiator, which is flanked on one side by an inlet tank and on the other by an outlet tank. To provide the cross-flow radiator with a reserve of coolant and a means of preventing aeration, a separate header tank is generally connected to the upper end of the outlet tank (Figure 5.17a). In some American applications, however, the separate header tank is dispensed with and the reserve of coolant is accommodated within the capacity of the outlet tank (Figure 5.17b). To prevent aeration, the filler cap and overflow arrangements are located at the upper end of the outlet tank, because it is in this region that any air entering the matrix tends to collect.

The pressurized cooling system

The advantages of pressurizing the cooling system may be summarized as follows:

- 1 It allows the coolant to circulate at a higher temperature without boiling, so that heat will be transferred more rapidly

from the radiator to the air by virtue of the greater temperature difference between them.

- 2 It compensates for the reduction in atmospheric pressure when motoring in high-altitude mountainous regions, where the boiling point of the coolant would otherwise be lower, with the consequent danger of overheating.
- 3 Since the heat transfer from the radiator is directly proportional to the temperature difference between the coolant and the air, a pressurized system allows the radiator to be reduced in size (although it must also be made stronger).
- 4 The increase in pressure at the coolant pump inlet reduces the possibility of cavitation damage, as earlier mentioned in connection with the separate header tank.
- 5 Because the system has to be sealed to pressurize it, instead of simply being vented to the atmosphere, coolant losses through evaporation and surging are minimized.

Construction and operation of the radiator pressure cap

The unsung hero of the modern vehicle cooling system must surely be John Karmazin of the Harrison Radiators Division of General Motors in America. His invention was the radiator pressure cap, which was first used in Buick cars in 1939, and has long since been taken for granted.

The radiator pressure cap is a combination of filler cap and pressure control valve. It is installed at the highest point in the cooling system and seals against a seating in the filler neck of the radiator. With the engine running and the cap in position, the cooling system is allowed to become pressurized. This occurs automatically because, as the temperature increases, the coolant is expanding in a closed system. The reason for using a radiator pressure cap is therefore to maintain the cooling system at a pressure above that of the atmosphere.

The cap consists essentially of a spring-loaded plate valve with a rubber facing, which is preloaded on to a fixed seating. According to whether an open or a closed type of construction is used, the fixed seating is provided in either the filler neck or the cap itself (Figure 5.18a and b). If the pressure in the system rises above the limit for which the control valve is rated, the spring loading on it is overcome so that the valve is lifted upwards off its seating and vents the system to atmosphere.

The pressure control valve is in turn fitted with a similar acting and concentric recuperation valve. This is arranged to open in the opposite direction against light spring loading. Air can therefore be admitted to the radiator to relieve any depression as the coolant temperature falls, otherwise at worst there could be a danger of the header tank collapsing.

The pressure control valve in the radiator cap is usually set to open at a specified value in the range of 28 to 105 kN/m² (4 to 15 lbf/in²). For each 7 kN/m² (1 lbf/in²) the system pressure is increased above atmospheric, the boiling point of the coolant is raised by about 1.5°C. The recuperation valve in the radiator cap is set to open whenever the system is subject to a depression higher than about 7 kN/m².

Caution must be observed in checking the coolant level of a hot engine. The radiator pressure cap should first be covered with a protective cloth and then part undone to release steam

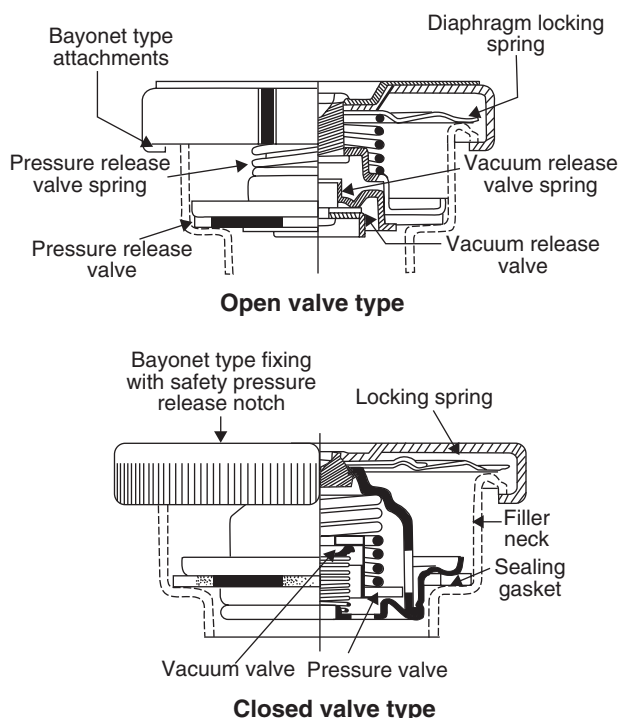


Figure 5.18 Types of radiator pressure cap and their actions: (a) open (b) closed construction (AC Delco)

and all pressure, before it is completely removed from the filler neck.

Sealed cooling systems

The attraction of the modern sealed cooling system is that it reduces both the need for frequent inspection of the coolant level and the risk of weakening the antifreeze solution by topping up. Although the advent of the radiator pressure cap provided what amounted to a semisealed cooling system, the modern so called fully sealed system constitutes in effect a coolant recovery system, and indeed is known as such by the American motor industry. It basically comprises a reserve tank that is independently mounted from the radiator and vented to the atmosphere (Figure 5.19). A flexible hose connects the overflow stub pipe of the radiator to a dip tube in the reserve tank, which is nominally one-third filled with a coolant or thereabouts.

The action of the system is such that when the coolant reaches a sufficiently high temperature and exerts enough pressure to open the control valve in the radiator cap, the excess coolant will flow into the reserve tank from which it merely displaces air. This is in contrast to the earlier non-sealed system where the coolant would simply drain to the ground. When the engine later cools and the reduced volume of coolant in the system loses its heat, the accompanying drop in pressure will cause the recuperation valve in the radiator cap to open. Since the coolant in the reserve tank is vented to the atmosphere and is now at a greater pressure than that existing in the cooling system, an appropriate amount will flow back to fill the radiator again.